HL7 FHIR R4 — Chapter 10: Diagnostic Medicine Module (sections 10 – 10.11.10)

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10 Diagnostic Medicine Module

URL: https://hl7.org/fhir/R4/diagnostics-module.html
Work Group Orders and Observations & Clinical Genomics | Standards Status: Informative

10.0 Diagnostic Medicine Module

10.0.1 Introduction

The Diagnostics Module provides an overview and guide to the FHIR content that addresses ordering and reporting of clinical diagnostics including laboratory testing, imaging and genomics.

10.0.2 Index

The Diagnostics module covers the following resources:

Observation DiagnosticReport ServiceRequest | Media ImagingStudy MolecularSequence | Specimen BodyStructure

The diagnostic resources and their relationships are shown below. The arrows represent the direction of the references between resources (for example, DiagnosticReport references ServiceRequest). See the Workflow Module for information about the coordination of activities such as ordering and fulfilling of diagnostics.

Implementation Note: See the Genomics Implementation Guidance for additional information about how to use the Diagnostic resources for Clinical Genomic Reporting and Analysis.

Name | Aliases | Description
Observation | Vital Signs, Measurement, Results | Measurements and simple assertions made about a patient, device or other subject.
DiagnosticReport | Report, Test, Result, Results, Labs | The findings and interpretation of diagnostic tests performed on patients, groups of patients, devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted representation of diagnostic reports.
ServiceRequest | diagnostic request, referral, referral request | A record of a request for service such as diagnostic investigations, treatments, or operations to be performed.
Media | | A photo, video, or audio recording acquired or used in healthcare. The actual content may be inline or provided by direct reference.
ImagingStudy | | Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data) acquired or produced in a common context. A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a study may have multiple series of different modalities.
MolecularSequence | | Raw data describing a biological sequence.
Specimen | | A sample to be used for analysis.
BodyStructure | anatomical location | Record details about an anatomical structure. This resource may be used when a coded concept does not provide the necessary detail needed for the use case.

10.0.3 Security and Privacy

The diagnostic resources often represent patient-specific data, and as such are susceptible to data breaching. Necessary privacy and security provisions must be in place when searching and fetching this information. For more general considerations, see the Security and Privacy module.

10.0.4 Common Use Cases

Diagnostic resources are commonly used to plan, recommend, order and report clinical diagnostics:

• Planning a diagnostic test such as in this Careplan-example

• Recommending diagnostics such as is done by Clinical Decision Support tools. See the Clinical Reasoning module for more information.

• Ordering a Laboratory Test such as a lipid panel

• Reporting a Laboratory Test such as a lipid panel

• Ordering an imaging study such as a CT study

• Reporting an imaging study such as a CT study

• Ordering a genetic test such as a gene mutation analysis

• Reporting a genetic test such as HLA genotyping

There are many ways to use these resources independently as well. The Observation resource, in particular, is central to capturing many measurements and events in healthcare and is often used outside the context of diagnostic orders and reports.

• recording vital signs such as temperature

• recording other observations of a patient's social history such as tobacco use

• recording other characteristics of a patient such as pregnancy status (note: The Condition resource may be used to represent these as well)

• recording clinical assessment tool scores such as an Apgar score.

• identifying and describing a specimen or body site

10.0.5 Related Resources and Modules

The resources that represent the basic information about a patient and a clinical encounter can be found in the Administration Module. Other resources that represent core clinical information generated by healthcare providers during the course of a patient encounter are detailed in the Clinical Summary Module and the Medications Module.

10.0.6 Developmental Roadmap

• The Observation and Diagnostic Report resources have been tested and used in production tooling and as such have reached a maturity level FMM=4 where changes become less likely and require implementer community approval. Future work will focus on developing core profiles and clarifying use cases and boundaries and providing more domain specific examples. We anticipate Observation and Diagnostic Report reaching normative status by the end of this cycle.

• The ImagingStudy resource has had some limited testing and use. Further use of this resource in production tooling is needed for it to reach a more stable FMM level.

• The ServiceRequest resources has undergone substantial changes since DSTU2. Foremost of which is the merging of DSTU2 DiagnosticOrder and ServiceRequest into a single resource. In addition, as a result of the addition of FHIR workflow for STU3, its structure and content have been updated to align with the general FHIR workflow "request" pattern. Because of these, it remains a relatively immature resource. No more sweeping changes to its structure are expected. However, its scope and boundaries vis a vis other resources will be the topic of future workgroup efforts. We anticipate its broad use by implementers and that it will mature as a resource throughout the upcoming cycle.

• The Specimen resource will be the focus of work group efforts for the upcoming cycle. A new SpecimenDefinition resource is planned for describing specimen types vs specimen instances and will take over some of the Specimen resource's current scope. We expect to see more widespread use of this resource and future updates to its FMM level

• The MolecularSequence resource remains draft and we expect to see more widespread use of this resource and future updates to its FMM level

• The BodyStructure resource is an immature resource with a FMM level of 1. To date there has been little implementation feedback.

• The Media resource is the newest member to the Diagnostics module having undergone substantial scope and structural changes to create a specialized observation resource focused on digital media such as images and audio files. Future work will focus on developing core profiles and providing more domain specific examples.

10.1 Resource Observation - Content

URL: https://hl7.org/fhir/R4/observation.html

See the section in the Observation resource operations page for more information and examples

Orders and Observations Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

| This page has been approved as part of an ANSI standard. See the Observation Package for further details.

Measurements and simple assertions made about a patient, device or other subject.

10.1.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow.

Observations are a central element in healthcare, used to support diagnosis, monitor progress, determine baselines and patterns and even capture demographic characteristics. Most observations are simple name/value pair assertions with some metadata, but some observations group other observations together logically, or even are multi-component observations. Note that the DiagnosticReport resource provides a clinical or workflow context for a set of observations and the Observation resource is referenced by DiagnosticReport to represent laboratory, imaging, and other clinical and diagnostic data to form a complete report.

Uses for the Observation resource include:

• Vital signs such as body weight, blood pressure, and temperature

• Laboratory Data like blood glucose, or an estimated GFR

• Imaging results like bone density or fetal measurements
• Clinical Findings* such as abdominal tenderness
• Device measurements such as EKG data or Pulse Oximetry data
• Clinical assessment tools such as APGAR or a Glasgow Coma Score
• Personal characteristics: such as eye-color
• Social history like tobacco use, family support, or cognitive status
• Core characteristics like pregnancy status, or a death assertion

*The boundaries between clinical findings and disorders remains a challenge in medical ontology. Refer the Boundaries section below and in Condition for general guidance. These boundaries can be clarified by profiling Observation for a particular use case.

10.1.1.1 Core Profiles for Observation Trial Use

The following core profiles for the Observation resource have been defined as well. If implementations use this Resource when expressing the profile-specific concepts as structured data, they SHALL conform to the following profiles:

Profile | Description
Vital signs | The FHIR Vital Signs profile sets minimum expectations for the Observation Resource to record, search and fetch the vital signs (e.g. temperature, blood pressure, respiration rate, etc.) associated with a patient

10.1.2 Boundaries and Relationships

At its core, Observation allows expressing a name-value pair or structured collection of name-value pairs. As such, it can support conveying any type of information desired. However, that is not its intent. Observation is intended for capturing measurements and subjective point-in-time assessments. It is not intended to be used for those specific contexts and use cases already covered by other FHIR resources. For example, the AllergyIntolerance resource represents a patient allergies, MedicationStatement resource: medications taken by a patient, FamilyMemberHistory resource: a patient's family history, Procedure resource: information about a procedure, and QuestionnaireResponse resource: a set of answers to a set of questions. The Observation resource should not be used to record clinical diagnosis about a patient or subject that are typically captured in the Condition resource or the ClinicalImpression resource. The Observation resource is often referenced by the Condition resource to provide specific subjective and objective data to support its assertions. There will however be situations of overlap. For example, a response to a question of "have you ever taken illicit drugs" could in principle be represented using MedicationStatement, but most systems would treat such an assertion as an Observation. In some cases, such as when source data is coming from an HL7 v2 feed, a system might not have information that allows it to distinguish diagnosis, allergy and other "specialized" types of observations from laboratory, vital sign and other observation types intended to be conveyed with this resource. In those circumstances, such specialized observations may also appear using this resource. Adhering to such convention is an appropriate use of Observation. If implementers are uncertain whether a proposed use of Observation is appropriate, they're encouraged to consult with implementers on chat.fhir.org implementer's stream

The Media resource captures a specific type of observation whose value is audio, video or image data. This resource is used instead of Observation to represent such forms of information as it exposes the metadata relevant for interpreting the information. See Media's boundaries section to see how Media (and Observation) differs from ImagingStudy and DocumentReference.

In contrast to the Observation resource, the DiagnosticReport resource typically includes additional clinical context and some mix of atomic results, images, imaging reports, textual and coded interpretation, and formatted representations. Laboratory reports, pathology reports, and imaging reports should be represented using the DiagnosticReport resource. The Observation resource is referenced by the DiagnosticReport to provide the atomic results for a particular investigation. "Laboratories routinely have a variable that is summative across a series of discrete variables - these are usually called 'impressions' or 'interpretations'. Sometimes they are algorithmically specified and sometimes they have the imprimatur of pathologists and they are conveyed in Observation or DiagnosticReport instead of the Clinical Impression resource. The Observation resource should not be used to record clinical diagnosis about a patient or subject as discussed above.

This resource is referenced by AdverseEvent, Appointment, CarePlan, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, Contract, DeviceRequest, DeviceUseStatement, DiagnosticReport, Encounter, FamilyMemberHistory, Goal, GuidanceResponse, ImagingStudy, Immunization, MedicationAdministration, MedicationRequest, MedicationStatement, MolecularSequence, itself, Procedure, QuestionnaireResponse, RequestGroup, RiskAssessment, ServiceRequest and SupplyRequest

10.1.3 Resource Content

• Structure
• UML

- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Observation | IN | | DomainResource | Measurements and simple assertions + Rule: dataAbsentReason SHALL only be present if Observation.value[x] is not present + Rule: If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for observation
basedOn | Σ | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | Σ | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!Σ | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
category | | 0..* | CodeableConcept | Classification of type of observation Observation Category Codes (Preferred)
code | Σ | 1..1 | CodeableConcept | Type of observation (code / type) LOINC Codes (Example)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣTU | 0..* | Reference(Any) | What the observation is about, when it is not about the subject of record
encounter | Σ | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | Σ | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | Σ | 0..1 | instant | Date/Time this version was made available
performer | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | 0..1 | | Actual result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing DataAbsentReason (Extensible)
interpretation | | 0..* | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)
note | | 0..* | Annotation | Comments about the observation
bodySite | | 0..1 | CodeableConcept | Observed body part SNOMED CT Body Structures (Example)
method | | 0..1 | CodeableConcept | How it was done Observation Methods (Example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation + Rule: Must have at least a low or a high or text
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Observation Reference Range Meaning Codes (Preferred)
appliesTo | | 0..* | CodeableConcept | Reference range population Observation Reference Range Applies To Codes (Example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
hasMember | Σ | 0..* | Reference(Observation | QuestionnaireResponse | MolecularSequence) | Related resource that belongs to the Observation group
derivedFrom | Σ | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence) | Related measurements the observation is made from
component | Σ | 0..* | BackboneElement | Component results
code | Σ | 1..1 | CodeableConcept | Type of component observation (code / type) LOINC Codes (Example)
value[x] | Σ | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData

valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing DataAbsentReason (Extensible)
interpretation | | 0..* | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)
referenceRange | | 0..* | see referenceRange | Provides guide for interpretation of component result
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Observation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for observation --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) Fulfills plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Immunization|ImagingStudy) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type) --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) Who and/or what the observation is about --></subject>
 <focus><!-- 0..* Reference(Any) What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!-- 0..1 Reference(Encounter) Healthcare event during which this observation is made --></encounter>
 <effective[x]><!-- 0..1 dateTime|Period|Timing|instant Clinically relevant time/time-period for observation
    --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this version was made available -->
 <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|RelatedPerson) Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|integer|Range|Ratio|
   SampledData|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing --></dataAbsentReason>
 <interpretation><!-- 0..* CodeableConcept High, low, normal, etc. --></interpretation>
 <note><!-- 0..* Annotation Comments about the observation --></note>
 <bodySite><!-- 0..1 CodeableConcept Observed body part --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device|DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  <!-- 0..* Provides guide for interpretation -->
  <low><!--  0..1 Quantity(SimpleQuantity) Low Range, if relevant --></low>
  <high><!--  0..1 Quantity(SimpleQuantity) High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population --></appliesTo>
  <age><!-- 0..1 Range Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!-- 0..* Reference(Observation|QuestionnaireResponse|
   MolecularSequence) Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!-- 0..* Reference(DocumentReference|ImagingStudy|Media|
   QuestionnaireResponse|Observation|MolecularSequence) Related measurements the observation is made from
    --></derivedFrom>
 <component>  <!-- 0..* Component results -->
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type) --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|boolean|integer|Range|
    Ratio|SampledData|time|dateTime|Period Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing --></dataAbsentReason>
  <interpretation><!-- 0..* CodeableConcept High, low, normal, etc. --></interpretation>
  <referenceRange><!-- 0..* Content as for Observation.referenceRange Provides guide for interpretation of component
    result --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{
  "resourceType" : "Observation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for observation
  "basedOn" : [{ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) }], // Fulfills plan, proposal or order
  "partOf" : [{ Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Immunization|ImagingStudy) }], // Part of referenced event
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "category" : [{ CodeableConcept }], // Classification of  type of observation
  "code" : { CodeableConcept }, // R!  Type of observation (code / type)
  "subject" : { Reference(Patient|Group|Device|Location) }, // Who and/or what the observation is about
  "focus" : [{ Reference(Any) }], // What the observation is about, when it is not about the subject of record
  "encounter" : { Reference(Encounter) }, // Healthcare event during which this observation is made
  // effective[x]: Clinically relevant time/time-period for observation. One of these 4:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "effectiveTiming" : { Timing },
  "effectiveInstant" : "<instant>",
  "issued" : "<instant>", // Date/Time this version was made available
  "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|RelatedPerson) }], // Who is responsible for the observation
  // value[x]: Actual result. One of these 11:
```

```
  "valueQuantity" : { Quantity },
  "valueCodeableConcept" : { CodeableConcept },
  "valueString" : "<string>",
  "valueBoolean" : <boolean>,
  "valueInteger" : <integer>,
  "valueRange" : { Range },
  "valueRatio" : { Ratio },
  "valueSampledData" : { SampledData },
  "valueTime" : "<time>",
  "valueDateTime" : "<dateTime>",
  "valuePeriod" : { Period },
  "dataAbsentReason" : { CodeableConcept }, // C? Why the result is missing
  "interpretation" : [{ CodeableConcept }], // High, low, normal, etc.
  "note" : [{ Annotation }], // Comments about the observation
  "bodySite" : { CodeableConcept }, // Observed body part
  "method" : { CodeableConcept }, // How it was done
  "specimen" : { Reference(Specimen) }, // Specimen used for this observation
  "device" : { Reference(Device|DeviceMetric) }, // (Measurement) Device
  "referenceRange" : [{ // Provides guide for interpretation
    "low" : { Quantity(SimpleQuantity) }, // C? Low Range, if relevant
    "high" : { Quantity(SimpleQuantity) }, // C? High Range, if relevant
    "type" : { CodeableConcept }, // Reference range qualifier
    "appliesTo" : [{ CodeableConcept }], // Reference range population
    "age" : { Range }, // Applicable age range, if relevant
    "text" : "<string>" // Text based reference range in an observation
  }],
  "hasMember" : [{ Reference(Observation|QuestionnaireResponse|
   MolecularSequence) }], // Related resource that belongs to the Observation group
  "derivedFrom" : [{ Reference(DocumentReference|ImagingStudy|Media|
   QuestionnaireResponse|Observation|MolecularSequence) }], // Related measurements the observation is made from
  "component" : [{ // Component results
    "code" : { CodeableConcept }, // R!  Type of component observation (code / type)
    // value[x]: Actual component result. One of these 11:
    "valueQuantity" : { Quantity },
    "valueCodeableConcept" : { CodeableConcept },
    "valueString" : "<string>",
    "valueBoolean" : <boolean>,
    "valueInteger" : <integer>,
    "valueRange" : { Range },
    "valueRatio" : { Ratio },
    "valueSampledData" : { SampledData },
    "valueTime" : "<time>",
    "valueDateTime" : "<dateTime>",
    "valuePeriod" : { Period },
    "dataAbsentReason" : { CodeableConcept }, // C? Why the component result is missing
    "interpretation" : [{ CodeableConcept }], // High, low, normal, etc.
    "referenceRange" : [{ Content as for Observation.referenceRange }] // Provides guide for interpretation of component
     result
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Observation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Observation.identifier [ Identifier ], ... ; # 0..* Business Identifier for observation
  fhir:Observation.basedOn [ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|MedicationRequest|
  NutritionOrder|ServiceRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:Observation.partOf [ Reference(MedicationAdministration|MedicationDispense|MedicationStatement|Procedure|
  Immunization|ImagingStudy) ], ... ; # 0..* Part of referenced event
  fhir:Observation.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:Observation.category [ CodeableConcept ], ... ; # 0..* Classification of  type of observation
  fhir:Observation.code [ CodeableConcept ]; # 1..1 Type of observation (code / type)
  fhir:Observation.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 Who and/or what the observation is about
  fhir:Observation.focus [ Reference(Any) ], ... ; # 0..* What the observation is about, when it is not about the
    subject of record
  fhir:Observation.encounter [ Reference(Encounter) ]; # 0..1 Healthcare event during which this observation is made
  # Observation.effective[x] : 0..1 Clinically relevant time/time-period for observation. One of these 4
    fhir:Observation.effectiveDateTime [ dateTime ]
    fhir:Observation.effectivePeriod [ Period ]
    fhir:Observation.effectiveTiming [ Timing ]
    fhir:Observation.effectiveInstant [ instant ]
  fhir:Observation.issued [ instant ]; # 0..1 Date/Time this version was made available
  fhir:Observation.performer [ Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|RelatedPerson) ],
    ... ; # 0..* Who is responsible for the observation
  # Observation.value[x] : 0..1 Actual result. One of these 11
    fhir:Observation.valueQuantity [ Quantity ]
    fhir:Observation.valueCodeableConcept [ CodeableConcept ]
    fhir:Observation.valueString [ string ]
    fhir:Observation.valueBoolean [ boolean ]
    fhir:Observation.valueInteger [ integer ]
    fhir:Observation.valueRange [ Range ]
    fhir:Observation.valueRatio [ Ratio ]
    fhir:Observation.valueSampledData [ SampledData ]
```

```
    fhir:Observation.valueTime [ time ]
    fhir:Observation.valueDateTime [ dateTime ]
    fhir:Observation.valuePeriod [ Period ]
  fhir:Observation.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the result is missing
  fhir:Observation.interpretation [ CodeableConcept ], ... ; # 0..* High, low, normal, etc.
  fhir:Observation.note [ Annotation ], ... ; # 0..* Comments about the observation
  fhir:Observation.bodySite [ CodeableConcept ]; # 0..1 Observed body part
  fhir:Observation.method [ CodeableConcept ]; # 0..1 How it was done
  fhir:Observation.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for this observation
  fhir:Observation.device [ Reference(Device|DeviceMetric) ]; # 0..1 (Measurement) Device
  fhir:Observation.referenceRange [ # 0..* Provides guide for interpretation
    fhir:Observation.referenceRange.low [ Quantity(SimpleQuantity) ]; # 0..1 Low Range, if relevant
    fhir:Observation.referenceRange.high [ Quantity(SimpleQuantity) ]; # 0..1 High Range, if relevant
    fhir:Observation.referenceRange.type [ CodeableConcept ]; # 0..1 Reference range qualifier
    fhir:Observation.referenceRange.appliesTo [ CodeableConcept ], ... ; # 0..* Reference range population
    fhir:Observation.referenceRange.age [ Range ]; # 0..1 Applicable age range, if relevant
    fhir:Observation.referenceRange.text [ string ]; # 0..1 Text based reference range in an observation
  ], ...;
  fhir:Observation.hasMember [ Reference(Observation|QuestionnaireResponse|MolecularSequence) ], ... ; # 0..* Related
    resource that belongs to the Observation group
  fhir:Observation.derivedFrom [ Reference(DocumentReference|ImagingStudy|Media|QuestionnaireResponse|Observation|
  MolecularSequence) ], ... ; # 0..* Related measurements the observation is made from
  fhir:Observation.component [ # 0..* Component results
    fhir:Observation.component.code [ CodeableConcept ]; # 1..1 Type of component observation (code / type)
    # Observation.component.value[x] : 0..1 Actual component result. One of these 11
      fhir:Observation.component.valueQuantity [ Quantity ]
      fhir:Observation.component.valueCodeableConcept [ CodeableConcept ]
      fhir:Observation.component.valueString [ string ]
      fhir:Observation.component.valueBoolean [ boolean ]
      fhir:Observation.component.valueInteger [ integer ]
      fhir:Observation.component.valueRange [ Range ]
      fhir:Observation.component.valueRatio [ Ratio ]
      fhir:Observation.component.valueSampledData [ SampledData ]
      fhir:Observation.component.valueTime [ time ]
      fhir:Observation.component.valueDateTime [ dateTime ]
      fhir:Observation.component.valuePeriod [ Period ]
    fhir:Observation.component.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the component result is missing
    fhir:Observation.component.interpretation [ CodeableConcept ], ... ; # 0..* High, low, normal, etc.
    fhir:Observation.component.referenceRange [ See Observation.referenceRange ], ... ; # 0..* Provides guide for
    interpretation of component result
  ], ...;
]
```

Changes since R3

Observation

Observation.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

Observation.partOf | Added Element

Observation.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1

Observation.focus | Added Element

Observation.encounter | Renamed from context to encounter Type Reference: Removed Target Type EpisodeOfCare

Observation.effective[x] | Add Types Timing, instant

Observation.performer | Type Reference: Added Target Types PractitionerRole, CareTeam

Observation.value[x] | Add Type integer Remove Type Attachment

Observation.dataAbsentReason | Change value set from http://hl7.org/fhir/ValueSet/observation-valueabsentreason to http://hl7.org/fhir/ValueSet/data-absent-reason Change code system for extensibly bound codes from "http://hl7.org/fhir/data-absent-reason" to "http://terminology.hl7.org/CodeSystem/data-absent-reason"

Observation.interpretation | Max Cardinality changed from 1 to * Change code system for extensibly bound codes from "http://terminology.hl7.org/CodeSystem/v2-0078" to "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"

Observation.note | Renamed from comment to note Max Cardinality changed from 1 to * Type changed from string to Annotation

Observation.referenceRange.type | Remove Binding http://hl7.org/fhir/ValueSet/referencerange-meaning (extensible)

Observation.hasMember | Added Element

Observation.derivedFrom | Added Element

Observation.component.value[x] | Add Types boolean, integer Remove Type Attachment

Observation.component.dataAbsentReason | Change value set from http://hl7.org/fhir/ValueSet/observation-valueabsentreason to http://hl7.org/fhir/ValueSet/data-absent-reason Change code system for extensibly bound codes from "http://hl7.org/fhir/data-absent-reason" to "http://terminology.hl7.org/CodeSystem/data-absent-reason"

Observation.component.interpretation | Max Cardinality changed from 1 to * Change code system for extensibly bound codes from "http://terminology.hl7.org/CodeSystem/v2-0078" to "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"

Observation.related | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 48 tests that all execute ok. All tests pass round-trip testing and 23 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Observation | IN | | DomainResource | Measurements and simple assertions + Rule: dataAbsentReason SHALL only be present if Observation.value[x] is not present + Rule: If Observation.code is the same as an Observation.component.code then the value

element associated with the code SHALL NOT be present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business Identifier for observation
basedOn | Σ | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | Σ | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!Σ | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
category | | 0..* | CodeableConcept | Classification of type of observation Observation Category Codes (Preferred)
code | Σ | 1..1 | CodeableConcept | Type of observation (code / type) LOINC Codes (Example)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣTU | 0..* | Reference(Any) | What the observation is about, when it is not about the subject of record
encounter | Σ | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | Σ | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | Σ | 0..1 | instant | Date/Time this version was made available
performer | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | 0..1 | | Actual result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing DataAbsentReason (Extensible)
interpretation | | 0..* | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)
note | | 0..* | Annotation | Comments about the observation
bodySite | | 0..1 | CodeableConcept | Observed body part SNOMED CT Body Structures (Example)
method | | 0..1 | CodeableConcept | How it was done Observation Methods (Example)
specimen | | 0..1 | Reference(Specimen) | Specimen used for this observation
device | | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation + Rule: Must have at least a low or a high or text
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | | 0..1 | CodeableConcept | Reference range qualifier Observation Reference Range Meaning Codes (Preferred)
appliesTo | | 0..* | CodeableConcept | Reference range population Observation Reference Range Applies To Codes (Example)
age | | 0..1 | Range | Applicable age range, if relevant
text | | 0..1 | string | Text based reference range in an observation
hasMember | Σ | 0..* | Reference(Observation | QuestionnaireResponse | MolecularSequence) | Related resource that belongs to the Observation group
derivedFrom | Σ | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence) | Related measurements the observation is made from
component | Σ | 0..* | BackboneElement | Component results
code | Σ | 1..1 | CodeableConcept | Type of component observation (code / type) LOINC Codes (Example)
value[x] | Σ | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing DataAbsentReason (Extensible)
interpretation | | 0..* | CodeableConcept | High, low, normal, etc. Observation Interpretation Codes (Extensible)
referenceRange | | 0..* | see referenceRange | Provides guide for interpretation of component result
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Observation xmlns="http://hl7.org/fhir">
```

```
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for observation --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) Fulfills plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Immunization|ImagingStudy) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <category><!-- 0..* CodeableConcept Classification of  type of observation --></category>
 <code><!-- 1..1 CodeableConcept Type of observation (code / type) --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) Who and/or what the observation is about --></subject>
 <focus><!-- 0..* Reference(Any) What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!-- 0..1 Reference(Encounter) Healthcare event during which this observation is made --></encounter>
 <effective[x]><!-- 0..1 dateTime|Period|Timing|instant Clinically relevant time/time-period for observation
    --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this version was made available -->
 <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|RelatedPerson) Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|integer|Range|Ratio|
   SampledData|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing --></dataAbsentReason>
 <interpretation><!-- 0..* CodeableConcept High, low, normal, etc. --></interpretation>
 <note><!-- 0..* Annotation Comments about the observation --></note>
 <bodySite><!-- 0..1 CodeableConcept Observed body part --></bodySite>
 <method><!-- 0..1 CodeableConcept How it was done --></method>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!-- 0..1 Reference(Device|DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  <!-- 0..* Provides guide for interpretation -->
  <low><!--  0..1 Quantity(SimpleQuantity) Low Range, if relevant --></low>
  <high><!--  0..1 Quantity(SimpleQuantity) High Range, if relevant --></high>
  <type><!-- 0..1 CodeableConcept Reference range qualifier --></type>
  <appliesTo><!-- 0..* CodeableConcept Reference range population --></appliesTo>
  <age><!-- 0..1 Range Applicable age range, if relevant --></age>
  <text value="[string]"/><!-- 0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!-- 0..* Reference(Observation|QuestionnaireResponse|
   MolecularSequence) Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!-- 0..* Reference(DocumentReference|ImagingStudy|Media|
   QuestionnaireResponse|Observation|MolecularSequence) Related measurements the observation is made from
    --></derivedFrom>
 <component>  <!-- 0..* Component results -->
  <code><!-- 1..1 CodeableConcept Type of component observation (code / type) --></code>
  <value[x]><!-- 0..1 Quantity|CodeableConcept|string|boolean|integer|Range|
    Ratio|SampledData|time|dateTime|Period Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing --></dataAbsentReason>
  <interpretation><!-- 0..* CodeableConcept High, low, normal, etc. --></interpretation>
  <referenceRange><!-- 0..* Content as for Observation.referenceRange Provides guide for interpretation of component
    result --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{
  "resourceType" : "Observation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for observation
  "basedOn" : [{ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) }], // Fulfills plan, proposal or order
  "partOf" : [{ Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Immunization|ImagingStudy) }], // Part of referenced event
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "category" : [{ CodeableConcept }], // Classification of  type of observation
  "code" : { CodeableConcept }, // R!  Type of observation (code / type)
  "subject" : { Reference(Patient|Group|Device|Location) }, // Who and/or what the observation is about
  "focus" : [{ Reference(Any) }], // What the observation is about, when it is not about the subject of record
  "encounter" : { Reference(Encounter) }, // Healthcare event during which this observation is made
  // effective[x]: Clinically relevant time/time-period for observation. One of these 4:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "effectiveTiming" : { Timing },
  "effectiveInstant" : "<instant>",
  "issued" : "<instant>", // Date/Time this version was made available
  "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|RelatedPerson) }], // Who is responsible for the observation
  // value[x]: Actual result. One of these 11:
  "valueQuantity" : { Quantity },
  "valueCodeableConcept" : { CodeableConcept },
  "valueString" : "<string>",
  "valueBoolean" : <boolean>,
  "valueInteger" : <integer>,
  "valueRange" : { Range },
  "valueRatio" : { Ratio },
  "valueSampledData" : { SampledData },
  "valueTime" : "<time>",
  "valueDateTime" : "<dateTime>",
  "valuePeriod" : { Period },
  "dataAbsentReason" : { CodeableConcept }, // C? Why the result is missing
  "interpretation" : [{ CodeableConcept }], // High, low, normal, etc.
```

```
  "note" : [{ Annotation }], // Comments about the observation
  "bodySite" : { CodeableConcept }, // Observed body part
  "method" : { CodeableConcept }, // How it was done
  "specimen" : { Reference(Specimen) }, // Specimen used for this observation
  "device" : { Reference(Device|DeviceMetric) }, // (Measurement) Device
  "referenceRange" : [{ // Provides guide for interpretation
    "low" : { Quantity(SimpleQuantity) }, // C? Low Range, if relevant
    "high" : { Quantity(SimpleQuantity) }, // C? High Range, if relevant
    "type" : { CodeableConcept }, // Reference range qualifier
    "appliesTo" : [{ CodeableConcept }], // Reference range population
    "age" : { Range }, // Applicable age range, if relevant
    "text" : "<string>" // Text based reference range in an observation
  }],
  "hasMember" : [{ Reference(Observation|QuestionnaireResponse|
   MolecularSequence) }], // Related resource that belongs to the Observation group
  "derivedFrom" : [{ Reference(DocumentReference|ImagingStudy|Media|
   QuestionnaireResponse|Observation|MolecularSequence) }], // Related measurements the observation is made from
  "component" : [{ // Component results
    "code" : { CodeableConcept }, // R!  Type of component observation (code / type)
    // value[x]: Actual component result. One of these 11:
    "valueQuantity" : { Quantity },
    "valueCodeableConcept" : { CodeableConcept },
    "valueString" : "<string>",
    "valueBoolean" : <boolean>,
    "valueInteger" : <integer>,
    "valueRange" : { Range },
    "valueRatio" : { Ratio },
    "valueSampledData" : { SampledData },
    "valueTime" : "<time>",
    "valueDateTime" : "<dateTime>",
    "valuePeriod" : { Period },
    "dataAbsentReason" : { CodeableConcept }, // C? Why the component result is missing
    "interpretation" : [{ CodeableConcept }], // High, low, normal, etc.
    "referenceRange" : [{ Content as for Observation.referenceRange }] // Provides guide for interpretation of component
     result
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Observation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Observation.identifier [ Identifier ], ... ; # 0..* Business Identifier for observation
  fhir:Observation.basedOn [ Reference(CarePlan|DeviceRequest|ImmunizationRecommendation|MedicationRequest|
  NutritionOrder|ServiceRequest) ], ... ; # 0..* Fulfills plan, proposal or order
  fhir:Observation.partOf [ Reference(MedicationAdministration|MedicationDispense|MedicationStatement|Procedure|
  Immunization|ImagingStudy) ], ... ; # 0..* Part of referenced event
  fhir:Observation.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:Observation.category [ CodeableConcept ], ... ; # 0..* Classification of  type of observation
  fhir:Observation.code [ CodeableConcept ]; # 1..1 Type of observation (code / type)
  fhir:Observation.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 Who and/or what the observation is about
  fhir:Observation.focus [ Reference(Any) ], ... ; # 0..* What the observation is about, when it is not about the
    subject of record
  fhir:Observation.encounter [ Reference(Encounter) ]; # 0..1 Healthcare event during which this observation is made
  # Observation.effective[x] : 0..1 Clinically relevant time/time-period for observation. One of these 4
    fhir:Observation.effectiveDateTime [ dateTime ]
    fhir:Observation.effectivePeriod [ Period ]
    fhir:Observation.effectiveTiming [ Timing ]
    fhir:Observation.effectiveInstant [ instant ]
  fhir:Observation.issued [ instant ]; # 0..1 Date/Time this version was made available
  fhir:Observation.performer [ Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|RelatedPerson) ],
    ... ; # 0..* Who is responsible for the observation
  # Observation.value[x] : 0..1 Actual result. One of these 11
    fhir:Observation.valueQuantity [ Quantity ]
    fhir:Observation.valueCodeableConcept [ CodeableConcept ]
    fhir:Observation.valueString [ string ]
    fhir:Observation.valueBoolean [ boolean ]
    fhir:Observation.valueInteger [ integer ]
    fhir:Observation.valueRange [ Range ]
    fhir:Observation.valueRatio [ Ratio ]
    fhir:Observation.valueSampledData [ SampledData ]
    fhir:Observation.valueTime [ time ]
    fhir:Observation.valueDateTime [ dateTime ]
    fhir:Observation.valuePeriod [ Period ]
  fhir:Observation.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the result is missing
  fhir:Observation.interpretation [ CodeableConcept ], ... ; # 0..* High, low, normal, etc.
  fhir:Observation.note [ Annotation ], ... ; # 0..* Comments about the observation
  fhir:Observation.bodySite [ CodeableConcept ]; # 0..1 Observed body part
  fhir:Observation.method [ CodeableConcept ]; # 0..1 How it was done
  fhir:Observation.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for this observation
  fhir:Observation.device [ Reference(Device|DeviceMetric) ]; # 0..1 (Measurement) Device
  fhir:Observation.referenceRange [ # 0..* Provides guide for interpretation
    fhir:Observation.referenceRange.low [ Quantity(SimpleQuantity) ]; # 0..1 Low Range, if relevant
    fhir:Observation.referenceRange.high [ Quantity(SimpleQuantity) ]; # 0..1 High Range, if relevant
```

```
    fhir:Observation.referenceRange.type [ CodeableConcept ]; # 0..1 Reference range qualifier
    fhir:Observation.referenceRange.appliesTo [ CodeableConcept ], ... ; # 0..* Reference range population
    fhir:Observation.referenceRange.age [ Range ]; # 0..1 Applicable age range, if relevant
    fhir:Observation.referenceRange.text [ string ]; # 0..1 Text based reference range in an observation
  ], ...;
  fhir:Observation.hasMember [ Reference(Observation|QuestionnaireResponse|MolecularSequence) ], ... ; # 0..* Related
    resource that belongs to the Observation group
  fhir:Observation.derivedFrom [ Reference(DocumentReference|ImagingStudy|Media|QuestionnaireResponse|Observation|
  MolecularSequence) ], ... ; # 0..* Related measurements the observation is made from
  fhir:Observation.component [ # 0..* Component results
    fhir:Observation.component.code [ CodeableConcept ]; # 1..1 Type of component observation (code / type)
    # Observation.component.value[x] : 0..1 Actual component result. One of these 11
      fhir:Observation.component.valueQuantity [ Quantity ]
      fhir:Observation.component.valueCodeableConcept [ CodeableConcept ]
      fhir:Observation.component.valueString [ string ]
      fhir:Observation.component.valueBoolean [ boolean ]
      fhir:Observation.component.valueInteger [ integer ]
      fhir:Observation.component.valueRange [ Range ]
      fhir:Observation.component.valueRatio [ Ratio ]
      fhir:Observation.component.valueSampledData [ SampledData ]
      fhir:Observation.component.valueTime [ time ]
      fhir:Observation.component.valueDateTime [ dateTime ]
      fhir:Observation.component.valuePeriod [ Period ]
    fhir:Observation.component.dataAbsentReason [ CodeableConcept ]; # 0..1 Why the component result is missing
    fhir:Observation.component.interpretation [ CodeableConcept ], ... ; # 0..* High, low, normal, etc.
    fhir:Observation.component.referenceRange [ See Observation.referenceRange ], ... ; # 0..* Provides guide for
    interpretation of component result
  ], ...;
]
```

Changes since Release 3

Observation

Observation.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

Observation.partOf | Added Element

Observation.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1

Observation.focus | Added Element

Observation.encounter | Renamed from context to encounter Type Reference: Removed Target Type EpisodeOfCare

Observation.effective[x] | Add Types Timing, instant

Observation.performer | Type Reference: Added Target Types PractitionerRole, CareTeam

Observation.value[x] | Add Type integer Remove Type Attachment

Observation.dataAbsentReason | Change value set from http://hl7.org/fhir/ValueSet/observation-valueabsentreason to http://hl7.org/fhir/ValueSet/data-absent-reason Change code system for extensibly bound codes from "http://hl7.org/fhir/data-absent-reason" to "http://terminology.hl7.org/CodeSystem/data-absent-reason"

Observation.interpretation | Max Cardinality changed from 1 to * Change code system for extensibly bound codes from "http://terminology.hl7.org/CodeSystem/v2-0078" to "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"

Observation.note | Renamed from comment to note Max Cardinality changed from 1 to * Type changed from string to Annotation

Observation.referenceRange.type | Remove Binding http://hl7.org/fhir/ValueSet/referencerange-meaning (extensible)

Observation.hasMember | Added Element

Observation.derivedFrom | Added Element

Observation.component.value[x] | Add Types boolean, integer Remove Type Attachment

Observation.component.dataAbsentReason | Change value set from http://hl7.org/fhir/ValueSet/observation-valueabsentreason to http://hl7.org/fhir/ValueSet/data-absent-reason Change code system for extensibly bound codes from "http://hl7.org/fhir/data-absent-reason" to "http://terminology.hl7.org/CodeSystem/data-absent-reason"

Observation.component.interpretation | Max Cardinality changed from 1 to * Change code system for extensibly bound codes from "http://terminology.hl7.org/CodeSystem/v2-0078" to "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"

Observation.related | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 48 tests that all execute ok. All tests pass round-trip testing and 23 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.1.3.1 Terminology Bindings

Path | Definition | Type | Reference

Observation.status | Codes providing the status of an observation. | Required | ObservationStatus

Observation.category | Codes for high level observation categories. | Preferred | ObservationCategoryCodes

Observation.code Observation.component.code | Codes identifying names of simple observations. | Example | LOINCCodes

Observation.dataAbsentReason Observation.component.dataAbsentReason | Codes specifying why the result (`Observation.value[x]`) is missing. | Extensible | DataAbsentReason

Observation.interpretation Observation.component.interpretation | Codes identifying interpretations of observations. | Extensible | ObservationInterpretationCodes

Observation.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures

Observation.method | Methods for simple observations. | Example | ObservationMethods

Observation.referenceRange.type | Code for the meaning of a reference range. | Preferred | ObservationReferenceRangeMeaningCodes
Observation.referenceRange.appliesTo | Codes identifying the population the reference range applies to. | Example | ObservationReferenceRangeAppliesToCodes

10.1.3.2 Constraints

id | Level | Location | Description | Expression
obs-3 | Rule | Observation.referenceRange | Must have at least a low or a high or text | low.exists() or high.exists() or text.exists()
obs-6 | Rule | (base) | dataAbsentReason SHALL only be present if Observation.value[x] is not present | dataAbsentReason.empty() or value.empty()
obs-7 | Rule | (base) | If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present | value.empty() or component.code.where(coding.intersect(%resource.code.coding).exists()).empty()

10.1.4 Notes:

10.1.4.1 Profiling Observation

At its simplest, a resource instance can consist of only a code and a value, and status flag. The relevance of other properties will vary based on the type of observation. Profiles are created to provide guidance on capturing certain types of observations for a given use case. The Observation resource focuses on the level of detail captured by most systems. However, for a given use case there may be additional constraints and additional information relevant in certain circumstances. As with other resources, extensions can be used to introduce this additional complexity.

10.1.4.2 Subject of an Observation

Typically, an observation is made about the subject - a patient, or group of patients, location, or device - and the distinction between the subject and what is directly measured for an observation is specified in the observation code itself (e.g., "Blood Glucose") and does not need to be represented separately. However, three attributes may be used for representing the focus of the observation if it is not the subject itself. The specimen and bodySite elements are used to represent measurements taken on subject samples or anatomic and morphological locations, and focus represents specific aspect of the subject that are the point of attention such as another observation or a device implanted in a patient.

10.1.4.3 Observation Grouping

Many observations have important relationships to other observations and need to be grouped together. These structures have been defined to do this: DiagnosticReport and DiagnosticReport.result, Observation and the elements: Observation.component, Observation.hasMember and Observation.derivedFrom. The sections below provides guidance around which structure to use. Because the idea of what to group together is often highly contextual and based upon the end user's point of view, the choice of which structure to use will be driven by jurisdiction, organizational practice and context. Profiling will normally be necessary for implementation.

10.1.4.3.1 DiagnosticReport.result

DiagnosticReport relates directly to an order (ServiceRequest). The DiagnosticReport.code names the panel and serves as the grouping element, which is traditionally referred to as a "panel" or "battery" by laboratories. The DiagnosticReport.result element references the individual observations. Several examples demonstrate observation grouping using DiagnosticReport as the grouping structure.

10.1.4.3.2 Observation.component

Observation.component is used for any supporting result that cannot reasonably be interpreted and used outside the scope of the Observation it is a component of. Component observations may make up the separate and individual parts of the observation or may provide qualifying information to Observation.code and may only be able to be understood in relation to the Observation.code (for example, see the $stats operation). Therefore all code-value and component.code-component.value pairs need to be taken into account to correctly understand the meaning of the observation. Components should only be used when there is only one method, one observation, one performer, one device, and one time. Some use cases for using this structure include:

• Observations that are commonly produced and interpreted together. For example, systolic and diastolic blood pressure are represented as a single Blood pressure panel.

• Assessment tool results that are commonly produced and interpreted together. For example, a newborn Apgar score that is a single Observation with five components.

• Representing multiple answers to a question (relationship and boundaries between Observation and Questionnaire/QuestionnaireResponse). For example, reporting the types of alcohol consumed by a patient

On the other hand, any observations that are clinically relevant outside the context of being a component of another observation should be represented by separate Observation resources. For example a Body Mass Index (BMI) Observation should not contain components for height and weight because they are clinically

relevant observations on their own and should be represented by separate Observation resources. See the section below on how to relate independent Observations.

10.1.4.3.3 Observation.hasMember of and Observation.derivedFrom

Observation.hasMember and Observation.derivedFrom and the core extensions: Observation-sequelTo and Observation-replaces are used for any supporting result that can be interpreted and used on its own and has one or more different values for method, observation, performer, device, time, and/or error conditions. Two common use cases for using this structure are:

• For grouping related observations such as for a "panel" or "battery". In this case the Observation.code represents the "panel" code, typically Observation.value[x] is not present, and the set of member Observations are listed in Observation.hasMember. This structure permits nested grouping when used with DiagnosticReport (e.g. complex micro isolate and sensitivities report).

• When linking to other Observations from which an Observation is derived. In this case both Observation.code and Observation.value[x] are present, and the linked observations are listed in Observation.derivedFrom. An example of this would be a Body Mass Index (BMI) Observation where the height and weight measurements are referenced.

10.1.4.4 Using codes in Observation

When a result value is a represented as a predefined concept using a code, valueCodeableConcept is used. This element is bound to a value set comprised of a standard nomenclature such as SNOMED CT or a source system ("local") coded result values.

10.1.4.4.1 Multiple Codings

Results may be coded in multiple value sets based on different code systems and these may be mapped using the ConceptMap resource and/or given as additional codings directly in the element as shown in the example below.

For example the LOINC 43304-5 Chlamydia trachomatis rRNA [Presence] in Unspecified specimen by Probe and target amplification method is typically associated with coded presence/absence concepts. Using the coded value for 'negative' with a standard code translation, valueCodeableConcept would be:

```
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "260385009",
        "display": "Negative"
      }, {
        "system": "https://acme.lab/resultcodes",
        "code": "NEG",
        "display": "Negative"
      }
    ],
    "text": "Negative for Chlamydia Trachomatis rRNA"
  }
```

10.1.4.4.2 Text values for coded results:

When the data element is usually coded or the type associated with the code element defines a coded value, use valueCodeableConcept even if there is no appropriate code and only free text is available. For example using text only, the valueCodeableConcept element would be:

```
  "valueCodeableConcept": {
    "text": "uncoded free text result"
  }
```

When a coded answer list includes a concept code for "other" and there is a free text description of the concept, the valueCodeableConcept.text element should be used to capture the full meaning of the source. In the example below, the answer code "Other" is provided in the valueCodeableConcept element and the text value supplied value in the CodeableConcept.text element.

```
{
  "resourceType": "Observation",
  ... snip ...
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "74076-1",
        "display": "Medication or substance involved"
      }
    ]
  },
  .. snip ...
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": " LA20343-2",
        "display": "Other substance: PLEASE SPECIFY"
```

```
      }
    ],
    "text": "Other: Blue pills I found under my couch"
  }
  .. snip ...
}
```

10.1.4.4.3 Interoperability Issues using code value pairs in FHIR

A recurring issue for many observation events, regardless of the particular pattern, is determining how to populate observation.code and observation.value. While this is typically straight-forward for laboratory observations, it can get blurry for other types of observations, such as findings and disorders, family history observations, etc. This discussion focuses on the way in which the coded representation of such statements is expressed using the Observation.code and Observation.value elements.

There are two distinct facets that are central to a FHIR Observations:

• The action taken to make the finding and/or the property about which the property was observed. For example: measurement of blood hemoglobin.

• The result of the observation. For example: 14 g/dl.

Several different ways of representing the same information exist using different combinations of the Observation.code and Observation.value. Unconstrained use of the alternatives presents a major challenge for computation of semantic equivalence and for safe interpretation of observations originating from different applications and users. The following four patterns could reasonably represent the same case. Considering that the Observation resource needs to support many use cases, the appropriate place to define the specific pattern is expected to be done through profiles and implementation guides as specified by the jurisdictions and/or organizations implementing FHIR:

• Observation.code represents the nature of the observation and the Observation.value a code represents the non-numeric result value. These are two distinct facets that are central to a FHIR Observations. For example: code=[Examination] value=[Abdomen tender]

• Observation.code is nearly identical to 1) above, but the level of granularity is shifted from the value to code. For example: code=[Abdominal examination] value=[Tenderness]

• The Observation.code is also expressed in a way that does not specify the observation action but indicates a statement about findings reduced to a single name (or term), as in the above item. In this example, the Observation.value is present and "qualifies" the finding typically confirming or refuting it. For example: code=[Abdominal tenderness] value=[found/true]

• in this example the Observation.code is expressed in a way that does not specify the observation action but indicates a statement about findings reduced to a single name (or term). In this particular example in that context, the Observation.value is omitted. For example: code=[Abdominal tenderness] value element is omitted

10.1.4.4.4 Guidance:

• Recommended rules for case 1 and 2 patterns: The Observation.code is preferably a LOINC concept code. If a SNOMED CT concept code is used, the expression SHOULD represent a 363787002 (Observable entity(Observable entity)) or 386053000 (Evaluation procedure(evaluation procedure)) For non-numeric values, the Observation.value is preferably a SNOMED CT concept code.

• Recommended rules for case 3 pattern: The Observation.code is preferably a LOINC or SNOMED CT concept code. If a SNOMED CT concept code is used, the expression SHOULD represent a 404684003 (Clinical finding (finding)) , 413350009 (Finding with explicit context(finding)), or 272379006 (Event(event)). The Observation.value is represented by either valueBoolean valueCodeableConcept preferably using: SNOMED CT where concept is-a 362981000 (Qualifier value (qualifier value)) v2 Yes/no Indicator v2 Expanded Yes/no Indicator (unfortunately is missing 'not given')

• Recommended rules for case 4 pattern: The Observation.code is preferably a SNOMED CT concept code where the concept is-a 404684003 (Clinical finding (finding)) , 413350009 (Finding with explicit context(finding)), or 272379006 (Event(event)). The Observation.value is omitted. The default interpretation is the concept (single code or expression) represented in Observation.code is present in the patient. An Observation.dataAbsentReason value of 'clinical-finding' SHOULD be used to indicate why the expected value is missing.

• SHOULD NOT use the Assertion pattern as described in HL7 Version 3 Implementation Guide: TermInfo - Using SNOMED CT in CDA R2 Models, Release 1 . (The code is 'ASSERTION' and the value is a SNOMED CT concept or expression)

10.1.4.5 Refining the interpretation of an Observation using additional codes or Observations

The following list provides guidance on using codes or other observations to provide additional context that may alter how an observation is interpreted.:

If possible, use the most specific code you can

e.g.:

```
{
"resourceType": "Observation",
... snip ...
"code": {
  "coding": [
    {
      "system": "http://loinc.org",
      "code": "6689-4",
      "display": "Glucose [Mass/volume] in Blood --2 hours post meal"
    }
  ]
},
... snip ...
}
```

Alternatively, use additional codes in Observation.code as described above.

e.g.: Observation.code = coding-1: 59408-5 Oxygen saturation in Arterial blood by Pulse oximetry, coding-2: 20564-1 Oxygen saturation in Blood

```
{
"resourceType": "Observation",
... snip ...
"code": {
  "coding": [
    {
      "system": "http://loinc.org",
      "code": "59408-5",
      "display": "Oxygen saturation in Arterial blood by Pulse oximetry"
    },
    {
      "system": "http://loinc.org",
      "code": "20564-1",
      "display": "Oxygen saturation in Blood"
    }
  ]
},
... snip ...
}
```

As described above, observations are typically grouped together to provide additional information needed for correctly understanding and interpreting the observation. As an alternative to grouping observations, extensions may be used to provide references to other observations needed for understanding and interpreting an observation.

Note:We are seeking input from the implementer community in evaluating existing Observation Extensions for this purpose

Feedback here .

10.1.4.6 Value[x] and Datatypes

• The element, Observation.value[x], has a variable name depending on the type as follows: valueQuantity valueCodeableConcept valueString valueBoolean valueInteger valueRange valueRatio valueSampledData valueTime valueDateTime valuePeriod

• See above section on Using codes for result values

• The Boolean data type is rarely used for value[x] because most observations result values are never truly Boolean due to exceptional values such as "unknown", therefore they should use the CodeableConcept data type instead and select codes from http://terminology.hl7.org/ValueSet/v2-0136 (these "yes/no" concepts can be mapped to the display name "true/false" or other mutually exclusive terms that may be needed")

valueQuantity

```
  "valueQuantity": {
    "value": 2.0,
    "comparator": "<",
    "unit": "mmol/l",
    "system": "http://unitsofmeasure.org",
    "code": "mmol/L"
  }
```

valueCodeableConcept

dataAbsentReason

```
  "dataAbsentReason": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
        "code": "NaN",
        "display": "Not a Number"
      }
    ]
  }
```

• Because there are multiple types allowed for the value element, multiple value search parameters are defined. There is no standard parameter for searching values of type Ratio

10.1.4.7 Physiologically Relevant Time of the Observation

The effectiveDateTime or effectivePeriod is the time that the observation is most relevant as an observation of the subject. For a biological subject (e.g. a human patient), this is the physiologically relevant time of the observation. In the case of an observation using a specimen, this represents the start and end of the specimen collection (e.g. 24-hour Urine Sodium), but if the collection time is sufficiently short, this is reported as a point in time value (e.g. normal venipuncture). In the case of an observation obtained directly from a subject (e.g. BP, Chest X-ray), this is the start and end time of the observation process, which again, is often reported as a single point in time.

10.1.4.8 Reference Range

Most common observations will only have one generic reference range. Reference ranges may be useful for laboratory tests and other measures like systolic blood pressure but will have little relevance for something like "pregnancy status". Systems MAY choose to restrict to only supplying the relevant reference range based on knowledge about the patient (e.g. specific to the patient's age, gender, weight and other factors), but this might not be possible or appropriate. Whenever more than one reference range is supplied, the differences between them SHOULD be provided in the reference range and/or age properties.

10.1.4.9 Canceled or Aborted Observations

If a measurement or test could not be completed (for example if the specimen is unsatisfactory or the provider cancelled the order), then the status value should be updated to "cancelled" and the specific details given - preferably as coded values in the dataAbsentReason or valueCodeableConcept element. Additional information may be provided in the `note` element as well. The specimen reject example demonstrates this using a coded value for unsatisfactory specimen in dataAbsentReason.

10.1.4.10 Genetic Observations

Genetic reporting makes heavy use of the DiagnosticReport and Observation resources. An implementation guide describing how to represent genetic results can be found here .

10.1.5 Operations defined for Observation

10.1.5.1 Searching for the Last N Observations

The lastn query operation meets the common need for searching for the most recent or "last known" Observations for a subject. Examples where this query could be used:

• Fetch the last 5 temperatures for a patient to view trends

• Get the most recent laboratory results for patient

• Fetch the last 3 results for all vitals for a patient

See the Last N Observations Query section in the Observation resource operations page for more information and examples

10.1.5.2 Retrieving Statistics for Laboratory Observations

The stats operation performs a set of statistical calculations on a set of clinical measurements such as a blood pressure as stored on the server. This operation is focused on Observation resources with valueQuantity elements that have UCUM unit codes. Examples where this operation could be used:

• Get the average, min, max and count of a series of BP measurements for a patient

• Determine 20th or 80th percentile on a set of measurements over a time period

Observation Statistics

10.1.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on TU | reference | Reference to the service request. | Observation.basedOn (CarePlan, MedicationRequest, NutritionOrder, DeviceRequest, ServiceRequest, ImmunizationRecommendation)
category TU | token | The classification of the type of observation | Observation.category
code TU | token | The code of the observation type | Observation.code | 13 Resources
code-value-concept TU | composite | Code and coded value parameter pair | On Observation: code: code value-concept: value.as(CodeableConcept)
code-value-date TU | composite | Code and date/time value parameter pair | On Observation: code: code value-date: value.as(DateTime) | value.as(Period)
code-value-quantity TU | composite | Code and quantity value parameter pair | On Observation: code: code value-quantity: value.as(Quantity)
code-value-string TU | composite | Code and string value parameter pair | On Observation: code: code value-string: value.as(string)

combo-code TU | token | The code of the observation type or component type | Observation.code | Observation.component.code
combo-code-value-concept TU | composite | Code and coded value parameter pair, including in components | On Observation | Observation.component: combo-code: code combo-value-concept: value.as(CodeableConcept)
combo-code-value-quantity TU | composite | Code and quantity value parameter pair, including in components | On Observation | Observation.component: combo-code: code combo-value-quantity: value.as(Quantity)
combo-data-absent-reason TU | token | The reason why the expected value in the element Observation.value[x] or Observation.component.value[x] is missing. | Observation.dataAbsentReason | Observation.component.dataAbsentReason
combo-value-concept TU | token | The value or component value of the observation, if the value is a CodeableConcept | (Observation.value as CodeableConcept) | (Observation.component.value as CodeableConcept)
combo-value-quantity TU | quantity | The value or component value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | (Observation.value as Quantity) | (Observation.value as SampledData) | (Observation.component.value as Quantity) | (Observation.component.value as SampledData)
component-code TU | token | The component code of the observation type | Observation.component.code
component-code-value-concept TU | composite | Component code and component coded value parameter pair | On Observation.component: component-code: code component-value-concept: value.as(CodeableConcept)
component-code-value-quantity TU | composite | Component code and component quantity value parameter pair | On Observation.component: component-code: code component-value-quantity: value.as(Quantity)
component-data-absent-reason TU | token | The reason why the expected value in the element Observation.component.value[x] is missing. | Observation.component.dataAbsentReason
component-value-concept TU | token | The value of the component observation, if the value is a CodeableConcept | (Observation.component.value as CodeableConcept)
component-value-quantity TU | quantity | The value of the component observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | (Observation.component.value as Quantity) | (Observation.component.value as SampledData)
data-absent-reason TU | token | The reason why the expected value in the element Observation.value[x] is missing. | Observation.dataAbsentReason
date TU | date | Obtained date/time. If the obtained element is a period, a date that falls in the period | Observation.effective | 17 Resources
derived-from TU | reference | Related measurements the observation is made from | Observation.derivedFrom (Media, Observation, ImagingStudy, MolecularSequence, QuestionnaireResponse, DocumentReference)
device TU | reference | The Device that generated the observation data. | Observation.device (Device, DeviceMetric)
encounter TU | reference | Encounter related to the observation | Observation.encounter (Encounter) | 12 Resources
focus TU | reference | The focus of an observation when the focus is not the patient of record. | Observation.focus (Any)
has-member TU | reference | Related resource that belongs to the Observation group | Observation.hasMember (Observation, MolecularSequence, QuestionnaireResponse)
identifier TU | token | The unique id for a particular observation | Observation.identifier | 30 Resources
method TU | token | The method used for the observation | Observation.method
part-of TU | reference | Part of referenced event | Observation.partOf (Immunization, MedicationDispense, MedicationAdministration, Procedure, ImagingStudy, MedicationStatement)
patient TU | reference | The subject that the observation is about (if patient) | Observation.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer TU | reference | Who performed the observation | Observation.performer (Practitioner, Organization, CareTeam, Patient, PractitionerRole, RelatedPerson)
specimen TU | reference | Specimen used for this observation | Observation.specimen (Specimen)
status TU | token | The status of the observation | Observation.status
subject TU | reference | The subject that the observation is about | Observation.subject (Group, Device, Patient, Location)
value-concept TU | token | The value of the observation, if the value is a CodeableConcept | (Observation.value as CodeableConcept)
value-date TU | date | The value of the observation, if the value is a date or period of time | (Observation.value as dateTime) | (Observation.value as Period)
value-quantity TU | quantity | The value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | (Observation.value as Quantity) | (Observation.value as SampledData)
value-string TU | string | The value of the observation, if the value is a string, and also searches in CodeableConcept.text | (Observation.value as string) | (Observation.value as CodeableConcept).text

10.1.7 Resource Observation - Examples

URL: https://hl7.org/fhir/R4/observation-examples.html

10.1.7 Resource Observation - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

10.1.7.1 Laboratory Results

The following Observations show examples of laboratory observations:

- Real-world patient - glucose
- Real-world patient - erythrocyte
- Real-world patient - hemoglobin
- Real-world patient -base excess
- Real-world patient - CO2
- Real-world patient - creatinine

• Real-world patient - estimated gfr
• Real-world patient - staphylococcus
• Specimen Reject Example
• An "ask at order" response using the datetime datetype for the result value

Simple blood typing example:

• A simple blood typing panel.
• An example of a blood group (ABO) measurement.
• An example of a blood Rh antigen measurement.

10.1.7.2 Exam Findings and Simple Assessments

The following Observations shows examples of physical exams finding and simple assessments:

• An exam finding of abdominal tenderness .Note that this example uses the pattern 3 described in the "Using Codes in Observation" Section.

• Observation to capture whether the mother is trained to change her child's tracheostomy tube. Note that in this example, the child is the patient of record and the mother is the focus.

10.1.7.3 Imaging Studies

The following Observations show examples of imaging results:

• Example of an encoded BMD measurements of left femur

10.1.7.4 Genomics

The following Observations show examples of genomics observations:

• Genetics example 1
• Genetics example 2
• Genetics example 3
• Genetics example 4
• Genetics example 5
• Example of diplotype data (with haplotypes observation as reference)
• Example of haplotype data that is basis of a diplotype data
• Example of another haplotype data that is basis of a diplotype data
• Example of phenotype data
• Example of a TPMT diplotype that link to two TPMT haplotype observations
• Example of a TPMT haplotype observation
• Example of another TPMT haplotype observation

10.1.7.5 Observations where subject is a Location

The following Observations show examples of observations about a location:

• Example of an Observations made about a location - an oyster farm

10.1.7.6 Observations where subject is a Group

The following Observations show examples of observations about a group:

• An example of a herd screening ELISA assay for Mycoplasmal Pneumonia in Pigs

10.1.7.7 Physical Traits

The following Observations show examples of physical trait observation:

• Example of a very simple personal characteristic observation

10.1.7.8 Assessment Tools Scores

The following Observations show examples of assessment tool scores:

• Clinical assessment tool example - GCS with individual score as components
• Clinical assessment tool example - GCS with individual scores as related QuestionnaireAnswer
• Example of a 1 minute Apgar score with individual score as components
• Example of a 2 minute Apgar score with individual score as components
• Example of a 5 minute Apgar score with individual score as components
• Example of a 10 minute Apgar score with individual score as components
• Example of a 20 minute Apgar score with individual score as components

10.1.7.9 Device Generated Data

The following Observations show examples of observations generated by a device:

• EKG Example using SampledData data type

• Oxygen Saturation by Pulse Oximetry Example (DeviceMetricObservation profile)

10.1.7.10 PHD Generated Data

The following Observations show examples of observations generated by a personal healthcare device:

• Spirometry Example referencing DeviceComponent

10.1.7.11 Social History Observations

The following Observations show examples of social history observations:

• An example of a finding of second hand smoke exposure

• Type of alcohol consumed

• Clinical gender observation. For a discussion of representing patient gender see the Patient resource.

10.1.7.12 Signs and Symptoms

The following Observations show examples of findings of signs and symptoms:

• An example of a finding of vomiting

10.1.7.13 Vital Signs

The following Observations show examples of vital signs measurements:

• Real-world patient - temperature

• Simple Weight Example

• Simple Respiratory Rate Example

• Simple Heart Rate Example

• Simple Body Temperature Example

• Simple Body Height Example

• Simple Body Length Example This example demonstrates the use of a code translation to allow the observation to be interpreted as Body Length (typically used for infants).

• Simple Head Circumference Example

• Simple BMI Example

• BMI with links to Height and Weight

• Oxygen Saturation by Pulse Oximetry Example (DeviceMetricObservation profile)

• Blood Pressure Measurement

• Blood Pressure Measurement with an absent Diastolic measurement

• Simple Mean Arterial Blood Pressure Example

• Simple Panel of Heart Rate, Repiratory Rate. Blood Pressure and Body Temperature

• Blood Pressure Measurement cancelled

• An example to demonstrate the bodyPosition and delta extensions

Example Name | id | Format
Simple Weight Example | example | XML | JSON | Turtle
Simple Respiratory Rate Example | respiratory-rate | XML | JSON | Turtle
Simple Heart Rate Example | heart-rate | XML | JSON | Turtle
Simple Body Temperature Example | body-temperature | XML | JSON | Turtle
Simple Body Height Example | body-height | XML | JSON | Turtle
Simple Body Length Example | body-length | XML | JSON | Turtle
Simple Head Circumference Example | head-circumference | XML | JSON | Turtle
Simple BMI Example | bmi | XML | JSON | Turtle
BMI with links to Height and Weight | bmi-using-related | XML | JSON | Turtle
Oxygen Saturation by Pulse Oximetry Example (DeviceMetricObservation profile) | satO2 | XML | JSON | Turtle
Blood Pressure Measurement | blood-pressure | XML | JSON | Turtle
Blood Pressure Measurement with an absent Diastolic measurement | blood-pressure-dar | XML | JSON | Turtle
Simple Mean Arterial Blood Pressure Example | mbp | XML | JSON | Turtle
Simple Panel of Heart Rate, Repiratory Rate. Blood Pressure and Body Temperature | vitals-panel | XML | JSON | Turtle
Blood Pressure Measurement cancelled | blood-pressure-cancel | XML | JSON | Turtle
Real-world patient - glucose | f001 | XML | JSON | Turtle
Specimen Reject Example | unsat | XML | JSON | Turtle
Real-world patient -base excess | f002 | XML | JSON | Turtle
Real-world patient - CO2 | f003 | XML | JSON | Turtle
Real-world patient - erythrocyte | f004 | XML | JSON | Turtle
Real-world patient - hemoglobin | f005 | XML | JSON | Turtle
An "ask at order" response using the datetime datetype for the result value | date-lastmp | XML | JSON | Turtle
Real-world patient - temperature | f202 | XML | JSON | Turtle

Real-world patient - CO2 | f203 | XML | JSON | Turtle
Real-world patient - creatinine | f204 | XML | JSON | Turtle
Real-world patient - estimated gfr | f205 | XML | JSON | Turtle
Real-world patient - staphylococcus | f206 | XML | JSON | Turtle
EKG Example using SampledData data type | ekg | XML | JSON | Turtle
Clinical assessment tool example - GCS with individual score as components | glasgow | XML | JSON | Turtle
Clinical assessment tool example - GCS with individual scores as related QuestionnaireAnswer | gcs-qa | XML | JSON | Turtle
Example of a 1 minute Apgar score with individual score as components | 1minute-apgar-score | XML | JSON | Turtle
Example of a 2 minute Apgar score with individual score as components | 2minute-apgar-score | XML | JSON | Turtle
Example of a 5 minute Apgar score with individual score as components | 5minute-apgar-score | XML | JSON | Turtle
Example of a 10 minute Apgar score with individual score as components | 10minute-apgar-score | XML | JSON | Turtle
Example of a 20 minute Apgar score with individual score as components | 20minute-apgar-score | XML | JSON | Turtle
Example of a clinical gender observation. For a discussion on representing a patient's gender the Patient Resource. | clinical-gender | XML | JSON | Turtle
Example of a very simple personal characteristic observation | eye-color | XML | JSON | Turtle
Example of an encoded BMD measurements of left femur | bmd | XML | JSON | Turtle
Spirometry Example referencing DeviceComponent | 656 | XML | JSON | Turtle
Example of a multi-select values/answers Observation with individual answers/values as components | alcohol-type | XML | JSON | Turtle
Example of an Observations made about a location - an oyster farm | vp-oyster | XML | JSON | Turtle
An example of a herd screening ELISA assay for Mycoplasmal Pneumonia in Pigs. | herd1 | XML | JSON | Turtle
An example of a finding of vomiting | vomiting | XML | JSON | Turtle
An example of a finding of second hand smoke exposure | secondsmoke | XML | JSON | Turtle
An example of capturing whether the mother is trained to change her child's tracheostomy tube. In this example, the child is the patient of record and the mother is the focus. | trachcare | XML | JSON | Turtle
An example of a simple blood typing panel. | bgpanel | XML | JSON | Turtle
An example of a blood group (ABO) measurement. | bloodgroup | XML | JSON | Turtle
An example of a blood Rh antigen measurement. | rhstatus | XML | JSON | Turtle
Explores the edge cases of the decimal data type | decimal | XML | JSON | Turtle
An example to demonstrate using the bodyPosition and delta extensions | map-sitting | XML | JSON | Turtle
An exam finding using pattern 3 described in the "Using Codes in Observation" Section | abdo-tender | XML | JSON | Turtle
Genetics example 1 | example-genetics-1 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 2 | example-genetics-2 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 3 | example-genetics-3 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 4 | example-genetics-4 | XML | JSON | Turtle | for Profile Observation-genetics
Genetics example 5 | example-genetics-5 | XML | JSON | Turtle | for Profile Observation-genetics
Example of diplotype data (with haplotypes observation as reference) | example-diplotype1 | XML | JSON | Turtle | for Profile Observation-genetics
Example of haplotype data that is basis of a diplotype data | example-haplotype1 | XML | JSON | Turtle | for Profile Observation-genetics
Example of another haplotype data that is basis of a diplotype data | example-haplotype2 | XML | JSON | Turtle | for Profile Observation-genetics
Example of phenotype data | example-phenotype | XML | JSON | Turtle | for Profile Observation-genetics
Example of a TPMT diplotype that link to two TPMT haplotype observations | example-TPMT-diplotype | XML | JSON | Turtle | for Profile Observation-genetics
Example of a TPMT haplotype observation | example-TPMT-haplotype-one | XML | JSON | Turtle | for Profile Observation-genetics
Example of another TPMT haplotype observation | example-TPMT-haplotype-two | XML | JSON | Turtle | for Profile Observation-genetics
Example of breast cancer genetic observation | example-genetics-brcapat | XML | JSON | Turtle | for Profile Observation-genetics

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.1.8 Resource Observation - Detailed Descriptions

URL: https://hl7.org/fhir/R4/observation-definitions.html

10.1.8 Resource Observation - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Observation resource.

Observation
Element Id | Observation
Definition | Measurements and simple assertions made about a patient, device or other subject.
Cardinality | 0..*
Type | DomainResource
Requirements | Observations are a key aspect of healthcare. This resource is used to capture those that do not require more sophisticated mechanisms.
Alternate Names | Vital Signs; Measurement; Results; Tests
Comments | Used for simple observations such as device measurements, laboratory atomic results, vital signs, height, weight, smoking status, comments, etc. Other resources are used to provide context for observations such as laboratory reports, etc.
Invariants | Defined on this element obs-6 Rule dataAbsentReason SHALL only be present if Observation.value[x] is not present dataAbsentReason.empty() or value.empty() obs-7 Rule If Observation.code is the same as an Observation.component.code then

the value element associated with the code SHALL NOT be present value.empty() or component.code.where(coding.intersect(%resource.code.coding).exists()).empty()
Defined on this element
obs-6 | Rule | dataAbsentReason SHALL only be present if Observation.value[x] is not present | dataAbsentReason.empty() or value.empty()
obs-7 | Rule | If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present | value.empty() or component.code.where(coding.intersect(%resource.code.coding).exists()).empty()
Observation.identifier
Element Id | Observation.identifier
Definition | A unique identifier assigned to this observation.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows observations to be distinguished and referenced.
Summary | true
Observation.basedOn
Element Id | Observation.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this event. For example, a MedicationRequest may require a patient to have laboratory test performed before it is dispensed.
Cardinality | 0..*
Type | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)
Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.
Alternate Names | Fulfills
Summary | true
Observation.partOf
Element Id | Observation.partOf
Definition | A larger event of which this particular Observation is a component or step. For example, an observation as part of a procedure.
Cardinality | 0..*
Type | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)
Alternate Names | Container
Summary | true
Comments | To link an Observation to an Encounter use encounter. See the Notes below for guidance on referencing another Observation.
Observation.status
Element Id | Observation.status
Definition | The status of the result value.
Cardinality | 1..1
Terminology Binding | ObservationStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | Need to track the status of individual results. Some results are finalized before the whole report is finalized.
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Observation.category
Element Id | Observation.category
Definition | A code that classifies the general type of observation being made.
Cardinality | 0..*
Terminology Binding | Observation Category Codes (Preferred)
Type | CodeableConcept
Requirements | Used for filtering what observations are retrieved and displayed.
Comments | In addition to the required category valueset, this element allows various categorization schemes based on the owner’s definition of the category and effectively multiple categories can be used at once. The level of granularity is defined by the category concepts in the value set.
Observation.code
Element Id | Observation.code
Definition | Describes what was observed. Sometimes this is called the observation "name".
Cardinality | 1..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Requirements | Knowing what kind of observation is being made is essential to understanding the observation.
Alternate Names | Name
Summary | true
Comments | All code-value and, if present, component.code-component.value pairs need to be taken into account to correctly understand the meaning of the observation.
Observation.subject
Element Id | Observation.subject
Definition | The patient, or group of patients, location, or device this observation is about and into whose record the observation is placed. If the actual focus of the observation is different from the subject (or a sample of, part, or region of the subject), the focus element or the code itself specifies the actual focus of the observation.

Cardinality | 0..1
Type | Reference(Patient | Group | Device | Location)
Requirements | Observations have no value if you don't know who or what they're about.
Summary | true
Comments | One would expect this element to be a cardinality of 1..1. The only circumstance in which the subject can be missing is when the observation is made by a device that does not know the patient. In this case, the observation SHALL be matched to a patient through some context/channel matching technique, and at this point, the observation should be updated.
Observation.focus
Element Id | Observation.focus
Definition | The actual focus of an observation when it is not the patient of record representing something or someone associated with the patient such as a spouse, parent, fetus, or donor. For example, fetus observations in a mother's record. The focus of an observation could also be an existing condition, an intervention, the subject's diet, another observation of the subject, or a body structure such as tumor or implanted device. An example use case would be using the Observation resource to capture whether the mother is trained to change her child's tracheostomy tube. In this example, the child is the patient of record and the mother is the focus.
Cardinality | 0..*
Type | Reference(Any)
Summary | true
Comments | Typically, an observation is made about the subject - a patient, or group of patients, location, or device - and the distinction between the subject and what is directly measured for an observation is specified in the observation code itself (e.g., "Blood Glucose") and does not need to be represented separately using this element. Use specimen if a reference to a specimen is required. If a code is required instead of a resource use either bodysite for bodysites or the standard extension focusCode.
Observation.encounter
Element Id | Observation.encounter
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) during which this observation is made.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | For some observations it may be important to know the link between an observation and a particular encounter.
Alternate Names | Context
Summary | true
Comments | This will typically be the encounter the event occurred within, but some events may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter (e.g. pre-admission laboratory tests).
Observation.effective[x]
Element Id | Observation.effective[x]
Definition | The time or time-period the observed value is asserted as being true. For biological subjects - e.g. human patients - this is usually called the "physiologically relevant time". This is usually either the time of the procedure or of specimen collection, but very often the source of the date/time is not known, only the date/time itself.
Cardinality | 0..1
Type | dateTime|Period|Timing|instant
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Knowing when an observation was deemed true is important to its relevance as well as determining trends.
Alternate Names | Occurrence
Summary | true
Comments | At least a date should be present unless this observation is a historical report. For recording imprecise or "fuzzy" times (For example, a blood glucose measurement taken "after breakfast") use the Timing datatype which allow the measurement to be tied to regular life events.
Observation.issued
Element Id | Observation.issued
Definition | The date and time this version of the observation was made available to providers, typically after the results have been reviewed and verified.
Cardinality | 0..1
Type | instant
Summary | true
Comments | For Observations that don’t require review and verification, it may be the same as the lastUpdated time of the resource itself. For Observations that do require review and verification for certain updates, it might not be the same as the lastUpdated time of the resource itself due to a non-clinically significant update that doesn’t require the new version to be reviewed and verified again.
Observation.performer
Element Id | Observation.performer
Definition | Who was responsible for asserting the observed value as "true".
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson)
Requirements | May give a degree of confidence in the observation and also indicates where follow-up questions should be directed.
Summary | true
Observation.value[x]
Element Id | Observation.value[x]
Definition | The information determined as a result of making the observation, if the information has a simple value.
Cardinality | 0..1
Type | Quantity|CodeableConcept|string|boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | An observation exists to have a value, though it might not if it is in error, or if it represents a group of observations.
Summary | true

Comments | An observation may have; 1) a single value here, 2) both a value and a set of related or component values, or 3) only a set of related or component values. If a value is present, the datatype for this element should be determined by Observation.code. A CodeableConcept with just a text would be used instead of a string if the field was usually coded, or if the type associated with the Observation.code defines a coded value. For additional guidance, see the Notes section below.
Invariants | Affect this element obs-7 Rule If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present value.empty() or component.code.where(coding.intersect(%resource.code.coding).exists()).empty()
Affect this element
obs-7 | Rule | If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present | value.empty() or component.code.where(coding.intersect(%resource.code.coding).exists()).empty()
Observation.dataAbsentReason
Element Id | Observation.dataAbsentReason
Definition | Provides a reason why the expected value in the element Observation.value[x] is missing.
Cardinality | 0..1
Terminology Binding | DataAbsentReason (Extensible)
Type | CodeableConcept
Requirements | For many results it is necessary to handle exceptional values in measurements.
Comments | Null or exceptional values can be represented two ways in FHIR Observations. One way is to simply include them in the value set and represent the exceptions in the value. For example, measurement values for a serology test could be "detected", "not detected", "inconclusive", or "specimen unsatisfactory". The alternate way is to use the value element for actual observations and use the explicit dataAbsentReason element to record exceptional values. For example, the dataAbsentReason code "error" could be used when the measurement was not completed. Note that an observation may only be reported if there are values to report. For example differential cell counts values may be reported only when > 0. Because of these options, use-case agreements are required to interpret general observations for null or exceptional values.
Invariants | Affect this element obs-6 Rule dataAbsentReason SHALL only be present if Observation.value[x] is not present dataAbsentReason.empty() or value.empty()
Affect this element
obs-6 | Rule | dataAbsentReason SHALL only be present if Observation.value[x] is not present | dataAbsentReason.empty() or value.empty()
Observation.interpretation
Element Id | Observation.interpretation
Definition | A categorical assessment of an observation value. For example, high, low, normal.
Cardinality | 0..*
Terminology Binding | Observation Interpretation Codes (Extensible)
Type | CodeableConcept
Requirements | For some results, particularly numeric results, an interpretation is necessary to fully understand the significance of a result.
Alternate Names | Abnormal Flag
Comments | Historically used for laboratory results (known as 'abnormal flag'), its use extends to other use cases where coded interpretations are relevant. Often reported as one or more simple compact codes this element is often placed adjacent to the result value in reports and flow sheets to signal the meaning/normalcy status of the result.
Observation.note
Element Id | Observation.note
Definition | Comments about the observation or the results.
Cardinality | 0..*
Type | Annotation
Requirements | Need to be able to provide free text additional information.
Comments | May include general statements about the observation, or statements about significant, unexpected or unreliable results values, or information about its source when relevant to its interpretation.
Observation.bodySite
Element Id | Observation.bodySite
Definition | Indicates the site on the subject's body where the observation was made (i.e. the target site).
Cardinality | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Comments | Only used if not implicit in code found in Observation.code. In many systems, this may be represented as a related observation instead of an inline component. If the use case requires BodySite to be handled as a separate resource (e.g. to identify and track separately) then use the standard extension bodySite.
Observation.method
Element Id | Observation.method
Definition | Indicates the mechanism used to perform the observation.
Cardinality | 0..1
Terminology Binding | Observation Methods (Example)
Type | CodeableConcept
Requirements | In some cases, method can impact results and is thus used for determining whether results can be compared or determining significance of results.
Comments | Only used if not implicit in code for Observation.code.
Observation.specimen
Element Id | Observation.specimen
Definition | The specimen that was used when this observation was made.
Cardinality | 0..1
Type | Reference(Specimen)
Comments | Should only be used if not implicit in code found in Observation.code. Observations are not made on specimens themselves; they are made on a subject, but in many cases by the means of a specimen. Note that although specimens are often

involved, they are not always tracked and reported explicitly. Also note that observation resources may be used in contexts that track the specimen explicitly (e.g. Diagnostic Report).
Observation.device
Element Id | Observation.device
Definition | The device used to generate the observation data.
Cardinality | 0..1
Type | Reference(Device | DeviceMetric)
Comments | Note that this is not meant to represent a device involved in the transmission of the result, e.g., a gateway. Such devices may be documented using the Provenance resource where relevant.
Observation.referenceRange
Element Id | Observation.referenceRange
Definition | Guidance on how to interpret the value by comparison to a normal or recommended range. Multiple reference ranges are interpreted as an "OR". In other words, to represent two distinct target populations, two referenceRange elements would be used.
Cardinality | 0..*
Requirements | Knowing what values are considered "normal" can help evaluate the significance of a particular result. Need to be able to provide multiple reference ranges for different contexts.
Comments | Most observations only have one generic reference range. Systems MAY choose to restrict to only supplying the relevant reference range based on knowledge about the patient (e.g., specific to the patient's age, gender, weight and other factors), but this might not be possible or appropriate. Whenever more than one reference range is supplied, the differences between them SHOULD be provided in the reference range and/or age properties.
Invariants | Defined on this element obs-3 Rule Must have at least a low or a high or text low.exists() or high.exists() or text.exists()
Defined on this element
obs-3 | Rule | Must have at least a low or a high or text | low.exists() or high.exists() or text.exists()
Observation.referenceRange.low
Element Id | Observation.referenceRange.low
Definition | The value of the low bound of the reference range. The low bound of the reference range endpoint is inclusive of the value (e.g. reference range is >=5 - <=9). If the low bound is omitted, it is assumed to be meaningless (e.g. reference range is <=2.3).
Cardinality | 0..1
Type | SimpleQuantity
Invariants | Affect this element obs-3 Rule Must have at least a low or a high or text low.exists() or high.exists() or text.exists()
Affect this element
obs-3 | Rule | Must have at least a low or a high or text | low.exists() or high.exists() or text.exists()
Observation.referenceRange.high
Element Id | Observation.referenceRange.high
Definition | The value of the high bound of the reference range. The high bound of the reference range endpoint is inclusive of the value (e.g. reference range is >=5 - <=9). If the high bound is omitted, it is assumed to be meaningless (e.g. reference range is >= 2.3).
Cardinality | 0..1
Type | SimpleQuantity
Invariants | Affect this element obs-3 Rule Must have at least a low or a high or text low.exists() or high.exists() or text.exists()
Affect this element
obs-3 | Rule | Must have at least a low or a high or text | low.exists() or high.exists() or text.exists()
Observation.referenceRange.type
Element Id | Observation.referenceRange.type
Definition | Codes to indicate the what part of the targeted reference population it applies to. For example, the normal or therapeutic range.
Cardinality | 0..1
Terminology Binding | Observation Reference Range Meaning Codes (Preferred)
Type | CodeableConcept
Requirements | Need to be able to say what kind of reference range this is - normal, recommended, therapeutic, etc., - for proper interpretation.
Comments | This SHOULD be populated if there is more than one range. If this element is not present then the normal range is assumed.
Observation.referenceRange.appliesTo
Element Id | Observation.referenceRange.appliesTo
Definition | Codes to indicate the target population this reference range applies to. For example, a reference range may be based on the normal population or a particular sex or race. Multiple appliesTo are interpreted as an "AND" of the target populations. For example, to represent a target population of African American females, both a code of female and a code for African American would be used.
Cardinality | 0..*
Terminology Binding | Observation Reference Range Applies To Codes (Example)
Type | CodeableConcept
Requirements | Need to be able to identify the target population for proper interpretation.
Comments | This SHOULD be populated if there is more than one range. If this element is not present then the normal population is assumed.
Observation.referenceRange.age
Element Id | Observation.referenceRange.age
Definition | The age at which this reference range is applicable. This is a neonatal age (e.g. number of weeks at term) if the meaning says so.
Cardinality | 0..1
Type | Range

Requirements | Some analytes vary greatly over age.
Observation.referenceRange.text
Element Id | Observation.referenceRange.text
Definition | Text based reference range in an observation which may be used when a quantitative range is not appropriate for an observation. An example would be a reference value of "Negative" or a list or table of "normals".
Cardinality | 0..1
Type | string
Observation.hasMember
Element Id | Observation.hasMember
Definition | This observation is a group observation (e.g. a battery, a panel of tests, a set of vital sign measurements) that includes the target as a member of the group.
Cardinality | 0..*
Type | Reference(Observation | QuestionnaireResponse | MolecularSequence)
Summary | true
Comments | When using this element, an observation will typically have either a value or a set of related resources, although both may be present in some cases. For a discussion on the ways Observations can assembled in groups together, see Notes below. Note that a system may calculate results from QuestionnaireResponse into a final score and represent the score as an Observation.
Observation.derivedFrom
Element Id | Observation.derivedFrom
Definition | The target resource that represents a measurement from which this observation value is derived. For example, a calculated anion gap or a fetal measurement based on an ultrasound image.
Cardinality | 0..*
Type | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence)
Summary | true
Comments | All the reference choices that are listed in this element can represent clinical observations and other measurements that may be the source for a derived value. The most common reference will be another Observation. For a discussion on the ways Observations can assembled in groups together, see Notes below.
Observation.component
Element Id | Observation.component
Definition | Some observations have multiple component observations. These component observations are expressed as separate code value pairs that share the same attributes. Examples include systolic and diastolic component observations for blood pressure measurement and multiple component observations for genetics observations.
Cardinality | 0..*
Requirements | Component observations share the same attributes in the Observation resource as the primary observation and are always treated a part of a single observation (they are not separable). However, the reference range for the primary observation value is not inherited by the component values and is required when appropriate for each component observation.
Summary | true
Comments | For a discussion on the ways Observations can be assembled in groups together see Notes below.
Observation.component.code
Element Id | Observation.component.code
Definition | Describes what was observed. Sometimes this is called the observation "code".
Cardinality | 1..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Requirements | Knowing what kind of observation is being made is essential to understanding the observation.
Summary | true
Comments | All code-value and component.code-component.value pairs need to be taken into account to correctly understand the meaning of the observation.
Observation.component.value[x]
Element Id | Observation.component.value[x]
Definition | The information determined as a result of making the observation, if the information has a simple value.
Cardinality | 0..1
Type | Quantity|CodeableConcept|string|boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | An observation exists to have a value, though it might not if it is in error, or if it represents a group of observations.
Summary | true
Comments | Used when observation has a set of component observations. An observation may have both a value (e.g. an Apgar score) and component observations (the observations from which the Apgar score was derived). If a value is present, the datatype for this element should be determined by Observation.code. A CodeableConcept with just a text would be used instead of a string if the field was usually coded, or if the type associated with the Observation.code defines a coded value. For additional guidance, see the Notes section below.
Observation.component.dataAbsentReason
Element Id | Observation.component.dataAbsentReason
Definition | Provides a reason why the expected value in the element Observation.component.value[x] is missing.
Cardinality | 0..1
Terminology Binding | DataAbsentReason (Extensible)
Type | CodeableConcept
Requirements | For many results it is necessary to handle exceptional values in measurements.
Comments | "Null" or exceptional values can be represented two ways in FHIR Observations. One way is to simply include them in the value set and represent the exceptions in the value. For example, measurement values for a serology test could be "detected", "not detected", "inconclusive", or "test not done". The alternate way is to use the value element for actual observations and use the explicit dataAbsentReason element to record exceptional values. For example, the dataAbsentReason

code "error" could be used when the measurement was not completed. Because of these options, use-case agreements are required to interpret general observations for exceptional values.
Invariants | Affect this element obs-6 Rule dataAbsentReason SHALL only be present if Observation.value[x] is not present dataAbsentReason.empty() or value.empty()
Affect this element
obs-6 | Rule | dataAbsentReason SHALL only be present if Observation.value[x] is not present | dataAbsentReason.empty() or value.empty()
Observation.component.interpretation
Element Id | Observation.component.interpretation
Definition | A categorical assessment of an observation value. For example, high, low, normal.
Cardinality | 0..*
Terminology Binding | Observation Interpretation Codes (Extensible)
Type | CodeableConcept
Requirements | For some results, particularly numeric results, an interpretation is necessary to fully understand the significance of a result.
Alternate Names | Abnormal Flag
Comments | Historically used for laboratory results (known as 'abnormal flag'), its use extends to other use cases where coded interpretations are relevant. Often reported as one or more simple compact codes this element is often placed adjacent to the result value in reports and flow sheets to signal the meaning/normalcy status of the result.
Observation.component.referenceRange
Element Id | Observation.component.referenceRange
Definition | Guidance on how to interpret the value by comparison to a normal or recommended range.
Cardinality | 0..*
Type | See Observation.referenceRange
Requirements | Knowing what values are considered "normal" can help evaluate the significance of a particular result. Need to be able to provide multiple reference ranges for different contexts.
Comments | Most observations only have one generic reference range. Systems MAY choose to restrict to only supplying the relevant reference range based on knowledge about the patient (e.g., specific to the patient's age, gender, weight and other factors), but this might not be possible or appropriate. Whenever more than one reference range is supplied, the differences between them SHOULD be provided in the reference range and/or age properties.

10.1.9 Resource Observation - Mappings

URL: https://hl7.org/fhir/R4/observation-mappings.html

10.1.9 Resource Observation - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

SNOMED CT Attribute Binding

SNOMED CT Concept Domain Binding

RIM Mapping

Mappings for the observation resource (see Mappings to Other Standards for further information & status).

10.1.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Observation
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
code | FiveWs.what[x]
subject | FiveWs.subject[x]
focus | FiveWs.subject[x]
encounter | FiveWs.context
effective[x] | FiveWs.done[x]
issued | FiveWs.recorded
performer | FiveWs.actor
code | FiveWs.what[x]

10.1.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Observation | Event
identifier | Event.identifier
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
code | Event.code
subject | Event.subject

encounter | Event.context
effective[x] | Event.occurrence[x]
performer | Event.performer.actor

10.1.9.3 HL7 v2 Mapping (http://hl7.org/v2)

Observation | OBX
identifier | OBX.21 For OBX segments from systems without OBX-21 support a combination of ORC/OBR and OBX must be negotiated between trading partners to uniquely identify the OBX segment. Depending on how V2 has been implemented each of these may be an option: 1) OBR-3 + OBX-3 + OBX-4 or 2) OBR-3 + OBR-4 + OBX-3 + OBX-4 or 2) some other way to uniquely ID the OBR/ORC + OBX-3 + OBX-4.
basedOn | ORC
partOf | Varies by domain
status | OBX-11
category
code | OBX-3
subject | PID-3
focus | OBX-3
encounter | PV1
effective[x] | OBX-14, and/or OBX-19 after v2.4 (depends on who observation made)
issued | OBR.22 (or MSH.7), or perhaps OBX-19 (depends on who observation made)
performer | OBX.15 / (Practitioner) OBX-16, PRT-5:PRT-4='RO' / (Device) OBX-18 , PRT-10:PRT-4='EQUIP' / (Organization) OBX-23, PRT-8:PRT-4='PO'
value[x] | OBX.2, OBX.5, OBX.6
dataAbsentReason | N/A
interpretation | OBX-8
note | NTE.3 (partner NTE to OBX, or sometimes another (child?) OBX)
bodySite | OBX-20
method | OBX-17
specimen | SPM segment
device | OBX-17 / PRT -10
referenceRange | OBX.7
low | OBX-7
high | OBX-7
type | OBX-10
appliesTo | OBX-10
age
text | OBX-7
hasMember | Relationships established by OBX-4 usage
derivedFrom | Relationships established by OBX-4 usage
component | containment by OBX-4?
code | OBX-3
value[x] | OBX.2, OBX.5, OBX.6
dataAbsentReason | N/A
interpretation | OBX-8
referenceRange | OBX.7

10.1.9.4 SNOMED CT Attribute Binding (http://snomed.org/attributebinding)

Attribute bindings link coded data elements in FHIR resources to a corresponding attribute in the SNOMED CT concept model. These bindings help to support:

• clarifying the intended meaning of the data element

• Quality checking the alignment between FHIR resource design and any coresponding SNOMED CT concept model

• Composition and decomposition of data instances by indicating the SNOMED CT concept model attribute whose value may be used to decompose a precoordinated concept into this data element

Observation
identifier
basedOn
partOf
status
category
code | 116680003 |Is a|
subject
focus
encounter
effective[x]
issued
performer
value[x] | 363714003 |Interprets|
dataAbsentReason
interpretation | 363713009 |Has interpretation|

note
bodySite | 718497002 |Inherent location|
method
specimen | 704319004 |Inherent in|
device | 424226004 |Using device|
referenceRange
low
high
type
appliesTo
age
text
hasMember
derivedFrom
component
code
value[x] | 363714003 |Interprets|
dataAbsentReason
interpretation | 363713009 |Has interpretation|
referenceRange

10.1.9.5 SNOMED CT Concept Domain Binding (http://snomed.info/conceptdomain)

Concept domain bindings link a resource or an element to a set of SNOMED CT concepts that represent the intended semantics of the instances (whether or not SNOMED CT is used to encode that data element). This set of concepts is represented using a SNOMED CT expression constraint. Note that the 'Concept domain binding' may be a superset of the 'value set binding'. These bindings help to support:

• Quality checking FHIR resources by ensuring that (a) the intended semantics of the instances matches the valid range of the corresponding SNOMED CT attribute, and (b) the intended value set is appropriate for the intended semantics of the instances

• Semantic checking of data instances by helping to detect potential inconsistencies caused by overlap between the semantics incorporated in two concept domains

Observation | < 363787002 |Observable entity|
identifier
basedOn
partOf
status | < 445584004 |Report by finality status|
category
code | < 363787002 |Observable entity| OR < 386053000 |Evaluation procedure|
subject
focus
encounter
effective[x]
issued
performer
value[x] | < 441742003 |Evaluation finding|
dataAbsentReason
interpretation | < 260245000 |Findings values|
note
bodySite | < 123037004 |Body structure|
method
specimen | < 123038009 |Specimen|
device | < 49062001 |Device|
referenceRange
low
high
type | < 260245000 |Findings values| OR < 365860008 |General clinical state finding| OR < 250171008 |Clinical history or observation findings| OR < 415229000 |Racial group| OR < 365400002 |Finding of puberty stage| OR < 443938003 |Procedure carried out on subject|
appliesTo | < 260245000 |Findings values| OR < 365860008 |General clinical state finding| OR < 250171008 |Clinical history or observation findings| OR < 415229000 |Racial group| OR < 365400002 |Finding of puberty stage| OR < 443938003 |Procedure carried out on subject|
age
text
hasMember
derivedFrom
component
code | < 363787002 |Observable entity| OR < 386053000 |Evaluation procedure|
value[x] | 363714003 |Interprets| < 441742003 |Evaluation finding|
dataAbsentReason
interpretation | < 260245000 |Findings values|

referenceRange

10.1.9.6 RIM Mapping (http://hl7.org/v3)

Observation | Observation[classCode=OBS, moodCode=EVN]
identifier | id
basedOn | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
partOf | .outboundRelationship[typeCode=FLFS].target
status | status Amended & Final are differentiated by whether it is the subject of a ControlAct event with a type of "revise"
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
code | code
subject | participation[typeCode=RTGT]
focus | participation[typeCode=SBJ]
encounter | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
effective[x] | effectiveTime
issued | participation[typeCode=AUT].time
performer | participation[typeCode=PRF]
value[x] | value
dataAbsentReason | value.nullFlavor
interpretation | interpretationCode
note | subjectOf.observationEvent[code="annotation"].value
bodySite | targetSiteCode
method | methodCode
specimen | participation[typeCode=SPC].specimen
device | participation[typeCode=DEV]
referenceRange | outboundRelationship[typeCode=REFV]/target[classCode=OBS, moodCode=EVN]
low | value:IVL_PQ.low
high | value:IVL_PQ.high
type | interpretationCode
appliesTo | interpretationCode
age | outboundRelationship[typeCode=PRCN].targetObservationCriterion[code="age"].value
text | value:ST
hasMember | outBoundRelationship
derivedFrom | .targetObservation
component | outBoundRelationship[typeCode=COMP]
code | code
value[x] | value
dataAbsentReason | value.nullFlavor
interpretation | interpretationCode
referenceRange | outboundRelationship[typeCode=REFV]/target[classCode=OBS, moodCode=EVN]

10.1.10 Resource Observation - Extensions & Profiles

URL: https://hl7.org/fhir/R4/observation-profiles.html

10.1.10 Resource Observation - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Observation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.1.10.1 Profiles

Profile | Description | Context
BMI | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body mass index. | observation-bmi
BP | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign blood pressure. | observation-bp
BodyHeight | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body height. | observation-bodyheight
BodyTemp | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body temperature. | observation-bodytemp
BodyWeight | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body weight. | observation-bodyweight
DeviceMetricObservation | This profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component. | Device Metric Observation Profile
HeadCircum | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign head circumference. | observation-headcircum
HeartRate | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign heart rate. | observation-heartrate
Observation-Genetics | Describes how the observation resource is used to report structured genetic test results | Observation-genetics

OxygenSat | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign oxygen saturation. | observation-oxygensat
RespRate | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign respiratory rate. | observation-resprate
VitalSigns | Defines constraints and extensions on the Observation resource for use in querying and retrieving Vital Signs information. | observation-vitalsigns
VitalsPanel | Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital signs panel. | observation-vitalspanel

10.1.10.2 Extensions

diagnosticReport-risk | Observation | risk | for DiagnosticReport HL7 Extensions
event-eventHistory | Observation | eventHistory | for Event Pattern HL7 Extensions
event-location | Observation | location | for Event Pattern HL7 Extensions
event-performerFunction | Observation.performer | performerFunction | for Event Pattern HL7 Extensions
event-statusReason | Observation | statusReason | for Event Pattern HL7 Extensions
observation-bodyPosition | Observation | bodyPosition | for Observation HL7 Extensions
observation-delta | Observation | delta | for Observation HL7 Extensions
observation-deviceCode | Observation | deviceCode | for Observation HL7 Extensions
observation-focusCode | Observation | focusCode | for Observation HL7 Extensions
observation-gatewayDevice | Observation | gatewayDevice | for Observation HL7 Extensions
observation-geneticsAllele | Observation | Allele | for Observation-genetics
observation-geneticsAminoAcidChange | Observation | AminoAcidChange | for Observation-genetics
observation-geneticsAncestry | Observation | Ancestry | for Observation-genetics
observation-geneticsCopyNumberEvent | Observation | CopyNumberEvent | for Observation-genetics
observation-geneticsDNARegionName | Observation | DNARegionName | for Observation-genetics
observation-geneticsGene | Observation | Gene | for Observation-genetics
observation-geneticsGenomicSourceClass | Observation | GenomicSourceClass | for Observation-genetics
observation-geneticsInterpretation | Observation | Interpretation | for Observation-genetics
observation-geneticsPhaseSet | Observation | PhaseSet | for Observation-genetics
observation-geneticsVariant | Observation | Variant | for Observation-genetics
observation-precondition | Observation | precondition | for Observation HL7 Extensions
observation-reagent | Observation | reagent | for Observation HL7 Extensions
observation-replaces | Observation | replaces | for Observation HL7 Extensions
observation-secondaryFinding | Observation | secondaryFinding | for Observation HL7 Extensions
observation-sequelTo | Observation | sequelTo | for Observation HL7 Extensions
observation-specimenCode | Observation | specimenCode | for Observation HL7 Extensions
observation-timeOffset | Observation.component | timeOffset | for Observation HL7 Extensions
workflow-episodeOfCare | Observation | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-instantiatesCanonical | Observation | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | Observation | instantiatesUri | for Workflow Pattern HL7 Extensions
workflow-reasonCode | Observation | reasonCode | for Workflow Pattern HL7 Extensions
workflow-reasonReference | Observation | reasonReference | for Workflow Pattern HL7 Extensions
workflow-relatedArtifact | Observation | relatedArtifact | for Workflow Pattern HL7 Extensions
workflow-researchStudy | Observation | researchStudy | for Workflow Pattern HL7 Extensions
workflow-supportingInfo | Observation | supportingInfo | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

family-member-history-genetics-observation | observation | for Family member history for genetics analysis
observation-geneticsInterpretation | Interpretation | for Observation-genetics
observation-precondition | precondition | for Observation HL7 Extensions
observation-replaces | replaces | for Observation HL7 Extensions
observation-sequelTo | sequelTo | for Observation HL7 Extensions
openEHR-test | test | for openEHR Archetype Profile
procedure-causedBy | causedBy | for Procedure HL7 Extensions
servicerequest-geneticsItem | Item | for ServiceRequest-Genetics
workflow-reasonReference | reasonReference | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.1.10.3 Search Extensions

amino-acid-change | string | HGVS Protein Change Expression: Observation.extension('http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName')

dna-variant | string | HGVS DNA variant Expression: Observation.extension('http://hl7.org/fhir/StructureDefinition/observation-geneticsDnaVariant')
gene-amino-acid-change | string | HGNC gene symbol and HGVS Protein change Expression: Observation.extension('http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChangeName')
gene-dnavariant | string | HGNC gene symbol and HGVS DNA Variant Expression: Observation.extension('http://hl7.org/fhir/StructureDefinition/observation-geneticsDnaVariant')
gene-identifier | token | HGNC gene symbol and identifier Expression: Observation.extension('http://hl7.org/fhir/StructureDefinition/observation-geneticsGene')

10.1.11 Resource Observation - Operations

URL: https://hl7.org/fhir/R4/observation-operations.html

10.1.11 Resource Observation - Operations

Orders and Observations Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This resource has 2 operations operations associated with it:

$stats | Observation Statistics | Trial Use
$lastn | Last N Observations Query | Trial Use

For more information about operations, including how they are invoked, see Operations.

10.1.11.1 Operation $stats on Observation

URL: https://hl7.org/fhir/R4/observation-operation-stats.html

10.1.11.1 Operation $stats on Observation

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

The Statistics operation performs a set of statistical calculations on a set of clinical measurements such as a blood pressure as stored on the server. This operation evaluates Observation resources having valueQuantity elements that have UCUM unit codes. Observations with a status of 'entered-in-error' will be excluded from the calculations.

The set of Observations is defined by 4 parameters:

• the subject of the observations for which the statistics are being generated (subject)

• which observations to generate statistics for (code and system, or coding)

• the time period over which to generate statistics 'durationorperiod`)

• the set of statistical analyses to return (statistic)

Possible statistical analyses (see StatisticsCode):

• average ("Average"): The mean of N measurements over the stated period.

• maximum ("Maximum"): The maximum value of N measurements over the stated period.

• minimum ("Minimum"): The minimum value of N measurements over the stated period.

• count ("Count"): The [number] of valid measurements over the stated period that contributed to the other statistical outputs.

• total-count ("Total Count"): The total [number] of valid measurements over the stated period, including observations that were ignored because they did not contain valid result values.

• median ("Median"): The median of N measurements over the stated period.

• std-dev ("Standard Deviation"): The standard deviation of N measurements over the stated period.

• sum ("Sum"): The sum of N measurements over the stated period.

• variance ("Variance"): The variance of N measurements over the stated period.

• 20-percent ("20th Percentile"): The 20th Percentile of N measurements over the stated period.

• 80-percent ("80th Percentile"): The 80th Percentile of N measurements over the stated period.

• 4-lower ("Lower Quartile"): The lower Quartile Boundary of N measurements over the stated period.

• 4-upper ("Upper Quartile"): The upper Quartile Boundary of N measurements over the stated period.

• 4-dev ("Quartile Deviation"): The difference between the upper and lower Quartiles is called the Interquartile range. (IQR = Q3-Q1) Quartile deviation or Semi-interquartile range is one-half the difference between the first and the third quartiles.

• 5-1 ("1st Quintile"): The lowest of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population.

• 5-2 ("2nd Quintile"): The second of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population.

• 5-3 ("3rd Quintile"): The third of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population.

• 5-4 ("4th Quintile"): The fourth of four values that divide the N measurements into a frequency distribution of five classes with each containing one fifth of the total population.

• skew ("Skew"): Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. The skewness value can be positive or negative, or even undefined. Source: Wikipedia .

• kurtosis ("Kurtosis"): Kurtosis is a measure of the "tailedness" of the probability distribution of a real-valued random variable. Source: Wikipedia .

• regression ("Regression"): Linear regression is an approach for modeling two-dimensional sample points with one independent variable and one dependent variable (conventionally, the x and y coordinates in a Cartesian coordinate system) and finds a linear function (a non-vertical straight line) that, as accurately as possible, predicts the dependent variable values as a function of the independent variables. Source: Wikipedia This Statistic code will return both a gradient and an intercept value.

If successful, the operation returns an Observation resource for each code with the results of the statistical calculations as component value pairs where the component code = the statistical code. The Observation also contains the input parameters patient,code and duration parameters. If unsuccessful, an OperationOutcome with an error message will be returned.

The client can request that all the observations on which the statistics are based be returned as well, using the include parameter. If an include parameter is specified, a limit may also be specified; the sources observations are subsetted at the server's discretion if count > limit. This functionality is included with the intent of supporting graphical presentation.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Observation-stats
```

Formal Definition (as a OperationDefinition).

URL: [base]/Observation/$stats

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

subject | 1..1 | uri | | | The subject of the relevant Observations, which has the value of the Observation.subject.reference. E.g. 'Patient/123'. Reference can be to an absolute URL, but servers only perform stats on their own observations

code | 0..* | string | | | The test code(s) upon which the statistics are being performed. Provide along with a system, or as a coding. For example, the LOINC code = 2339-0 (Glucose [Mass/volume] in Blood) will evaluate all relevant Observations with this code in Observation.code and Observation.component.code. For LOINC codes that are panels, e.g., 85354-9(Blood pressure panel with all children optional), the stats operation returns statistics for each of the individual panel measurements. That means it will include and evaluate all values grouped by code for all the individual observations that are: 1) referenced in .related for .related.type = 'has-member' and 2) component observations in Observation.component.

system | 0..1 | uri | | | The system for the code(s). Or provide a coding instead

coding | 0..* | Coding | | | The test code upon which the statistics are being performed, as a Coding

duration | 0..1 | decimal | | | The time period of interest given as hours. For example, the duration = "1" represents the last hour - the time period from on hour ago to now

period | 0..1 | Period | | | The time period over which the calculations to be performed, if a duration is not provided

statistic | 1..* | code | | | average|max|min|count The statistical operations to be performed on the relevant operations. Multiple statistics operations can be specified. These codes are defined here

include | 0..1 | boolean | | | Whether to return the observations on which the statistics are based

limit | 0..1 | positiveInt | | | If an include parameter is specified, a limit may also be specified to limit the number of source Observations returned. If the include paramter is absent or equal to "false" the limit parameter SHALL be ignored by the server

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

statistics | 1..* | Observation | | | A set of observations, one observation for each code, each containing one component for each statistic. The Observation.component.code contains the statistic, and is relative to the Observation.code and cannot be interpreted independently. The Observation will also contain a subject, effectivePeriod, and code reflecting the input parameters. The status is fixed to final.

source | 0..* | Observation | | | Source observations on which the statistics are based

If modifier extensions are present in the Observation, they must be accounted for by implementers. A modifier extension may affect the observation.value in a way that it should be excluded from the from the calculations.

This operation cannot be performed on observations that the user is not authorized to see. It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single patient, or determine whether the context has the rights to the nominated patient, if there is one. If there is no nominated patient (e.g. the operation is invoked at the system level) and the context is not associated with a single patient record, then the server should return an error. Specifying the relationship between the context, a user and patient records is outside the scope of this specification.

10.1.11.1.1 Examples

Request: Get the average, min, max and count of a series of BP measurements for a patient

```
GET [base]/Observation/$stats?subject=Patient/123&code=85354-
    9&system=http://loinc.org&duration=1&statistic=average&statistic=min&statistic=max&statistic=count
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  The outcome for this particular operation is 2 Observation resources - one for the diastolic (8462-4)
  and one for the systolic components (8480-6), because the requested LOINC code is a panel code
-->
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="statistics"/>
    <resource>
      <Observation>
        <status value="final"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="8480-6"/>
            <display value="Systolic blood pressure"/>
          </coding>
        </code>
        <subject>
           <reference value="Patient/123"/>
        </subject>
        <effectivePeriod>
           <start value="2016-05-25T19:26:16Z"/>
           <end value="2016-05-25T20:26:16Z"/>
        </effectivePeriod>
        <!--The Observation.component.code is relative to the Observation.code and cannot be interpreted
    independently.-->
        <component>
           <code>
              <!--   the average parameter code -->
              <coding>
                 <system value="http://hl7.org/fhir/observation-paramcode"/>
                 <code value="average"/>
                 <display value="Average"/>
              </coding>
           </code>
           <valueQuantity>
              <value value="121.8"/>
              <unit value="mmHg"/>
              <system value="http://unitsofmeasure.org"/>
              <code value="mm[Hg]"/>
           </valueQuantity>
        </component>
        <component>
           <!--  the max parameter code -->
           <code>
              <coding>
                 <system value="http://hl7.org/fhir/observation-paramcode"/>
                 <code value="max"/>
                 <display value="Maximum"/>
              </coding>
           </code>
           <valueQuantity>
              <value value="136"/>
              <unit value="mmHg"/>
              <system value="http://unitsofmeasure.org"/>
              <code value="mm[Hg]"/>
           </valueQuantity>
        </component>
        <component>
           <!--  the min parameter code -->
           <code>
              <coding>
                 <system value="http://hl7.org/fhir/observation-paramcode"/>
                 <code value="min"/>
                 <display value="Minimum"/>
              </coding>
           </code>
           <valueQuantity>
              <value value="110"/>
              <unit value="mmHg"/>
              <system value="http://unitsofmeasure.org"/>
              <code value="mm[Hg]"/>
           </valueQuantity>
        </component>
        <component>
           <!--  the count parameter code -->
           <code>
              <coding>
                <system value="http://hl7.org/fhir/observation-paramcode"/>
                  <code value="count"/>
                <display value="Count"/>
              </coding>
           </code>
           <valueQuantity>
```

```
            <value value="7"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="{observations}"/>
         </valueQuantity>
      </component>
   </Observation>
  </resource>
</parameter>
<parameter>
  <name value="statistics"/>
  <resource>
    <Observation>
      <status value="final"/>
      <code>
         <coding>
            <system value="http://loinc.org"/>
            <code value="8462-4"/>
            <display value="Diastolic blood pressure"/>
         </coding>
      </code>
      <subject>
         <reference value="Patient/123"/>
      </subject>
      <effectivePeriod>
         <start value="2016-05-25T19:26:16Z"/>
         <end value="2016-05-25T20:26:16Z"/>
      </effectivePeriod>
      <!--The Observation.component.code is relative to the Observation.code and cannot be interpreted
  independently.-->
      <component>
         <code>
            <!--   the average parameter code -->
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="average"/>
               <display value="Average"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="79.7"/>
            <unit value="mmHg"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="mm[Hg]"/>
         </valueQuantity>
      </component>
      <component>
         <!--  the max parameter code -->
         <code>
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="max"/>
               <display value="Maximum"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="110"/>
            <unit value="mmHg"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="mm[Hg]"/>
         </valueQuantity>
      </component>
      <component>
         <!--  the min parameter code -->
         <code>
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="min"/>
               <display value="Minimum"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="56"/>
            <unit value="mmHg"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="mm[Hg]"/>
         </valueQuantity>
      </component>
      <component>
         <!--  the count parameter code -->
         <code>
            <coding>
               <system value="http://hl7.org/fhir/observation-paramcode"/>
               <code value="count"/>
               <display value="Count"/>
            </coding>
         </code>
         <valueQuantity>
            <value value="7"/>
            <system value="http://unitsofmeasure.org"/>
```

```
            <code value="{observations}"/>
          </valueQuantity>
        </component>
      </Observation>
    </resource>
  </parameter>
</Parameters>
```

For more information about operations, including how they are invoked, see Operations.

10.1.11.2 Operation $lastn on Observation

URL: https://hl7.org/fhir/R4/observation-operation-lastn.html

10.1.11.2 Operation $lastn on Observation

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

The lastn query meets the common need for searching for the most recent or last n=number of observations for a subject. For example, retrieving the last 5 temperatures for a patient to view trends or fetching the most recent laboratory results or vitals signs. To ask a server to return the last n=number of observations, the lastn query uses the normal search parameters defined for the Observation resource. However, rather than their normal use, they are interpreted as inputs - i.e.. instead of requiring that the resources literally contain the search parameters, they are passed to a server algorithm of some kind that uses them to determine the most appropriate matches.

The request for a lastn query SHALL include:

• A $lastn operation parameter

• A subject using either the patient or subject search parameter

• A category parameter and/or a search parameter that contains a code element in its FHIRpath expression. (e.g., code or code-value-concept)

The request for a lastn query MAY include:

• Other Observation search parameters and modifiers

The response from a lastn query is a set of observations:

• Filtered by additional parameters If not explicitly filtered by status then will include statuses of 'entered-in-error'

• 'GROUP BY' Observation.code Codes SHALL be considered equivalent if the coding.value and coding.system are the same. Text only codes SHALL be treated and grouped based on the text. For codes with translations (multiple codings), the code translations are assumed to be equal and the grouping by code SHALL follow the transitive property of equality.

for example:

Observation.code for observation a | Observation.code for observation b | Observation.code for observation c | number of groups [codes/text in each group]
a | b | c | 3 [a],[b],[c]
a | b | a,c | 2 [a.c],[b]
a | b | a,b | 1 [a,b]
'textM' | 'Text' | 't e x t' | 3 ['text'],['Text'],['t e x t']

• Sorted from most recent to the oldest

• Limited to the number of requested responses per group specified by the optional max query parameter In case of a tie - when the effective times for >1 Observations are equal - both will be returned. Therefore, more Observations may be returned than is specified in max. For example, 4 Observations instead of 3 if the 3rd and 4th most recent observation had the same effective time.

• If no maximum number is given then only the most recent Observation in each group is returned.

The set of returned observations should represent distinct real world observations and not the same observation with changes in status or versions. If there are no matches, the lastn query SHALL return an empty search set with no error, but may include an operation outcome with further advice.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Observation-lastn
```

Formal Definition (as a OperationDefinition).

URL: [base]/Observation/$lastn

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
max | 0..1 | positiveInt | | | max is an optional input parameter to the lastn query operation. It is used to specify the maximum number of Observations to return from each group. For example for the query "Fetch the last 3 results for all vitals for a patient" max = 3.
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The set of most recent N Observations that match the lastn query search criteria. Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this query operation and simply searching Observation using the combination of _count and _sort parameters are:

• The lastn query returns only the last N resource grouped by code. Using the _count query method doesn't restrict the total matches so you may need to page through several "A" Observations before getting to Observation "B".

• The server is responsible for grouping the observations by codes. This frees the client from needing to determine which codes she should ask for.

This operation cannot be performed on observations that the user is not authorized to see. It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single patient, or determine whether the context has the rights to the nominated patient, if there is one. If there is no nominated patient (e.g. the operation is invoked at the system level) and the context is not associated with a single patient record, then the server should return an error. Specifying the relationship between the context, a user and patient records is outside the scope of this specification.

10.1.11.2.1 Examples

Request: 1): Fetch the last 3 results for all vitals for a patient 2): Fetch the last laboratory results for a patient 3): Fetch the last 3 blood pressures, respiratory rate and heart rates for a patient

```
1) GET [base]/Observation/$lastn?max=3&patient=Patient/123&category=vital-signs

-------------------------------------------

2) GET [base]/Observation/$lastn?patient=Patient/123&category=laboratory

-------------------------------------------

3) GET [base]/Observation/$lastn?max=3&patient=Patient/123&category=vital-signs&code=9279-1,8867-4,85354-9
```

Response: 1): A Server returns from an imaginary Last N algorithm the last 3 of 9 different vital signs observations on a patient. e.g. body temperature, body weight, body height... (truncated for readability) 2): A Server returns from an imaginary Last N algorithm the last laboratory observations for a patient. e.g. CBC Panel, UA, Chem Panel, Lipid Panel components and HbA1c, (truncated for readability) 3): A Server returns from an imaginary Last N algorithm the last 3 blood pressures, respiratory rate and heart rates for a patient as specified by code from the request. (truncated for readability)

```
========================== Example 1 ============================

HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2017-02-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 27,
  // the patient temperature was being taken several times a day
  "entry": [{
    "fullUrl": "http://server/path/Observation/20170219-06temp",
    "resource": {
      "resourceType": "Observation",
      "id": "20170219temp",
      .. snip ...
  ,{
   "fullUrl": "http://server/path/Observation/20170219-05temp",
       "resource": {
         "resourceType": "Observation",
         "id": "20170218temp",
      .. snip ...
    }
      ,{
       "fullUrl": "http://server/path/Observation/20170219-04temp",
           "resource": {
             "resourceType": "Observation",
             "id": "20170217temp",
          .. snip ...
    },
    // the patient was being weighed daily
    {
    "fullUrl": "http://server/path/Observation/20170219weight",
    "resource": {
      "resourceType": "Observation",
      "id": "20170219weight",
      .. snip ...
  ,{
   "fullUrl": "http://server/path/Observation/20170218weight",
       "resource": {
```

```
          "resourceType": "Observation",
          "id": "20170218weight",
        .. snip ...
    }
      ,{
       "fullUrl": "http://server/path/Observation/20170217weight",
           "resource": {
             "resourceType": "Observation",
             "id": "20170217weight",
          .. snip ...
    },
    // height was recorded occasionally over the years
        {
        "fullUrl": "http://server/path/Observation/20170101height",
        "resource": {
          "resourceType": "Observation",
          "id": "20170219height",
          .. snip ...
      ,{
       "fullUrl": "http://server/path/Observation/20160101height",
           "resource": {
             "resourceType": "Observation",
             "id": "20160101height",
          .. snip ...
        }
          ,{
          "fullUrl": "http://server/path/Observation/20131001height",
              "resource": {
                "resourceType": "Observation",
                "id": "20131001height",
             .. snip ...
    }
  ]
}

---------------------------------

========================== Example 2 ============================


HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2017-02-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 45,
  // the patient's last CBC Panel component observations
  "entry": [{
    "fullUrl": "http://server/path/Observation/20170219-CBCPanel-RBCs",
    "resource": {
      "resourceType": "Observation",
      "id": "0170219-CBCPanel-RBCs",
      ... snip ...
      "code" : {
    "coding" : [
      {
        "system" : "http://loinc.org",
        "code" : "789-8",
        "display" : "RBC # Bld Auto"
      }
    ],
    "text" : "RBC # Bld Auto"
  },
  ...snip....
  },
  // the rest of the patient CBC Panel components observations
  { ...snip...
  },
  // the patient's last UA Panel component observations
    {
    "fullUrl": "http://server/path/Observation/20160610-UA-Color",
    "resource": {
      "resourceType": "Observation",
      "id": "20160610-UA-Color",
      .. snip ...
      "code" : {
   "coding" : [
     {
       "system" : "http://loinc.org",
       "code" : "5778-6",
       "display" : "Color of Urine"
     }
   ],
   "text" : "COLOR"
```

```
 },
   ...snip....
   },
   // the rest of the patient UA Panel component observations
   { ...snip...
   },
   // the patient's last Metabolic Panel component observations
  {
   "fullUrl": "http://server/path/Observation/20170219MetPanel-Sodium",
       "resource": {
         "resourceType": "Observation",
         "id": "20170219MetPanel-Sodium",
      ... snip ...
        "code" : {
          "coding" : [
          {
            "system" : "http://loinc.org",
            "code" : "2951-2",
            "display" : "Sodium SerPl-sCnc"
            }
          ],
          "text" : "Sodium SerPl-sCnc"
          },
      ...snip....
      },
      // the rest of the patient Metabolic Panel component observations
      { ...snip...
      },
    // the patient Lipase Panel components observations
        {
        "fullUrl": "http://server/path/Observation/20170219LipidPanel-Chol",
        "resource": {
          "resourceType": "Observation",
          "id": "20170219LipidPanel-Chol",
          ... snip ...
          "code": {
                 "coding": [
                   {
                     "system": "http://loinc.org",
                     "code": "14647-2",
                     "display": "Cholesterol [Moles/□volume] in Serum or Plasma"
                   }
                 ],
                 "text": "Cholesterol"
          },
          ...snip....
          },
          // the patient HbA1c observation
          {
          "fullUrl": "http://server/path/Observation/20170219HbA1c",
          "resource": {
            "resourceType": "Observation",
            "id": "20170219HbA1c",
            ... snip ...
            "code": {
                   "coding": [
                     {
                       "system": "http://loinc.org",
                       "code": "4548-4",
                       "display": "Hemoglobin A1c/□Hemoglobin.total in Blood"
                     }
                   ],
                   "text": "HbA1c"
            },
            ...snip....
    }
  ]
}

---------------------------------


========================== Example 3 ============================


HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2017-02-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 9,
  // Returns the patient's last 3 code = 9279-1 Respiratory Rate observations All from 20170219
  "entry": [{
    "fullUrl": "http://server/path/Observation/20170219-06RespRate",
```

```
   "resource": {
     "resourceType": "Observation",
     "id": "20170219-06RespRate",
     .. snip ...
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "9279-1",
          "display": "Respiratory rate"
        }
      ],
      "text": "Respiratory rate"
},
    .. snip ...
 ,{
  "fullUrl": "http://server/path/Observation/20170219-05RespRate",
      "resource": {
        "resourceType": "Observation",
        "id": "20170219-05RespRate",
     .. snip ...
   }
     ,{
      "fullUrl": "http://server/path/Observation/20170219-04RespRate",
          "resource": {
            "resourceType": "Observation",
            "id": "20170217-04RespRate",
         .. snip ...
   },
   // Returns the patient's last 3 code =  8867-4 Heart Rate observations All from 20170219
   // the patient was being weighed daily
   {
   "fullUrl": "http://server/path/Observation/20170219-06HeartRate",
   "resource": {
     "resourceType": "Observation",
     "id": "20170219-06HeartRate",
     .. snip ...
     ],
   "code": {
     "coding": [
       {
         "system": "http://loinc.org",
         "code": "8867-4",
         "display": "Heart rate"
       }
     ],
     "text": "Heart rate"
 ...snip...
 ,{
  "fullUrl": "http://server/path/Observation/20170219-05HeartRate",
      "resource": {
        "resourceType": "Observation",
        "id": "20170219-05HeartRate",
     .. snip ...
   }
     ,{
      "fullUrl": "http://server/path/Observation/20170219-04HeartRate",
          "resource": {
            "resourceType": "Observation",
            "id": "20170219-04HeartRate",
         .. snip ...
   },
   // Returns the patient's last 3 code =  85354-9 Blood pressure systolic and diastolic All from 20170219
       {
       "fullUrl": "http://server/path/Observation/20170219-06BloodPressure",
       "resource": {
         "resourceType": "Observation",
         "id": "20170219-06BloodPressure",
         .. snip ...
         "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "85354-9",
              "display": "Bood pressure panel with all children optional"
            }
          ],
          "text": "Blood pressure systolic & diastolic"
        },
        ...snip...
                "component": [
        {
          "code": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "8480-6",
                "display": "Systolic blood pressure"
              },
```

```
              {
           ...snip...
                      {
             "code": {
               "coding": [
                 {
                   "system": "http://loinc.org",
                   "code": "8462-4",
                   "display": "Diastolic blood pressure"
                 }
               ]
             },
             ...snip....
      ,{
       "fullUrl": "http://server/path/Observation/20170219-05BloodPressure",
           "resource": {
             "resourceType": "Observation",
             "id": "20170219-05BloodPressure",
          .. snip ...
        }
          ,{
           "fullUrl": "http://server/path/Observation/20170219-04BloodPressure",
               "resource": {
                 "resourceType": "Observation",
                 "id": "20170219-04BloodPressure",
             .. snip ...
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

10.1.12 Resource Observation - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/observation-version-maps.html

No validation errors - all conversions are clean

10.1.12 Resource Observation - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Observation.

Functional status for this map: 48 tests that all execute ok. All tests pass round-trip testing and 23 r3 resources are invalid (0 errors). (see documentation)

10.1.12.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Observation3to4" = "R3 to R4 Conversions for Observation"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Observation" alias ObservationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Observation" alias Observation as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Observation(source src : ObservationR3, target tgt : Observation) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.issued -> tgt.issued;
  src.performer -> tgt.performer;
  src.value : Quantity as vs -> tgt.value = create('Quantity') as vt then Quantity(vs, vt);
  src.value : CodeableConcept as vs -> tgt.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.value : string as vs -> tgt.value = create('string') as vt then string(vs, vt);
  src.value : boolean as vs -> tgt.value = create('boolean') as vt then boolean(vs, vt);
  src.value : Range as vs -> tgt.value = create('Range') as vt then Range(vs, vt);
  src.value : Ratio as vs -> tgt.value = create('Ratio') as vt then Ratio(vs, vt);
  src.value : SampledData as vs -> tgt.value = create('SampledData') as vt then SampledData(vs, vt);
  src.value : time as vs -> tgt.value = create('time') as vt then time(vs, vt);
  src.value : dateTime as vs -> tgt.value = create('dateTime') as vt then dateTime(vs, vt);
  src.value : Period as vs -> tgt.value = create('Period') as vt then Period(vs, vt);
  src.dataAbsentReason -> tgt.dataAbsentReason;
  src.interpretation -> tgt.interpretation;
  src.comment as vs0 -> tgt.note as vt0 then string2Annotation(vs0, vt0);
  src.bodySite -> tgt.bodySite;
  src.method -> tgt.method;
  src.specimen -> tgt.specimen;
  src.device -> tgt.device;
  src.referenceRange as vs0 -> tgt.referenceRange as vt0 then referenceRange(vs0, vt0);
  src.related as vs0 where type = 'has-member' -> tgt.hasMember as vt0 then related(vs0, vt0) "related1";
  src.related as vs0 where type = 'derived-from' -> tgt.derivedFrom as vt0 then related(vs0, vt0) "related2";
```

```
  src.related as vs0 where type = 'sequel-to' ->  tgt.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/Observation.sequelTo',  ext.value = create('Reference') as vt0 then
    related(vs0, vt0) "related3";
  src.related as vs0 where type = 'replaces' ->  tgt.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/Observation.replaces',  ext.value = create('Reference') as vt0 then
    related(vs0, vt0) "related4";
  src.related as vs0 where type = 'qualified-by' ->  tgt.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/Observation.qualifiedBy',  ext.value = create('Reference') as vt0 then
    related(vs0, vt0) "related5";
  src.related as vs0 where type = 'interfered-by' ->  tgt.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/Observation.interferedBy',  ext.value = create('Reference') as vt0 then
     related(vs0, vt0) "related6";
  src.component as vs0 -> tgt.component as vt0 then component(vs0, vt0);
}

group referenceRange(source src, target tgt) extends BackboneElement {
  src.low -> tgt.low;
  src.high -> tgt.high;
  src.type -> tgt.type;
  src.appliesTo -> tgt.appliesTo;
  src.age -> tgt.age;
  src.text -> tgt.text;
}

group related(source src, target tgt) extends BackboneElement {
  src.target as vs0 then Reference(vs0, tgt);
}

group component(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value : Quantity as vs -> tgt.value = create('Quantity') as vt then Quantity(vs, vt);
  src.value : CodeableConcept as vs -> tgt.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.value : string as vs -> tgt.value = create('string') as vt then string(vs, vt);
  src.value : Range as vs -> tgt.value = create('Range') as vt then Range(vs, vt);
  src.value : Ratio as vs -> tgt.value = create('Ratio') as vt then Ratio(vs, vt);
  src.value : SampledData as vs -> tgt.value = create('SampledData') as vt then SampledData(vs, vt);
  src.value : time as vs -> tgt.value = create('time') as vt then time(vs, vt);
  src.value : dateTime as vs -> tgt.value = create('dateTime') as vt then dateTime(vs, vt);
  src.value : Period as vs -> tgt.value = create('Period') as vt then Period(vs, vt);
  src.dataAbsentReason -> tgt.dataAbsentReason;
  src.interpretation -> tgt.interpretation;
  src.referenceRange as vs0 -> tgt.referenceRange as vt0 then referenceRange(vs0, vt0);
}
```

10.1.12.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Observation4to3" = "R4 to R3 Conversion for Observation"

uses "http://hl7.org/fhir/StructureDefinition/Observation" alias Observation as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Observation" alias ObservationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Observation(source src : Observation, target tgt : ObservationR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.issued -> tgt.issued;
  src.performer -> tgt.performer;
  src.value : Quantity as vs -> tgt.value = create('Quantity') as vt then Quantity(vs, vt);
  src.value : CodeableConcept as vs -> tgt.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.value : string as vs -> tgt.value = create('string') as vt then string(vs, vt);
  src.value : boolean as vs -> tgt.value = create('boolean') as vt then boolean(vs, vt);
  src.value : Range as vs -> tgt.value = create('Range') as vt then Range(vs, vt);
  src.value : Ratio as vs -> tgt.value = create('Ratio') as vt then Ratio(vs, vt);
  src.value : SampledData as vs -> tgt.value = create('SampledData') as vt then SampledData(vs, vt);
  src.value : time as vs -> tgt.value = create('time') as vt then time(vs, vt);
  src.value : dateTime as vs -> tgt.value = create('dateTime') as vt then dateTime(vs, vt);
  src.value : Period as vs -> tgt.value = create('Period') as vt then Period(vs, vt);
  src.dataAbsentReason -> tgt.dataAbsentReason;
  src.interpretation -> tgt.interpretation;
  src.note first as vs0 -> tgt.comment as vt0 then Annotation2string(vs0, vt0);
  src.bodySite -> tgt.bodySite;
  src.method -> tgt.method;
  src.specimen -> tgt.specimen;
  src.device -> tgt.device;
  src.referenceRange as vs0 -> tgt.referenceRange as vt0 then referenceRange(vs0, vt0);
  src.hasMember as vs0 ->  tgt.related as vt0,  vt0.type = 'has-member',  vt0.target as vt2 then Reference(vs0, vt2);
  src.derivedFrom as vs0 ->  tgt.related as vt0,  vt0.type = 'derived-from',  vt0.target as vt2 then Reference(vs0,
    vt2);
  src.extension as vs0 where url = 'http://hl7.org/fhir/3.0/StructureDefinition/Observation.sequelTo' ->  tgt.related as
     rel,  rel.type = 'sequel-to',  rel.target as vt0 then relatedExt(vs0, vt0) "related3";
  src.extension as vs0 where url = 'http://hl7.org/fhir/3.0/StructureDefinition/Observation.replaces' ->  tgt.related as
     rel,  rel.type = 'replaces',  rel.target as vt0 then relatedExt(vs0, vt0) "related4";
```

```
  src.extension as vs0 where url = 'http://hl7.org/fhir/3.0/StructureDefinition/Observation.qualifiedBy' ->  tgt.related
     as rel,  rel.type = 'qualifiedBy',  rel.target as vt0 then relatedExt(vs0, vt0) "related5";
  src.extension as vs0 where url = 'http://hl7.org/fhir/3.0/StructureDefinition/Observation.interferedBy' ->
    tgt.related as rel,  rel.type = 'interfered-by',  rel.target as vt0 then relatedExt(vs0, vt0) "related6";
  src.component as vs0 -> tgt.component as vt0 then component(vs0, vt0);
}

group relatedExt(source src, target tgt) {
  src.value : Reference as vs0 then Reference(vs0, tgt);
}

group referenceRange(source src, target tgt) extends BackboneElement {
  src.low -> tgt.low;
  src.high -> tgt.high;
  src.type -> tgt.type;
  src.appliesTo -> tgt.appliesTo;
  src.age -> tgt.age;
  src.text -> tgt.text;
}

group component(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value : Quantity as vs -> tgt.value = create('Quantity') as vt then Quantity(vs, vt);
  src.value : CodeableConcept as vs -> tgt.value = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.value : string as vs -> tgt.value = create('string') as vt then string(vs, vt);
  src.value : Range as vs -> tgt.value = create('Range') as vt then Range(vs, vt);
  src.value : Ratio as vs -> tgt.value = create('Ratio') as vt then Ratio(vs, vt);
  src.value : SampledData as vs -> tgt.value = create('SampledData') as vt then SampledData(vs, vt);
  src.value : time as vs -> tgt.value = create('time') as vt then time(vs, vt);
  src.value : dateTime as vs -> tgt.value = create('dateTime') as vt then dateTime(vs, vt);
  src.value : Period as vs -> tgt.value = create('Period') as vt then Period(vs, vt);
  src.dataAbsentReason -> tgt.dataAbsentReason;
  src.interpretation -> tgt.interpretation;
  src.referenceRange as vs0 -> tgt.referenceRange as vt0 then referenceRange(vs0, vt0);
}
```

10.1.12.3 R4 Validation Errors

10.1.13 Device Metric Observation Profile

URL: https://hl7.org/fhir/R4/observation-devicemetricobservation.html

For the initial scope, this DeviceMetricObservation profile is only applicable to describe a measurement that is produced by any medical device that implements or derives from the ISO/IEEE 11073 standard.

10.1.13 Device Metric Observation Profile

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.13.1 Scope and Usage

The DeviceMetricObservation profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component.

10.1.13.2 Content

Profiles:

DeviceMetricObservation | This profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component.

10.1.13.3 StructureDefinition: DeviceMetricObservation

URL: https://hl7.org/fhir/R4/devicemetricobservation.html

10.1.13.3 StructureDefinition: DeviceMetricObservation

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/devicemetricobservation`

This profile describes the direct or derived, qualitative or quantitative physiological measurement, setting, or calculation data produced by a medical device or a device component.

This profile was published on Mon, Nov 17, 2014 00:00+1100 as a draft by Health Level Seven International (Health Care Devices).

10.1.13.3.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table

- XML Template
- JSON Template
- All

This structure is derived from Observation.

Summary

Mandatory: 5 elements Must-Support: 12 elements Prohibited: 4 elements

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		
identifier	S	0..*	Identifier	
status	S	1..1	code	
code	S	1..1	CodeableConcept	
subject	S	1..1	Reference(Patient | Device)	
encounter		0..0		
effectiveDateTime	S	1..1	dateTime	
issued		0..0		
value[x]	S	0..1		
valueQuantity			Quantity	
valueCodeableConcept			CodeableConcept	
valueString			string	
valueRange			Range	
valueRatio			Ratio	
valueSampledData			SampledData	
valueTime			time	
valueDateTime			dateTime	
valuePeriod			Period	
dataAbsentReason		0..0		
interpretation		0..1	CodeableConcept	
note		0..*	Annotation	
bodySite	S	0..1	CodeableConcept	
method	S	0..1	CodeableConcept	
specimen		0..0		
device	S	1..1	Reference(DeviceMetric)	
referenceRange	S	0..1	BackboneElement	
hasMember	S	0..*	Reference(Observation)	
derivedFrom	S	0..*	Reference(Observation)	

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		Measurements and simple assertions
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	SΣI	0..*	Identifier	Business Identifier for observation
basedOn	ΣI	0..*	Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	Fulfills plan, proposal or order
partOf	ΣI	0..*	Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)	Part of referenced event
status	?!SΣI	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	I	0..*	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred)
code	SΣI	1..1	CodeableConcept	Type of observation (code / type) Binding: LOINC Codes (example)
subject	SΣI	1..1	Reference(Patient | Device)	Who and/or what the observation is about
focus	ΣI	0..*	Reference(Resource)	What the observation is about, when it is not about the subject of record
effective[x]	ΣI		(Slice Definition)	Clinically relevant time/time-period for observation Slice: Unordered, Closed by type:$this
effective[x]:effectiveDateTime	SΣI	1..1	dateTime	Clinically relevant time/time-period for observation
performer	ΣI	0..*	Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson)	Who is responsible for the observation
value[x]	SΣI	0..1		Actual result
valueQuantity			Quantity	
valueCodeableConcept			CodeableConcept	
valueString			string	
valueRange			Range	
valueRatio			Ratio	
valueSampledData			SampledData	
valueTime			time	

valueDateTime | | | dateTime
valuePeriod | | | Period
interpretation | I | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | SI | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | SI | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
device | SI | 1..1 | Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 0..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | SΣI | 0..* | Reference(Observation) | Related resource that belongs to the Observation group
derivedFrom | SΣI | 0..* | Reference(Observation) | Related measurements the observation is made from
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- Device Metric Observation Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Type of observation (code / type)   --></code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..0 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <-- effectiveDateTime sliced by type:$this  in the specified orderClosed-->
 <effectiveDateTime value="[dateTime]"/><!--  1..1 Clinically relevant time/time-period for observation -->
 <issued value="[instant]"/><!--  0..0 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|Range|Ratio|
   SampledData|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..0 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
```

```
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..0 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  1..1 Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..1 BackboneElement  <!--  0..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..* Reference(Observation)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Device Metric Observation Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! Type of observation (code / type)
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "effectiveDateTime" : [ //  sliced by type:$this  in the specified order, Closed
      { // Clinically relevant time/time-period for observation // C? R!
        // from Element: extension
      }
    ],
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Actual result. One of these 9:
    "valueQuantity" : { Quantity }, // C?Actual result
    "valueCodeableConcept" : { CodeableConcept }, // C?Actual result
    "valueString" : "<string>", // C?Actual result
    "valueRange" : { Range }, // C?Actual result
    "valueRatio" : { Ratio }, // C?Actual result
    "valueSampledData" : { SampledData }, // C?Actual result
    "valueTime" : "<time>", // C?Actual result
    "valueDateTime" : "<dateTime>", // C?Actual result
    "valuePeriod" : { Period }, // C?Actual result
    "interpretation" : { CodeableConcept }, // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept } // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "device" : { Reference(DeviceMetric) }, // C? R! (Measurement) Device
    "referenceRange" : { BackboneElement }, // C?Provides guide for interpretation
    "hasMember" : [{ Reference(Observation) }], // C?Related resource that belongs to the Observation group
    "derivedFrom" : [{ Reference(Observation) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 5 elements Must-Support: 12 elements Prohibited: 4 elements

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		
identifier	S	0..*	Identifier	
status	S	1..1	code	

code | S | 1..1 | CodeableConcept
subject | S | 1..1 | Reference(Patient | Device)
encounter | | 0..0
effectiveDateTime | S | 1..1 | dateTime
issued | | 0..0
value[x] | S | 0..1
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | | 0..0
interpretation | | 0..1 | CodeableConcept
note | | 0..* | Annotation
bodySite | S | 0..1 | CodeableConcept
method | S | 0..1 | CodeableConcept
specimen | | 0..0
device | S | 1..1 | Reference(DeviceMetric)
referenceRange | S | 0..1 | BackboneElement
hasMember | S | 0..* | Reference(Observation)
derivedFrom | S | 0..* | Reference(Observation)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | SΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | SΣI | 1..1 | Reference(Patient | Device) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
effective[x] | ΣI | | (Slice Definition) | Clinically relevant time/time-period for observation Slice: Unordered, Closed by type:$this
effective[x]:effectiveDateTime | SΣI | 1..1 | dateTime | Clinically relevant time/time-period for observation
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | 0..1 | | Actual result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
interpretation | I | 0..1 | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | SI | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | SI | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
device | SI | 1..1 | Reference(DeviceMetric) | (Measurement) Device
referenceRange | SI | 0..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant

type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | SΣI | 0..* | Reference(Observation) | Related resource that belongs to the Observation group
derivedFrom | SΣI | 0..* | Reference(Observation) | Related measurements the observation is made from
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Device Metric Observation Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Type of observation (code / type)   --></code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..0 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <-- effectiveDateTime sliced by type:$this  in the specified orderClosed-->
 <effectiveDateTime value="[dateTime]"/><!--  1..1 Clinically relevant time/time-period for observation -->
 <issued value="[instant]"/><!--  0..0 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|Range|Ratio|
   SampledData|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..0 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..0 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  1..1 Reference(DeviceMetric) (Measurement) Device --></device>
 <referenceRange>  0..1 BackboneElement  <!--  0..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
```

```
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..* Reference(Observation)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Device Metric Observation Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! Type of observation (code / type)
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "effectiveDateTime" : [ //  sliced by type:$this  in the specified order, Closed
      { // Clinically relevant time/time-period for observation // C? R!
        // from Element: extension
      }
    ],
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Actual result. One of these 9:
    "valueQuantity" : { Quantity }, // C?Actual result
    "valueCodeableConcept" : { CodeableConcept }, // C?Actual result
    "valueString" : "<string>", // C?Actual result
    "valueRange" : { Range }, // C?Actual result
    "valueRatio" : { Ratio }, // C?Actual result
    "valueSampledData" : { SampledData }, // C?Actual result
    "valueTime" : "<time>", // C?Actual result
    "valueDateTime" : "<dateTime>", // C?Actual result
    "valuePeriod" : { Period }, // C?Actual result
    "interpretation" : { CodeableConcept }, // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept } // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "device" : { Reference(DeviceMetric) }, // C? R! (Measurement) Device
    "referenceRange" : { BackboneElement }, // C?Provides guide for interpretation
    "hasMember" : [{ Reference(Observation) }], // C?Related resource that belongs to the Observation group
    "derivedFrom" : [{ Reference(Observation) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.13.3.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | LOINC Codes | example | LOINC Codes
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | LOINC Codes | example | LOINC Codes
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.14 Observation HL7 Extensions

URL: https://hl7.org/fhir/R4/observation-extensions.html

Defines common extensions used with or related to the Observation resource

10.1.14 Observation HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.14.1 Content

Extensions:

observation-bodyPosition | bodyPosition : The position of the body when the observation was done, e.g. standing, sitting. To be used only when the body position in not precoordinated in the observation code.

observation-delta | delta : The qualitative change in the value relative to the previous measurement. Usually only recorded if the change is clinically significant.

observation-focusCode | focusCode : A code representing the focus of an observation when the focus is not the patient of record. In other words, the focus of the observation is different from Observation.subject. An example use case would be using the Observation resource to capture whether the mother is trained to change her child's tracheostomy tube. In this example, the child is the patient of record and the mother is focal subject referenced using this extension. Other example focal subjects include spouses, related persons, feti, or donors.

observation-timeOffset | timeOffset : A specific offset time in milliseconds from the stated time in the Observation.appliesDateTime to allow for representation of sequential recording of sampled data from the same lead or data stream. For example, an ECG recorder may record sequentially 3 leads four time to receive 12-lead ECG, see ISO 22077 .

observation-sequelTo | sequelTo : This observation follows the target observation (e.g. timed tests such as Glucose Tolerance Test).

observation-replaces | replaces : This observation replaces a previous observation (i.e. a revised value).

observation-gatewayDevice | gatewayDevice : The Provenance/AuditEvent resources can represent the same information. Note that the Provenance/AuditEvent resources can represent the same information.

observation-specimenCode | specimenCode : A code representing the the type of specimen used for this observation. Should only be used if not implicit in the code found in Observation.code.

observation-deviceCode | deviceCode : A code representing the the type of device used for this observation. Should only be used if not implicit in the code found in Observation.code.

observation-precondition | precondition : Other preceding or concurrent observations that must be known to correctly interpret the the observation. For example an fiO2 measure taken alongside of a SpO2 measurement. See the Observation notes section for additional guidance.

observation-reagent | reagent : Reference to reagents used to generate this observation. This is intended for this for in-lab transactions between instruments and Laboratory Information Systems (LIS).

observation-secondaryFinding | secondaryFinding : Secondary findings are genetic test results that provide information about variants in a gene unrelated to the primary purpose for the testing, most often discovered when Whole Exome Sequencing (WES) or Whole Genome Sequencing (WGS) is performed. This extension should be used to denote when a genetic finding is being shared as a secondary finding, and ideally refer to a corresponding guideline or policy statement. For more detail, please see: https://ghr.nlm.nih.gov/primer/testing/secondaryfindings.

10.1.15 Observation-genetics

URL: https://hl7.org/fhir/R4/observation-genetic.html

10.1.15 Observation-genetics

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

The material on this page is currently undergoing work to be refactored or potentially removed in a future release as further analysis is done as part of the Genomics Reporting Implementation Guide .

10.1.15.1 Overview

Observation-genetics-profile extends Observation resource to enable reporting of structured genetic test results. In addition, the genetics profile contextualizes well established standards from the field of clinical genetics into the standards of healthcare (e.g. HGNC - HUGO Gene Nomenclature Committee's international standard for gene names, symbols, and identifiers).

For use cases, details on how this resource interact with other Clinical Genomics resources or profiles, please refer to implementation guidance document here .

10.1.15.1.1 Genetic Standards and Resources include:

• Variant Databases: dbSNP , ClinVar , and COSMIC

• Reference Sequences: RefSeq and ENSEMBL

• Gene Symbols and Identifiers: HGNC - Human Genee Nomenclature Committee

• Variant Nomenclature: HGVS nomenclature from the Human Genome Variantion Society

• Locus: Gene

10.1.15.2 Scope and Usage

The Observation-genetics profile supports reporting of a DNA variant at the genomic, cDNA, and protein change level. In addition, a condition context may be provided, as AssessedCondition. For large genomic tests, a condition may be used as an input into the analytic pipeline to aid in the identification of clinically relevant variants related to the test order. It is strongly encouraged to provide all available information in this

profile for any reported variants, because receiving systems (e.g. discovery research, outcomes analysis, and public health reporting) may use this information to normalize variants over time or across sources. However, these data should not be used to dynamically correct/change variant representations for clinical use outside of the laboratory , due to insufficient information.

Implementers should be aware that semantic equivalency of results of genetic variants cannot be guaranteed unless there is an agreed upon standard between sending and receiving systems.

This FHIR genomics work is based on work of the HL7 Clinical Genomics Workgroup and modeled based on the Domain Analysis Model and SMART on FHIR Genomics as published in JAMIA 2015 (http://jamia.oxfordjournals.org/content/early/2015/07/21/jamia.ocv045.long).

The HL7 Clinical Genomics Work Group emphasizes the importance of transmitting structured genetic findings within the clinical, translational, and research environments fully integrated with other clinical data, in order to drive outcomes analysis, operational decision making, discovery research, and public health reporting. The standard doesn't currently cover the reporting of clinically relevant negative or wild type results within genetic data portion of the message.

Here is the document of HL7 Version 3 Domain Analysis Model where the examples used in genetics profile are from (Page 5).

10.1.15.3 Interpretation

Extension - geneticsInterpretation points to an Observation entity. In this Observation entity, Observation.component is used for recording clinical interpretations for variation.

Here is a LOINC panel that could be supported by Observation.component:

LOINC # | Component | Description/Comments
51963-7 | Medication Assessed | A coded medication accessed in a pharmacogenetic test (recommend RxNorm).
51964-5 | Drug Efficacy Analysis Overall Interpretation | Overall predicted phenotype for drug efficacy for all DNA Sequence Variations identified in a single case. LOINC Answer List values can be seen in table below.
51967-8 | Genetic disease assessed | A coded disease which is associated with the region of DNA covered by the genetic test (recommend SNOMED).
51969-4 | Genetic analysis summary report | Narrative report in disease diagnostic-based format, which is used for pharmacogenomic reporting as well and disease risk or diagnosis. These reports currently follow the same formatting recommendations.
51971-0 | Drug metabolism analysis overall interpretation | Overall predicted phenotype for drug metabolism for all DNA Sequence Variations identified in a single case. LOINC Answer List values can be seen in table below.
53039-4 | Genetic Disease Analysis Overall Carrier Interpretation | Carrier Identification interpretation of all identified DNA Sequence Variations along with any known clinical information for the benefit of aiding clinicians in understanding the results overall. LOINC Answer List values can be seen in table below.

LOINC Code | Sequence | Answer text | LOINC answer code
51964-5 | 1 | Responsive | LA6677-4
2 | Resistant | LA6676-6
3 | Negative | LA6577-6
4 | Inconclusive | LA9663-1
5 | Failure | LA9664-9
51971-0 | 1 | Ultrarapid metabolizer | LA10315-2
2 | Extensive metabolizer | LA10316-0
3 | Intermediate metabolizer | LA10317-8
4 | Poor metabolizer | LA9657-3
5 | Inconclusive | LA9663-1
53039-4 | 1 | Carrier | LA10314-5
2 | Negative | LA6577-6
3 | Inconclusive | LA9663-1
4 | Failure | LA9664-9

In addition, in this Observation entity, Observation.related is used to link itself with Observation-genetics profile. Observation.related.type is 'derived-from' as it provides clinical interpretation for variation recoreded in Observation-genetics entity.

10.1.15.4 Content

Profiles:
Observation-Genetics | Describes how the observation resource is used to report structured genetic test results
Extensions:
observation-geneticsGene | Gene : A region (or regions) that includes all of the sequence elements necessary to encode a functional transcript. A gene may include regulatory regions, transcribed regions and/or other functional sequence regions (SO:0000704). This element is the official gene symbol approved by the HGNC, which is a short abbreviated form of the gene name (HGNC). LOINC Code: (48018-6).
observation-geneticsDNARegionName | DNARegionName : A human readable name for the region of interest. Typically Exon #, Intron # or other. NOTE: This is not standardized and is mainly for convenience and display purposes. LOINC Code: (47999-8).
observation-geneticsCopyNumberEvent | CopyNumberEvent : A variation that increases or decreases the copy number of a given region (SO:0001019). Values: amplification/deletion/LOH.
observation-geneticsGenomicSourceClass | GenomicSourceClass : Source of sample used to determine the sequence in sequencing lab -- germline, somatic, prenatal. LOINC Code: (48002-0).
observation-geneticsInterpretation | Interpretation : Clinical Interpretations for variant. It's a reference to an Observation resource.

observation-geneticsVariant | Variant : Variant information.
observation-geneticsAminoAcidChange | AminoAcidChange : AminoAcidChange information.
observation-geneticsAllele | Allele : Allele information.
observation-geneticsAncestry | Ancestry : Ancestry information.
observation-geneticsPhaseSet | PhaseSet : Phase set information.
Examples:
Genetics 1 | Genetics example 1
Genetics 2 | Genetics example 2
Genetics 3 | Genetics example 3
Genetics 4 | Genetics example 4
Genetics 5 | Genetics example 5
Diplotype(with Haplotypes Ref) | Example of diplotype data (with haplotypes observation as reference)
Haplotype | Example of haplotype data that is basis of a diplotype data
Haplotype-2 | Example of another haplotype data that is basis of a diplotype data
Phenotype | Example of phenotype data
TPMT-diplotype | Example of a TPMT diplotype that link to two TPMT haplotype observations
TPMT-haplotype-one | Example of a TPMT haplotype observation
TPMT-haplotype-two | Example of another TPMT haplotype observation
Breast Cancer genetic test | Example of breast cancer genetic observation

10.1.15.5 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
amino-acid-change | string | HGVS Protein Change | | XML / JSON
dna-variant | string | HGVS DNA variant | | XML / JSON
gene-amino-acid-change | string | HGNC gene symbol and HGVS Protein change | | XML / JSON
gene-dnavariant | string | HGNC gene symbol and HGVS DNA Variant | | XML / JSON
gene-identifier | token | HGNC gene symbol and identifier | | XML / JSON

10.1.15.6 StructureDefinition: Observation-Genetics

URL: https://hl7.org/fhir/R4/observation-genetics.html

10.1.15.6 StructureDefinition: Observation-Genetics

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/observation-genetics`

Describes how the observation resource is used to report structured genetic test results

This profile was published on Mon, Mar 14, 2016 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

10.1.15.6.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Observation.

Summary

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/observation-geneticsGene
- http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
- http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
- http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
- http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
- http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
- http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
- http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele

- http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
- http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..*
observation-geneticsGene | | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: H g n c (preferred)
observation-geneticsDNARegionName | | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsCopyNumberEvent | | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsInterpretation | | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
observation-geneticsVariant | | 0..1 | (Complex) | Variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
observation-geneticsAminoAcidChange | | 0..1 | (Complex) | AminoAcidChange URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
observation-geneticsAllele | | 0..1 | (Complex) | Allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele
observation-geneticsAncestry | | 0..1 | (Complex) | Ancestry URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
observation-geneticsPhaseSet | | 0..* | (Complex) | Phase set URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
observation-geneticsGene | I | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: H g n c (preferred)
observation-geneticsDNARegionName | I | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsCopyNumberEvent | I | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | I | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsInterpretation | I | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
observation-geneticsVariant | I | 0..1 | (Complex) | Variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
observation-geneticsAminoAcidChange | I | 0..1 | (Complex) | AminoAcidChange URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
observation-geneticsAllele | I | 0..1 | (Complex) | Allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele
observation-geneticsAncestry | I | 0..1 | (Complex) | Ancestry URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
observation-geneticsPhaseSet | I | 0..* | (Complex) | Phase set URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | ΣI | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | 0..1 | | Actual result
valueQuantity | | | Quantity

valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(Observation | QuestionnaireResponse | MolecularSequence) | Related resource that belongs to the Observation group
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence) | Related measurements the observation is made from
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- Observation-genetics -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Type of observation (code / type)   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
```

```
    What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
    Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
    Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
    Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|integer|
   Range|Ratio|SampledData|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
     Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
     High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
     Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(Observation)
    Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
    Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
     Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Observation-genetics
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // HGNC gene symbol // C?
        // from Element: extension
      },
      { // DNA region name // C?
        // from Element: extension
      },
      { // Copy number variation // C?
        // from Element: extension
      },
      { // Genomic source class // C?
        // from Element: extension
      },
      { // Clinical interpretations for variant // C?
        // from Element: extension
      },
      { // Variant // C?
        // from Element: extension
      },
      { // AminoAcidChange // C?
        // from Element: extension
      },
      { // Allele // C?
        // from Element: extension
      },
      { // Ancestry // C?
        // from Element: extension
      },
      { // Phase set // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
```

```
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! Type of observation (code / type)
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Actual result. One of these 11:
    "valueQuantity" : { Quantity }, // C?Actual result
    "valueCodeableConcept" : { CodeableConcept }, // C?Actual result
    "valueString" : "<string>", // C?Actual result
    "valueBoolean" : <boolean>, // C?Actual result
    "valueInteger" : <integer>, // C?Actual result
    "valueRange" : { Range }, // C?Actual result
    "valueRatio" : { Ratio }, // C?Actual result
    "valueSampledData" : { SampledData }, // C?Actual result
    "valueTime" : "<time>", // C?Actual result
    "valueDateTime" : "<dateTime>", // C?Actual result
    "valuePeriod" : { Period }, // C?Actual result
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }] // C?Provides guide for interpretation
    "hasMember" : [{ Reference(Observation) }], // C?Related resource that belongs to the Observation group
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/observation-geneticsGene
- http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
- http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
- http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
- http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
- http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
- http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
- http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele
- http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
- http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		
observation-geneticsGene		0..1	CodeableConcept	HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: H g n c (preferred)
observation-geneticsDNARegionName		0..1	string	DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsCopyNumberEvent		0..1	CodeableConcept	Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass		0..1	CodeableConcept	Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsInterpretation		0..1	Reference(Observation)	Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
observation-geneticsVariant		0..1	(Complex)	Variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
observation-geneticsAminoAcidChange		0..1	(Complex)	AminoAcidChange URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
observation-geneticsAllele		0..1	(Complex)	Allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele
observation-geneticsAncestry		0..1	(Complex)	Ancestry URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
observation-geneticsPhaseSet		0..*	(Complex)	Phase set URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet

Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | Measurements and simple assertions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
observation-geneticsGene | I | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: H g n c (preferred)
observation-geneticsDNARegionName | I | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsCopyNumberEvent | I | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | I | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsInterpretation | I | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
observation-geneticsVariant | I | 0..1 | (Complex) | Variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
observation-geneticsAminoAcidChange | I | 0..1 | (Complex) | AminoAcidChange URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
observation-geneticsAllele | I | 0..1 | (Complex) | Allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele
observation-geneticsAncestry | I | 0..1 | (Complex) | Ancestry URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
observation-geneticsPhaseSet | I | 0..* | (Complex) | Phase set URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | ΣI | 1..1 | CodeableConcept | Type of observation (code / type) Binding: LOINC Codes (example)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | 0..1 | | Actual result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant

high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(Observation | QuestionnaireResponse | MolecularSequence) | Related resource that belongs to the Observation group
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence) | Related measurements the observation is made from
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Observation-genetics -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Type of observation (code / type)   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|integer|
   Range|Ratio|SampledData|time|dateTime|Period Actual result --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
```

```
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Observation-genetics
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // HGNC gene symbol // C?
        // from Element: extension
      },
      { // DNA region name // C?
        // from Element: extension
      },
      { // Copy number variation // C?
        // from Element: extension
      },
      { // Genomic source class // C?
        // from Element: extension
      },
      { // Clinical interpretations for variant // C?
        // from Element: extension
      },
      { // Variant // C?
        // from Element: extension
      },
      { // AminoAcidChange // C?
        // from Element: extension
      },
      { // Allele // C?
        // from Element: extension
      },
      { // Ancestry // C?
        // from Element: extension
      },
      { // Phase set // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! Type of observation (code / type)
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Actual result. One of these 11:
    "valueQuantity" : { Quantity }, // C?Actual result
    "valueCodeableConcept" : { CodeableConcept }, // C?Actual result
    "valueString" : "<string>", // C?Actual result
    "valueBoolean" : <boolean>, // C?Actual result
    "valueInteger" : <integer>, // C?Actual result
    "valueRange" : { Range }, // C?Actual result
    "valueRatio" : { Ratio }, // C?Actual result
    "valueSampledData" : { SampledData }, // C?Actual result
```

```
  "valueTime" : "<time>", // C?Actual result
  "valueDateTime" : "<dateTime>", // C?Actual result
  "valuePeriod" : { Period }, // C?Actual result
  "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
  "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
  "note" : [{ Annotation }], // C?Comments about the observation
  "bodySite" : { CodeableConcept }, // C?Observed body part
  "method" : { CodeableConcept }, // C?How it was done
  "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
  "device" : { Reference(Device) }, // C?(Measurement) Device
  "referenceRange" : [{ BackboneElement }] // C?Provides guide for interpretation
  "hasMember" : [{ Reference(Observation) }], // C?Related resource that belongs to the Observation group
  "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
  "component" : [{ BackboneElement }], // C?Component results
}
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.15.6.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | LOINC Codes | example | LOINC Codes
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | LOINC Codes | example | LOINC Codes
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.15.6.3 Constraints

Id | Path | Details | Requirements
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()

10.1.15.13 Observation-genetic-Observation-gene-identifier

URL: https://hl7.org/fhir/R4/observation-genetic-Observation-gene-identifier.html

10.1.15.13 Observation-genetic-Observation-gene-identifier

Search Parameter gene-identifier (type: token) defined as part of the observation-genetic Profile.

Orders and Observations | Maturity Level: 0 | Informative

HGNC gene symbol and identifier

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-gene-identifier
• Paths :
• Targets :

10.1.15.14 Observation-genetic-Observation-dna-variant

URL: https://hl7.org/fhir/R4/observation-genetic-Observation-dna-variant.html

10.1.15.14 Observation-genetic-Observation-dna-variant

Search Parameter dna-variant (type: string) defined as part of the observation-genetic Profile.

Orders and Observations | Maturity Level: 0 | Informative

HGVS DNA variant

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-dna-variant
• Paths :
• Targets :

10.1.15.15 Observation-genetic-Observation-amino-acid-change

URL: https://hl7.org/fhir/R4/observation-genetic-Observation-amino-acid-change.html

10.1.15.15 Observation-genetic-Observation-amino-acid-change

Search Parameter amino-acid-change (type: string) defined as part of the observation-genetic Profile.

Orders and Observations | Maturity Level: 0 | Informative

HGVS Protein Change

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-amino-acid-change

• Paths :

• Targets :

10.1.15.16 Observation-genetic-Observation-gene-dnavariant

URL: https://hl7.org/fhir/R4/observation-genetic-Observation-gene-dnavariant.html

10.1.15.16 Observation-genetic-Observation-gene-dnavariant

Search Parameter gene-dnavariant (type: string) defined as part of the observation-genetic Profile.

Orders and Observations | Maturity Level: 0 | Informative

HGNC gene symbol and HGVS DNA Variant

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-gene-dnavariant

• Paths :

• Targets :

10.1.15.17 Observation-genetic-Observation-gene-amino-acid-change

URL: https://hl7.org/fhir/R4/observation-genetic-Observation-gene-amino-acid-change.html

10.1.15.17 Observation-genetic-Observation-gene-amino-acid-change

Search Parameter gene-amino-acid-change (type: string) defined as part of the observation-genetic Profile.

Orders and Observations | Maturity Level: 0 | Informative

HGNC gene symbol and HGVS Protein change

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/observation-genetic-Observation-gene-amino-acid-change

• Paths :

• Targets :

10.1.16 observation-vitalsigns

URL: https://hl7.org/fhir/R4/observation-vitalsigns.html

10.1.16 observation-vitalsigns

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.17 Introduction

Vital signs will be one of the first areas where there is a need for a single, global vocabulary to allow for ubiquitous access and re-use. Particularly with the use of wearables by patients where they want to/need to share information from those devices. To meet this need there must be a consistent vocabulary and a common syntax to achieve semantic interoperability. The FHIR Vital Signs profile sets minimum expectations for the Observation resource to record, search and fetch the vital signs associated with a patient that include the primary vital signs plus additional measurements such as height, weight and BMI. Support for basic mandatory searching of resources is defined below in the Quick Start section. When a FHIR implementation supports any of the vital signs listed below, the implementation SHALL conform to this profile for the vital sign observation.

These requirements were originally developed, balloted, and published in FHIR DSTU2 as part of the ONC sponsored Data Access Framework (DAF) project and were subsequently updated to define the minimum mandatory conformance requirements needed for accessing patient data as defined by the Argonaut pilot implementations.

10.1.18 Scope and Usage

Example Usage Scenarios:

The following are example usage scenarios for this profile:

• Query for vital signs of a particular patient

10.1.18.0.0.1 Mandatory Data Elements and Terminology

The following data-elements are mandatory (i.e. data SHALL be present). These are presented below in a simple human-readable explanation. Profile-specific guidance and valid examples are provided as well. Note that many of the examples capture more than the minimum required. The links to the Profile Definitions provide the formal views of the profile content, descriptions, mappings and the StructureDefinitions in JSON and XML.

Each Observation must have:

• a status

• a category code of 'vital-signs'

• a "magic value" which tells you what is being measured LOINC was chosen for the "magic values" because this aligns with the most countries, but it can be treated as simply a fixed core set of common codes to communicate basic vital signs. Implementers that need to use a different code system can still map accordingly.

• a patient

• a time indicating when the measurement was taken

• a numeric result value and standard UCUM unit which is taken from the Unit Code column in the table below. note: if there is no numeric result then you have to supply a reason

10.1.19 Formal View of Profile Content

Vital Signs Profile : Link to the formal definition views for the vital signs listed in this table.

• The table below represents a minimum set of vital sign concepts, the required codes ("magic values"), and UCUM units of measure codes used for representing vital signs observations. These are extensible bindings and require that when a system supports any of these vital signs concepts, they must represent them using these codes. In addition, if you have a blood pressure observation, you must have both a systolic and a diastolic component, though one or both may have dataAbsentReason instead of a value.

• The first column of this table links to the formal views of the individual profile for each vital sign.

• If a more specific code or another code system is recorded or required, implementers must support both the values (LOINC) listed below and the translated code - e.g. method specific LOINC codes, SNOMED CT concepts, system specific (local) codes.

• In addition the implementer may choose to provide alternate codes in addition to the standard codes defined here. The examples illustrate using other codes as translations.

• Other profiles may make rules about which vital sign must be present or must be present as part of a panel or expand the list to include other vital signs. For implementers using LOINC, optional qualifier codes are provided in the notes below.

Profile Name | "Magic Value" (LOINC) | LOINC Name and Comments | UCUM Unit Code | Examples

Vital Signs Panel | 85353-1 | Vital signs, weight, height, head circumference, oxygen saturation and BMI panel - It represent a panel of vital signs listed in this table. All members of the panel are optional and note that querying for the panel may miss individual results that are not part of the actual panel. When used, Observation.valueQuantity is not present; instead, related links (with type=has-member) reference the vital signs observations (e.g. respiratory rate, heart rate, BP, etc.). This code replaces the deprecated code 8716-3 - Vital signs which is used in the Argonaut Data Query Implementation Guide. | - | Vital Signs Panel Example

Respiratory Rate | 9279-1 | Respiratory Rate | /min | Respiratory Rate Example

Heart rate | 8867-4 | Heart rate - To supplement this vital sign observation, 8887-2 - Heart rate device type MAY be included as an additional observation. | /min | Heart Rate Example

Oxygen saturation | 2708-6 | Oxygen saturation in Arterial blood - This code replaces 59408-5 Oxygen saturation in Arterial blood by Pulse oximetry which MAY be included as an additional observation code. | % | Oxygen Saturation Example

Body temperature | 8310-5 | Body temperature - To supplement this vital sign observation, 8327-9 - Body temperature measurement site (oral, forehead, rectal, etc.) and 8326-1 -Type of body temperature device MAY be used as additional observations. | Cel, [degF] | Body Temperature Example

Body height | 8302-2 | Body height - To supplement this vital sign observation, 8306-3 -Body height - lying (i.e., body length - typically used for infants) MAY be included as an additional observation code. | cm, [in_i] | Body height Example

Head circumference | 9843-4 | Head Occipital-frontal circumference | cm, [in_i] | Head Circumference Example

Body weight | 29463-7 | Body weight - To supplement this vital sign observation, 8352-7 - Clothing worn during measure and 8361-8 - Body position with respect to gravity MAY be included as additional observation codes. | g, kg,[lb_av] | Body Weight Example

Body mass index | 39156-5 | Body mass index (BMI) [Ratio] | kg/m2 | Body Mass Example

Blood pressure systolic and diastolic | 85354-9 | Blood pressure panel with all children optional - This is a component observation. It has no value in Observation.valueQuantity and contains at least one component (systolic and/or diastolic). To supplement this vital sign observation, 8478-0 - Mean blood pressure, 8357-6 - Blood pressure method, 41904-4 - Blood pressure measurement site, 8358-4 - Blood pressure device cuff size, 41901-0 - Type of blood pressure device MAY be used as additional observations. | - | Blood Pressure Example, Blood Pressure Example with missing Diastolic measurement

Systolic blood pressure | 8480-6 | Systolic blood pressure - Observation.component code for a blood pressure Observation | mm[Hg] | Blood Pressure Example
Diastolic blood pressure | 8462-4 | Diastolic blood pressure - Observation.component code for a blood pressure Observation | mm[Hg] | Blood Pressure Example

10.1.20 Quick Start

Below is an overview of required search and read operations

Clients

• A client has connected to a server and fetched all of a patient's vital signs by searching by category using GET [base]/Observation?patient=[id]&category=vital-signs.

• A client has connected to a server and fetched all of a patient's vital signs searching by category code and date range using GET [base]/Observation?patient=[id]&category=vital-signs&date=[date]{&date=[date]}.

• A client has connected to a server and fetched any of a patient's vital signs by searching by one or more of the codes listed above using GET [base]/Observation?patient=[id]&code[vital sign LOINC{,LOINC2,LOINC3,...}].

• A client SHOULD be capable of connecting to a server and fetching any of a patient's vital signs searching by one or more of the codes listed above and date range using GET [base]/Observation?patient=[id]&code=[LOINC{,LOINC2...}]vital-signs&date=[date]{&date=[date]}.

Servers

• A server is capable of returning all of a patient's vital signs that it supports using GET [base]/Observation?patient=[id]&category=vital-signs.

• A server is capable of returning all of a patient's vital signs queried by date range using GET [base]/Observation?patient=[id]&category=vital-signs&date=[date]{&date=[date]}.

• A server is capable of returning any of a patient's vital signs queried by one or more of the codes listed above using GET [base]/Observation?patient=[id]&code[vital sign LOINC{,LOINC2,LOINC3,...}].

• A server SHOULD be capable of returning any of a patient's vital signs queried by one or more of the codes listed above and date range using GET [base]/Observation?patient=[id]&code=[LOINC{,LOINC2...}]vital-signs&date=[date]{&date=[date]}.

• A server has ensured that every API request includes a valid Authorization token, supplied via:Authorization: Bearer {server-specific-token-here}

• A server has rejected any unauthorized requests by returning an HTTP 401 Unauthorized response code.

10.1.20.0.1 GET [base]/Observation?patient=[id]&category=vital-signs

Example: Search for all Vital Signs measurements for a patient

GET [base]/Observation?patient=1186747&category=vital-signs

Support: Mandatory to support search by category code.

Implementation Notes: Search based on vital sign category code. This search fetches a bundle of all Observation resources with category 'vital-signs' for the specified patient (how to search by reference) and (how to search by token). The table above is the minimum set, additional vital signs are allowed.

Response Class:

• (Status 200): successful operation

• (Status 400): invalid parameter

• (Status 401/4xx): unauthorized request

• (Status 403): insufficient scope

10.1.20.0.2 GET [base]/Observation?patient=[id]&code=[vital sign LOINC{,LOINC2,LOINC3,...}]

Example: Search for all heart rate observations for a patient:

GET [base]/Observation?patient=1186747&code=8867-4

Example: Search for all heart rate, respiratory rate and blood pressure observations for a patient:

GET [base]/Observation?patient=1186747&code=8867-4,9279-1,85354-9

Support: Mandatory to support search by vital sign LOINC(s) listed above.

Implementation Notes: 1)Search based on vital sign LOINC code(s). This fetches a bundle of all Observation resources for specific vital sign(s) listed in the table above for the specified patient (how to search by reference) and [how to search by token)]. 2) The "code" parameter searches only Observation.code. For example when fetching blood pressures the resource will be only be returned when the search is based on 85354-9(Systolic and Diastolic BP). Using the component codes 8480-6(Systolic BP) or 8462-4 (Diastolic BP) will not return the resource . In order to search both Observation.code and Observation.component.code in a single query, use the "combo-code" search parameter.

Response Class:

• (Status 200): successful operation

• (Status 400): invalid parameter

• (Status 401/4xx): unauthorized request

• (Status 403): insufficient scope

10.1.20.0.3 GET [base]/Observation?patient=[id]&category=vital-signs&date=[date]{&date=[date]}

Example: Find all the blood pressures after 2015-01-14

GET [base]/Observation?patient=555580&code=85354-9&date=ge2015-01-14

Support: Mandatory to support search by category code and date

Implementation Notes: Search based on vital sign category code and date. This fetches a bundle of all Observation resources with category 'vital-signs' for the specified patient for a specified time period (how to search by reference) and (how to search by token).

Response Class:

• (Status 200): successful operation

• (Status 400): invalid parameter

• (Status 401/4xx): unauthorized request

• (Status 403): insufficient scope

10.1.20.1 Content

Profiles:
VitalSigns | FHIR Vital Signs Profile

10.1.20.2 StructureDefinition: VitalSigns

URL: https://hl7.org/fhir/R4/vitalsigns.html

10.1.20.2 StructureDefinition: VitalSigns

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/vitalsigns`

FHIR Vital Signs Profile

This profile was published on Fri, Mar 25, 2016 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.20.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary

• Differential Table

• Snapshot Table

• XML Template

• JSON Template

• All

This structure is derived from Observation.

Summary

"#### Complete Summary of the Mandatory Requirements

One status in Observation.statuswhich has a required binding to:

• ObservationStatus value set.

A category in Observation.category which must have:

• a fixed Observation.category.coding.system="http://terminology.hl7.org/CodeSystem/observation-category"

• a fixed Observation.category.coding.code= "vital-signs"

A code in Observation.code

• a fixed Observation.code.coding.system= "http://loinc.org""

• a LOINC code in Observation.code.coding.code which has an extensible binding to: Vital Signs value set.

One patient in Observation.subject

A date and time in effectiveDateTime or effectivePeriod

Either one Observation.value[x] or, if there is no value, one code in Observation.DataAbsentReason

• if a vital sign measure then: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code which has an required binding to the Vital Signs Units value set.

• Observation.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group component observations (Note: See the comments regarding blood pressure in the table above), one or more Observation.component.code each of which must have:

• a fixed Observation.component.code.coding.system =""http://loinc.org"" a LOINC code in Observation.code.coding.code which has an extensible binding to: Vital Signs Units value set.

Either one Observation.component.valueQuantity or, if there is no value, one code in Observation.component.DataAbsentReason

• Observation.component.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group observations, one or more reference to Vitals Signs Observations in Observation.related.target

• a fixed Observation.related.type= "has-member""

Mandatory: 9 elements (+1 nested mandatory element) Must-Support: 15 elements Fixed Value: 2 elements

Slices

This structure defines the following Slices:

• The element Observation.category is sliced based on the values of value:coding.code, value:coding.system

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or hasMember element then either a value[x] or a data absent reason must be present.
status | S | 1..1 | code | Binding: ObservationStatus (required)
category | S | | (Slice Definition) | Slice: Unordered, Open by value:coding.code, value:coding.system
category:VSCat | S | 1..1 | CodeableConcept
coding | S | 1..* | Coding
system | S | 1..1 | uri | Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
code | S | 1..1 | code | Fixed Value: vital-signs
code | S | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | S | 1..1 | Reference(Patient)
effective[x] | SI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
value[x] | SI | 0..1 | Quantity, CodeableConcept, string, boolean, integer, Range, Ratio, SampledData, time, dateTime, Period | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
dataAbsentReason | SI | 0..1 | CodeableConcept
hasMember | | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile)
component | SI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
code | S | 1..1 | CodeableConcept | Binding: Vital Signs (extensible)
value[x] | SI | 0..1 | Quantity, CodeableConcept, string, boolean, integer, Range, Ratio, SampledData, time, dateTime, Period | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or hasMember element then either a value[x] or a data absent reason must be present.
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)

category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | 0..1 | | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean

valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-vitalsigns -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code><!--  1..1 CodeableConcept Coded Responses from C-CDA Vital Sign Results   --></code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|integer|
   Range|Ratio|SampledData|time|dateTime|Period
     Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could
    use other datatypes such as CodeableConcept. --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
```

```
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-vitalsigns
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Coded Responses from C-CDA Vital Sign Results
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept.. One of these 11:
    "valueQuantity" : { Quantity }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueCodeableConcept" : { CodeableConcept }, // C?Vital Signs value are recorded using the Quantity data type. For
    supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueString" : "<string>", // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueBoolean" : <boolean>, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueInteger" : <integer>, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueRange" : { Range }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueRatio" : { Ratio }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueSampledData" : { SampledData }, // C?Vital Signs value are recorded using the Quantity data type. For
    supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueTime" : "<time>", // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueDateTime" : "<dateTime>", // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valuePeriod" : { Period }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "dataAbsentReason" : { CodeableConcept } // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from Observation.

Summary

"#### Complete Summary of the Mandatory Requirements

One status in Observation.statuswhich has a required binding to:

• ObservationStatus value set.

A category in Observation.category which must have:

• a fixed Observation.category.coding.system="http://terminology.hl7.org/CodeSystem/observation-category"

• a fixed Observation.category.coding.code= "vital-signs"

A code in Observation.code

• a fixed Observation.code.coding.system= "http://loinc.org""

• a LOINC code in Observation.code.coding.code which has an extensible binding to: Vital Signs value set.

One patient in Observation.subject

A date and time in effectiveDateTime or effectivePeriod

Either one Observation.value[x] or, if there is no value, one code in Observation.DataAbsentReason

• if a vital sign measure then: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code which has an required binding to the Vital Signs Units value set.

• Observation.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group component observations (Note: See the comments regarding blood pressure in the table above), one or more Observation.component.code each of which must have:

• a fixed Observation.component.code.coding.system =""http://loinc.org"" a LOINC code in Observation.code.coding.code which has an extensible binding to: Vital Signs Units value set.

Either one Observation.component.valueQuantity or, if there is no value, one code in Observation.component.DataAbsentReason

• Observation.component.DataAbsentReason is bound to Observation Value Absent Reason value set.

When using a panel code to group observations, one or more reference to Vitals Signs Observations in Observation.related.target

• a fixed Observation.related.type= "has-member""

Mandatory: 9 elements (+1 nested mandatory element) Must-Support: 15 elements Fixed Value: 2 elements

Slices

This structure defines the following Slices:

• The element Observation.category is sliced based on the values of value:coding.code, value:coding.system

Differential View

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or hasMember element then either a value[x] or a data absent reason must be present.
status | S | 1..1 | code | Binding: ObservationStatus (required)
category | S | | (Slice Definition) | Slice: Unordered, Open by value:coding.code, value:coding.system
category:VSCat | S | 1..1 | CodeableConcept
coding | S | 1..* | Coding
system | S | 1..1 | uri | Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
code | S | 1..1 | code | Fixed Value: vital-signs
code | S | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | S | 1..1 | Reference(Patient)
effective[x] | SI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
value[x] | SI | 0..1 | Quantity, CodeableConcept, string, boolean, integer, Range, Ratio, SampledData, time, dateTime, Period | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
dataAbsentReason | SI | 0..1 | CodeableConcept
hasMember | | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile)
component | SI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
code | S | 1..1 | CodeableConcept | Binding: Vital Signs (extensible)
value[x] | SI | 0..1 | Quantity, CodeableConcept, string, boolean, integer, Range, Ratio, SampledData, time, dateTime, Period | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Profile vs-2: If there is no component or hasMember element then either a value[x] or a data absent reason must be present.
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources

extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Coded Responses from C-CDA Vital Sign Results Binding: Vital Signs (extensible)
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | 0..1 | | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing

extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-vitalsigns -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code><!--  1..1 CodeableConcept Coded Responses from C-CDA Vital Sign Results   --></code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|boolean|integer|
   Range|Ratio|SampledData|time|dateTime|Period
     Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could
    use other datatypes such as CodeableConcept. --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
```

```
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-vitalsigns
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Coded Responses from C-CDA Vital Sign Results
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept.. One of these 11:
    "valueQuantity" : { Quantity }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueCodeableConcept" : { CodeableConcept }, // C?Vital Signs value are recorded using the Quantity data type. For
    supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueString" : "<string>", // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueBoolean" : <boolean>, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueInteger" : <integer>, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueRange" : { Range }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueRatio" : { Ratio }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueSampledData" : { SampledData }, // C?Vital Signs value are recorded using the Quantity data type. For
    supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueTime" : "<time>", // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valueDateTime" : "<dateTime>", // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "valuePeriod" : { Period }, // C?Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept.
    "dataAbsentReason" : { CodeableConcept } // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.20.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.20.2.3 Constraints

Id | Path | Details | Requirements
vs-2 | Observation | If there is no component or hasMember element then either a value[x] or a data absent reason must be present. Expression : (component.empty() and hasMember.empty()) implies (dataAbsentReason.exists() or value.exists())
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.21 observation-bodyweight

URL: https://hl7.org/fhir/R4/observation-bodyweight.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body weight.

10.1.21 observation-bodyweight

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.21.1 Content

Profiles:
BodyWeight | FHIR Body Weight Profile

10.1.21.2 StructureDefinition: BodyWeight

URL: https://hl7.org/fhir/R4/bodyweight.html

10.1.21.2 StructureDefinition: BodyWeight

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodyweight`

FHIR Body Weight Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.21.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from observation-vitalsigns.

Summary

10.1.21.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '29463-7' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'kg', 'g', or '[lb_av]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Weight Profile
code | | 1..1 | CodeableConcept | Body Weight
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BodyWeightCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 29463-7
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Weight Units (required)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Weight Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Weight Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BodyWeightCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url

system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 29463-7
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Weight Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-bodyweight -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Weight -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
```

```
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-bodyweight
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Weight
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.21.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '29463-7' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'kg', 'g', or '[lb_av]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints

Observation | | 0..* | | FHIR Body Weight Profile
code | | 1..1 | CodeableConcept | Body Weight
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BodyWeightCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 29463-7
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Weight Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Weight Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Weight Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BodyWeightCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 29463-7
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this

value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Weight Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-bodyweight -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
```

```
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Weight -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
     --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
     --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
     --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-bodyweight
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
```

```
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Weight
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.21.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.value[x].code | Body Weight Units | required | Body Weight Units
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.21.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.22 observation-vitalspanel

URL: https://hl7.org/fhir/R4/observation-vitalspanel.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital signs panel.

10.1.22 observation-vitalspanel

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.22.1 Content

Profiles:
VitalsPanel | FHIR Vital Signs Panel Profile

10.1.22.2 StructureDefinition: VitalsPanel

URL: https://hl7.org/fhir/R4/vitalspanel.html

10.1.22.2 StructureDefinition: VitalsPanel

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/vitalspanel`

FHIR Vital Signs Panel Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.22.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from observation-vitalsigns.

Summary

10.1.22.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '85353-1' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value
• No value in Observation.value[x]
• At Least one Vitals Sign Observation reference in Observation.hasMember

Mandatory: 1 element Must-Support: 1 element Fixed Value: 2 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Vital Signs Panel Profile
code		1..1	CodeableConcept	Vital Signs Panel
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:VitalsPanelCode		0..*	Coding	
system		0..1	uri	Fixed Value: http://loinc.org
code		0..1	code	Fixed Value: 85353-1
value[x]		0..0		
hasMember	S	1..*	Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile)	

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Vital Signs Panel Profile
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation

contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Vital Signs Panel Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:VitalsPanelCode | ΣI | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 85353-1
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | SΣI | 1..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing

extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-vitalspanel -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Vital Signs Panel -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  0..* Coding  <!--  0..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  0..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..0 Quantity|CodeableConcept|string|boolean|integer|
   Range|Ratio|SampledData|time|dateTime|Period
     Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could
    use other datatypes such as CodeableConcept. --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
```

```
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  1..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-vitalspanel
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Vital Signs Panel
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
// value[x]: Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept.. One of these 11:
    "dataAbsentReason" : { CodeableConcept } // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C? R! Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.22.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '85353-1' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• No value in Observation.value[x]

• At Least one Vitals Sign Observation reference in Observation.hasMember

Mandatory: 1 element Must-Support: 1 element Fixed Value: 2 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Vital Signs Panel Profile
code | | 1..1 | CodeableConcept | Vital Signs Panel
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:VitalsPanelCode | | 0..* | Coding
system | | 0..1 | uri | Fixed Value: http://loinc.org
code | | 0..1 | code | Fixed Value: 85353-1
value[x] | | 0..0
hasMember | S | 1..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Vital Signs Panel Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Vital Signs Panel Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:VitalsPanelCode | ΣI | 0..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 0..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 0..1 | code | Symbol in syntax defined by the system Fixed Value: 85353-1
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)

interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | SΣI | 1..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-vitalspanel -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Vital Signs Panel -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  0..* Coding  <!--  0..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  0..1 Identity of the terminology system -->
```

```
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  0..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <value[x]><!--  0..0 Quantity|CodeableConcept|string|boolean|integer|
   Range|Ratio|SampledData|time|dateTime|Period
     Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could
    use other datatypes such as CodeableConcept. --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  1..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-vitalspanel
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Vital Signs Panel
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
```

```
// value[x]: Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept.. One of these 11:
    "dataAbsentReason" : { CodeableConcept } // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C? R! Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.22.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.22.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.23 observation-bodyheight

URL: https://hl7.org/fhir/R4/observation-bodyheight.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body height.

10.1.23 observation-bodyheight

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.23.1 Content

Profiles:
BodyHeight | FHIR Body Height Profile

10.1.23.2 StructureDefinition: BodyHeight

URL: https://hl7.org/fhir/R4/bodyheight.html

10.1.23.2 StructureDefinition: BodyHeight

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodyheight`

FHIR Body Height Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.23.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary

• Differential Table

• Snapshot Table

• XML Template

• JSON Template

• All

This structure is derived from observation-vitalsigns.

Summary

10.1.23.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '8302-2' 8306-3 -Body height - lying (i.e., body length - typically used for infants) MAY be included as an additional observation code - Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'cm', or '[in_i]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Body Height Profile
code		1..1	CodeableConcept	Body Height
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:BodyHeightCode		1..1	Coding	
system		1..1	uri	Fixed Value: http://loinc.org
code		1..1	code	Fixed Value: 8302-2
valueQuantity		0..1	Quantity	
value	S	1..1	decimal	
unit	S	1..1	string	
system	S	1..1	uri	Fixed Value: http://unitsofmeasure.org
code	S	1..1	code	Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Body Height Profile
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..*	Identifier	Business Identifier for observation
basedOn	ΣI	0..*	Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	Fulfills plan, proposal or order
partOf	ΣI	0..*	Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)	Part of referenced event
status	?!SΣI	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI		(Slice Definition)	Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat	SI	1..1	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred)
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations Slice: Unordered, Open by value:url
coding	SΣI	1..*	Coding	Code defined by a terminology system
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations Slice: Unordered, Open by value:url
system	SΣI	1..1	uri	Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category

version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Height Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BodyHeightCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8302-2
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!SΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean

valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-bodyheight -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Height -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
```

```
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-bodyheight
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Height
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.23.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '8302-2' 8306-3 -Body height - lying (i.e., body length - typically used for infants) MAY be included as an additional observation code - Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'cm', or '[in_i]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Height Profile
code | | 1..1 | CodeableConcept | Body Height
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BodyHeightCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8302-2
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Height Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Height Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BodyHeightCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8302-2
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user

text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-bodyheight -->

<Observation xmlns="http://hl7.org/fhir"
>
```

```
 <!-- from Element: extension -->
<id><!-- 0..1 string Logical id of this artifact --></id>
<meta><!--  0..1 Meta Metadata about the resource --></meta>
<implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
<language value="[code]"/><!--  0..1 Language of the resource content   -->
<text><!--  0..1 Narrative
    Text summary of the resource, for human interpretation --></text>
<contained><!-- 0..* Resource Contained, inline Resources --></contained>
<modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
<identifier><!--  0..* Identifier
    Business Identifier for observation --></identifier>
<basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
<partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
<status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
<-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
<category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
 <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
  <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
  <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
  <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
  <display value="[string]"/><!--  0..1 Representation defined by the system -->
  <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
 </coding>
 <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
</category>
<code>  1..1 CodeableConcept  <!--  1..1 Body Height -->
 <-- coding sliced by value:code, value:system  in the specified orderOpen-->
 <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
  <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
  <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
  <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
  <display value="[string]"/><!--  0..1 Representation defined by the system -->
  <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
 </coding>
 <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
</code>
<subject><!--  1..1 Reference(Patient)
    Who and/or what the observation is about --></subject>
<focus><!--  0..* Reference(Resource)
    What the observation is about, when it is not about the subject of record --></focus>
<encounter><!--  0..1 Reference(Encounter)
    Healthcare event during which this observation is made --></encounter>
<effective[x]><!--  1..1 dateTime|Period
    Often just a dateTime for Vital Signs --></effective[x]>
<issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
<performer><!--  0..* Reference(Practitioner)
    Who is responsible for the observation --></performer>
<-- valueQuantity sliced by type:$this  in the specified orderClosed-->
<valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
   observations such as Cuff size could use other datatypes such as CodeableConcept. -->
 <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
 <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
 <unit value="[string]"/><!--  1..1 Unit representation -->
 <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
 <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
</valueQuantity>
<dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
<interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
<note><!--  0..* Annotation Comments about the observation --></note>
<bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
<method><!--  0..1 CodeableConcept How it was done   --></method>
<specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
<device><!--  0..1 Reference(Device) (Measurement) Device --></device>
<referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <low><!--  0..1 Quantity
     Low Range, if relevant --></low>
 <high><!--  0..1 Quantity
     High Range, if relevant --></high>
 <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
 <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
 <age><!--  0..1 Range
     Applicable age range, if relevant --></age>
 <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
</referenceRange>
<hasMember><!--  0..* Reference(QuestionnaireResponse)
    Used when reporting vital signs panel components --></hasMember>
<derivedFrom><!--  0..* Reference(DocumentReference)
    Related measurements the observation is made from --></derivedFrom>
<component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|
   boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
```

```
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-bodyheight
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Height
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.23.2.4 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	preferred	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	Vital Signs	extensible	Vital Signs
Observation.value[x].comparator	QuantityComparator	required	QuantityComparator
Observation.value[x].code	Body Length Units	required	Body Length Units
Observation.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.referenceRange.type	Observation Reference Range Meaning Codes	preferred	Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo	Observation Reference Range Applies To Codes	example	Observation Reference Range Applies To Codes
Observation.component.code	Vital Signs	extensible	Vital Signs
Observation.component.value[x]	Vital Signs Units	required	Vital Signs Units
Observation.component.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.1.23.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.24 observation-resprate

URL: https://hl7.org/fhir/R4/observation-resprate.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign respiratory rate.

10.1.24 observation-resprate

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.24.1 Content

Profiles:
RespRate | FHIR Respiratory Rate Profile

10.1.24.2 StructureDefinition: RespRate

URL: https://hl7.org/fhir/R4/resprate.html

10.1.24.2 StructureDefinition: RespRate

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/resprate`

FHIR Respiratory Rate Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.24.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from observation-vitalsigns.

Summary

10.1.24.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '9279-1' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = '/min'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Respiratory Rate Profile
code | | 1..1 | CodeableConcept | Respiratory Rate
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:RespRateCode | | 1..1 | Coding

system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 9279-1
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Respiratory Rate Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Respiratory Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:RespRateCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 9279-1
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org

code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-resprate -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Respiratory Rate -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
```

```
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-resprate
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Respiratory Rate
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
```

```
  "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
  "issued" : "<instant>", // C?Date/Time this version was made available
  "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
  "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
    { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
  could use other datatypes such as CodeableConcept. // C?
      // from Element: extension
      "extension" : [ //  sliced by value:url  in the specified order, Open ]
      "value" : <decimal>, // C? R! Numerical value (with implicit precision)
      "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
      "unit" : "<string>", // C? R! Unit representation
      "system" : "http://unitsofmeasure.org", // C? R!
      "code" : "/min" // C? R!
    }
  ],
  "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
  "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
  "note" : [{ Annotation }], // C?Comments about the observation
  "bodySite" : { CodeableConcept }, // C?Observed body part
  "method" : { CodeableConcept }, // C?How it was done
  "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
  "device" : { Reference(Device) }, // C?(Measurement) Device
  "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
  "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
  "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
  "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
}
```

This structure is derived from observation-vitalsigns.

Summary

10.1.24.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '9279-1' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = '/min'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Respiratory Rate Profile
code		1..1	CodeableConcept	Respiratory Rate
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:RespRateCode		1..1	Coding	
system		1..1	uri	Fixed Value: http://loinc.org
code		1..1	code	Fixed Value: 9279-1
valueQuantity		0..1	Quantity	
value	S	1..1	decimal	
unit	S	1..1	string	
system	S	1..1	uri	Fixed Value: http://unitsofmeasure.org
code	S	1..1	code	Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Respiratory Rate Profile
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..*	Identifier	Business Identifier for observation
basedOn	ΣI	0..*	Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	Fulfills plan, proposal or order

partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Respiratory Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:RespRateCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 9279-1
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components

derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-resprate -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Respiratory Rate -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
```

```
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-resprate
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Respiratory Rate
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "/min" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
```

```
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.24.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.24.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.25 observation-heartrate

URL: https://hl7.org/fhir/R4/observation-heartrate.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign heart rate.

10.1.25 observation-heartrate

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.25.1 Content

Profiles:
HeartRate | FHIR Heart Rate Profile

10.1.25.2 StructureDefinition: HeartRate

URL: https://hl7.org/fhir/R4/heartrate.html

10.1.25.2 StructureDefinition: HeartRate

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/heartrate`

FHIR Heart Rate Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.25.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template

• JSON Template

• All

This structure is derived from observation-vitalsigns.

Summary

10.1.25.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '8867-4' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = '/min'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Heart Rate Profile
code		1..1	CodeableConcept	Heart Rate
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:HeartRateCode		1..1	Coding	
system		1..1	uri	Fixed Value: http://loinc.org
code		1..1	code	Fixed Value: 8867-4
valueQuantity		0..1	Quantity	
value	S	1..1	decimal	
unit	S	1..1	string	
system	S	1..1	uri	Fixed Value: http://unitsofmeasure.org
code	S	1..1	code	Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Heart Rate Profile
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..*	Identifier	Business Identifier for observation
basedOn	ΣI	0..*	Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	Fulfills plan, proposal or order
partOf	ΣI	0..*	Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)	Part of referenced event
status	?!SΣI	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI		(Slice Definition)	Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat	SI	1..1	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred)
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations Slice: Unordered, Open by value:url
coding	SΣI	1..*	Coding	Code defined by a terminology system
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations Slice: Unordered, Open by value:url
system	SΣI	1..1	uri	Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version	ΣI	0..1	string	Version of the system - if relevant
code	SΣI	1..1	code	Symbol in syntax defined by the system Fixed Value: vital-signs
display	ΣI	0..1	string	Representation defined by the system
userSelected	ΣI	0..1	boolean	If this coding was chosen directly by the user
text	ΣI	0..1	string	Plain text representation of the concept
code	SΣI	1..1	CodeableConcept	Heart Rate Binding: Vital Signs (extensible)
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations Slice: Unordered, Open by value:url
coding	ΣI		(Slice Definition)	Code defined by a terminology system Slice: Unordered, Open by value:code, value:system

coding:HeartRateCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8867-4
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)

referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- observation-heartrate -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Heart Rate -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
```

```
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-heartrate
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Heart Rate
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "/min" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.25.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '8867-4' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = '/min'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Heart Rate Profile
code | | 1..1 | CodeableConcept | Heart Rate
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:HeartRateCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8867-4
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Heart Rate Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Heart Rate Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:HeartRateCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8867-4
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation

value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: /min
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-heartrate -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
```

```
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Heart Rate -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-heartrate
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
```

```
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Heart Rate
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "/min" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.25.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.25.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.26 observation-bodytemp

URL: https://hl7.org/fhir/R4/observation-bodytemp.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body temperature.

10.1.26 observation-bodytemp

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.26.1 Content

Profiles:
BodyTemp | FHIR Body Temperature Profile

10.1.26.2 StructureDefinition: BodyTemp

URL: https://hl7.org/fhir/R4/bodytemp.html

10.1.26.2 StructureDefinition: BodyTemp

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bodytemp`

FHIR Body Temperature Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.26.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from observation-vitalsigns.

Summary

10.1.26.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '8310-5' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'Cel', or '[degF]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Temperature Profile
code | | 1..1 | CodeableConcept | Body Temperature
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BodyTempCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8310-5
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Temperature Units (required)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Temperature Profile

id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Temperature Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BodyTempCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8310-5
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Temperature Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text

id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-bodytemp -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Temperature -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
```

```
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-bodytemp
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Temperature
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
```

```
      "system" : "http://unitsofmeasure.org", // C? R!
      "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
    }
  ],
  "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
  "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
  "note" : [{ Annotation }], // C?Comments about the observation
  "bodySite" : { CodeableConcept }, // C?Observed body part
  "method" : { CodeableConcept }, // C?How it was done
  "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
  "device" : { Reference(Device) }, // C?(Measurement) Device
  "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
  "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
  "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
  "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
}
```

This structure is derived from observation-vitalsigns.

Summary

10.1.26.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '8310-5' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'Cel', or '[degF]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Temperature Profile
code | | 1..1 | CodeableConcept | Body Temperature
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BodyTempCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8310-5
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Temperature Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Temperature Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing

extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Temperature Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BodyTempCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8310-5
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Temperature Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized

code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-bodytemp -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Temperature -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
```

```
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-bodytemp
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Temperature
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.26.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.value[x].code | Body Temperature Units | required | Body Temperature Units
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.26.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.27 observation-headcircum

URL: https://hl7.org/fhir/R4/observation-headcircum.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign head circumference.

10.1.27 observation-headcircum

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.27.1 Content

Profiles:
HeadCircum | FHIR Head Circumference Profile

10.1.27.2 StructureDefinition: HeadCircum

URL: https://hl7.org/fhir/R4/headcircum.html

10.1.27.2 StructureDefinition: HeadCircum

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/headcircum`

FHIR Head Circumference Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.27.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from observation-vitalsigns.

Summary

10.1.27.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '9843-4' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'cm', or '[in_i]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Head Circumference Profile
code | | 1..1 | CodeableConcept | Head Circumference
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:HeadCircumCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 9843-4
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Head Circumference Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Head Circumference Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:HeadCircumCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url

system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 9843-4
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-headcircum -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Head Circumference -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
```

```
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-headcircum
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Head Circumference
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.27.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '9843-4' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'cm', or '[in_i]'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 3 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints

Observation | | 0..* | | FHIR Head Circumference Profile
code | | 1..1 | CodeableConcept | Head Circumference
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:HeadCircumCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 9843-4
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Head Circumference Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Head Circumference Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:HeadCircumCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 9843-4
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this

value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Binding: Body Length Units (required)
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-headcircum -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
```

```
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Head Circumference -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set.   -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
     --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
     --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
     --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-headcircum
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
```

```
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Head Circumference
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "<code>" // C? R! Coded responses from the common UCUM units for vital signs value set.
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.27.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.value[x].code | Body Length Units | required | Body Length Units
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.27.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.28 observation-oxygensat

URL: https://hl7.org/fhir/R4/observation-oxygensat.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign oxygen saturation.

10.1.28 observation-oxygensat

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.28.1 Content

Profiles:
OxygenSat | FHIR Oxygen Saturation Profile

10.1.28.2 StructureDefinition: OxygenSat

URL: https://hl7.org/fhir/R4/oxygensat.html

10.1.28.2 StructureDefinition: OxygenSat

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/oxygensat`

FHIR Oxygen Saturation Profile

This profile was published on Tue, Oct 23, 2018 00:00+1100 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.28.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from observation-vitalsigns.

Summary

"#### Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '2708-6' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = '%'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Oxygen Saturation Profile
code | | 1..1 | CodeableConcept | Oxygen Saturation
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:OxygenSatCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 2708-6
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: %
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Oxygen Saturation Profile
id | Σ | 0..1 | string | Logical id of this artifact

meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Oxygen Saturation Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:OxygenSatCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 2708-6
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: %
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations

modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-oxygensat -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Oxygen Saturation -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
```

```
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-oxygensat
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Oxygen Saturation
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "%" // C? R!
```

```
    }
  ],
  "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
  "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
  "note" : [{ Annotation }], // C?Comments about the observation
  "bodySite" : { CodeableConcept }, // C?Observed body part
  "method" : { CodeableConcept }, // C?How it was done
  "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
  "device" : { Reference(Device) }, // C?(Measurement) Device
  "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
  "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
  "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
  "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
}
```

This structure is derived from observation-vitalsigns.

Summary

"#### Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '2708-6' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = '%'

Mandatory: 3 elements (+4 nested mandatory elements) Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Oxygen Saturation Profile
code		1..1	CodeableConcept	Oxygen Saturation
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:OxygenSatCode		1..1	Coding	
system		1..1	uri	Fixed Value: http://loinc.org
code		1..1	code	Fixed Value: 2708-6
valueQuantity		0..1	Quantity	
value	S	1..1	decimal	
unit	S	1..1	string	
system	S	1..1	uri	Fixed Value: http://unitsofmeasure.org
code	S	1..1	code	Coded responses from the common UCUM units for vital signs value set. Fixed Value: %

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	0..*		FHIR Oxygen Saturation Profile
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..*	Identifier	Business Identifier for observation
basedOn	ΣI	0..*	Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	Fulfills plan, proposal or order
partOf	ΣI	0..*	Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)	Part of referenced event
status	?!SΣI	1..1	code	registered | preliminary | final | amended + Binding: ObservationStatus (required)
category	SI		(Slice Definition)	Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat	SI	1..1	CodeableConcept	Classification of type of observation Binding: Observation Category Codes (preferred)
id		0..1	string	Unique id for inter-element referencing
extension	I	0..*	Extension	Additional content defined by implementations Slice: Unordered, Open by value:url
coding	SΣI	1..*	Coding	Code defined by a terminology system
id		0..1	string	Unique id for inter-element referencing

extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Oxygen Saturation Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:OxygenSatCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 2708-6
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: %
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept

valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-oxygensat -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Oxygen Saturation -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
```

```
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-oxygensat
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Oxygen Saturation
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "%" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.28.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	preferred	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus

Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.28.2.3 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.29 observation-bmi

URL: https://hl7.org/fhir/R4/observation-bmi.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign body mass index.

10.1.29 observation-bmi

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.29.1 Content

Profiles:
BMI | FHIR Body Mass Index (BMI) Profile

10.1.29.2 StructureDefinition: BMI

URL: https://hl7.org/fhir/R4/bmi.html

10.1.29.2 StructureDefinition: BMI

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bmi`

FHIR Body Mass Index (BMI) Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.29.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from observation-vitalsigns.

Summary

10.1.29.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '39156-5' Other additional Codings are allowed in Observation.code-

e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'kg/m2'

Mandatory: 8 elements Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Body Mass Index (BMI) Profile
code | | 1..1 | CodeableConcept | Body Mass Index (BMI)
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BMICode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 39156-5
valueQuantity | | 1..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: kg/m2
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Mass Index (BMI) Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Mass Index (BMI) Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BMICode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 39156-5
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept

subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 1..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: kg/m2
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- observation-bmi -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
```

```
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Mass Index (BMI) -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  1..1 Quantity  <!--  1..1 Vital Signs value are recorded using the Quantity data type. For supporting
    observations such as Cuff size could use other datatypes such as CodeableConcept. -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-bmi
```

```
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Mass Index (BMI)
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "kg/m2" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.29.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= '39156-5' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'kg/m2'

Mandatory: 8 elements Must-Support: 4 elements Fixed Value: 4 elements

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Body Mass Index (BMI) Profile
code		1..1	CodeableConcept	Body Mass Index (BMI)
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:BMICode		1..1	Coding	
system		1..1	uri	Fixed Value: http://loinc.org
code		1..1	code	Fixed Value: 39156-5
valueQuantity		1..1	Quantity	

value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: kg/m2
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Body Mass Index (BMI) Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Body Mass Index (BMI) Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BMICode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 39156-5
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 1..1 | Quantity | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: kg/m2

dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | 0..* | BackboneElement | Used when reporting systolic and diastolic blood pressure. vs-3: If there is no a value a data absent reason must be present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- observation-bmi -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Body Mass Index (BMI) -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
```

```
  <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
  <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
  <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
  <display value="[string]"/><!--  0..1 Representation defined by the system -->
  <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
 </coding>
 <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
</code>
<subject><!--  1..1 Reference(Patient)
    Who and/or what the observation is about --></subject>
<focus><!--  0..* Reference(Resource)
    What the observation is about, when it is not about the subject of record --></focus>
<encounter><!--  0..1 Reference(Encounter)
    Healthcare event during which this observation is made --></encounter>
<effective[x]><!--  1..1 dateTime|Period
    Often just a dateTime for Vital Signs --></effective[x]>
<issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
<performer><!--  0..* Reference(Practitioner)
    Who is responsible for the observation --></performer>
<-- valueQuantity sliced by type:$this  in the specified orderClosed-->
<valueQuantity>  1..1 Quantity  <!--  1..1 Vital Signs value are recorded using the Quantity data type. For supporting
   observations such as Cuff size could use other datatypes such as CodeableConcept. -->
 <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
 <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
 <unit value="[string]"/><!--  1..1 Unit representation -->
 <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
 <code value="[code]"/><!--  1..1 Coded responses from the common UCUM units for vital signs value set. -->
</valueQuantity>
<dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
<interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
<note><!--  0..* Annotation Comments about the observation --></note>
<bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
<method><!--  0..1 CodeableConcept How it was done   --></method>
<specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
<device><!--  0..1 Reference(Device) (Measurement) Device --></device>
<referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <low><!--  0..1 Quantity
     Low Range, if relevant --></low>
 <high><!--  0..1 Quantity
     High Range, if relevant --></high>
 <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
 <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
 <age><!--  0..1 Range
     Applicable age range, if relevant --></age>
 <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
</referenceRange>
<hasMember><!--  0..* Reference(QuestionnaireResponse)
    Used when reporting vital signs panel components --></hasMember>
<derivedFrom><!--  0..* Reference(DocumentReference)
    Related measurements the observation is made from --></derivedFrom>
<component>  0..* BackboneElement  <!--  0..* Used when reporting systolic and diastolic blood pressure. -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
 <value[x]><!--  0..1 Quantity|CodeableConcept|string|
   boolean|integer|Range|Ratio|SampledData|time|dateTime|Period Vital Sign Value recorded with UCUM   --></value[x]>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
   --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-bmi
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Body Mass Index (BMI)
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
```

```
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C? R! Numerical value (with implicit precision)
        "comparator" : "<code>", // C?< | <= | >= | > - how to understand the value
        "unit" : "<string>", // C? R! Unit representation
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "kg/m2" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [{ BackboneElement }], // C?Used when reporting systolic and diastolic blood pressure.
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.29.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.29.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()
vs-3 | Observation.component | If there is no a value a data absent reason must be present Expression : value.exists() or dataAbsentReason.exists()

10.1.30 observation-bp

URL: https://hl7.org/fhir/R4/observation-bp.html

Defines constraints and extensions on the Observation resource for use in querying and retrieving the vital sign blood pressure.

10.1.30 observation-bp

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.1.30.1 Content

Profiles:
BP | FHIR Blood Pressure Profile

10.1.30.2 StructureDefinition: BP

URL: https://hl7.org/fhir/R4/bp.html

10.1.30.2 StructureDefinition: BP

Orders and Observations Work Group | Maturity Level: 5 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/bp`

FHIR Blood Pressure Profile

This profile was published on Sat, Aug 11, 2018 00:00+1000 as a draft by Health Level Seven International (Orders and Observations Workgroup).

10.1.30.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary

• Differential Table

• Snapshot Table

• XML Template

• JSON Template

• All

This structure is derived from observation-vitalsigns.

Summary

10.1.30.2.2 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= 85354-9' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• One Observation.component.code which must have a fixed Observation.component.code.coding.system='http://loinc.org' fixed Observation.component.code.coding.code= '8480-6' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• One Observation.component.code which must have a fixed Observation.component.code.coding.system='http://loinc.org' fixed Observation.component.code.coding.code= '8462-4' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'mm[Hg]'

Mandatory: 11 elements (+8 nested mandatory elements) Must-Support: 8 elements Fixed Value: 10 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

• The element Observation.component is sliced based on the values of value:code.coding.code, value:code.coding.system

• The element Observation.component.code.coding is sliced based on the values of value:code, value:system

This structure is derived from observation-vitalsigns.

Name	Flags	Card.	Type	Description & Constraints
Observation		0..*		FHIR Blood Pressure Profile
code		1..1	CodeableConcept	Blood Pressure
coding			(Slice Definition)	Slice: Unordered, Open by value:code, value:system
coding:BPCode		1..1	Coding	
system		1..1	uri	Fixed Value: http://loinc.org
code		1..1	code	Fixed Value: 85354-9
valueQuantity		0..0		
component			(Slice Definition)	Slice: Unordered, Open by value:code.coding.code, value:code.coding.system
component:SystolicBP		1..1	BackboneElement	
code		1..1	CodeableConcept	

coding | | | (Slice Definition) | Systolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:SBPCode | | 1..1 | Coding | Systolic Blood Pressure
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8480-6
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
component:DiastolicBP | | 1..1 | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | (Slice Definition) | Diastolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:DBPCode | | 1..1 | Coding | Diastolic Blood Pressure
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8462-4
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Blood Pressure Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Blood Pressure Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BPCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 85354-9
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime

effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | | (Slice Definition) | Used when reporting systolic and diastolic blood pressure. Slice: Unordered, Open by value:code.coding.code, value:code.coding.system
(All Slices) | | | | Content/Rules for all slices
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
component:SystolicBP | SΣI | 1..1 | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Systolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:SBPCode | ΣI | 1..1 | Coding | Systolic Blood Pressure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8480-6
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
value[x] | SΣI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)

unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
component:DiastolicBP | SΣI | 1..1 | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Diastolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:DBPCode | ΣI | 1..1 | Coding | Diastolic Blood Pressure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8462-4
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
value[x] | SΣI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- observation-bp -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Blood Pressure -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
```

```
    What the observation is about, when it is not about the subject of record --></focus>
<encounter><!--  0..1 Reference(Encounter)
    Healthcare event during which this observation is made --></encounter>
<effective[x]><!--  1..1 dateTime|Period
    Often just a dateTime for Vital Signs --></effective[x]>
<issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
<performer><!--  0..* Reference(Practitioner)
    Who is responsible for the observation --></performer>
<-- valueQuantity sliced by type:$this  in the specified orderClosed-->
<valueQuantity><!--  0..0 Quantity
    Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could
   use other datatypes such as CodeableConcept. --></valueQuantity>
<dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
<interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
<note><!--  0..* Annotation Comments about the observation --></note>
<bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
<method><!--  0..1 CodeableConcept How it was done   --></method>
<specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
<device><!--  0..1 Reference(Device) (Measurement) Device --></device>
<referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <low><!--  0..1 Quantity
     Low Range, if relevant --></low>
 <high><!--  0..1 Quantity
     High Range, if relevant --></high>
 <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
 <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
 <age><!--  0..1 Range
     Applicable age range, if relevant --></age>
 <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
</referenceRange>
<hasMember><!--  0..* Reference(QuestionnaireResponse)
    Used when reporting vital signs panel components --></hasMember>
<derivedFrom><!--  0..* Reference(DocumentReference)
    Related measurements the observation is made from --></derivedFrom>
<-- component sliced by value:code.coding.code, value:code.coding.system  in the specified orderOpen-->
<component>  1..1 BackboneElement  <!--  1..1 Used when reporting systolic and diastolic blood pressure. -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <code>  1..1 CodeableConcept  <!--  1..1 Type of component observation (code / type) -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Systolic Blood Pressure -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
   --></referenceRange>
</component>
<component>  1..1 BackboneElement  <!--  1..1 Used when reporting systolic and diastolic blood pressure. -->
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
   --></modifierExtension>
 <code>  1..1 CodeableConcept  <!--  1..1 Type of component observation (code / type) -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Diastolic Blood Pressure -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Sign Value recorded with UCUM -->
  <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
  <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1
      Coded responses from the common UCUM units for vital signs value set. -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
```

```
    --></referenceRange>
 </component>
</Observation>
```

```
{ // observation-bp
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Blood Pressure
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [ //  sliced by value:code.coding.code, value:code.coding.system  in the specified order, Open
      { // Used when reporting systolic and diastolic blood pressure. // C? R!
        // from Element: extension
        "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored even if unrecognized
        "code" : { CodeableConcept }, // C? R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, // C?Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, // C?Why the component result is missing
        "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      },
      { // Used when reporting systolic and diastolic blood pressure. // C? R!
        // from Element: extension
        "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored even if unrecognized
        "code" : { CodeableConcept }, // C? R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, // C?Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, // C?Why the component result is missing
        "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      }
    ],
  }
```

This structure is derived from observation-vitalsigns.

Summary

10.1.30.2.3 Complete Summary of the Mandatory Requirements

• One code in Observation.code which must have a fixed Observation.code.coding.system='http ://loinc.org' a fixed Observation.code.coding.code= 85354-9' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• One Observation.component.code which must have a fixed Observation.component.code.coding.system='http://loinc.org' fixed Observation.component.code.coding.code= '8480-6' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• One Observation.component.code which must have a fixed Observation.component.code.coding.system='http://loinc.org' fixed Observation.component.code.coding.code= '8462-4' Other additional Codings are allowed in Observation.code- e.g. more specific LOINC Codes, SNOMED CT concepts, system specific codes. All codes SHALL have an system value

• Either one Observation.valueQuantity or, if there is no value, one code in Observation.DataAbsentReason Each Observation.valueQuantity must have: One numeric value in Observation.valueQuantity.value a fixed Observation.valueQuantity.system="http://unitsofmeasure.org" a UCUM unit code in Observation.valueQuantity.code = 'mm[Hg]'

Mandatory: 11 elements (+8 nested mandatory elements) Must-Support: 8 elements Fixed Value: 10 elements Prohibited: 1 element

Slices

This structure defines the following Slices:

• The element Observation.code.coding is sliced based on the values of value:code, value:system

• The element Observation.component is sliced based on the values of value:code.coding.code, value:code.coding.system

• The element Observation.component.code.coding is sliced based on the values of value:code, value:system

Differential View

This structure is derived from observation-vitalsigns.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..* | | FHIR Blood Pressure Profile
code | | 1..1 | CodeableConcept | Blood Pressure
coding | | | (Slice Definition) | Slice: Unordered, Open by value:code, value:system
coding:BPCode | | 1..1 | Coding
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 85354-9
valueQuantity | | 0..0
component | | | (Slice Definition) | Slice: Unordered, Open by value:code.coding.code, value:code.coding.system
component:SystolicBP | | 1..1 | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | (Slice Definition) | Systolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:SBPCode | | 1..1 | Coding | Systolic Blood Pressure
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8480-6
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
component:DiastolicBP | | 1..1 | BackboneElement
code | | 1..1 | CodeableConcept
coding | | | (Slice Definition) | Diastolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:DBPCode | | 1..1 | Coding | Diastolic Blood Pressure
system | | 1..1 | uri | Fixed Value: http://loinc.org
code | | 1..1 | code | Fixed Value: 8462-4
valueQuantity | | 0..1 | Quantity
value | S | 1..1 | decimal
unit | S | 1..1 | string
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 0..* | | FHIR Blood Pressure Profile
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event

status | ?!SΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | SI | | (Slice Definition) | Classification of type of observation Slice: Unordered, Open by value:coding.code, value:coding.system Binding: Observation Category Codes (preferred)
category:VSCat | SI | 1..1 | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | SΣI | 1..* | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | SΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/observation-category
version | ΣI | 0..1 | string | Version of the system - if relevant
code | SΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: vital-signs
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
code | SΣI | 1..1 | CodeableConcept | Blood Pressure Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Code defined by a terminology system Slice: Unordered, Open by value:code, value:system
coding:BPCode | ΣI | 1..1 | Coding | Code defined by a terminology system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 85354-9
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
subject | SΣI | 1..1 | Reference(Patient) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | SΣI | 1..1 | | Often just a dateTime for Vital Signs vs-1: if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | SΣI | | (Slice Definition) | Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could use other datatypes such as CodeableConcept. Slice: Unordered, Closed by type:$this
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
note | I | 0..* | Annotation | Comments about the observation
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | I | 0..* | BackboneElement | Provides guide for interpretation obs-3: Must have at least a low or a high or text
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 0..1 | SimpleQuantity | Low Range, if relevant
high | I | 0..1 | SimpleQuantity | High Range, if relevant
type | I | 0..1 | CodeableConcept | Reference range qualifier Binding: Observation Reference Range Meaning Codes (preferred)
appliesTo | I | 0..* | CodeableConcept | Reference range population Binding: Observation Reference Range Applies To Codes (example)
age | I | 0..1 | Range | Applicable age range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
hasMember | ΣI | 0..* | Reference(QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Used when reporting vital signs panel components
derivedFrom | ΣI | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | MolecularSequence | Vital Signs Profile) | Related measurements the observation is made from
component | SΣI | | (Slice Definition) | Used when reporting systolic and diastolic blood pressure. Slice: Unordered, Open by value:code.coding.code, value:code.coding.system
(All Slices) | | | | Content/Rules for all slices
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
value[x] | SΣI | 0..1 | | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
valueQuantity | | | Quantity

valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
component:SystolicBP | SΣI | 1..1 | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Systolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:SBPCode | ΣI | 1..1 | Coding | Systolic Blood Pressure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8480-6
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
value[x] | SΣI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
component:DiastolicBP | SΣI | 1..1 | BackboneElement | Used when reporting systolic and diastolic blood pressure.
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: Vital Signs (extensible)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
coding | ΣI | | (Slice Definition) | Diastolic Blood Pressure Slice: Unordered, Open by value:code, value:system
coding:DBPCode | ΣI | 1..1 | Coding | Diastolic Blood Pressure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
system | ΣI | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | ΣI | 0..1 | string | Version of the system - if relevant
code | ΣI | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 8462-4
display | ΣI | 0..1 | string | Representation defined by the system
userSelected | ΣI | 0..1 | boolean | If this coding was chosen directly by the user
text | ΣI | 0..1 | string | Plain text representation of the concept
value[x] | SΣI | 0..1 | Quantity | Vital Sign Value recorded with UCUM Binding: Vital Signs Units (required)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 1..1 | decimal | Numerical value (with implicit precision)
comparator | ?!ΣI | 0..1 | code | < | <= | >= | > - how to understand the value Binding: QuantityComparator (required)
unit | SΣI | 1..1 | string | Unit representation
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded responses from the common UCUM units for vital signs value set. Fixed Value: mm[Hg]
dataAbsentReason | SI | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

XML Template

```
<!-- observation-bp -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <-- category sliced by value:coding.code, value:coding.system  in the specified orderOpen-->
 <category>  1..1 CodeableConcept  <!--  1..1 Classification of  type of observation -->
  <coding>  1..* Coding  <!--  1..* Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </category>
 <code>  1..1 CodeableConcept  <!--  1..1 Blood Pressure -->
  <-- coding sliced by value:code, value:system  in the specified orderOpen-->
  <coding>  1..1 Coding  <!--  1..1 Code defined by a terminology system -->
   <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
   <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
   <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
   <display value="[string]"/><!--  0..1 Representation defined by the system -->
   <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
  </coding>
  <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
 </code>
 <subject><!--  1..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  1..1 dateTime|Period
     Often just a dateTime for Vital Signs --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..0 Quantity
     Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size could
    use other datatypes such as CodeableConcept. --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
 <note><!--  0..* Annotation Comments about the observation --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  0..* BackboneElement  <!--  0..* Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..1 Quantity
      Low Range, if relevant --></low>
  <high><!--  0..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..1 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..* CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..1 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..* Reference(QuestionnaireResponse)
     Used when reporting vital signs panel components --></hasMember>
 <derivedFrom><!--  0..* Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <-- component sliced by value:code.coding.code, value:code.coding.system  in the specified orderOpen-->
 <component>  1..1 BackboneElement  <!--  1..1 Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code>  1..1 CodeableConcept  <!--  1..1 Type of component observation (code / type) -->
   <-- coding sliced by value:code, value:system  in the specified orderOpen-->
   <coding>  1..1 Coding  <!--  1..1 Systolic Blood Pressure -->
```

```
    <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
    <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
    <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
    <display value="[string]"/><!--  0..1 Representation defined by the system -->
    <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
   </coding>
   <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
  </code>
  <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
   <unit value="[string]"/><!--  1..1 Unit representation -->
   <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
   <code value="[code]"/><!--  1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
 <component>  1..1 BackboneElement  <!--  1..1 Used when reporting systolic and diastolic blood pressure. -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code>  1..1 CodeableConcept  <!--  1..1 Type of component observation (code / type) -->
   <-- coding sliced by value:code, value:system  in the specified orderOpen-->
   <coding>  1..1 Coding  <!--  1..1 Diastolic Blood Pressure -->
    <system value="[uri]"/><!--  1..1 Identity of the terminology system -->
    <version value="[string]"/><!--  0..1 Version of the system - if relevant -->
    <code value="[code]"/><!--  1..1 Symbol in syntax defined by the system -->
    <display value="[string]"/><!--  0..1 Representation defined by the system -->
    <userSelected value="[boolean]"/><!--  0..1 If this coding was chosen directly by the user -->
   </coding>
   <text value="[string]"/><!--  0..1 Plain text representation of the concept -->
  </code>
  <valueQuantity>  0..1 Quantity  <!--  0..1 Vital Sign Value recorded with UCUM -->
   <value value="[decimal]"/><!--  1..1 Numerical value (with implicit precision) -->
   <comparator value="[code]"/><!--  0..1 < | <= | >= | > - how to understand the value   -->
   <unit value="[string]"/><!--  1..1 Unit representation -->
   <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
   <code value="[code]"/><!--  1..1
       Coded responses from the common UCUM units for vital signs value set. -->
  </valueQuantity>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // observation-bp
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [ //  sliced by value:coding.code, value:coding.system  in the specified order, Open
      { // Classification of  type of observation // C? R!
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "coding" : [{ Coding }], // C? R! Code defined by a terminology system
        "text" : "<string>" // C?Plain text representation of the concept
      }
    ],
    "code" : { CodeableConcept }, // C? R! Blood Pressure
    "subject" : { Reference(Patient) }, // C? R! Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Often just a dateTime for Vital Signs. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C? R! Often just a dateTime for Vital Signs
    "effectivePeriod" : { Period }, // C? R! Often just a dateTime for Vital Signs
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Vital Signs value are recorded using the Quantity data type. For supporting observations such as Cuff size
    could use other datatypes such as CodeableConcept. // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
```

```
    "note" : [{ Annotation }], // C?Comments about the observation
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : [{ BackboneElement }], // C?Provides guide for interpretation
    "hasMember" : [{ Reference(QuestionnaireResponse) }], // C?Used when reporting vital signs panel components
    "derivedFrom" : [{ Reference(DocumentReference) }], // C?Related measurements the observation is made from
    "component" : [ //  sliced by value:code.coding.code, value:code.coding.system  in the specified order, Open
      { // Used when reporting systolic and diastolic blood pressure. // C? R!
        // from Element: extension
        "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored even if unrecognized
        "code" : { CodeableConcept }, // C? R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, // C?Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, // C?Why the component result is missing
        "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      },
      { // Used when reporting systolic and diastolic blood pressure. // C? R!
        // from Element: extension
        "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored even if unrecognized
        "code" : { CodeableConcept }, // C? R! Type of component observation (code / type)
        "valueQuantity" : { Quantity }, // C?Vital Sign Value recorded with UCUM
        "dataAbsentReason" : { CodeableConcept }, // C?Why the component result is missing
        "interpretation" : [{ CodeableConcept }], // C?High, low, normal, etc.
// value[x]: Provides guide for interpretation of component result. One of these 0:
      }
    ],
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.1.30.2.4 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | Vital Signs | extensible | Vital Signs
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.referenceRange.type | Observation Reference Range Meaning Codes | preferred | Observation Reference Range Meaning Codes
Observation.referenceRange.appliesTo | Observation Reference Range Applies To Codes | example | Observation Reference Range Applies To Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.component.code | Vital Signs | extensible | Vital Signs
Observation.component.value[x] | Vital Signs Units | required | Vital Signs Units
Observation.component.value[x].comparator | QuantityComparator | required | QuantityComparator
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.1.30.2.5 Constraints

Id | Path | Details | Requirements
vs-1 | Observation.effective[x] | if Observation.effective[x] is dateTime and has a value then that value shall be precise to the day Expression : ($this as dateTime).toString().length() >= 8
obs-3 | Observation.referenceRange | Must have at least a low or a high or text Expression : low.exists() or high.exists() or text.exists()

10.1.31 JSON Schema for Observation

URL: https://hl7.org/fhir/R4/observation.schema.json.html

10.1.31 JSON Schema for Observation

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Observation, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Observation",
  "$ref": "#/definitions/Observation",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Observation": {
      "description": "Measurements and simple assertions made about a patient, device or other subject.",
      "properties": {
        "resourceType": {
          "description": "This is a Observation resource",
          "const": "Observation"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
          "description": "A unique identifier assigned to this observation.",
          "items": {
            "$ref": "Identifier.schema.json#/definitions/Identifier"
          },
          "type": "array"
        },
        "basedOn": {
          "description": "A plan, proposal or order that is fulfilled in whole or in part by this event.  For example, a
```

```
 MedicationRequest may require a patient to have laboratory test performed before  it is dispensed.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger event of which this particular Observation is a component or step.  For example,  an
observation as part of a procedure.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of the result value.",
      "enum": [
        "registered",
        "preliminary",
        "final",
        "amended",
        "corrected",
        "cancelled",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "A code that classifies the general type of observation being made.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "code": {
      "description": "Describes what was observed. Sometimes this is called the observation \"name\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The patient, or group of patients, location, or device this observation is about and into
whose record the observation is placed. If the actual focus of the observation is different from the subject (or a
sample of, part, or region of the subject), the `focus` element or the `code` itself specifies the actual focus of
the observation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "focus": {
      "description": "The actual focus of an observation when it is not the patient of record representing something
 or someone associated with the patient such as a spouse, parent, fetus, or donor. For example, fetus observations
in a mother\u0027s record.  The focus of an observation could also be an existing condition,  an intervention, the
subject\u0027s diet,  another observation of the subject,  or a body structure such as tumor or implanted device.
An example use case would be using the Observation resource to capture whether the mother is trained to change her
child\u0027s tracheostomy tube. In this example, the child is the patient of record and the mother is the focus.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "encounter": {
      "description": "The healthcare event  (e.g. a patient and healthcare provider interaction) during which this
observation is made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "effectiveDateTime": {
      "description": "The time or time-period the observed value is asserted as being true. For biological subjects
- e.g. human patients - this is usually called the \"physiologically relevant time\". This is usually either the
time of the procedure or of specimen collection, but very often the source of the date/time is not known, only the
date/time itself.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_effectiveDateTime": {
      "description": "Extensions for effectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The time or time-period the observed value is asserted as being true. For biological subjects
- e.g. human patients - this is usually called the \"physiologically relevant time\". This is usually either the
time of the procedure or of specimen collection, but very often the source of the date/time is not known, only the
date/time itself.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "effectiveTiming": {
      "description": "The time or time-period the observed value is asserted as being true. For biological subjects
- e.g. human patients - this is usually called the \"physiologically relevant time\". This is usually either the
```

```
time of the procedure or of specimen collection, but very often the source of the date/time is not known, only the
date/time itself.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "effectiveInstant": {
      "description": "The time or time-period the observed value is asserted as being true. For biological subjects
- e.g. human patients - this is usually called the \"physiologically relevant time\". This is usually either the
time of the procedure or of specimen collection, but very often the source of the date/time is not known, only the
date/time itself.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)-(0[1-9]|1[0-2])-(0[1-9]|[1-2][0-9]|3[0-
1])T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))$",
      "type": "string"
    },
    "_effectiveInstant": {
      "description": "Extensions for effectiveInstant",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "issued": {
      "description": "The date and time this version of the observation was made available to providers, typically
after the results have been reviewed and verified.",
      "$ref": "#/definitions/instant"
    },
    "_issued": {
      "description": "Extensions for issued",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "performer": {
      "description": "Who was responsible for asserting the observed value as \"true\".",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "valueQuantity": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "valueCodeableConcept": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueString": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueRange": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "valueRatio": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "valueSampledData": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "SampledData.schema.json#/definitions/SampledData"
    },
    "valueTime": {
```

```
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
      "type": "string"
    },
    "_valueTime": {
      "description": "Extensions for valueTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valuePeriod": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "dataAbsentReason": {
      "description": "Provides a reason why the expected value in the element Observation.value[x] is missing.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "interpretation": {
      "description": "A categorical assessment of an observation value.  For example, high, low, normal.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "note": {
      "description": "Comments about the observation or the results.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "bodySite": {
      "description": "Indicates the site on the subject\u0027s body where the observation was made (i.e. the target
site).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "method": {
      "description": "Indicates the mechanism used to perform the observation.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "specimen": {
      "description": "The specimen that was used when this observation was made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "device": {
      "description": "The device used to generate the observation data.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "referenceRange": {
      "description": "Guidance on how to interpret the value by comparison to a normal or recommended range.
Multiple reference ranges are interpreted as an \"OR\".   In other words, to represent two distinct target
populations, two `referenceRange` elements would be used.",
      "items": {
        "$ref": "#/definitions/Observation_ReferenceRange"
      },
      "type": "array"
    },
    "hasMember": {
      "description": "This observation is a group observation (e.g. a battery, a panel of tests, a set of vital sign
 measurements) that includes the target as a member of the group.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "derivedFrom": {
      "description": "The target resource that represents a measurement from which this observation value is
derived. For example, a calculated anion gap or a fetal measurement based on an ultrasound image.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "component": {
      "description": "Some observations have multiple component observations.  These component observations are
expressed as separate code value pairs that share the same attributes.  Examples include systolic and diastolic
component observations for blood pressure measurement and multiple component observations for genetics
```

```
observations.",
      "items": {
        "$ref": "#/definitions/Observation_Component"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "code",
    "resourceType"
  ]
},
"Observation_ReferenceRange": {
  "description": "Measurements and simple assertions made about a patient, device or other subject.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "low": {
      "description": "The value of the low bound of the reference range.  The low bound of the reference range
endpoint is inclusive of the value (e.g.  reference range is \u003e\u003d5 - \u003c\u003d9). If the low bound is
omitted,  it is assumed to be meaningless (e.g. reference range is \u003c\u003d2.3).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "high": {
      "description": "The value of the high bound of the reference range.  The high bound of the reference range
endpoint is inclusive of the value (e.g.  reference range is \u003e\u003d5 - \u003c\u003d9). If the high bound is
omitted,  it is assumed to be meaningless (e.g. reference range is \u003e\u003d 2.3).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "type": {
      "description": "Codes to indicate the what part of the targeted reference population it applies to. For
example, the normal or therapeutic range.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "appliesTo": {
      "description": "Codes to indicate the target population this reference range applies to.  For example, a
reference range may be based on the normal population or a particular sex or race.  Multiple `appliesTo`  are
interpreted as an \"AND\" of the target populations.  For example, to represent a target population of African
American females, both a code of female and a code for African American would be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "age": {
      "description": "The age at which this reference range is applicable. This is a neonatal age (e.g. number of
weeks at term) if the meaning says so.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "text": {
      "description": "Text based reference range in an observation which may be used when a quantitative range is
not appropriate for an observation.  An example would be a reference value of \"Negative\" or a list or table of
\"normals\".",
      "$ref": "#/definitions/string"
    },
    "_text": {
      "description": "Extensions for text",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
```

```
},
"Observation_Component": {
  "description": "Measurements and simple assertions made about a patient, device or other subject.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Describes what was observed. Sometimes this is called the observation \"code\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueQuantity": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "valueCodeableConcept": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueString": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueRange": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "valueRatio": {
      "description": "The information determined as a result of making the observation, if the information has a
simple value.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "valueSampledData": {
      "description": "The information determined as a result of making the observation, if the information has a
```

```
    simple value.",
          "$ref": "SampledData.schema.json#/definitions/SampledData"
        },
        "valueTime": {
          "description": "The information determined as a result of making the observation, if the information has a
    simple value.",
          "pattern": "^([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?$",
          "type": "string"
        },
        "_valueTime": {
          "description": "Extensions for valueTime",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "valueDateTime": {
          "description": "The information determined as a result of making the observation, if the information has a
    simple value.",
          "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
    1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))?)?)?$",
          "type": "string"
        },
        "_valueDateTime": {
          "description": "Extensions for valueDateTime",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "valuePeriod": {
          "description": "The information determined as a result of making the observation, if the information has a
    simple value.",
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "dataAbsentReason": {
          "description": "Provides a reason why the expected value in the element Observation.component.value[x] is
    missing.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "interpretation": {
          "description": "A categorical assessment of an observation value.  For example, high, low, normal.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "referenceRange": {
          "description": "Guidance on how to interpret the value by comparison to a normal or recommended range.",
          "items": {
            "$ref": "#/definitions/Observation_ReferenceRange"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "code"
      ]
    }
  }
}
```


10.1.32 Resource Observation - Maturity Dependency View

URL: https://hl7.org/fhir/R4/observation-dependencies.html

10.1.32 Resource Observation - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

| Some of the content on this page (marked clearly) has been approved as part of an ANSI standard. See the Observation Package for further details.

Dependency Graph for Observation FMM level 5

Name | Card. | Type | Dependency Analysis
Observation | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (DeviceRequest = FMM1-Trial Use vs. Element = FMM5-Normative) Type Warning: (ImmunizationRecommendation = FMM1-Trial Use vs. Element = FMM5-Normative) Type Warning: (MedicationRequest = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (NutritionOrder = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM5-Normative)
partOf | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Type Warning: (MedicationAdministration = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (MedicationDispense = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (MedicationStatement = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (Procedure = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (Immunization = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (ImagingStudy = FMM3-Trial Use vs. Element = FMM5-Normative)

status | 1..1 | code
category | 0..* | CodeableConcept
code | 1..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group | Device | Location) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM5-Normative) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (Location = FMM3-Trial Use vs. Element = FMM5-Normative)
focus TU | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM5-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM5-Normative)
effective[x] | 0..1
effectiveDateTime | | dateTime
effectivePeriod | | Period
effectiveTiming | | Timing
effectiveInstant | | instant
issued | 0..1 | instant
performer | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Type Warning: (Practitioner = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (CareTeam = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM5-Normative)
value[x] | 0..1
valueQuantity | | Quantity
valueCodeableConcept | | CodeableConcept
valueString | | string
valueBoolean | | boolean
valueInteger | | integer
valueRange | | Range
valueRatio | | Ratio
valueSampledData | | SampledData | Type Error: (SampledData = FMM5-Trial Use vs. Element = FMM5-Normative)
valueTime | | time
valueDateTime | | dateTime
valuePeriod | | Period
dataAbsentReason | 0..1 | CodeableConcept
interpretation | 0..* | CodeableConcept
note | 0..* | Annotation
bodySite | 0..1 | CodeableConcept
method | 0..1 | CodeableConcept
specimen | 0..1 | Reference(Specimen) | Type Warning: (Specimen = FMM2-Trial Use vs. Element = FMM5-Normative)
device | 0..1 | Reference(Device | DeviceMetric) | Type Warning: (Device = FMM2-Trial Use vs. Element = FMM5-Normative) Type Warning: (DeviceMetric = FMM1-Trial Use vs. Element = FMM5-Normative)
referenceRange | 0..* | BackboneElement
low | 0..1 | SimpleQuantity
high | 0..1 | SimpleQuantity
type | 0..1 | CodeableConcept
appliesTo | 0..* | CodeableConcept
age | 0..1 | Range
text | 0..1 | string
hasMember | 0..* | Reference(Observation | QuestionnaireResponse | MolecularSequence) | Type Warning: (QuestionnaireResponse = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (MolecularSequence = FMM1-Trial Use vs. Element = FMM5-Normative)
derivedFrom | 0..* | Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence) | Type Warning: (DocumentReference = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (ImagingStudy = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (Media = FMM1-Trial Use vs. Element = FMM5-Normative) Type Warning: (QuestionnaireResponse = FMM3-Trial Use vs. Element = FMM5-Normative) Type Warning: (MolecularSequence = FMM1-Trial Use vs. Element = FMM5-Normative)
component | 0..* | BackboneElement
code | 1..1 | CodeableConcept
value[x] | 0..1
valueQuantity | | Quantity
valueCodeableConcept | | CodeableConcept
valueString | | string
valueBoolean | | boolean
valueInteger | | integer
valueRange | | Range
valueRatio | | Ratio
valueSampledData | | SampledData | Type Error: (SampledData = FMM5-Trial Use vs. Element = FMM5-Normative)
valueTime | | time
valueDateTime | | dateTime
valuePeriod | | Period
dataAbsentReason | 0..1 | CodeableConcept
interpretation | 0..* | CodeableConcept
referenceRange | 0..* | see referenceRange

10.2 Resource Media - Content

URL: https://hl7.org/fhir/R4/media.html
Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

A photo, video, or audio recording acquired or used in healthcare. The actual content may be inline or provided by direct reference.

10.2.1 Scope and Usage

Trial-Use Note: The Orders and Observations workgroups wants to make the balloter aware than observations that require the attachment data types are represented by the Media resource instead of Observation. There have been several significant changes to the Media resource in order to align it with Observation and the workflow event pattern including:

• The addition of the partOf and issued elements.

• The required type was changed an optional category and the subtype element was renamed to modality.

• Additional reference targets added to the subject,operator, and basedOn elements

• The additions of several standard event extensions.

The Media resource contains photos, videos, and audio recordings. It is used with media acquired or used as part of the healthcare process. Here are some typical usages:

• Photos and videos of diagnostic or care provision procedures for recording purposes

• Images contained in diagnostic reports

10.2.2 Boundaries and Relationships

This resource captures a specific type of Observation - an Observation whose value is audio, video or image data. This resource is the preferred representation of such forms of information as it exposes the metadata relevant for interpreting the information. However, in some legacy environments, media information may occasionally appear in Observation instead. Systems should be aware of this possibility.

The Media resource is able to contain medical images in a DICOM format. These images may also be made accessible through an ImagingStudy resource, which provides a direct reference to the image to a WADO-RS server.

For such images, the WADO-RS framework is a preferred method for representing the images - the WADO-RS service may include rendering the image with annotations and display parameters from an associated DICOM presentation state, for instance.

On the other hand, the media resource allows for a robust transfer of an image across boundaries where the WADO-RS service is not available. For this reason, medical images can also be represented in a Media resource, but the Media.content.url should provide a reference to a source WADO-RS service for the image.

This resource is referenced by AdverseEvent, ClinicalImpression, DeviceUseStatement, DiagnosticReport, ImagingStudy, MedicationKnowledge and Observation

10.2.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Media | TU | | DomainResource | A photo, video, or audio recording acquired or used in healthcare. The actual content may be inline or provided by direct reference Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifier(s) for the image
basedOn | Σ | 0..* | Reference(ServiceRequest | CarePlan) | Procedure that caused this media to be created
partOf | Σ | 0..* | Reference(Any) | Part of referenced event
status | ?!Σ | 1..1 | code | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown EventStatus (Required)
type | Σ | 0..1 | CodeableConcept | Classification of media as image, video, or audio Media Type (Extensible)
modality | Σ | 0..1 | CodeableConcept | The type of acquisition equipment/process Media Modality (Example)
view | Σ | 0..1 | CodeableConcept | Imaging view, e.g. Lateral or Antero-posterior Media Collection View/Projection (Example)
subject | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Group | Device | Specimen | Location) | Who/What this Media is a record of
encounter | Σ | 0..1 | Reference(Encounter) | Encounter associated with media
created[x] | Σ | 0..1 | | When Media was collected

createdDateTime | | | dateTime
createdPeriod | | | Period
issued | Σ | 0..1 | instant | Date/Time this version was made available
operator | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | The person who generated the image
reasonCode | Σ | 0..* | CodeableConcept | Why was event performed? Procedure Reason Codes (Example)
bodySite | Σ | 0..1 | CodeableConcept | Observed body part SNOMED CT Body Structures (Example)
deviceName | Σ | 0..1 | string | Name of the device/manufacturer
device | Σ | 0..1 | Reference(Device | DeviceMetric | Device) | Observing Device
height | Σ | 0..1 | positiveInt | Height of the image in pixels (photo/video)
width | Σ | 0..1 | positiveInt | Width of the image in pixels (photo/video)
frames | Σ | 0..1 | positiveInt | Number of frames if > 1 (photo)
duration | Σ | 0..1 | decimal | Length in seconds (audio / video)
content | Σ | 1..1 | Attachment | Actual Media - reference or data
note | | 0..* | Annotation | Comments made about the media
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Media xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier(s) for the image --></identifier>
 <basedOn><!-- 0..* Reference(ServiceRequest|CarePlan) Procedure that caused this media to be created --></basedOn>
 <partOf><!-- 0..* Reference(Any) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown -->
 <type><!-- 0..1 CodeableConcept Classification of media as image, video, or audio --></type>
 <modality><!-- 0..1 CodeableConcept The type of acquisition equipment/process --></modality>
 <view><!-- 0..1 CodeableConcept Imaging view, e.g. Lateral or Antero-posterior --></view>
 <subject><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|Group|Device|
   Specimen|Location) Who/What this Media is a record of --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter associated with media --></encounter>
 <created[x]><!-- 0..1 dateTime|Period When Media was collected --></created[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this version was made available -->
 <operator><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|Device|RelatedPerson) The person who generated the image --></operator>
 <reasonCode><!-- 0..* CodeableConcept Why was event performed? --></reasonCode>
 <bodySite><!-- 0..1 CodeableConcept Observed body part --></bodySite>
 <deviceName value="[string]"/><!-- 0..1 Name of the device/manufacturer -->
 <device><!-- 0..1 Reference(Device|DeviceMetric|Device) Observing Device --></device>
 <height value="[positiveInt]"/><!-- 0..1 Height of the image in pixels (photo/video) -->
 <width value="[positiveInt]"/><!-- 0..1 Width of the image in pixels (photo/video) -->
 <frames value="[positiveInt]"/><!-- 0..1 Number of frames if > 1 (photo) -->
 <duration value="[decimal]"/><!-- 0..1 Length in seconds (audio / video) -->
 <content><!-- 1..1 Attachment Actual Media - reference or data --></content>
 <note><!-- 0..* Annotation Comments made about the media --></note>
</Media>
```

JSON Template

```
{
  "resourceType" : "Media",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier(s) for the image
  "basedOn" : [{ Reference(ServiceRequest|CarePlan) }], // Procedure that caused this media to be created
  "partOf" : [{ Reference(Any) }], // Part of referenced event
  "status" : "<code>", // R!  preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error |
    unknown
  "type" : { CodeableConcept }, // Classification of media as image, video, or audio
  "modality" : { CodeableConcept }, // The type of acquisition equipment/process
  "view" : { CodeableConcept }, // Imaging view, e.g. Lateral or Antero-posterior
  "subject" : { Reference(Patient|Practitioner|PractitionerRole|Group|Device|
   Specimen|Location) }, // Who/What this Media is a record of
  "encounter" : { Reference(Encounter) }, // Encounter associated with media
  // created[x]: When Media was collected. One of these 2:
  "createdDateTime" : "<dateTime>",
  "createdPeriod" : { Period },
  "issued" : "<instant>", // Date/Time this version was made available
  "operator" : { Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|Device|RelatedPerson) }, // The person who generated the image
  "reasonCode" : [{ CodeableConcept }], // Why was event performed?
  "bodySite" : { CodeableConcept }, // Observed body part
  "deviceName" : "<string>", // Name of the device/manufacturer
  "device" : { Reference(Device|DeviceMetric|Device) }, // Observing Device
  "height" : "<positiveInt>", // Height of the image in pixels (photo/video)
  "width" : "<positiveInt>", // Width of the image in pixels (photo/video)
  "frames" : "<positiveInt>", // Number of frames if > 1 (photo)
  "duration" : <decimal>, // Length in seconds (audio / video)
  "content" : { Attachment }, // R!  Actual Media - reference or data
  "note" : [{ Annotation }] // Comments made about the media
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Media;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Media.identifier [ Identifier ], ... ; # 0..* Identifier(s) for the image
  fhir:Media.basedOn [ Reference(ServiceRequest|CarePlan) ], ... ; # 0..* Procedure that caused this media to be created
  fhir:Media.partOf [ Reference(Any) ], ... ; # 0..* Part of referenced event
  fhir:Media.status [ code ]; # 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown
  fhir:Media.type [ CodeableConcept ]; # 0..1 Classification of media as image, video, or audio
  fhir:Media.modality [ CodeableConcept ]; # 0..1 The type of acquisition equipment/process
  fhir:Media.view [ CodeableConcept ]; # 0..1 Imaging view, e.g. Lateral or Antero-posterior
  fhir:Media.subject [ Reference(Patient|Practitioner|PractitionerRole|Group|Device|Specimen|Location) ]; # 0..1
    Who/What this Media is a record of
  fhir:Media.encounter [ Reference(Encounter) ]; # 0..1 Encounter associated with media
  # Media.created[x] : 0..1 When Media was collected. One of these 2
    fhir:Media.createdDateTime [ dateTime ]
    fhir:Media.createdPeriod [ Period ]
  fhir:Media.issued [ instant ]; # 0..1 Date/Time this version was made available
  fhir:Media.operator [ Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson) ]; #
     0..1 The person who generated the image
  fhir:Media.reasonCode [ CodeableConcept ], ... ; # 0..* Why was event performed?
  fhir:Media.bodySite [ CodeableConcept ]; # 0..1 Observed body part
  fhir:Media.deviceName [ string ]; # 0..1 Name of the device/manufacturer
  fhir:Media.device [ Reference(Device|DeviceMetric|Device) ]; # 0..1 Observing Device
  fhir:Media.height [ positiveInt ]; # 0..1 Height of the image in pixels (photo/video)
  fhir:Media.width [ positiveInt ]; # 0..1 Width of the image in pixels (photo/video)
  fhir:Media.frames [ positiveInt ]; # 0..1 Number of frames if > 1 (photo)
  fhir:Media.duration [ decimal ]; # 0..1 Length in seconds (audio / video)
  fhir:Media.content [ Attachment ]; # 1..1 Actual Media - reference or data
  fhir:Media.note [ Annotation ], ... ; # 0..* Comments made about the media
]
```

Changes since R3

Media
Media.basedOn | Type Reference: Added Target Types ServiceRequest, CarePlan Type Reference: Removed Target Type ProcedureRequest
Media.partOf | Added Element
Media.status | Added Mandatory Element
Media.type | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change binding strength from required to extensible Change value set from http://hl7.org/fhir/ValueSet/digital-media-type to http://hl7.org/fhir/ValueSet/media-type
Media.modality | Renamed from subtype to modality
Media.subject | Type Reference: Added Target Types PractitionerRole, Location
Media.encounter | Added Element
Media.created[x] | Renamed from occurrence[x] to created[x]
Media.issued | Added Element
Media.operator | Type Reference: Added Target Types PractitionerRole, Organization, CareTeam, Patient, Device, RelatedPerson
Media.deviceName | Added Element
Media.frames | Default Value "1" removed
Media.duration | Type changed from unsignedInt to decimal
Media.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Media | TU | | DomainResource | A photo, video, or audio recording acquired or used in healthcare. The actual content may be inline or provided by direct reference Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifier(s) for the image
basedOn | Σ | 0..* | Reference(ServiceRequest | CarePlan) | Procedure that caused this media to be created
partOf | Σ | 0..* | Reference(Any) | Part of referenced event
status | ?!Σ | 1..1 | code | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown EventStatus (Required)
type | Σ | 0..1 | CodeableConcept | Classification of media as image, video, or audio Media Type (Extensible)
modality | Σ | 0..1 | CodeableConcept | The type of acquisition equipment/process Media Modality (Example)
view | Σ | 0..1 | CodeableConcept | Imaging view, e.g. Lateral or Antero-posterior Media Collection View/Projection (Example)
subject | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Group | Device | Specimen | Location) | Who/What this Media is a record of
encounter | Σ | 0..1 | Reference(Encounter) | Encounter associated with media
created[x] | Σ | 0..1 | | When Media was collected
createdDateTime | | | dateTime

createdPeriod | | | Period
issued | Σ | 0..1 | instant | Date/Time this version was made available
operator | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | The person who generated the image
reasonCode | Σ | 0..* | CodeableConcept | Why was event performed? Procedure Reason Codes (Example)
bodySite | Σ | 0..1 | CodeableConcept | Observed body part SNOMED CT Body Structures (Example)
deviceName | Σ | 0..1 | string | Name of the device/manufacturer
device | Σ | 0..1 | Reference(Device | DeviceMetric | Device) | Observing Device
height | Σ | 0..1 | positiveInt | Height of the image in pixels (photo/video)
width | Σ | 0..1 | positiveInt | Width of the image in pixels (photo/video)
frames | Σ | 0..1 | positiveInt | Number of frames if > 1 (photo)
duration | Σ | 0..1 | decimal | Length in seconds (audio / video)
content | Σ | 1..1 | Attachment | Actual Media - reference or data
note | | 0..* | Annotation | Comments made about the media
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Media xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier(s) for the image --></identifier>
 <basedOn><!-- 0..* Reference(ServiceRequest|CarePlan) Procedure that caused this media to be created --></basedOn>
 <partOf><!-- 0..* Reference(Any) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown -->
 <type><!-- 0..1 CodeableConcept Classification of media as image, video, or audio --></type>
 <modality><!-- 0..1 CodeableConcept The type of acquisition equipment/process --></modality>
 <view><!-- 0..1 CodeableConcept Imaging view, e.g. Lateral or Antero-posterior --></view>
 <subject><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|Group|Device|
   Specimen|Location) Who/What this Media is a record of --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter associated with media --></encounter>
 <created[x]><!-- 0..1 dateTime|Period When Media was collected --></created[x]>
 <issued value="[instant]"/><!-- 0..1 Date/Time this version was made available -->
 <operator><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|Device|RelatedPerson) The person who generated the image --></operator>
 <reasonCode><!-- 0..* CodeableConcept Why was event performed? --></reasonCode>
 <bodySite><!-- 0..1 CodeableConcept Observed body part --></bodySite>
 <deviceName value="[string]"/><!-- 0..1 Name of the device/manufacturer -->
 <device><!-- 0..1 Reference(Device|DeviceMetric|Device) Observing Device --></device>
 <height value="[positiveInt]"/><!-- 0..1 Height of the image in pixels (photo/video) -->
 <width value="[positiveInt]"/><!-- 0..1 Width of the image in pixels (photo/video) -->
 <frames value="[positiveInt]"/><!-- 0..1 Number of frames if > 1 (photo) -->
 <duration value="[decimal]"/><!-- 0..1 Length in seconds (audio / video) -->
 <content><!-- 1..1 Attachment Actual Media - reference or data --></content>
 <note><!-- 0..* Annotation Comments made about the media --></note>
</Media>
```

JSON Template

```
{
  "resourceType" : "Media",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier(s) for the image
  "basedOn" : [{ Reference(ServiceRequest|CarePlan) }], // Procedure that caused this media to be created
  "partOf" : [{ Reference(Any) }], // Part of referenced event
  "status" : "<code>", // R!  preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error |
    unknown
  "type" : { CodeableConcept }, // Classification of media as image, video, or audio
  "modality" : { CodeableConcept }, // The type of acquisition equipment/process
  "view" : { CodeableConcept }, // Imaging view, e.g. Lateral or Antero-posterior
  "subject" : { Reference(Patient|Practitioner|PractitionerRole|Group|Device|
   Specimen|Location) }, // Who/What this Media is a record of
  "encounter" : { Reference(Encounter) }, // Encounter associated with media
  // created[x]: When Media was collected. One of these 2:
  "createdDateTime" : "<dateTime>",
  "createdPeriod" : { Period },
  "issued" : "<instant>", // Date/Time this version was made available
  "operator" : { Reference(Practitioner|PractitionerRole|Organization|
   CareTeam|Patient|Device|RelatedPerson) }, // The person who generated the image
  "reasonCode" : [{ CodeableConcept }], // Why was event performed?
  "bodySite" : { CodeableConcept }, // Observed body part
  "deviceName" : "<string>", // Name of the device/manufacturer
  "device" : { Reference(Device|DeviceMetric|Device) }, // Observing Device
  "height" : "<positiveInt>", // Height of the image in pixels (photo/video)
  "width" : "<positiveInt>", // Width of the image in pixels (photo/video)
  "frames" : "<positiveInt>", // Number of frames if > 1 (photo)
  "duration" : <decimal>, // Length in seconds (audio / video)
  "content" : { Attachment }, // R!  Actual Media - reference or data
  "note" : [{ Annotation }] // Comments made about the media
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:Media;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Media.identifier [ Identifier ], ... ; # 0..* Identifier(s) for the image
  fhir:Media.basedOn [ Reference(ServiceRequest|CarePlan) ], ... ; # 0..* Procedure that caused this media to be created
  fhir:Media.partOf [ Reference(Any) ], ... ; # 0..* Part of referenced event
  fhir:Media.status [ code ]; # 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown
  fhir:Media.type [ CodeableConcept ]; # 0..1 Classification of media as image, video, or audio
  fhir:Media.modality [ CodeableConcept ]; # 0..1 The type of acquisition equipment/process
  fhir:Media.view [ CodeableConcept ]; # 0..1 Imaging view, e.g. Lateral or Antero-posterior
  fhir:Media.subject [ Reference(Patient|Practitioner|PractitionerRole|Group|Device|Specimen|Location) ]; # 0..1
    Who/What this Media is a record of
  fhir:Media.encounter [ Reference(Encounter) ]; # 0..1 Encounter associated with media
  # Media.created[x] : 0..1 When Media was collected. One of these 2
    fhir:Media.createdDateTime [ dateTime ]
    fhir:Media.createdPeriod [ Period ]
  fhir:Media.issued [ instant ]; # 0..1 Date/Time this version was made available
  fhir:Media.operator [ Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson) ]; #
     0..1 The person who generated the image
  fhir:Media.reasonCode [ CodeableConcept ], ... ; # 0..* Why was event performed?
  fhir:Media.bodySite [ CodeableConcept ]; # 0..1 Observed body part
  fhir:Media.deviceName [ string ]; # 0..1 Name of the device/manufacturer
  fhir:Media.device [ Reference(Device|DeviceMetric|Device) ]; # 0..1 Observing Device
  fhir:Media.height [ positiveInt ]; # 0..1 Height of the image in pixels (photo/video)
  fhir:Media.width [ positiveInt ]; # 0..1 Width of the image in pixels (photo/video)
  fhir:Media.frames [ positiveInt ]; # 0..1 Number of frames if > 1 (photo)
  fhir:Media.duration [ decimal ]; # 0..1 Length in seconds (audio / video)
  fhir:Media.content [ Attachment ]; # 1..1 Actual Media - reference or data
  fhir:Media.note [ Annotation ], ... ; # 0..* Comments made about the media
]
```

Changes since Release 3

Media
Media.basedOn | Type Reference: Added Target Types ServiceRequest, CarePlan Type Reference: Removed Target Type ProcedureRequest
Media.partOf | Added Element
Media.status | Added Mandatory Element
Media.type | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change binding strength from required to extensible Change value set from http://hl7.org/fhir/ValueSet/digital-media-type to http://hl7.org/fhir/ValueSet/media-type
Media.modality | Renamed from subtype to modality
Media.subject | Type Reference: Added Target Types PractitionerRole, Location
Media.encounter | Added Element
Media.created[x] | Renamed from occurrence[x] to created[x]
Media.issued | Added Element
Media.operator | Type Reference: Added Target Types PractitionerRole, Organization, CareTeam, Patient, Device, RelatedPerson
Media.deviceName | Added Element
Media.frames | Default Value "1" removed
Media.duration | Type changed from unsignedInt to decimal
Media.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.2.3.1 Terminology Bindings

Path | Definition | Type | Reference
Media.status | Codes identifying the lifecycle stage of an event. | Required | EventStatus
Media.type | Codes for high level media categories. | Extensible | MediaType
Media.modality | Detailed information about the type of the image - its kind, purpose, or the kind of equipment used to generate it. | Example | MediaModality
Media.view | Imaging view (projection) used when collecting an image. | Example | MediaCollectionView/Projection
Media.reasonCode | The reason for the media. | Example | ProcedureReasonCodes
Media.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures

10.2.3.2 Implementation Notes

The media resource contains several date/times:

• Media.created[x] - The date(/time) of collection, or the period over which collection occurred

• Media.duration - The duration of the media. The duration might differ from createdPeriod if recording was paused

• Media.content.creation - This should be consistent with the Media.created[x] but might be different due to partial / edited recordings

This resource can embed the image information directly through the attachment.data element. However, good practice is generally to use the attachment.url element to point to a Binary resource. Servers will frequently be able to persist Binary resources in purpose-dedicated repositories more suitable to potentially large artifacts.

10.2.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | Procedure that caused this media to be created | Media.basedOn (CarePlan, ServiceRequest)
created | date | When Media was collected | Media.created
device | reference | Observing Device | Media.device (Device, DeviceMetric)
encounter | reference | Encounter associated with media | Media.encounter (Encounter)
identifier | token | Identifier(s) for the image | Media.identifier
modality | token | The type of acquisition equipment/process | Media.modality
operator | reference | The person who generated the image | Media.operator (Practitioner, Organization, CareTeam, Device, Patient, PractitionerRole, RelatedPerson)
patient | reference | Who/What this Media is a record of | Media.subject.where(resolve() is Patient) (Patient)
site | token | Observed body part | Media.bodySite
status | token | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown | Media.status
subject | reference | Who/What this Media is a record of | Media.subject (Practitioner, Group, Specimen, Device, Patient, PractitionerRole, Location)
type | token | Classification of media as image, video, or audio | Media.type
view | token | Imaging view, e.g. Lateral or Antero-posterior | Media.view

10.2.5 Resource Media - Examples

URL: https://hl7.org/fhir/R4/media-examples.html

10.2.5 Resource Media - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Example Name | id | Format
Simple photo | example | XML | JSON | Turtle
Example Ultrasound Image | 1.2.840.11361907579238403408700.3.1.04.19970327150033 | XML | JSON | Turtle
Example Sound Recording | sound | XML | JSON | Turtle
Example hand x-ray | xray | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.2.6 Resource Media - Detailed Descriptions

URL: https://hl7.org/fhir/R4/media-definitions.html

10.2.6 Resource Media - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the Media resource.

Media
Element Id | Media
Definition | A photo, video, or audio recording acquired or used in healthcare. The actual content may be inline or provided by direct reference.
Cardinality | 0..*
Type | DomainResource
Media.identifier
Element Id | Media.identifier
Definition | Identifiers associated with the image - these may include identifiers for the image itself, identifiers for the context of its collection (e.g. series ids) and context ids such as accession numbers or other workflow identifiers.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Comments | The identifier label and use can be used to determine what kind of identifier it is.
Media.basedOn
Element Id | Media.basedOn
Definition | A procedure that is fulfilled in whole or in part by the creation of this media.

Cardinality | 0..*
Type | Reference(ServiceRequest | CarePlan)
Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
Summary | true
Media.partOf
Element Id | Media.partOf
Definition | A larger event of which this particular event is a component or step.
Cardinality | 0..*
Type | Reference(Any)
Requirements | E.g. Drug administration as part of a procedure, procedure as part of observation, etc.
Alternate Names | container
Summary | true
Comments | Not to be used to link an event to an Encounter - use Media.encounter for that. [The allowed reference resources may be adjusted as appropriate for the event resource].
Media.status
Element Id | Media.status
Definition | The current state of the {{title}}.
Cardinality | 1..1
Terminology Binding | EventStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | A nominal state-transition diagram can be found in the [[event.html#statemachine | Event pattern]] documentation Unknown does not represent "other" - one of the defined statuses must apply. Unknown is used when the authoring system is not sure what the current status is.
Media.type
Element Id | Media.type
Definition | A code that classifies whether the media is an image, video or audio recording or some other media category.
Cardinality | 0..1
Terminology Binding | Media Type (Extensible)
Type | CodeableConcept
Requirements | Used for filtering what observations are retrieved and displayed.
Summary | true
Media.modality
Element Id | Media.modality
Definition | Details of the type of the media - usually, how it was acquired (what type of device). If images sourced from a DICOM system, are wrapped in a Media resource, then this is the modality.
Cardinality | 0..1
Terminology Binding | Media Modality (Example)
Type | CodeableConcept
Summary | true
Media.view
Element Id | Media.view
Definition | The name of the imaging view e.g. Lateral or Antero-posterior (AP).
Cardinality | 0..1
Terminology Binding | Media Collection View/Projection (Example)
Type | CodeableConcept
Summary | true
Media.subject
Element Id | Media.subject
Definition | Who/What this Media is a record of.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | Group | Device | Specimen | Location)
Summary | true
Media.encounter
Element Id | Media.encounter
Definition | The encounter that establishes the context for this media.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | Links the Media to the Encounter context.
Alternate Names | context
Summary | true
Comments | This will typically be the encounter the media occurred within.
Media.created[x]
Element Id | Media.created[x]
Definition | The date and time(s) at which the media was collected.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | timing
Summary | true

Media.issued
Element Id | Media.issued
Definition | The date and time this version of the media was made available to providers, typically after having been reviewed.
Cardinality | 0..1
Type | instant
Summary | true
Comments | It may be the same as the lastUpdated time of the resource itself. For Observations that do require review and verification for certain updates, it might not be the same as the lastUpdated time of the resource itself due to a non-clinically significant update that does not require the new version to be reviewed and verified again.
Media.operator
Element Id | Media.operator
Definition | The person who administered the collection of the image.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson)
Summary | true
Media.reasonCode
Element Id | Media.reasonCode
Definition | Describes why the event occurred in coded or textual form.
Cardinality | 0..*
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be captured using reasonCode.text.
Media.bodySite
Element Id | Media.bodySite
Definition | Indicates the site on the subject's body where the observation was made (i.e. the target site).
Cardinality | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Comments | Only used if not implicit in code found in Observation.code. In many systems, this may be represented as a related observation instead of an inline component. If the use case requires BodySite to be handled as a separate resource (e.g. to identify and track separately) then use the standard extension bodySite.
Media.deviceName
Element Id | Media.deviceName
Definition | The name of the device / manufacturer of the device that was used to make the recording.
Cardinality | 0..1
Type | string
Summary | true
Media.device
Element Id | Media.device
Definition | The device used to collect the media.
Cardinality | 0..1
Type | Reference(Device | DeviceMetric | Device)
Summary | true
Comments | An extension should be used if further typing of the device is needed. Secondary devices used to support collecting a media can be represented using either extension or through the Observation.related element.
Media.height
Element Id | Media.height
Definition | Height of the image in pixels (photo/video).
Cardinality | 0..1
Type | positiveInt
Summary | true
Media.width
Element Id | Media.width
Definition | Width of the image in pixels (photo/video).
Cardinality | 0..1
Type | positiveInt
Summary | true
Media.frames
Element Id | Media.frames
Definition | The number of frames in a photo. This is used with a multi-page fax, or an imaging acquisition context that takes multiple slices in a single image, or an animated gif. If there is more than one frame, this SHALL have a value in order to alert interface software that a multi-frame capable rendering widget is required.
Cardinality | 0..1
Type | positiveInt
Summary | true
Comments | if the number of frames is not supplied, the value may be unknown. Applications should not assume that there is only one frame unless it is explicitly stated.
Media.duration
Element Id | Media.duration
Definition | The duration of the recording in seconds - for audio and video.
Cardinality | 0..1

Type | decimal
Summary | true
Comments | The duration might differ from occurrencePeriod if recording was paused.
Media.content
Element Id | Media.content
Definition | The actual content of the media - inline or by direct reference to the media source file.
Cardinality | 1..1
Type | Attachment
Summary | true
Comments | Recommended content types: image/jpeg, image/png, image/tiff, video/mpeg, audio/mp4, application/dicom. Application/dicom can contain the transfer syntax as a parameter. For media that covers a period of time (video/sound), the content.creationTime is the end time. Creation time is used for tracking, organizing versions and searching.
Media.note
Element Id | Media.note
Definition | Comments made about the media by the performer, subject or other participants.
Cardinality | 0..*
Type | Annotation
Comments | Not to be used for observations, conclusions, etc. Instead use an Observation based on the Media/ImagingStudy resource.

10.2.7 Resource Media - Mappings

URL: https://hl7.org/fhir/R4/media-mappings.html

10.2.7 Resource Media - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

SNOMED CT Attribute Binding

SNOMED CT Concept Domain Binding

RIM Mapping

DICOM Tag Mapping

Mappings for the media resource (see Mappings to Other Standards for further information & status).

10.2.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Media
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class
modality | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
created[x] | FiveWs.done[x]
issued | FiveWs.recorded
operator | FiveWs.actor
reasonCode | FiveWs.why[x]

10.2.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Media | Event
identifier | Event.identifier
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
modality | Event.code
subject | Event.subject
encounter | Event.encounter
created[x] | Event.occurrence[x]
operator | Event.performer.actor
reasonCode | Event.reasonCode
note | Event.note

10.2.7.3 HL7 v2 Mapping (http://hl7.org/v2)

Media
identifier
basedOn

partOf
status | Varies by domain
type
modality
view
subject
encounter
created[x]
issued | OBR.22 (or MSH.7), or perhaps OBX-19 (depends on who observation made)
operator
reasonCode
bodySite
deviceName
device
height
width
frames
duration
content
note | NTE

10.2.7.4 SNOMED CT Attribute Binding (http://snomed.org/attributebinding)

Attribute bindings link coded data elements in FHIR resources to a corresponding attribute in the SNOMED CT concept model. These bindings help to support:

• clarifying the intended meaning of the data element

• Quality checking the alignment between FHIR resource design and any coresponding SNOMED CT concept model

• Composition and decomposition of data instances by indicating the SNOMED CT concept model attribute whose value may be used to decompose a precoordinated concept into this data element

Media
identifier
basedOn
partOf
status
type
modality
view
subject
encounter
created[x]
issued
operator
reasonCode
bodySite | 718497002 |Inherent location|
deviceName
device | 424226004 |Using device|
height
width
frames
duration
content
note

10.2.7.5 SNOMED CT Concept Domain Binding (http://snomed.info/conceptdomain)

Concept domain bindings link a resource or an element to a set of SNOMED CT concepts that represent the intended semantics of the instances (whether or not SNOMED CT is used to encode that data element). This set of concepts is represented using a SNOMED CT expression constraint. Note that the 'Concept domain binding' may be a superset of the 'value set binding'. These bindings help to support:

• Quality checking FHIR resources by ensuring that (a) the intended semantics of the instances matches the valid range of the corresponding SNOMED CT attribute, and (b) the intended value set is appropriate for the intended semantics of the instances

• Semantic checking of data instances by helping to detect potential inconsistencies caused by overlap between the semantics incorporated in two concept domains

Media
identifier
basedOn
partOf
status

type
modality
view
subject
encounter
created[x]
issued
operator
reasonCode
bodySite | < 123037004 |Body structure|
deviceName
device | < 49062001 |Device|
height
width
frames
duration
content
note

10.2.7.6 RIM Mapping (http://hl7.org/v3)

Media | Act[classCode="DGIMG", moodCode="EVN"]
identifier | .id
basedOn | .outboundRelationship[typeCode=FLFS].target
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
status | .status
type | .code
modality | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
view | DiagnosticImage.subjectOrientationCode
subject | .participation[typeCode="SUBJ"]???
encounter | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
created[x] | .effectiveTime
issued | participation[typeCode=AUT].time
operator | .participation[typeCode="PERF"]???
reasonCode | .reasonCode
bodySite | targetSiteCode
deviceName | .participation[typeCode="DEV"].role.player.Entity[classCode="DEV"].name
device | participation[typeCode=DEV]
height | .outboundRelationship[typeCode="COMP].target[classCode="OBS", moodCode="EVN",code="<CODE>"].value
width | .outboundRelationship[typeCode="COMP].target[classCode="OBS", moodCode="EVN",code="<CODE>"].value
frames | .outboundRelationship[typeCode="COMP].target[classCode="OBS", moodCode="EVN",code="<CODE>"].value
duration | .outboundRelationship[typeCode="COMP].target[classCode="OBS", moodCode="EVN",code="<CODE>"].value
content | .text
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]

10.2.7.7 DICOM Tag Mapping (http://nema.org/dicom)

Media
identifier | 0008,0018 | ~0008,0050 | 0020,000D | 0020,000E
basedOn
partOf
status
type
modality | 0008,0060 | 0008,1032
view
subject | ~0010,0020
encounter
created[x]
issued
operator | 0008,1060?
reasonCode
bodySite
deviceName | 0008,0070
device
height | 0028,0010
width | 0028,0011
frames
duration | ~0028,0002
content
note

10.2.8 Resource Media - Extensions & Profiles

URL: https://hl7.org/fhir/R4/media-profiles.html

10.2.8 Resource Media - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

This table lists profiles and extensions for the Media resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.2.8.1 Profiles

No Profiles defined for this resource

10.2.8.2 Extensions

event-location | Media | location | for Event Pattern HL7 Extensions
event-performerFunction | Media.operator | performerFunction | for Event Pattern HL7 Extensions
event-statusReason | Media | statusReason | for Event Pattern HL7 Extensions
workflow-episodeOfCare | Media | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-instantiatesCanonical | Media | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | Media | instantiatesUri | for Workflow Pattern HL7 Extensions
workflow-reasonCode | Media | reasonCode | for Workflow Pattern HL7 Extensions
workflow-reasonReference | Media | reasonReference | for Workflow Pattern HL7 Extensions
workflow-researchStudy | Media | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-reasonReference | reasonReference | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.2.8.3 Search Extensions

No Search Extensions defined for this resource

10.2.9 Resource Media - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/media-version-maps.html

No validation errors - all conversions are clean

10.2.9 Resource Media - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

R3 : R4 Conversion maps for Media.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (0 errors). (see documentation)

10.2.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Media3to4" = "R3 to R4 Conversions for Media"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Media" alias MediaR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Media" alias Media as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Media(source src : MediaR3, target tgt : Media) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.type as vs0 ->  tgt.type as vt0,  vt0.coding as c,  c.system = 'http://terminology.hl7.org/CodeSystem/media-type',
      c.code = vs0;
  src.subtype -> tgt.modality;
  src.view -> tgt.view;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.occurrence : dateTime as vs -> tgt.created = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.created = create('Period') as vt then Period(vs, vt);
  src.operator -> tgt.operator;
  src.reasonCode -> tgt.reasonCode;
```

```
  src.bodySite -> tgt.bodySite;
  src.device -> tgt.device;
  src.height -> tgt.height;
  src.width -> tgt.width;
  src.frames -> tgt.frames;
  src.duration -> tgt.duration;
  src.content -> tgt.content;
  src.note -> tgt.note;
}
```

10.2.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Media4to3" = "R4 to R3 Conversion for Media"

uses "http://hl7.org/fhir/StructureDefinition/Media" alias Media as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Media" alias MediaR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Media(source src : MediaR3, target tgt : Media) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.type as vs0 then {
    vs0.coding as c where system = 'http://terminology.hl7.org/CodeSystem/media-type' then {
      c.code -> tgt.type;
    };
  };
  src.modality -> tgt.subtype;
  src.view -> tgt.view;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.created : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.created : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.operator -> tgt.operator;
  src.reasonCode -> tgt.reasonCode;
  src.bodySite -> tgt.bodySite;
  src.device -> tgt.device;
  src.height -> tgt.height;
  src.width -> tgt.width;
  src.frames -> tgt.frames;
  src.duration -> tgt.duration;
  src.content -> tgt.content;
  src.note -> tgt.note;
}
```

10.2.9.3 R4 Validation Errors

10.2.10 JSON Schema for Media

URL: https://hl7.org/fhir/R4/media.schema.json.html

10.2.10 JSON Schema for Media

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

This is a representation of the json schema for Media, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Media",
  "$ref": "#/definitions/Media",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Media": {
      "description": "A photo, video, or audio recording acquired or used in healthcare. The actual content may be
    inline or provided by direct reference.",
      "properties": {
        "resourceType": {
          "description": "This is a Media resource",
          "const": "Media"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
```

```
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers associated with the image - these may include identifiers for the image itself, identifiers for the context of its collection (e.g. series ids) and context ids such as accession numbers or other workflow identifiers.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A procedure that is fulfilled in whole or in part by the creation of this media.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger event of which this particular event is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the {{title}}.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "A code that classifies whether the media is an image, video or audio recording or some other media category.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "modality": {
      "description": "Details of the type of the media - usually, how it was acquired (what type of device). If images sourced from a DICOM system, are wrapped in a Media resource, then this is the modality.",
```

```
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "view": {
      "description": "The name of the imaging view e.g. Lateral or Antero-posterior (AP).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "Who/What this Media is a record of.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The encounter that establishes the context for this media.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "createdDateTime": {
      "description": "The date and time(s) at which the media was collected.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_createdDateTime": {
      "description": "Extensions for createdDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "createdPeriod": {
      "description": "The date and time(s) at which the media was collected.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "issued": {
      "description": "The date and time this version of the media was made available to providers, typically after
having been reviewed.",
      "$ref": "#/definitions/instant"
    },
    "_issued": {
      "description": "Extensions for issued",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "operator": {
      "description": "The person who administered the collection of the image.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "Describes why the event occurred in coded or textual form.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "bodySite": {
      "description": "Indicates the site on the subject\u0027s body where the observation was made (i.e. the target
site).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "deviceName": {
      "description": "The name of the device / manufacturer of the device  that was used to make the recording.",
      "$ref": "#/definitions/string"
    },
    "_deviceName": {
      "description": "Extensions for deviceName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "device": {
      "description": "The device used to collect the media.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "height": {
      "description": "Height of the image in pixels (photo/video).",
      "$ref": "#/definitions/positiveInt"
    },
    "_height": {
      "description": "Extensions for height",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "width": {
      "description": "Width of the image in pixels (photo/video).",
      "$ref": "#/definitions/positiveInt"
    },
    "_width": {
      "description": "Extensions for width",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "frames": {
      "description": "The number of frames in a photo. This is used with a multi-page fax, or an imaging acquisition
 context that takes multiple slices in a single image, or an animated gif. If there is more than one frame, this
SHALL have a value in order to alert interface software that a multi-frame capable rendering widget is required.",
      "$ref": "#/definitions/positiveInt"
    },
    "_frames": {
      "description": "Extensions for frames",
```

```
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "duration": {
          "description": "The duration of the recording in seconds - for audio and video.",
          "$ref": "#/definitions/decimal"
        },
        "_duration": {
          "description": "Extensions for duration",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "content": {
          "description": "The actual content of the media - inline or by direct reference to the media source file.",
          "$ref": "Attachment.schema.json#/definitions/Attachment"
        },
        "note": {
          "description": "Comments made about the media by the performer, subject or other participants.",
          "items": {
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "content",
        "resourceType"
      ]
    }
  }
}
```


10.2.11 Resource Media - Maturity Dependency View

URL: https://hl7.org/fhir/R4/media-dependencies.html

10.2.11 Resource Media - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner

Dependency Graph for Media FMM level 1

Name | Card. | Type | Dependency Analysis
Media | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(ServiceRequest | CarePlan)
partOf | 0..* | Reference(Any)
status | 1..1 | code
type | 0..1 | CodeableConcept
modality | 0..1 | CodeableConcept
view | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Group | Device | Specimen | Location)
encounter | 0..1 | Reference(Encounter)
created[x] | 0..1
createdDateTime | | dateTime
createdPeriod | | Period
issued | 0..1 | instant
operator | 0..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson)
reasonCode | 0..* | CodeableConcept
bodySite | 0..1 | CodeableConcept
deviceName | 0..1 | string
device | 0..1 | Reference(Device | DeviceMetric | Device)
height | 0..1 | positiveInt
width | 0..1 | positiveInt
frames | 0..1 | positiveInt
duration | 0..1 | decimal
content | 1..1 | Attachment
note | 0..* | Annotation

10.3 Resource DiagnosticReport - Content

URL: https://hl7.org/fhir/R4/diagnosticreport.html

The datatype has a element that may be used to distinguish the identifiers assigned by the requester and the performer of the request (known as the 'Placer' and 'Filler' in the HL7 Version 2 Messaging Standard). Use the identifier type code "PLAC" for the Placer Identifier and "FILL" for the Filler identifier as is shown in the example below:

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

The findings and interpretation of diagnostic tests performed on patients, groups of patients, devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted representation of diagnostic reports.

10.3.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow. It is the intent of the Orders and Observation Workgroup to align this resource with the workflow pattern for event resources.

A diagnostic report is the set of information that is typically provided by a diagnostic service when investigations are complete. The information includes a mix of atomic results, text reports, images, and codes. The mix varies depending on the nature of the diagnostic procedure, and sometimes on the nature of the outcomes for a particular investigation. In FHIR, the report can be conveyed in a variety of ways including a Document, RESTful API, or Messaging framework. Included within each of these, would be the DiagnosticReport resource itself.

The DiagnosticReport resource has information about the diagnostic report itself, and about the subject and, in the case of laboratory tests, the specimen of the report. It can also refer to the request details and atomic observations details or image instances. Report conclusions can be expressed as a simple text blob, structured coded data or as an attached fully formatted report such as a PDF.

The DiagnosticReport resource is suitable for the following kinds of diagnostic reports:

• Laboratory (Clinical Chemistry, Hematology, Microbiology, etc.)

• Pathology / Histopathology / related disciplines

• Imaging Investigations (x-ray, CT, MRI etc.)

• Other diagnostics - Cardiology, Gastroenterology etc.

The DiagnosticReport resource is not intended to support cumulative result presentation (tabular presentation of past and present results in the resource). The DiagnosticReport resource does not yet provide full support for detailed structured reports of sequencing; this is planned for a future release.

10.3.2 Background and Context

Diagnostic Report Names

The words "tests", "results", "observations", "panels" and "batteries" are often used interchangeably when describing the various parts of a diagnostic report. This leads to much confusion. The naming confusion is worsened because of the wide variety of forms that the result of a diagnostic investigation can take, as described above. Languages other than English have their own variations on this theme.

This resource uses one particular set of terms. A practitioner "requests" a set of "tests". The diagnostic service returns a "report" which may contain a "narrative" - a written summary of the outcomes, and/or "results" - the individual pieces of atomic data which each are "observations". The results are assembled in "groups" which are nested structures of Observations (traditionally referred to as "panels" or " batteries" by laboratories) that can be used to represent relationships between the individual data items.

10.3.3 Boundaries and Relationships

Note that many diagnostic processes are procedures that generate observations and diagnostic reports. In many cases, such an observation does not require an explicit representation of the procedure used to create the observation, but where there are details of interest about how the diagnostic procedure was performed, the Procedure resource is used to describe the activity.

In contrast to the Observation resource, the DiagnosticReport resource typically includes additional clinical context and some mix of atomic results, images, imaging reports, textual and coded interpretation, and formatted representations. Laboratory reports, pathology reports, and imaging reports should be represented using the DiagnosticReport resource. The Observation resource is referenced by the DiagnosticReport to provide the atomic results for a particular investigation.

If you have a highly structured report, then use DiagnosticReport - it has data and workflow support. Details about the request for a diagnostic investigation are captured in the various "request" resources (e.g., the ServiceRequest) and allow the report to connect to clinical workflows. For more narrative driven reports with less work flow (histology/mortuary, etc.), the Composition resource would be more appropriate.

Image and media representations of the report and supporting images are referenced in the DiagnosticReport resource. The details and actual image instances can be referenced directly in Diagnostic report using the "imaging" element or by indirect reference through the ImagingStudy resources which represent the content produced in a DICOM imaging study or set of DICOM Instances for a patient.

This resource is referenced by CarePlan, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, Contract, DeviceRequest, DeviceUseStatement, FamilyMemberHistory, GuidanceResponse, ImagingStudy, Immunization, MedicationAdministration, MedicationStatement, Procedure, RequestGroup, RiskAssessment, ServiceRequest and SupplyRequest

10.3.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | TU | | DomainResource | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | What was requested
status | ?!Σ | 1..1 | code | registered | partial | preliminary | final + DiagnosticReportStatus (Required)
category | Σ | 0..* | CodeableConcept | Service category Diagnostic Service Section Codes (Example)
code | Σ | 1..1 | CodeableConcept | Name/Code for this diagnostic report LOINC Diagnostic Report Codes (Preferred)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | Σ | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | Σ | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | Σ | 0..1 | instant | DateTime this version was made
performer | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations
imagingStudy | | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | Σ | 0..* | BackboneElement | Key images associated with this report
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | Σ | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical conclusion (interpretation) of test results
conclusionCode | | 0..* | CodeableConcept | Codes for the clinical conclusion of test results SNOMED CT Clinical Findings (Example)
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DiagnosticReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for report --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final + -->
 <category><!-- 0..* CodeableConcept Service category --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) The subject of the report - usually, but not always, the
    patient --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Health care event when test ordered --></encounter>
 <effective[x]><!-- 0..1 dateTime|Period Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was made -->
 <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
   CareTeam) Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!-- 0..* Reference(Practitioner|PractitionerRole|
   Organization|CareTeam) Primary result interpreter --></resultsInterpreter>
 <specimen><!-- 0..* Reference(Specimen) Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation) Observations --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy) Reference to full details of imaging associated with the diagnostic
    report --></imagingStudy>
 <media>  <!-- 0..* Key images associated with this report -->
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media) Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!-- 0..1 Clinical conclusion (interpretation) of test results -->
 <conclusionCode><!-- 0..* CodeableConcept Codes for the clinical conclusion of test results --></conclusionCode>
 <presentedForm><!-- 0..* Attachment Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{
  "resourceType" : "DiagnosticReport",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for report
  "basedOn" : [{ Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) }], // What was requested
  "status" : "<code>", // R!  registered | partial | preliminary | final +
  "category" : [{ CodeableConcept }], // Service category
  "code" : { CodeableConcept }, // R!  Name/Code for this diagnostic report
  "subject" : { Reference(Patient|Group|Device|Location) }, // The subject of the report - usually, but not always, the
    patient
  "encounter" : { Reference(Encounter) }, // Health care event when test ordered
  // effective[x]: Clinically relevant time/time-period for report. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "issued" : "<instant>", // DateTime this version was made
  "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
   CareTeam) }], // Responsible Diagnostic Service
  "resultsInterpreter" : [{ Reference(Practitioner|PractitionerRole|
   Organization|CareTeam) }], // Primary result interpreter
  "specimen" : [{ Reference(Specimen) }], // Specimens this report is based on
  "result" : [{ Reference(Observation) }], // Observations
  "imagingStudy" : [{ Reference(ImagingStudy) }], // Reference to full details of imaging associated with the diagnostic
     report
  "media" : [{ // Key images associated with this report
    "comment" : "<string>", // Comment about the image (e.g. explanation)
    "link" : { Reference(Media) } // R!  Reference to the image source
  }],
  "conclusion" : "<string>", // Clinical conclusion (interpretation) of test results
  "conclusionCode" : [{ CodeableConcept }], // Codes for the clinical conclusion of test results
  "presentedForm" : [{ Attachment }] // Entire report as issued
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DiagnosticReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DiagnosticReport.identifier [ Identifier ], ... ; # 0..* Business identifier for report
  fhir:DiagnosticReport.basedOn [ Reference(CarePlan|ImmunizationRecommendation|MedicationRequest|NutritionOrder|
  ServiceRequest) ], ... ; # 0..* What was requested
  fhir:DiagnosticReport.status [ code ]; # 1..1 registered | partial | preliminary | final +
  fhir:DiagnosticReport.category [ CodeableConcept ], ... ; # 0..* Service category
  fhir:DiagnosticReport.code [ CodeableConcept ]; # 1..1 Name/Code for this diagnostic report
  fhir:DiagnosticReport.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 The subject of the report -
    usually, but not always, the patient
  fhir:DiagnosticReport.encounter [ Reference(Encounter) ]; # 0..1 Health care event when test ordered
  # DiagnosticReport.effective[x] : 0..1 Clinically relevant time/time-period for report. One of these 2
    fhir:DiagnosticReport.effectiveDateTime [ dateTime ]
    fhir:DiagnosticReport.effectivePeriod [ Period ]
  fhir:DiagnosticReport.issued [ instant ]; # 0..1 DateTime this version was made
  fhir:DiagnosticReport.performer [ Reference(Practitioner|PractitionerRole|Organization|CareTeam) ], ... ; # 0..*
    Responsible Diagnostic Service
  fhir:DiagnosticReport.resultsInterpreter [ Reference(Practitioner|PractitionerRole|Organization|CareTeam) ], ... ; #
    0..* Primary result interpreter
  fhir:DiagnosticReport.specimen [ Reference(Specimen) ], ... ; # 0..* Specimens this report is based on
  fhir:DiagnosticReport.result [ Reference(Observation) ], ... ; # 0..* Observations
  fhir:DiagnosticReport.imagingStudy [ Reference(ImagingStudy) ], ... ; # 0..* Reference to full details of imaging
    associated with the diagnostic report
  fhir:DiagnosticReport.media [ # 0..* Key images associated with this report
    fhir:DiagnosticReport.media.comment [ string ]; # 0..1 Comment about the image (e.g. explanation)
    fhir:DiagnosticReport.media.link [ Reference(Media) ]; # 1..1 Reference to the image source
  ], ...;
  fhir:DiagnosticReport.conclusion [ string ]; # 0..1 Clinical conclusion (interpretation) of test results
  fhir:DiagnosticReport.conclusionCode [ CodeableConcept ], ... ; # 0..* Codes for the clinical conclusion of test
    results
  fhir:DiagnosticReport.presentedForm [ Attachment ], ... ; # 0..* Entire report as issued
]
```

Changes since R3

DiagnosticReport

DiagnosticReport.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

DiagnosticReport.status | Change value set from http://hl7.org/fhir/ValueSet/diagnostic-report-status to http://hl7.org/fhir/ValueSet/diagnostic-report-status|4.0.1

DiagnosticReport.category | Max Cardinality changed from 1 to *

DiagnosticReport.encounter | Added Element

DiagnosticReport.performer | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | CareTeam)

DiagnosticReport.resultsInterpreter | Added Element

DiagnosticReport.imagingStudy | Type Reference: Removed Target Type ImagingManifest

DiagnosticReport.media | Renamed from image to media

DiagnosticReport.media.comment | Moved from DiagnosticReport.image to DiagnosticReport.media

DiagnosticReport.media.link | Moved from DiagnosticReport.image to DiagnosticReport.media

DiagnosticReport.conclusionCode | Renamed from codedDiagnosis to conclusionCode
DiagnosticReport.context | deleted
DiagnosticReport.performer.role | deleted
DiagnosticReport.performer.actor | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and 9 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | TU | | DomainResource | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier for report
basedOn | | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | What was requested
status | ?!Σ | 1..1 | code | registered | partial | preliminary | final + DiagnosticReportStatus (Required)
category | Σ | 0..* | CodeableConcept | Service category Diagnostic Service Section Codes (Example)
code | Σ | 1..1 | CodeableConcept | Name/Code for this diagnostic report LOINC Diagnostic Report Codes (Preferred)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | Σ | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | Σ | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | Σ | 0..1 | instant | DateTime this version was made
performer | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | Σ | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | | 0..* | Reference(Specimen) | Specimens this report is based on
result | | 0..* | Reference(Observation) | Observations
imagingStudy | | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | Σ | 0..* | BackboneElement | Key images associated with this report
comment | | 0..1 | string | Comment about the image (e.g. explanation)
link | Σ | 1..1 | Reference(Media) | Reference to the image source
conclusion | | 0..1 | string | Clinical conclusion (interpretation) of test results
conclusionCode | | 0..* | CodeableConcept | Codes for the clinical conclusion of test results SNOMED CT Clinical Findings (Example)
presentedForm | | 0..* | Attachment | Entire report as issued
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DiagnosticReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for report --></identifier>
 <basedOn><!-- 0..* Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) What was requested --></basedOn>
 <status value="[code]"/><!-- 1..1 registered | partial | preliminary | final + -->
 <category><!-- 0..* CodeableConcept Service category --></category>
 <code><!-- 1..1 CodeableConcept Name/Code for this diagnostic report --></code>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Location) The subject of the report - usually, but not always, the
    patient --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Health care event when test ordered --></encounter>
 <effective[x]><!-- 0..1 dateTime|Period Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!-- 0..1 DateTime this version was made -->
 <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
   CareTeam) Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!-- 0..* Reference(Practitioner|PractitionerRole|
   Organization|CareTeam) Primary result interpreter --></resultsInterpreter>
 <specimen><!-- 0..* Reference(Specimen) Specimens this report is based on --></specimen>
 <result><!-- 0..* Reference(Observation) Observations --></result>
 <imagingStudy><!-- 0..* Reference(ImagingStudy) Reference to full details of imaging associated with the diagnostic
    report --></imagingStudy>
 <media>  <!-- 0..* Key images associated with this report -->
  <comment value="[string]"/><!-- 0..1 Comment about the image (e.g. explanation) -->
  <link><!-- 1..1 Reference(Media) Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!-- 0..1 Clinical conclusion (interpretation) of test results -->
 <conclusionCode><!-- 0..* CodeableConcept Codes for the clinical conclusion of test results --></conclusionCode>
 <presentedForm><!-- 0..* Attachment Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{
  "resourceType" : "DiagnosticReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for report
```

```
  "basedOn" : [{ Reference(CarePlan|ImmunizationRecommendation|
   MedicationRequest|NutritionOrder|ServiceRequest) }], // What was requested
  "status" : "<code>", // R!  registered | partial | preliminary | final +
  "category" : [{ CodeableConcept }], // Service category
  "code" : { CodeableConcept }, // R!  Name/Code for this diagnostic report
  "subject" : { Reference(Patient|Group|Device|Location) }, // The subject of the report - usually, but not always, the
    patient
  "encounter" : { Reference(Encounter) }, // Health care event when test ordered
  // effective[x]: Clinically relevant time/time-period for report. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "issued" : "<instant>", // DateTime this version was made
  "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
   CareTeam) }], // Responsible Diagnostic Service
  "resultsInterpreter" : [{ Reference(Practitioner|PractitionerRole|
   Organization|CareTeam) }], // Primary result interpreter
  "specimen" : [{ Reference(Specimen) }], // Specimens this report is based on
  "result" : [{ Reference(Observation) }], // Observations
  "imagingStudy" : [{ Reference(ImagingStudy) }], // Reference to full details of imaging associated with the diagnostic
     report
  "media" : [{ // Key images associated with this report
    "comment" : "<string>", // Comment about the image (e.g. explanation)
    "link" : { Reference(Media) } // R!  Reference to the image source
  }],
  "conclusion" : "<string>", // Clinical conclusion (interpretation) of test results
  "conclusionCode" : [{ CodeableConcept }], // Codes for the clinical conclusion of test results
  "presentedForm" : [{ Attachment }] // Entire report as issued
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DiagnosticReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DiagnosticReport.identifier [ Identifier ], ... ; # 0..* Business identifier for report
  fhir:DiagnosticReport.basedOn [ Reference(CarePlan|ImmunizationRecommendation|MedicationRequest|NutritionOrder|
  ServiceRequest) ], ... ; # 0..* What was requested
  fhir:DiagnosticReport.status [ code ]; # 1..1 registered | partial | preliminary | final +
  fhir:DiagnosticReport.category [ CodeableConcept ], ... ; # 0..* Service category
  fhir:DiagnosticReport.code [ CodeableConcept ]; # 1..1 Name/Code for this diagnostic report
  fhir:DiagnosticReport.subject [ Reference(Patient|Group|Device|Location) ]; # 0..1 The subject of the report -
    usually, but not always, the patient
  fhir:DiagnosticReport.encounter [ Reference(Encounter) ]; # 0..1 Health care event when test ordered
  # DiagnosticReport.effective[x] : 0..1 Clinically relevant time/time-period for report. One of these 2
    fhir:DiagnosticReport.effectiveDateTime [ dateTime ]
    fhir:DiagnosticReport.effectivePeriod [ Period ]
  fhir:DiagnosticReport.issued [ instant ]; # 0..1 DateTime this version was made
  fhir:DiagnosticReport.performer [ Reference(Practitioner|PractitionerRole|Organization|CareTeam) ], ... ; # 0..*
    Responsible Diagnostic Service
  fhir:DiagnosticReport.resultsInterpreter [ Reference(Practitioner|PractitionerRole|Organization|CareTeam) ], ... ; #
    0..* Primary result interpreter
  fhir:DiagnosticReport.specimen [ Reference(Specimen) ], ... ; # 0..* Specimens this report is based on
  fhir:DiagnosticReport.result [ Reference(Observation) ], ... ; # 0..* Observations
  fhir:DiagnosticReport.imagingStudy [ Reference(ImagingStudy) ], ... ; # 0..* Reference to full details of imaging
    associated with the diagnostic report
  fhir:DiagnosticReport.media [ # 0..* Key images associated with this report
    fhir:DiagnosticReport.media.comment [ string ]; # 0..1 Comment about the image (e.g. explanation)
    fhir:DiagnosticReport.media.link [ Reference(Media) ]; # 1..1 Reference to the image source
  ], ...;
  fhir:DiagnosticReport.conclusion [ string ]; # 0..1 Clinical conclusion (interpretation) of test results
  fhir:DiagnosticReport.conclusionCode [ CodeableConcept ], ... ; # 0..* Codes for the clinical conclusion of test
    results
  fhir:DiagnosticReport.presentedForm [ Attachment ], ... ; # 0..* Entire report as issued
]
```

Changes since Release 3

DiagnosticReport

DiagnosticReport.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

DiagnosticReport.status | Change value set from http://hl7.org/fhir/ValueSet/diagnostic-report-status to http://hl7.org/fhir/ValueSet/diagnostic-report-status|4.0.1

DiagnosticReport.category | Max Cardinality changed from 1 to *

DiagnosticReport.encounter | Added Element

DiagnosticReport.performer | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | CareTeam)

DiagnosticReport.resultsInterpreter | Added Element

DiagnosticReport.imagingStudy | Type Reference: Removed Target Type ImagingManifest

DiagnosticReport.media | Renamed from image to media

DiagnosticReport.media.comment | Moved from DiagnosticReport.image to DiagnosticReport.media

DiagnosticReport.media.link | Moved from DiagnosticReport.image to DiagnosticReport.media

DiagnosticReport.conclusionCode | Renamed from codedDiagnosis to conclusionCode

DiagnosticReport.context | deleted
DiagnosticReport.performer.role | deleted
DiagnosticReport.performer.actor | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and 9 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.3.4.1 Terminology Bindings

Path | Definition | Type | Reference
DiagnosticReport.status | The status of the diagnostic report. | Required | DiagnosticReportStatus
DiagnosticReport.category | Codes for diagnostic service sections. | Example | DiagnosticServiceSectionCodes
DiagnosticReport.code | Codes that describe Diagnostic Reports. | Preferred | LOINCDiagnosticReportCodes
DiagnosticReport.conclusionCode | Diagnosis codes provided as adjuncts to the report. | Example | SNOMEDCTClinicalFindings

10.3.5 Notes:

10.3.5.1 Identifiers

identifier

type

```
<!-- Placer identifier-->
<identifier>
<type>
<coding>
  <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
  <code value="PLAC"/>
</coding>
<text value="Placer"/>
</type>
<system value="urn:oid:1.3.4.5.6.7"/>
<value value="2345234234234"/>
</identifier>
<!-- Filler identifier-->
<identifier>
<type>
<coding>
  <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
  <code value="PLAC"/>
</coding>
<text value="Placer"/>
</type>
<system value=" http://terminology.hl7.org/CodeSystem/v2-0203"/>
<value value="567890"/>
</identifier>
```

10.3.5.2 Clinically Relevant Time

If the diagnostic procedure was performed on the patient directly, the effective[x] element is a dateTime, the time it was performed. If specimens were taken, the clinically relevant time of the report can be derived from the specimen collection times, but since detailed specimen information is not always available, and nor is the clinically relevant time always exactly the specimen collection time (e.g. complex timed tests), the reports SHALL always include an effective[x] element. Note that HL7 v2 messages often carry a diagnostically relevant time without carrying any specimen information.

10.3.5.3 Associated Observations

• The DiagnosticReport.code always contains the name of the report itself. The report can also contain a set of Observations in the DiagnosticReport.result element. These Observations can be simple observations (e.g. atomic results) or groups/panels of other observations. The Observation.code indicates the nature of the observation or panel (e.g. individual measure, organism isolate/sensitivity or antibody functional testing). When relevant, the observation can specify a particular specimen from which the result comes. Examples of nesting groups/panels within an observation include reporting a "profile" consisting of several panels as is shown in this example, a group of antibiotic isolate/sensitivities for a bacterial culture, or a set of perinatal measurements on a single fetus. There is rarely a need for more than two levels of nesting in the Observation tree. One known use is for organism sensitivities - an example of a complex Micro Isolate and Sensitivities is provided.

10.3.5.4 Associated Images

ImagingStudy and ImageObjectStudy and the media element are somewhat overlapping - typically, the list of image references in the image element will also be found in one of the imaging study resources. However, each caters to different types of displays for different types of purposes. Neither, either, or both may be provided.

10.3.5.5 Diagnostic Report Status

• Applications consuming diagnostic reports must take careful note of updated (revised) reports and ensure that retracted reports are appropriately handled.

• For applications providing diagnostic reports, a report should not be final until all the individual data items reported with it are final or appended.

• If the report has been withdrawn following a previous final release, the DiagnosticReport and associated Observations should be retracted by replacing the status codes with the concept "entered-in-error" and setting the conclusion/comment (if provided) and the text narrative to some text like "This report has been withdrawn" in the appropriate language. A reason for retraction may be provided in the narrative.

10.3.5.6 Report Content

Typically, a report is either: all data, no narrative (e.g. Core lab) or a mix of data with some concluding narrative (e.g. Structured Pathology Report, Bone Density), or all narrative (for example a typical imaging report, histopathology). This resource provides for these 3 different presentations:

• As atomic data items: a hierarchical set of nested references to Observation resources often including pathologist/radiologist interpretation(s), one or more images, and possibly with a conclusion and/or one or more coded diagnoses

• As narrative: an XHTML presentation in the standard resource narrative

• As a "presented form": A rich text representation of the report - typically a PDF

Note that the conclusion and the coded diagnoses are part of the atomic data and SHOULD be duplicated in the narrative and in the presented form if the latter is present. The narrative and the presented form serve the same function: a representation of the report for a human. The presented form is included since diagnostic service reports often contain presentation features that are not easy to reproduce in the HTML narrative. Whether or not the presented form is included, the narrative must be a clinically safe view of the diagnostic report; at a minimum, this could be fulfilled by a note indicating that the narrative is not proper representation of the report, and that the presented form must be used, or a generated view from the atomic data. However, consumers of the report will best be served if the narrative contains clinically relevant data from the form. Commonly, the following patterns are used:

• Simple Laboratory Reports: A single set of atomic observations, and a tabular presentation in narrative. This is typically encountered in high volume areas such as Biochemistry and Hematology

• Histopathology Report: A document report in a presented form and the narrative. Possibly a few key images, and some coded diagnoses for registries. If the service is creating a structured report, some atomic data may be included

• imaging Report: A document report in a presented form and the narrative, with an imaging study reference and possibly some key images. Some imaging reports such as a Bone Density Scan may include some atomic data

Note that the nature of reports from the various disciplines that provide diagnostic reports are changing quickly, as expert systems provide improved narrative reporting in high volume reports, structured reporting brings additional data to areas that have classically been narrative based, and the nature of the imaging and laboratory procedures are merging. Therefore, these patterns described above are only examples of how a diagnostic report can be used.

10.3.5.7 Genetic Observations

Genetic reporting makes heavy use of the DiagnosticReport and Observation resources. An implementation guide describing how to represent genetic results can be found here .

10.3.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | Reference to the service request. | DiagnosticReport.basedOn (CarePlan, MedicationRequest, NutritionOrder, ServiceRequest, ImmunizationRecommendation)
category | token | Which diagnostic discipline/department created the report | DiagnosticReport.category
code | token | The code for the report, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result | DiagnosticReport.code | 13 Resources
conclusion | token | A coded conclusion (interpretation/impression) on the report | DiagnosticReport.conclusionCode
date | date | The clinically relevant time of the report | DiagnosticReport.effective | 17 Resources
encounter | reference | The Encounter when the order was made | DiagnosticReport.encounter (Encounter) | 12 Resources
identifier | token | An identifier for the report | DiagnosticReport.identifier | 30 Resources
issued | date | When the report was issued | DiagnosticReport.issued
media | reference | A reference to the image source. | DiagnosticReport.media.link (Media)
patient | reference | The subject of the report if a patient | DiagnosticReport.subject.where(resolve() is Patient) (Patient) | 33 Resources

performer | reference | Who is responsible for the report | DiagnosticReport.performer (Practitioner, Organization, CareTeam, PractitionerRole)
result | reference | Link to an atomic result (observation resource) | DiagnosticReport.result (Observation)
results-interpreter | reference | Who was the source of the report | DiagnosticReport.resultsInterpreter (Practitioner, Organization, CareTeam, PractitionerRole)
specimen | reference | The specimen details | DiagnosticReport.specimen (Specimen)
status | token | The status of the report | DiagnosticReport.status
subject | reference | The subject of the report | DiagnosticReport.subject (Group, Device, Patient, Location)

10.3.7 Resource DiagnosticReport - Examples

URL: https://hl7.org/fhir/R4/diagnosticreport-examples.html

10.3.7 Resource DiagnosticReport - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

10.3.7.1 Laboratory Results

The following examples of DiagnosticReport resources show examples of laboratory reports:

- General Lab Report Example
- Various reports gathered from v2 messages
- Real-world patient example
- Real-world patient example
- Lipid profile
- An Example of a report with multiple specimen and panels

10.3.7.1.1 Micro Isolate(Culture) and Susceptibility

The following 2 examples of DiagnosticReport resources show example of a complex Micro Isolate and Sensitivities with both 1 and 2 levels of nested Organism and Susceptibilty panels using the related Observation Structure

Micro Isolate and Sensitivities-1: An example of a complex Micro Isolate and Sensitivities with 2 level nested Organism and Susceptibilty panels using the related Observation Structure. The following diagram is provided above to help teh reader navigate the nested structures.

Micro Isolate and Sensitivities-2: An collection Bundle showing an example of a complex Micro Isolate and Sensitivities based on the HL7 V2 Lab Reports Interface (LRI) example. This example has a single level nested Organism panels using the related Observation Structure. The following diagram is provided to help the reader navigate the nested structures.

10.3.7.2 Imaging Studies

The following examples of DiagnosticReport resources show examples of various imaging reports:

- Real-world patient example
- Imaging Example (Bone Density DXA)
- Abdominal Ultrasound example

10.3.7.3 Genomics

The following examples of DiagnosticReport resources show examples of genomics studies:

- An example of a genetics test report with Family member history
- An example of a HLA genotyping results report
- An example of a comprehensive bone marrow report
- An example of a PGx haplotype report

10.3.7.4 Anatomic Pathology

The following examples of DiagnosticReport resources show examples of anatomic pathology reports:

- An example of a presented form anatomic pathology report
- An example of a principally narrative anatomic pathology report

Example Name | id | Format
General Laboratory Report Example | 101 | XML | JSON | Turtle
Various reports gathered from v2 messages | 72ac8493-52ac-41bd-8d5d-7258c289b5ea | XML | JSON | Turtle
Real-world patient example | f001 | XML | JSON | Turtle
Real-world patient example | f201 | XML | JSON | Turtle
Real-world patient example | f202 | XML | JSON | Turtle
Imaging Example (Bone Density DXA) | 102 | XML | JSON | Turtle
An example of a complex Micro Isolate and Sensitivities. | micro | XML | JSON | Turtle

An collection Bundle showing an example of a complex Micro Isolate and Sensitivities | lri-example | XML | JSON | Turtle
Abdominal Ultrasound example | ultrasound | XML | JSON | Turtle
An Example of a report with multiple specimen and panels | ghp | XML | JSON | Turtle
An example of a principally narrative anatomic pathology report | pap | XML | JSON | Turtle
An example of a presented form anatomic pathology report | gingival-mass | XML | JSON | Turtle
Lipid profile | lipids | XML | JSON | Turtle | for Profile Example Lipid Profile
An example of a genetics test report with Family member history | dg2 | XML | JSON | Turtle | for Profile DiagnosticReport-Genetics
An example of a comprehensive bone marrow report | report | XML | JSON | Turtle | for Profile DiagnosticReport-Genetics
An example of a PGx haplotype report | example-pgx | XML | JSON | Turtle | for Profile DiagnosticReport-Genetics
An example of a HLA genotyping results report | hla-1 | XML | JSON | Turtle | for Profile Profile for HLA Genotyping Results

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.3.8 Resource DiagnosticReport - Detailed Descriptions

URL: https://hl7.org/fhir/R4/diagnosticreport-definitions.html

10.3.8 Resource DiagnosticReport - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the DiagnosticReport resource.

DiagnosticReport
Element Id | DiagnosticReport
Definition | The findings and interpretation of diagnostic tests performed on patients, groups of patients, devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted representation of diagnostic reports.
Cardinality | 0..*
Type | DomainResource
Requirements | To support reporting for any diagnostic report into a clinical data repository.
Alternate Names | Report; Test; Result; Results; Labs; Laboratory
Comments | This is intended to capture a single report and is not suitable for use in displaying summary information that covers multiple reports. For example, this resource has not been designed for laboratory cumulative reporting formats nor detailed structured reports for sequencing.
DiagnosticReport.identifier
Element Id | DiagnosticReport.identifier
Definition | Identifiers assigned to this report by the performer or other systems.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Need to know what identifier to use when making queries about this report from the source laboratory, and for linking to the report outside FHIR context.
Alternate Names | ReportID; Filler ID; Placer ID
Summary | true
Comments | Usually assigned by the Information System of the diagnostic service provider (filler id).
DiagnosticReport.basedOn
Element Id | DiagnosticReport.basedOn
Definition | Details concerning a service requested.
Cardinality | 0..*
Type | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)
Requirements | This allows tracing of authorization for the report and tracking whether proposals/recommendations were acted upon.
Alternate Names | Request
Comments | Note: Usually there is one test request for each result, however in some circumstances multiple test requests may be represented using a single test result resource. Note that there are also cases where one request leads to multiple reports.
DiagnosticReport.status
Element Id | DiagnosticReport.status
Definition | The status of the diagnostic report.
Cardinality | 1..1
Terminology Binding | DiagnosticReportStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | Diagnostic services routinely issue provisional/incomplete reports, and sometimes withdraw previously released reports.
Summary | true
DiagnosticReport.category
Element Id | DiagnosticReport.category
Definition | A code that classifies the clinical discipline, department or diagnostic service that created the report (e.g. cardiology, biochemistry, hematology, MRI). This is used for searching, sorting and display purposes.
Cardinality | 0..*
Terminology Binding | Diagnostic Service Section Codes (Example)
Type | CodeableConcept

Alternate Names | Department; Sub-department; Service; Discipline
Summary | true
Comments | Multiple categories are allowed using various categorization schemes. The level of granularity is defined by the category concepts in the value set. More fine-grained filtering can be performed using the metadata and/or terminology hierarchy in DiagnosticReport.code.
DiagnosticReport.code
Element Id | DiagnosticReport.code
Definition | A code or name that describes this diagnostic report.
Cardinality | 1..1
Terminology Binding | LOINC Diagnostic Report Codes (Preferred)
Type | CodeableConcept
Alternate Names | Type
Summary | true
DiagnosticReport.subject
Element Id | DiagnosticReport.subject
Definition | The subject of the report. Usually, but not always, this is a patient. However, diagnostic services also perform analyses on specimens collected from a variety of other sources.
Cardinality | 0..1
Type | Reference(Patient | Group | Device | Location)
Requirements | SHALL know the subject context.
Alternate Names | Patient
Summary | true
DiagnosticReport.encounter
Element Id | DiagnosticReport.encounter
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) which this DiagnosticReport is about.
Cardinality | 0..1
Type | Reference(Encounter)
Requirements | Links the request to the Encounter context.
Alternate Names | Context
Summary | true
Comments | This will typically be the encounter the event occurred within, but some events may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter (e.g. pre-admission laboratory tests).
DiagnosticReport.effective[x]
Element Id | DiagnosticReport.effective[x]
Definition | The time or time-period the observed values are related to. When the subject of the report is a patient, this is usually either the time of the procedure or of specimen collection(s), but very often the source of the date/time is not known, only the date/time itself.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Need to know where in the patient history to file/present this report.
Alternate Names | Observation time; Effective Time; Occurrence
Summary | true
Comments | If the diagnostic procedure was performed on the patient, this is the time it was performed. If there are specimens, the diagnostically relevant time can be derived from the specimen collection times, but the specimen information is not always available, and the exact relationship between the specimens and the diagnostically relevant time is not always automatic.
DiagnosticReport.issued
Element Id | DiagnosticReport.issued
Definition | The date and time that this version of the report was made available to providers, typically after the report was reviewed and verified.
Cardinality | 0..1
Type | instant
Requirements | Clinicians need to be able to check the date that the report was released.
Alternate Names | Date published; Date Issued; Date Verified
Summary | true
Comments | May be different from the update time of the resource itself, because that is the status of the record (potentially a secondary copy), not the actual release time of the report.
DiagnosticReport.performer
Element Id | DiagnosticReport.performer
Definition | The diagnostic service that is responsible for issuing the report.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam)
Requirements | Need to know whom to contact if there are queries about the results. Also may need to track the source of reports for secondary data analysis.
Alternate Names | Laboratory; Service; Practitioner; Department; Company; Authorized by; Director
Summary | true
Comments | This is not necessarily the source of the atomic data items or the entity that interpreted the results. It is the entity that takes responsibility for the clinical report.
DiagnosticReport.resultsInterpreter
Element Id | DiagnosticReport.resultsInterpreter
Definition | The practitioner or organization that is responsible for the report's conclusions and interpretations.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam)

Requirements | Need to know whom to contact if there are queries about the results. Also may need to track the source of reports for secondary data analysis.
Alternate Names | Analyzed by; Reported by
Summary | true
Comments | Might not be the same entity that takes responsibility for the clinical report.
DiagnosticReport.specimen
Element Id | DiagnosticReport.specimen
Definition | Details about the specimens on which this diagnostic report is based.
Cardinality | 0..*
Type | Reference(Specimen)
Requirements | Need to be able to report information about the collected specimens on which the report is based.
Comments | If the specimen is sufficiently specified with a code in the test result name, then this additional data may be redundant. If there are multiple specimens, these may be represented per observation or group.
DiagnosticReport.result
Element Id | DiagnosticReport.result
Definition | Observations that are part of this diagnostic report.
Cardinality | 0..*
Type | Reference(Observation)
Requirements | Need to support individual results, or groups of results, where the result grouping is arbitrary, but meaningful.
Alternate Names | Data; Atomic Value; Result; Atomic result; Data; Test; Analyte; Battery; Organizer
Comments | Observations can contain observations.
DiagnosticReport.imagingStudy
Element Id | DiagnosticReport.imagingStudy
Definition | One or more links to full details of any imaging performed during the diagnostic investigation. Typically, this is imaging performed by DICOM enabled modalities, but this is not required. A fully enabled PACS viewer can use this information to provide views of the source images.
Cardinality | 0..*
Type | Reference(ImagingStudy)
Comments | ImagingStudy and the image element are somewhat overlapping - typically, the list of image references in the image element will also be found in one of the imaging study resources. However, each caters to different types of displays for different types of purposes. Neither, either, or both may be provided.
DiagnosticReport.media
Element Id | DiagnosticReport.media
Definition | A list of key images associated with this report. The images are generally created during the diagnostic process, and may be directly of the patient, or of treated specimens (i.e. slides of interest).
Cardinality | 0..*
Requirements | Many diagnostic services include images in the report as part of their service.
Alternate Names | DICOM; Slides; Scans
Summary | true
DiagnosticReport.media.comment
Element Id | DiagnosticReport.media.comment
Definition | A comment about the image. Typically, this is used to provide an explanation for why the image is included, or to draw the viewer's attention to important features.
Cardinality | 0..1
Type | string
Requirements | The provider of the report should make a comment about each image included in the report.
Comments | The comment should be displayed with the image. It would be common for the report to include additional discussion of the image contents in other sections such as the conclusion.
DiagnosticReport.media.link
Element Id | DiagnosticReport.media.link
Definition | Reference to the image source.
Cardinality | 1..1
Type | Reference(Media)
Summary | true
DiagnosticReport.conclusion
Element Id | DiagnosticReport.conclusion
Definition | Concise and clinically contextualized summary conclusion (interpretation/impression) of the diagnostic report.
Cardinality | 0..1
Type | string
Requirements | Need to be able to provide a conclusion that is not lost among the basic result data.
Alternate Names | Report
DiagnosticReport.conclusionCode
Element Id | DiagnosticReport.conclusionCode
Definition | One or more codes that represent the summary conclusion (interpretation/impression) of the diagnostic report.
Cardinality | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
DiagnosticReport.presentedForm
Element Id | DiagnosticReport.presentedForm
Definition | Rich text representation of the entire result as issued by the diagnostic service. Multiple formats are allowed but they SHALL be semantically equivalent.
Cardinality | 0..*
Type | Attachment

Requirements | Gives laboratory the ability to provide its own fully formatted report for clinical fidelity.
Comments | "application/pdf" is recommended as the most reliable and interoperable in this context.

10.3.9 Resource DiagnosticReport - Mappings

URL: https://hl7.org/fhir/R4/diagnosticreport-mappings.html

10.3.9 Resource DiagnosticReport - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the diagnosticreport resource (see Mappings to Other Standards for further information & status).

10.3.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DiagnosticReport
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
effective[x] | FiveWs.done[x]
issued | FiveWs.recorded
performer | FiveWs.actor
resultsInterpreter | FiveWs.actor

10.3.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

DiagnosticReport | Event
identifier | Event.identifier
basedOn | Event.basedOn
status | Event.status
code | Event.code
subject | Event.subject
encounter | Event.encounter
effective[x] | Event.occurrence[x]
performer | Event.performer.actor
resultsInterpreter | Event.performer.actor

10.3.9.3 HL7 v2 Mapping (http://hl7.org/v2)

DiagnosticReport | ORU -> OBR
identifier | OBR-51/ for globally unique filler ID - OBR-3 , For non-globally unique filler-id the fllller/placer number must be combined with the universal service Id - OBR-2(if present)+OBR-3+OBR-4
basedOn | ORC? OBR-2/3?
status | OBR-25 (not 1:1 mapping)
category | OBR-24
code | OBR-4 (HL7 v2 doesn't provide an easy way to indicate both the ordered test and the performed panel)
subject | PID-3 (no HL7 v2 mapping for Group or Device)
encounter | PV1-19
effective[x] | OBR-7
issued | OBR-22
performer | PRT-8 (where this PRT-4-Participation = "PO")
resultsInterpreter | OBR-32, PRT-8 (where this PRT-4-Participation = "PI")
specimen | SPM
result | OBXs
imagingStudy
media | OBX?
comment
link
conclusion | OBX
conclusionCode | OBX
presentedForm | OBX

10.3.9.4 RIM Mapping (http://hl7.org/v3)

DiagnosticReport | Observation[classCode=OBS, moodCode=EVN]
identifier | id
basedOn | outboundRelationship[typeCode=FLFS].target
status | statusCode Note: final and amended are distinguished by whether observation is the subject of a ControlAct event of type "revise"
category | inboundRelationship[typeCode=COMP].source[classCode=LIST, moodCode=EVN, code < LabService].code
code | code
subject | participation[typeCode=SBJ]
encounter | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
effective[x] | effectiveTime
issued | participation[typeCode=VRF or AUT].time
performer | .participation[typeCode=PRF]
resultsInterpreter | .participation[typeCode=PRF]
specimen | participation[typeCode=SBJ]
result | outboundRelationship[typeCode=COMP].target
imagingStudy | outboundRelationship[typeCode=COMP].target[classsCode=DGIMG, moodCode=EVN]
media | outboundRelationship[typeCode=COMP].target
comment | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="annotation"].value
link | .value.reference
conclusion | inboundRelationship[typeCode="SPRT"].source[classCode=OBS, moodCode=EVN, code=LOINC:48767-8].value (type=ST)
conclusionCode | inboundRelationship[typeCode=SPRT].source[classCode=OBS, moodCode=EVN, code=LOINC:54531-9].value (type=CD)
presentedForm | text (type=ED)

10.3.10 Resource DiagnosticReport - Extensions & Profiles

URL: https://hl7.org/fhir/R4/diagnosticreport-profiles.html

10.3.10 Resource DiagnosticReport - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

This table lists profiles and extensions for the DiagnosticReport resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.3.10.1 Profiles

Profile | Description | Context
DiagnosticReport-Genetics | Describes how the DiagnosticReport resource is used to report structured genetic test results | DiagnosticReport-Genetics
HLAResult | Describes how the HLA genotyping results | Profile for HLA Genotyping Results
LipidProfile | Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes | Example Lipid Profile

10.3.10.2 Extensions

DiagnosticReport-geneticsAnalysis | DiagnosticReport | Analysis | for DiagnosticReport-Genetics
DiagnosticReport-geneticsAssessedCondition | DiagnosticReport | AssessedCondition | for DiagnosticReport-Genetics
DiagnosticReport-geneticsFamilyMemberHistory | DiagnosticReport | FamilyMemberHistory | for DiagnosticReport-Genetics
DiagnosticReport-geneticsReferences | DiagnosticReport | References | for DiagnosticReport-Genetics
diagnosticReport-addendumOf | DiagnosticReport | addendumOf | for DiagnosticReport HL7 Extensions
diagnosticReport-extends | DiagnosticReport | extends | for DiagnosticReport HL7 Extensions
diagnosticReport-locationPerformed | DiagnosticReport | locationPerformed | for DiagnosticReport HL7 Extensions
diagnosticReport-replaces | DiagnosticReport | replaces | for DiagnosticReport HL7 Extensions
diagnosticReport-risk | DiagnosticReport | risk | for DiagnosticReport HL7 Extensions
diagnosticReport-summaryOf | DiagnosticReport | summaryOf | for DiagnosticReport HL7 Extensions
event-eventHistory | DiagnosticReport | eventHistory | for Event Pattern HL7 Extensions
event-location | DiagnosticReport | location | for Event Pattern HL7 Extensions
event-partOf | DiagnosticReport | partOf | for Event Pattern HL7 Extensions
event-performerFunction | DiagnosticReport.performer | performerFunction | for Event Pattern HL7 Extensions
event-statusReason | DiagnosticReport | statusReason | for Event Pattern HL7 Extensions
hla-genotyping-results-allele-database | DiagnosticReport | allele-database | for Profile for HLA Genotyping Results
hla-genotyping-results-glstring | DiagnosticReport | glstring | for Profile for HLA Genotyping Results
hla-genotyping-results-haploid | DiagnosticReport | haploid | for Profile for HLA Genotyping Results
hla-genotyping-results-method | DiagnosticReport | method | for Profile for HLA Genotyping Results
servicerequest-geneticsItem | DiagnosticReport | Item | for ServiceRequest-Genetics
workflow-episodeOfCare | DiagnosticReport | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-instantiatesCanonical | DiagnosticReport | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | DiagnosticReport | instantiatesUri | for Workflow Pattern HL7 Extensions
workflow-reasonCode | DiagnosticReport | reasonCode | for Workflow Pattern HL7 Extensions
workflow-reasonReference | DiagnosticReport | reasonReference | for Workflow Pattern HL7 Extensions
workflow-relatedArtifact | DiagnosticReport | relatedArtifact | for Workflow Pattern HL7 Extensions

workflow-researchStudy | DiagnosticReport | researchStudy | for Workflow Pattern HL7 Extensions
workflow-supportingInfo | DiagnosticReport | supportingInfo | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

diagnosticReport-addendumOf | addendumOf | for DiagnosticReport HL7 Extensions
diagnosticReport-extends | extends | for DiagnosticReport HL7 Extensions
diagnosticReport-replaces | replaces | for DiagnosticReport HL7 Extensions
diagnosticReport-summaryOf | summaryOf | for DiagnosticReport HL7 Extensions
openEHR-test | test | for openEHR Archetype Profile
procedure-causedBy | causedBy | for Procedure HL7 Extensions
workflow-reasonReference | reasonReference | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.3.10.3 Search Extensions

assessed-condition | reference | Condition assessed by genetic test Expression: DiagnosticReport.extension('http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition')

10.3.11 Resource DiagnosticReport - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/diagnosticreport-version-maps.html

No validation errors - all conversions are clean

10.3.11 Resource DiagnosticReport - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner

R3 : R4 Conversion maps for DiagnosticReport.

Functional status for this map: 14 tests that all execute ok. All tests pass round-trip testing and 9 r3 resources are invalid (0 errors). (see documentation)

10.3.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DiagnosticReport3to4" = "R3 to R4 Conversions for DiagnosticReport"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DiagnosticReport" alias DiagnosticReportR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DiagnosticReport" alias DiagnosticReport as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DiagnosticReport(source src : DiagnosticReportR3, target tgt : DiagnosticReport) extends DomainResource <<type+>>
    {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.issued -> tgt.issued;
  src.performer as vs0 -> tgt.performer as vt0 then performer(vs0, vt0);
  src.specimen -> tgt.specimen;
  src.result -> tgt.result;
  src.imagingStudy -> tgt.imagingStudy;
  src.image as vs0 -> tgt.media as vt0 then image(vs0, vt0);
  src.conclusion -> tgt.conclusion;
  src.codedDiagnosis -> tgt.conclusionCode;
  src.presentedForm -> tgt.presentedForm;
}

group performer(source src, target tgt) {
  src.actor as v then Reference(v, tgt);
  src.role as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    DiagnosticReport.performer.role',  vt.value = v;
}

group image(source src, target tgt) extends BackboneElement {
  src.comment -> tgt.comment;
  src.link -> tgt.link;
```

```
}
```

10.3.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DiagnosticReport4to3" = "R4 to R3 Conversion for DiagnosticReport"

uses "http://hl7.org/fhir/StructureDefinition/DiagnosticReport" alias DiagnosticReport as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DiagnosticReport" alias DiagnosticReportR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DiagnosticReport(source src : DiagnosticReport, target tgt : DiagnosticReportR3) extends DomainResource <<type+>>
    {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.issued -> tgt.issued;
  src.performer as vs0 -> tgt.performer as vt0 then performer(vs0, vt0);
  src.specimen -> tgt.specimen;
  src.result -> tgt.result;
  src.imagingStudy -> tgt.imagingStudy;
  src.media as vs0 -> tgt.image as vt0 then image(vs0, vt0);
  src.conclusion -> tgt.conclusion;
  src.conclusionCode -> tgt.codedDiagnosis;
  src.presentedForm -> tgt.presentedForm;
}

group performer(source src, target tgt) {
  src -> tgt.actor as vt then Reference(src, vt) "actor";
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-DiagnosticReport.performer.role'
     then {
    e.value -> tgt.role;
  };
}

group image(source src, target tgt) extends BackboneElement {
  src.comment -> tgt.comment;
  src.link -> tgt.link;
}
```

10.3.11.3 R4 Validation Errors

10.3.12 Example Lipid Profile

URL: https://hl7.org/fhir/R4/lipid-report.html

Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes

10.3.12 Example Lipid Profile

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.3.12.1 Content

Profiles:
LipidProfile | Lipid Lab Report
Cholesterol | Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes
Triglyceride | Triglyceride Result
HDLCholesterol | HDL Cholesterol Result
LDLCholesterol | LDL Cholesterol Result
Examples:
Lipid profile | Lipid profile

10.3.12.2 StructureDefinition: LipidProfile

URL: https://hl7.org/fhir/R4/lipidprofile.html

10.3.12.2 StructureDefinition: LipidProfile

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Not yet ready for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/lipidprofile`

Lipid Lab Report

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.3.12.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from DiagnosticReport.

Summary

This is a test summary lipids

Mandatory: 4 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 1 element

Structures

This structure refers to these other structures:

• http://hl7.org/fhir/StructureDefinition/cholesterol
• http://hl7.org/fhir/StructureDefinition/triglyceride
• http://hl7.org/fhir/StructureDefinition/hdlcholesterol
• http://hl7.org/fhir/StructureDefinition/ldlcholesterol

Slices

This structure defines the following Slices:

• The element DiagnosticReport.result is sliced based on the value of value:resolve().code (ordered, Closed)

This structure is derived from DiagnosticReport.

Name	Flags	Card.	Type	Description & Constraints
DiagnosticReport		1..1		Lipid Lab Report
code		1..1	CodeableConcept	LOINC Code for Lipid Panel with LDL Fixed Value: As shown
coding		1..1	Coding	Code defined by a terminology system Fixed Value: (complex)
system		1..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
code		1..1	code	Symbol in syntax defined by the system Fixed Value: 57698-3
display		1..1	string	Representation defined by the system Fixed Value: Lipid panel with direct LDL - Serum or Plasma
result			(Slice Definition)	Slice: Ordered, Closed by value:resolve().code
result:Cholesterol	S	1..1	Reference(Example Lipid Profile)	Cholesterol Result
result:Triglyceride	S	1..1	Reference(Example Lipid Profile)	Triglyceride Result
result:HDLCholesterol	S	1..1	Reference(Example Lipid Profile)	HDL Cholesterol Result
result:LDLCholesterol	S	0..1	Reference(Example Lipid Profile)	LDL Cholesterol result, if reported
conclusion	S	0..1	string	Clinical Interpretation of Lipid Panel
conclusionCode		0..0		

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
DiagnosticReport	I	1..1		Lipid Lab Report
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
identifier	ΣI	0..*	Identifier	Business identifier for report
basedOn	I	0..*	Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	What was requested
status	?!ΣI	1..1	code	registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category	ΣI	0..*	CodeableConcept	Service category Binding: Diagnostic Service Section Codes (example)
code	ΣI	1..1	CodeableConcept	LOINC Code for Lipid Panel with LDL Binding: LOINC Diagnostic Report Codes (preferred) Fixed Value: As shown
id		0..0	string	Unique id for inter-element referencing
extension		0..0	Extension	Additional content defined by implementations
coding		1..1	Coding	Code defined by a terminology system Fixed Value: (complex)
id		0..0	string	Unique id for inter-element referencing
extension		0..0	Extension	Additional content defined by implementations
system		1..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
version		0..0	string	Version of the system - if relevant
code		1..1	code	Symbol in syntax defined by the system Fixed Value: 57698-3
display		1..1	string	Representation defined by the system Fixed Value: Lipid panel with direct LDL - Serum or Plasma
userSelected		0..0	boolean	If this coding was chosen directly by the user

text | | 0..0 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | ΣI | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | DateTime this version was made
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | I | 0..* | Reference(Specimen) | Specimens this report is based on
result | I | | (Slice Definition) | Observations Slice: Ordered, Closed by value:resolve().code
result:Cholesterol | SI | 1..1 | Reference(Example Lipid Profile) | Cholesterol Result
result:Triglyceride | SI | 1..1 | Reference(Example Lipid Profile) | Triglyceride Result
result:HDLCholesterol | SI | 1..1 | Reference(Example Lipid Profile) | HDL Cholesterol Result
result:LDLCholesterol | SI | 0..1 | Reference(Example Lipid Profile) | LDL Cholesterol result, if reported
imagingStudy | I | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | ΣI | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
comment | I | 0..1 | string | Comment about the image (e.g. explanation)
link | ΣI | 1..1 | Reference(Media) | Reference to the image source
conclusion | SI | 0..1 | string | Clinical Interpretation of Lipid Panel
presentedForm | I | 0..* | Attachment | Entire report as issued
Documentation for this format

```
<!-- Example Lipid Profile -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) What was requested --></basedOn>
 <status value="[code]"/><!--  1..1 registered | partial | preliminary | final +   -->
 <category><!--  0..* CodeableConcept Service category   --></category>
 <code><!--  1..1 CodeableConcept LOINC Code for Lipid Panel with LDL   --></code>
 <subject><!--  0..1 Reference(Patient)
     The subject of the report - usually, but not always, the patient --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Health care event when test ordered --></encounter>
 <effective[x]><!--  0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!--  0..1 DateTime this version was made -->
 <performer><!--  0..* Reference(Practitioner)
     Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!--  0..* Reference(Practitioner)
     Primary result interpreter --></resultsInterpreter>
 <specimen><!--  0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <-- result sliced by value:resolve().code  in any order-->
 <result><!--  1..1 Reference(Example Lipid Profile) Cholesterol Result --></result>
 <result><!--  1..1 Reference(Example Lipid Profile) Triglyceride Result --></result>
 <result><!--  1..1 Reference(Example Lipid Profile) HDL Cholesterol Result --></result>
 <result><!--  0..1 Reference(Example Lipid Profile)
     LDL Cholesterol result, if reported --></result>
 <imagingStudy><!--  0..* Reference(ImagingStudy)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <media>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <comment value="[string]"/><!--  0..1 Comment about the image (e.g. explanation) -->
  <link><!--  1..1 Reference(Media)
      Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!--  0..1 Clinical Interpretation of Lipid Panel -->
 <conclusionCode><!--  0..0 CodeableConcept No codes for a lipid panel   --></conclusionCode>
 <presentedForm><!--  0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
```

```
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business identifier for report
    "basedOn" : [{ Reference(CarePlan) }], // C?What was requested
    "status" : "<code>", // C? R! registered | partial | preliminary | final +
    "category" : [{ CodeableConcept }], // C?Service category
    "code" : { CodeableConcept }, // C? R!
    "subject" : { Reference(Patient) }, // C?The subject of the report - usually, but not always, the patient
    "encounter" : { Reference(Encounter) }, // C?Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for report
    "issued" : "<instant>", // C?DateTime this version was made
    "performer" : [{ Reference(Practitioner) }], // C?Responsible Diagnostic Service
    "resultsInterpreter" : [{ Reference(Practitioner) }], // C?Primary result interpreter
    "specimen" : [{ Reference(Specimen) }], // C?Specimens this report is based on
    "result" : [ //  sliced by value:resolve().code  in any order,
      { // Cholesterol Result // C? R!
        // from Element: extension
      },
      { // Triglyceride Result // C? R!
        // from Element: extension
      },
      { // HDL Cholesterol Result // C? R!
        // from Element: extension
      },
      { // LDL Cholesterol result, if reported // C?
        // from Element: extension
      }
    ],
    "imagingStudy" : [{ Reference(ImagingStudy) }], // C?Reference to full details of imaging associated with the
    diagnostic report
    "media" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", // C?Clinical Interpretation of Lipid Panel
    "presentedForm" : [{ Attachment }], // C?Entire report as issued
  }
```

This structure is derived from DiagnosticReport.

Summary

This is a test summary lipids

Mandatory: 4 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 1 element

Structures

This structure refers to these other structures:

• http://hl7.org/fhir/StructureDefinition/cholesterol

• http://hl7.org/fhir/StructureDefinition/triglyceride

• http://hl7.org/fhir/StructureDefinition/hdlcholesterol

• http://hl7.org/fhir/StructureDefinition/ldlcholesterol

Slices

This structure defines the following Slices:

• The element DiagnosticReport.result is sliced based on the value of value:resolve().code (ordered, Closed)

Differential View

This structure is derived from DiagnosticReport.

Name	Flags	Card.	Type	Description & Constraints
DiagnosticReport		1..1		Lipid Lab Report
code		1..1	CodeableConcept	LOINC Code for Lipid Panel with LDL Fixed Value: As shown
coding		1..1	Coding	Code defined by a terminology system Fixed Value: (complex)
system		1..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
code		1..1	code	Symbol in syntax defined by the system Fixed Value: 57698-3
display		1..1	string	Representation defined by the system Fixed Value: Lipid panel with direct LDL - Serum or Plasma
result			(Slice Definition)	Slice: Ordered, Closed by value:resolve().code
result:Cholesterol	S	1..1	Reference(Example Lipid Profile)	Cholesterol Result
result:Triglyceride	S	1..1	Reference(Example Lipid Profile)	Triglyceride Result
result:HDLCholesterol	S	1..1	Reference(Example Lipid Profile)	HDL Cholesterol Result
result:LDLCholesterol	S	0..1	Reference(Example Lipid Profile)	LDL Cholesterol result, if reported
conclusion	S	0..1	string	Clinical Interpretation of Lipid Panel
conclusionCode		0..0		

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
DiagnosticReport	I	1..1		Lipid Lab Report
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource

implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business identifier for report
basedOn | I | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | What was requested
status | ?!ΣI | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ΣI | 0..* | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ΣI | 1..1 | CodeableConcept | LOINC Code for Lipid Panel with LDL Binding: LOINC Diagnostic Report Codes (preferred) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 57698-3
display | | 1..1 | string | Representation defined by the system Fixed Value: Lipid panel with direct LDL - Serum or Plasma
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | ΣI | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | DateTime this version was made
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | I | 0..* | Reference(Specimen) | Specimens this report is based on
result | I | | (Slice Definition) | Observations Slice: Ordered, Closed by value:resolve().code
result:Cholesterol | SI | 1..1 | Reference(Example Lipid Profile) | Cholesterol Result
result:Triglyceride | SI | 1..1 | Reference(Example Lipid Profile) | Triglyceride Result
result:HDLCholesterol | SI | 1..1 | Reference(Example Lipid Profile) | HDL Cholesterol Result
result:LDLCholesterol | SI | 0..1 | Reference(Example Lipid Profile) | LDL Cholesterol result, if reported
imagingStudy | I | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | ΣI | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
comment | I | 0..1 | string | Comment about the image (e.g. explanation)
link | ΣI | 1..1 | Reference(Media) | Reference to the image source
conclusion | SI | 0..1 | string | Clinical Interpretation of Lipid Panel
presentedForm | I | 0..* | Attachment | Entire report as issued
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) What was requested --></basedOn>
 <status value="[code]"/><!--  1..1 registered | partial | preliminary | final +   -->
 <category><!--  0..* CodeableConcept Service category   --></category>
 <code><!--  1..1 CodeableConcept LOINC Code for Lipid Panel with LDL   --></code>
 <subject><!--  0..1 Reference(Patient)
     The subject of the report - usually, but not always, the patient --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Health care event when test ordered --></encounter>
 <effective[x]><!--  0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!--  0..1 DateTime this version was made -->
 <performer><!--  0..* Reference(Practitioner)
```

```
    Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!--  0..* Reference(Practitioner)
    Primary result interpreter --></resultsInterpreter>
 <specimen><!--  0..* Reference(Specimen)
    Specimens this report is based on --></specimen>
 <-- result sliced by value:resolve().code  in any order-->
 <result><!--  1..1 Reference(Example Lipid Profile) Cholesterol Result --></result>
 <result><!--  1..1 Reference(Example Lipid Profile) Triglyceride Result --></result>
 <result><!--  1..1 Reference(Example Lipid Profile) HDL Cholesterol Result --></result>
 <result><!--  0..1 Reference(Example Lipid Profile)
    LDL Cholesterol result, if reported --></result>
 <imagingStudy><!--  0..* Reference(ImagingStudy)
    Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <media>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <comment value="[string]"/><!--  0..1 Comment about the image (e.g. explanation) -->
  <link><!--  1..1 Reference(Media)
     Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!--  0..1 Clinical Interpretation of Lipid Panel -->
 <conclusionCode><!--  0..0 CodeableConcept No codes for a lipid panel   --></conclusionCode>
 <presentedForm><!--  0..* Attachment
    Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business identifier for report
    "basedOn" : [{ Reference(CarePlan) }], // C?What was requested
    "status" : "<code>", // C? R! registered | partial | preliminary | final +
    "category" : [{ CodeableConcept }], // C?Service category
    "code" : { CodeableConcept }, // C? R!
    "subject" : { Reference(Patient) }, // C?The subject of the report - usually, but not always, the patient
    "encounter" : { Reference(Encounter) }, // C?Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for report
    "issued" : "<instant>", // C?DateTime this version was made
    "performer" : [{ Reference(Practitioner) }], // C?Responsible Diagnostic Service
    "resultsInterpreter" : [{ Reference(Practitioner) }], // C?Primary result interpreter
    "specimen" : [{ Reference(Specimen) }], // C?Specimens this report is based on
    "result" : [ //  sliced by value:resolve().code  in any order,
      { // Cholesterol Result // C? R!
        // from Element: extension
      },
      { // Triglyceride Result // C? R!
        // from Element: extension
      },
      { // HDL Cholesterol Result // C? R!
        // from Element: extension
      },
      { // LDL Cholesterol result, if reported // C?
        // from Element: extension
      }
    ],
    "imagingStudy" : [{ Reference(ImagingStudy) }], // C?Reference to full details of imaging associated with the
    diagnostic report
    "media" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", // C?Clinical Interpretation of Lipid Panel
    "presentedForm" : [{ Attachment }], // C?Entire report as issued
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.12.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
DiagnosticReport.language | Common Languages | preferred | Common Languages
DiagnosticReport.status | DiagnosticReportStatus | required | DiagnosticReportStatus
DiagnosticReport.category | Diagnostic Service Section Codes | example | Diagnostic Service Section Codes
DiagnosticReport.code | LOINC Diagnostic Report Codes | preferred | LOINC Diagnostic Report Codes

10.3.12.7 StructureDefinition: Cholesterol

URL: https://hl7.org/fhir/R4/cholesterol.html

10.3.12.7 StructureDefinition: Cholesterol

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Not yet ready for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cholesterol`

Describes how the lab report is used for a standard Lipid Profile - Cholesterol, Triglyceride and Cholesterol fractions. Uses LOINC codes

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.3.12.7.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Observation.

Summary

Mandatory: 3 elements (+3 nested mandatory elements) Must-Support: 9 elements Fixed Value: 5 elements Prohibited: 7 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1
code | S | 1..1 | CodeableConcept | Cholesterol Fixed Value: As shown
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35200-5
display | | 1..1 | string | Representation defined by the system Fixed Value: Cholesterol [Moles/volume] in Serum or Plasma
valueQuantity | S | 0..1 | Quantity | Cholesterol value
value | S | 0..1 | decimal | Actual numerical value
comparator | | 0..0
unit | S | 1..1 | string | Fixed Value: mmol/L
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Fixed Value: mmol/L
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
note | S | 0..* | Annotation | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: As shown
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 4.5
type | | 0..0
appliesTo | | 0..0
age | | 0..0
hasMember | | 0..0
derivedFrom | | 0..0

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Measurements and simple assertions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | Cholesterol Binding: LOINC Codes (example) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations

system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35200-5
display | | 1..1 | string | Representation defined by the system Fixed Value: Cholesterol [Moles/volume] in Serum or Plasma
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Cholesterol value
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 0..1 | decimal | Actual numerical value
unit | SΣI | 1..1 | string | Unit representation Fixed Value: mmol/L
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded form of the unit Fixed Value: mmol/L
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 4.5
comparator | | 0..0 | code | < | <= | >= | > - how to understand the value
unit | | 0..0 | string | Unit representation
system | | 0..0 | uri | System that defines coded unit form
code | | 0..0 | code | Coded form of the unit
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
```

```
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Cholesterol   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Cholesterol value -->
  <value value="[decimal]"/><!--  0..1 Actual numerical value -->
  <comparator value="[code]"/><!--  0..0 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded form of the unit -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R!
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
```

```
  "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
  "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
  "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
  "issued" : "<instant>", // C?Date/Time this version was made available
  "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
  "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
    { // Cholesterol value // C?
      // from Element: extension
      "extension" : [ //  sliced by value:url  in the specified order, Open ]
      "value" : <decimal>, // C?Actual numerical value
      "unit" : "mmol/L", // C? R!
      "system" : "http://unitsofmeasure.org", // C? R!
      "code" : "mmol/L" // C? R!
    }
  ],
  "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
  "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
  "note" : [{ Annotation }], // C?Comments about result
  "bodySite" : { CodeableConcept }, // C?Observed body part
  "method" : { CodeableConcept }, // C?How it was done
  "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
  "device" : { Reference(Device) }, // C?(Measurement) Device
  "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
  "component" : [{ BackboneElement }], // C?Component results
}
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements (+3 nested mandatory elements) Must-Support: 9 elements Fixed Value: 5 elements Prohibited: 7 elements

Differential View

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1
code | S | 1..1 | CodeableConcept | Cholesterol Fixed Value: As shown
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35200-5
display | | 1..1 | string | Representation defined by the system Fixed Value: Cholesterol [Moles/volume] in Serum or Plasma
valueQuantity | S | 0..1 | Quantity | Cholesterol value
value | S | 0..1 | decimal | Actual numerical value
comparator | | 0..0
unit | S | 1..1 | string | Fixed Value: mmol/L
system | S | 1..1 | uri | Fixed Value: http://unitsofmeasure.org
code | S | 1..1 | code | Fixed Value: mmol/L
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
note | S | 0..* | Annotation | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: As shown
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 4.5
type | | 0..0
appliesTo | | 0..0
age | | 0..0
hasMember | | 0..0
derivedFrom | | 0..0
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Measurements and simple assertions
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)

code | SΣI | 1..1 | CodeableConcept | Cholesterol Binding: LOINC Codes (example) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35200-5
display | | 1..1 | string | Representation defined by the system Fixed Value: Cholesterol [Moles/volume] in Serum or Plasma
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Cholesterol value
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations Slice: Unordered, Open by value:url
value | SΣI | 0..1 | decimal | Actual numerical value
unit | SΣI | 1..1 | string | Unit representation Fixed Value: mmol/L
system | SΣI | 1..1 | uri | System that defines coded unit form Fixed Value: http://unitsofmeasure.org
code | SΣI | 1..1 | code | Coded form of the unit Fixed Value: mmol/L
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 4.5
comparator | | 0..0 | code | < | <= | >= | > - how to understand the value
unit | | 0..0 | string | Unit representation
system | | 0..0 | uri | System that defines coded unit form
code | | 0..0 | code | Coded form of the unit
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Cholesterol   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity>  0..1 Quantity  <!--  0..1 Cholesterol value -->
  <value value="[decimal]"/><!--  0..1 Actual numerical value -->
  <comparator value="[code]"/><!--  0..0 < | <= | >= | > - how to understand the value   -->
  <unit value="[string]"/><!--  1..1 Unit representation -->
  <system value="[uri]"/><!--  1..1 System that defines coded unit form -->
  <code value="[code]"/><!--  1..1 Coded form of the unit -->
 </valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
```

```
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R!
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Cholesterol value // C?
        // from Element: extension
        "extension" : [ //  sliced by value:url  in the specified order, Open ]
        "value" : <decimal>, // C?Actual numerical value
        "unit" : "mmol/L", // C? R!
        "system" : "http://unitsofmeasure.org", // C? R!
        "code" : "mmol/L" // C? R!
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.12.7.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	preferred	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	LOINC Codes	example	LOINC Codes
Observation.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.component.code	LOINC Codes	example	LOINC Codes
Observation.component.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.3.12.14 StructureDefinition: Triglyceride

URL: https://hl7.org/fhir/R4/triglyceride.html

10.3.12.14 StructureDefinition: Triglyceride

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Not yet ready for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/triglyceride`

Triglyceride Result

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.3.12.14.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Prohibited: 6 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | Triglyceride Result
code | S | 1..1 | CodeableConcept | Triglyceride Required Pattern: At least the following
coding | | 1..* | Coding | Code defined by a terminology system Fixed Value: (complex)
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35217-9
display | | 1..1 | string | Representation defined by the system Fixed Value: Triglyceride [Moles/volume] in Serum or Plasma
valueQuantity | S | 0..1 | Quantity | Triglyceride value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
note | S | 0..* | Annotation | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity
type | | 0..0
appliesTo | | 0..0
age | | 0..0
hasMember | | 0..0
derivedFrom | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | Triglyceride Result
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | Triglyceride Binding: LOINC Codes (example) Required Pattern: At least the following
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
coding | | 1..* | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..1 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35217-9
display | | 1..1 | string | Representation defined by the system Fixed Value: Triglyceride [Moles/volume] in Serum or Plasma
userSelected | | 0..1 | boolean | If this coding was chosen directly by the user
text | | 0..1 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Triglyceride value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation

id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
high | I | 1..1 | SimpleQuantity | High Range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Triglyceride   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..1 Quantity Triglyceride value --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
```

```
    Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! Triglyceride
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Triglyceride value // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Prohibited: 6 elements

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		1..1		Triglyceride Result
code	S	1..1	CodeableConcept	Triglyceride Required Pattern: At least the following
coding		1..*	Coding	Code defined by a terminology system Fixed Value: (complex)
system		1..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
code		1..1	code	Symbol in syntax defined by the system Fixed Value: 35217-9
display		1..1	string	Representation defined by the system Fixed Value: Triglyceride [Moles/volume] in Serum or Plasma
valueQuantity	S	0..1	Quantity	Triglyceride value
interpretation	S	0..1	CodeableConcept	+ | ++ | +++ | - | -- | ---
note	S	0..*	Annotation	Comments about result
referenceRange	S	1..1	BackboneElement	
low		0..0		
high		1..1	SimpleQuantity	
type		0..0		
appliesTo		0..0		
age		0..0		
hasMember		0..0		
derivedFrom		0..0		
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints

Observation | I | 1..1 | | Triglyceride Result
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | Triglyceride Binding: LOINC Codes (example) Required Pattern: At least the following
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
coding | | 1..* | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..1 | string | Unique id for inter-element referencing
extension | | 0..* | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..1 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 35217-9
display | | 1..1 | string | Representation defined by the system Fixed Value: Triglyceride [Moles/volume] in Serum or Plasma
userSelected | | 0..1 | boolean | If this coding was chosen directly by the user
text | | 0..1 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | Triglyceride value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
high | I | 1..1 | SimpleQuantity | High Range, if relevant
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)

interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept Triglyceride   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..1 Quantity Triglyceride value --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      High Range, if relevant --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
```

```
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! Triglyceride
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // Triglyceride value // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.12.14.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	preferred	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	LOINC Codes	example	LOINC Codes
Observation.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.component.code	LOINC Codes	example	LOINC Codes
Observation.component.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.3.12.21 StructureDefinition: HDLCholesterol

URL: https://hl7.org/fhir/R4/hdlcholesterol.html

10.3.12.21 StructureDefinition: HDLCholesterol

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Not yet ready for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/hdlcholesterol`

HDL Cholesterol Result

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.3.12.21.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 2 elements Prohibited: 6 elements

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		1..1		HDL Cholesterol Result
code	S	1..1	CodeableConcept	HDL Cholesterol Fixed Value: As shown

coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 2085-9
display | | 1..1 | string | Representation defined by the system Fixed Value: HDL Cholesterol
valueQuantity | S | 0..1 | Quantity | HDL Cholesterol value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
note | S | 0..* | Annotation | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 1..1 | SimpleQuantity | Fixed Value: As shown
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 1.5
high | | 0..0
type | | 0..0
appliesTo | | 0..0
age | | 0..0
hasMember | | 0..0
derivedFrom | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | HDL Cholesterol Result
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | HDL Cholesterol Binding: LOINC Codes (example) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 2085-9
display | | 1..1 | string | Representation defined by the system Fixed Value: HDL Cholesterol
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | HDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing

extension | | 0..0 | Extension | Additional content defined by implementations
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 1.5
comparator | | 0..0 | code | < | <= | >= | > - how to understand the value
unit | | 0..0 | string | Unit representation
system | | 0..0 | uri | System that defines coded unit form
code | | 0..0 | code | Coded form of the unit
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result

Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept HDL Cholesterol   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..1 Quantity HDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  1..1 Quantity
      A fixed quantity (no comparator) --></low>
  <high><!--  0..0 Quantity
      High Range, if relevant --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
```

```
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R!
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // HDL Cholesterol value // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 2 elements Prohibited: 6 elements

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		1..1		HDL Cholesterol Result
code	S	1..1	CodeableConcept	HDL Cholesterol Fixed Value: As shown
coding		1..1	Coding	Code defined by a terminology system Fixed Value: (complex)
system		1..1	uri	Identity of the terminology system Fixed Value: http://loinc.org
code		1..1	code	Symbol in syntax defined by the system Fixed Value: 2085-9
display		1..1	string	Representation defined by the system Fixed Value: HDL Cholesterol
valueQuantity	S	0..1	Quantity	HDL Cholesterol value
interpretation	S	0..1	CodeableConcept	+ | ++ | +++ | - | -- | ---
note	S	0..*	Annotation	Comments about result
referenceRange	S	1..1	BackboneElement	
low		1..1	SimpleQuantity	Fixed Value: As shown
value		1..1	decimal	Numerical value (with implicit precision) Fixed Value: 1.5
high		0..0		
type		0..0		
appliesTo		0..0		
age		0..0		
hasMember		0..0		
derivedFrom		0..0		

Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | HDL Cholesterol Result
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | HDL Cholesterol Binding: LOINC Codes (example) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://loinc.org
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: 2085-9
display | | 1..1 | string | Representation defined by the system Fixed Value: HDL Cholesterol
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | HDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
low | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 1.5
comparator | | 0..0 | code | < | <= | >= | > - how to understand the value
unit | | 0..0 | string | Unit representation
system | | 0..0 | uri | System that defines coded unit form
code | | 0..0 | code | Coded form of the unit
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept

valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept HDL Cholesterol   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..1 Quantity HDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  1..1 Quantity
      A fixed quantity (no comparator) --></low>
  <high><!--  0..0 Quantity
      High Range, if relevant --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R!
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // HDL Cholesterol value // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.12.21.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Observation.language	Common Languages	preferred	Common Languages
Observation.status	ObservationStatus	required	ObservationStatus
Observation.category	Observation Category Codes	preferred	Observation Category Codes
Observation.code	LOINC Codes	example	LOINC Codes
Observation.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes
Observation.bodySite	SNOMED CT Body Structures	example	SNOMED CT Body Structures
Observation.method	Observation Methods	example	Observation Methods
Observation.component.code	LOINC Codes	example	LOINC Codes
Observation.component.dataAbsentReason	DataAbsentReason	extensible	DataAbsentReason
Observation.component.interpretation	Observation Interpretation Codes	extensible	Observation Interpretation Codes

10.3.12.28 StructureDefinition: LDLCholesterol

URL: https://hl7.org/fhir/R4/ldlcholesterol.html

10.3.12.28 StructureDefinition: LDLCholesterol

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Not yet ready for Production use

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/ldlcholesterol`

LDL Cholesterol Result

This profile was published on Sat, May 12, 2012 00:00+1000 as a draft by Grahame Grieve.

10.3.12.28.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template

• All

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 6 elements

This structure is derived from Observation.

Name | Flags | Card. | Type | Description & Constraints
Observation | | 1..1 | | LDL Cholesterol Result
code | S | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: L d l codes (required)
valueQuantity | S | 0..1 | Quantity | LDL Cholesterol value
interpretation | S | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | ---
note | S | 0..* | Annotation | Comments about result
referenceRange | S | 1..1 | BackboneElement
low | | 0..0
high | | 1..1 | SimpleQuantity | Fixed Value: As shown
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 3.0
type | | 0..0
appliesTo | | 0..0
age | | 0..0
hasMember | | 0..0
derivedFrom | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Observation | I | 1..1 | | LDL Cholesterol Result
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: L d l codes (required)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | LDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 3.0
comparator | | 0..0 | code | < | <= | >= | > - how to understand the value
unit | | 0..0 | string | Unit representation
system | | 0..0 | uri | System that defines coded unit form

code | | 0..0 | code | Coded form of the unit
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept LDL Cholesterol -measured or calculated per code   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..1 Quantity LDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
```

```
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! LDL Cholesterol -measured or calculated per code
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // LDL Cholesterol value // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

This structure is derived from Observation.

Summary

Mandatory: 3 elements Must-Support: 5 elements Fixed Value: 1 element Prohibited: 6 elements

Differential View

This structure is derived from Observation.

Name	Flags	Card.	Type	Description & Constraints
Observation		1..1		LDL Cholesterol Result
code	S	1..1	CodeableConcept	LDL Cholesterol -measured or calculated per code Binding: L d l codes (required)
valueQuantity	S	0..1	Quantity	LDL Cholesterol value
interpretation	S	0..1	CodeableConcept	+ | ++ | +++ | - | -- | ---
note	S	0..*	Annotation	Comments about result
referenceRange	S	1..1	BackboneElement	
low		0..0		
high		1..1	SimpleQuantity	Fixed Value: As shown
value		1..1	decimal	Numerical value (with implicit precision) Fixed Value: 3.0
type		0..0		
appliesTo		0..0		
age		0..0		
hasMember		0..0		
derivedFrom		0..0		
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Observation	I	1..1		LDL Cholesterol Result
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages

text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business Identifier for observation
basedOn | ΣI | 0..* | Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Fulfills plan, proposal or order
partOf | ΣI | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy) | Part of referenced event
status | ?!ΣI | 1..1 | code | registered | preliminary | final | amended + Binding: ObservationStatus (required)
category | I | 0..* | CodeableConcept | Classification of type of observation Binding: Observation Category Codes (preferred)
code | SΣI | 1..1 | CodeableConcept | LDL Cholesterol -measured or calculated per code Binding: L d l codes (required)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | Who and/or what the observation is about
focus | ΣI | 0..* | Reference(Resource) | What the observation is about, when it is not about the subject of record
encounter | ΣI | 0..1 | Reference(Encounter) | Healthcare event during which this observation is made
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for observation
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
effectiveTiming | | | Timing
effectiveInstant | | | instant
issued | ΣI | 0..1 | instant | Date/Time this version was made available
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson) | Who is responsible for the observation
value[x] | ΣI | | (Slice Definition) | Actual result Slice: Unordered, Closed by type:$this
value[x]:valueQuantity | SΣI | 0..1 | Quantity | LDL Cholesterol value
dataAbsentReason | I | 0..1 | CodeableConcept | Why the result is missing Binding: DataAbsentReason (extensible)
interpretation | SI | 0..1 | CodeableConcept | + | ++ | +++ | - | -- | --- Binding: Observation Interpretation Codes (extensible)
note | SI | 0..* | Annotation | Comments about result
bodySite | I | 0..1 | CodeableConcept | Observed body part Binding: SNOMED CT Body Structures (example)
method | I | 0..1 | CodeableConcept | How it was done Binding: Observation Methods (example)
specimen | I | 0..1 | Reference(Specimen) | Specimen used for this observation
device | I | 0..1 | Reference(Device | DeviceMetric) | (Measurement) Device
referenceRange | SI | 1..1 | BackboneElement | Provides guide for interpretation
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
high | I | 1..1 | SimpleQuantity | A fixed quantity (no comparator) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
value | | 1..1 | decimal | Numerical value (with implicit precision) Fixed Value: 3.0
comparator | | 0..0 | code | < | <= | >= | > - how to understand the value
unit | | 0..0 | string | Unit representation
system | | 0..0 | uri | System that defines coded unit form
code | | 0..0 | code | Coded form of the unit
text | I | 0..1 | string | Text based reference range in an observation
component | ΣI | 0..* | BackboneElement | Component results
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | CodeableConcept | Type of component observation (code / type) Binding: LOINC Codes (example)
value[x] | ΣI | 0..1 | | Actual component result
valueQuantity | | | Quantity
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueRange | | | Range
valueRatio | | | Ratio
valueSampledData | | | SampledData
valueTime | | | time
valueDateTime | | | dateTime
valuePeriod | | | Period
dataAbsentReason | I | 0..1 | CodeableConcept | Why the component result is missing Binding: DataAbsentReason (extensible)
interpretation | I | 0..* | CodeableConcept | High, low, normal, etc. Binding: Observation Interpretation Codes (extensible)
referenceRange | I | 0..* | | Unknown reference to #Observation.referenceRange Provides guide for interpretation of component result
Documentation for this format

XML Template

```
<!-- Example Lipid Profile -->

<Observation xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
```

```
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business Identifier for observation --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) Fulfills plan, proposal or order --></basedOn>
 <partOf><!--  0..* Reference(MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!--  1..1 registered | preliminary | final | amended +   -->
 <category><!--  0..* CodeableConcept Classification of  type of observation   --></category>
 <code><!--  1..1 CodeableConcept LDL Cholesterol -measured or calculated per code   --></code>
 <subject><!--  0..1 Reference(Patient)
     Who and/or what the observation is about --></subject>
 <focus><!--  0..* Reference(Resource)
     What the observation is about, when it is not about the subject of record --></focus>
 <encounter><!--  0..1 Reference(Encounter)
     Healthcare event during which this observation is made --></encounter>
 <effective[x]><!--  0..1 dateTime|Period|Timing|instant
     Clinically relevant time/time-period for observation --></effective[x]>
 <issued value="[instant]"/><!--  0..1 Date/Time this version was made available -->
 <performer><!--  0..* Reference(Practitioner)
     Who is responsible for the observation --></performer>
 <-- valueQuantity sliced by type:$this  in the specified orderClosed-->
 <valueQuantity><!--  0..1 Quantity LDL Cholesterol value --></valueQuantity>
 <dataAbsentReason><!--  0..1 CodeableConcept Why the result is missing   --></dataAbsentReason>
 <interpretation><!--  0..1 CodeableConcept + | ++ | +++ | - | -- | ---   --></interpretation>
 <note><!--  0..* Annotation Comments about result --></note>
 <bodySite><!--  0..1 CodeableConcept Observed body part   --></bodySite>
 <method><!--  0..1 CodeableConcept How it was done   --></method>
 <specimen><!--  0..1 Reference(Specimen) Specimen used for this observation --></specimen>
 <device><!--  0..1 Reference(Device) (Measurement) Device --></device>
 <referenceRange>  1..1 BackboneElement  <!--  1..1 Provides guide for interpretation -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <low><!--  0..0 Quantity
      Low Range, if relevant --></low>
  <high><!--  1..1 Quantity
      A fixed quantity (no comparator) --></high>
  <type><!--  0..0 CodeableConcept Reference range qualifier   --></type>
  <appliesTo><!--  0..0 CodeableConcept Reference range population   --></appliesTo>
  <age><!--  0..0 Range
      Applicable age range, if relevant --></age>
  <text value="[string]"/><!--  0..1 Text based reference range in an observation -->
 </referenceRange>
 <hasMember><!--  0..0 Reference(Observation)
     Related resource that belongs to the Observation group --></hasMember>
 <derivedFrom><!--  0..0 Reference(DocumentReference)
     Related measurements the observation is made from --></derivedFrom>
 <component>  0..* BackboneElement  <!--  0..* Component results -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Type of component observation (code / type)   --></code>
  <value[x]><!--  0..1 Quantity|CodeableConcept|string|
    boolean|integer|Range|Ratio|SampledData|time|dateTime|Period
      Actual component result --></value[x]>
  <dataAbsentReason><!--  0..1 CodeableConcept Why the component result is missing   --></dataAbsentReason>
  <interpretation><!--  0..* CodeableConcept High, low, normal, etc.   --></interpretation>
  <referenceRange><!-- See #Observation.referenceRange  Provides guide for interpretation of component result
    --></referenceRange>
 </component>
</Observation>
```

JSON Template

```
{ // Example Lipid Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business Identifier for observation
    "basedOn" : [{ Reference(CarePlan) }], // C?Fulfills plan, proposal or order
    "partOf" : [{ Reference(MedicationAdministration) }], // C?Part of referenced event
    "status" : "<code>", // C? R! registered | preliminary | final | amended +
    "category" : [{ CodeableConcept }], // C?Classification of  type of observation
    "code" : { CodeableConcept }, // C? R! LDL Cholesterol -measured or calculated per code
    "subject" : { Reference(Patient) }, // C?Who and/or what the observation is about
    "focus" : [{ Reference(Resource) }], // C?What the observation is about, when it is not about the subject of record
    "encounter" : { Reference(Encounter) }, // C?Healthcare event during which this observation is made
// value[x]: Clinically relevant time/time-period for observation. One of these 4:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for observation
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for observation
    "effectiveTiming" : { Timing }, // C?Clinically relevant time/time-period for observation
    "effectiveInstant" : "<instant>", // C?Clinically relevant time/time-period for observation
```

```
    "issued" : "<instant>", // C?Date/Time this version was made available
    "performer" : [{ Reference(Practitioner) }], // C?Who is responsible for the observation
    "valueQuantity" : [ //  sliced by type:$this  in the specified order, Closed
      { // LDL Cholesterol value // C?
        // from Element: extension
      }
    ],
    "dataAbsentReason" : { CodeableConcept }, // C?Why the result is missing
    "interpretation" : { CodeableConcept }, // C?+ | ++ | +++ | - | -- | ---
    "note" : [{ Annotation }], // C?Comments about result
    "bodySite" : { CodeableConcept }, // C?Observed body part
    "method" : { CodeableConcept }, // C?How it was done
    "specimen" : { Reference(Specimen) }, // C?Specimen used for this observation
    "device" : { Reference(Device) }, // C?(Measurement) Device
    "referenceRange" : { BackboneElement }, // C? R! Provides guide for interpretation
    "component" : [{ BackboneElement }], // C?Component results
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.12.28.2 Terminology Bindings

Path | Name | Conformance | ValueSet
Observation.language | Common Languages | preferred | Common Languages
Observation.status | ObservationStatus | required | ObservationStatus
Observation.category | Observation Category Codes | preferred | Observation Category Codes
Observation.code | L d l codes | required | L d l codes
Observation.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes
Observation.bodySite | SNOMED CT Body Structures | example | SNOMED CT Body Structures
Observation.method | Observation Methods | example | Observation Methods
Observation.component.code | LOINC Codes | example | LOINC Codes
Observation.component.dataAbsentReason | DataAbsentReason | extensible | DataAbsentReason
Observation.component.interpretation | Observation Interpretation Codes | extensible | Observation Interpretation Codes

10.3.13 DiagnosticReport HL7 Extensions

URL: https://hl7.org/fhir/R4/diagnosticreport-extensions.html

Defines common extensions used with or related to the DiagnosticReport resource

10.3.13 DiagnosticReport HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.3.13.1 Content

Extensions:
diagnosticReport-locationPerformed | locationPerformed : Facility location where this report was prepared.
diagnosticReport-summaryOf | summaryOf : A summary report that points to subordinate target reports.
diagnosticReport-replaces | replaces : The report replaces the target resource. For example, when a final anatomic pathology report replaces a preliminary anatomic pathology report replaces where the subsequent observation of case and report may be on more or different material (specimen). Note that this is not same concept asDiagnosticReport.status = preliminary of final, but industry definition of preliminary and final.
diagnosticReport-extends | extends : The report references related ("sibling") reports.
diagnosticReport-addendumOf | addendumOf : The supplements or provides additional information for the target report.
diagnosticReport-risk | risk : Provides a link to an assessment of prognosis or risk as informed by the diagnostic results (For example, genetic results and possibly by patient genetic family history information). This extension is used when need RiskAssessment as an alternate choice for Observation.hasMember or DiagnosticReport.result.

10.3.14 DiagnosticReport-Genetics

URL: https://hl7.org/fhir/R4/diagnosticreport-genetic.html

10.3.14 DiagnosticReport-Genetics

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

Trial-Use Note:

The material on this page is currently undergoing work to be refactored or potentially removed in a future release as further analysis is done as part of the Genomics Reporting Implementation Guide .

10.3.14.1 Content

Profiles:
DiagnosticReport-Genetics | Describes how the DiagnosticReport resource is used to report structured genetic test results
Extensions:
DiagnosticReport-geneticsAssessedCondition | AssessedCondition : Used to denote condition context for genetic testing, which may influence reported variants and interpretation for large genomic testing panels e.g. lung cancer or familial breast cancer.

DiagnosticReport-geneticsFamilyMemberHistory | FamilyMemberHistory : Significant health events and conditions for a person related to the patient relevant in the context of care for the patient.
DiagnosticReport-geneticsAnalysis | Analysis : Knowledge-based comments on the effect of the sequence on patient's condition/medication reaction.
DiagnosticReport-geneticsReferences | References : Additional bibliographic reference information about genetics, medications, clinical trials, etc. associated with knowledge-based information on genetics/genetic condition.
Examples:
Genetics -with family | An example of a genetics test report with Family member history
Comprehensive report | An example of a comprehensive bone marrow report
PGx Example | An example of a PGx haplotype report

10.3.14.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
assessed-condition | reference | Condition assessed by genetic test | | XML / JSON

10.3.14.3 StructureDefinition: DiagnosticReport-Genetics

URL: https://hl7.org/fhir/R4/diagnosticreport-genetics.html

10.3.14.3 StructureDefinition: DiagnosticReport-Genetics

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetics`

Describes how the DiagnosticReport resource is used to report structured genetic test results

This profile was published on Fri, Oct 9, 2015 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

10.3.14.3.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from DiagnosticReport.

Summary

Prohibited: 1 element

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
• http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
• http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
• http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*
DiagnosticReport-geneticsAssessedCondition | | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
DiagnosticReport-geneticsReferences | | 0..* | (Complex) | References URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences
conclusionCode | | 0..0
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | Σ | 0..1 | string | Logical id of this artifact

meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
DiagnosticReport-geneticsAssessedCondition | I | 0..* | Reference(Condition) | AssessedCondition URL:
http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | I | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL:
http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | I | 0..* | (Complex) | Analysis URL:
http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
DiagnosticReport-geneticsReferences | I | 0..* | (Complex) | References URL:
http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business identifier for report
basedOn | I | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) |
What was requested
status | ?!ΣI | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ΣI | 0..* | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ΣI | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | ΣI | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | DateTime this version was made
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | I | 0..* | Reference(Specimen) | Specimens this report is based on
result | I | 0..* | Reference(Observation) | Observations
imagingStudy | I | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | ΣI | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
comment | I | 0..1 | string | Comment about the image (e.g. explanation)
link | ΣI | 1..1 | Reference(Media) | Reference to the image source
conclusion | I | 0..1 | string | Clinical conclusion (interpretation) of test results
presentedForm | I | 0..* | Attachment | Entire report as issued
Documentation for this format

```
<!-- DiagnosticReport-Genetics -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) What was requested --></basedOn>
 <status value="[code]"/><!--  1..1 registered | partial | preliminary | final +   -->
 <category><!--  0..* CodeableConcept Service category   --></category>
 <code><!--  1..1 CodeableConcept Name/Code for this diagnostic report   --></code>
 <subject><!--  0..1 Reference(Patient)
     The subject of the report - usually, but not always, the patient --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Health care event when test ordered --></encounter>
 <effective[x]><!--  0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!--  0..1 DateTime this version was made -->
 <performer><!--  0..* Reference(Practitioner)
     Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!--  0..* Reference(Practitioner)
     Primary result interpreter --></resultsInterpreter>
 <specimen><!--  0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <result><!--  0..* Reference(Observation) Observations --></result>
 <imagingStudy><!--  0..* Reference(ImagingStudy)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <media>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
```

```
  <comment value="[string]"/><!--  0..1 Comment about the image (e.g. explanation) -->
  <link><!--  1..1 Reference(Media)
      Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!--  0..1 Clinical conclusion (interpretation) of test results -->
 <conclusionCode><!--  0..0 CodeableConcept Codes for the clinical conclusion of test results   --></conclusionCode>
 <presentedForm><!--  0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

```
{ // DiagnosticReport-Genetics
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // AssessedCondition // C?
        // from Element: extension
      },
      { // FamilyHistory // C?
        // from Element: extension
      },
      { // Analysis // C?
        // from Element: extension
      },
      { // References // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business identifier for report
    "basedOn" : [{ Reference(CarePlan) }], // C?What was requested
    "status" : "<code>", // C? R! registered | partial | preliminary | final +
    "category" : [{ CodeableConcept }], // C?Service category
    "code" : { CodeableConcept }, // C? R! Name/Code for this diagnostic report
    "subject" : { Reference(Patient) }, // C?The subject of the report - usually, but not always, the patient
    "encounter" : { Reference(Encounter) }, // C?Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for report
    "issued" : "<instant>", // C?DateTime this version was made
    "performer" : [{ Reference(Practitioner) }], // C?Responsible Diagnostic Service
    "resultsInterpreter" : [{ Reference(Practitioner) }], // C?Primary result interpreter
    "specimen" : [{ Reference(Specimen) }], // C?Specimens this report is based on
    "result" : [{ Reference(Observation) }], // C?Observations
    "imagingStudy" : [{ Reference(ImagingStudy) }], // C?Reference to full details of imaging associated with the
    diagnostic report
    "media" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", // C?Clinical conclusion (interpretation) of test results
    "presentedForm" : [{ Attachment }], // C?Entire report as issued
  }
```

This structure is derived from DiagnosticReport.

Summary

Prohibited: 1 element

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
- http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
- http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
- http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences

Differential View

This structure is derived from DiagnosticReport.

Name	Flags	Card.	Type	Description & Constraints
DiagnosticReport		0..*		
DiagnosticReport-geneticsAssessedCondition		0..*	Reference(Condition)	AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory		0..*	Reference(FamilyMemberHistory)	FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis		0..*	(Complex)	Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
DiagnosticReport-geneticsReferences		0..*	(Complex)	References URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences
conclusionCode		0..0		
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints

DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
DiagnosticReport-geneticsAssessedCondition | I | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | I | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | I | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
DiagnosticReport-geneticsReferences | I | 0..* | (Complex) | References URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business identifier for report
basedOn | I | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | What was requested
status | ?!ΣI | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ΣI | 0..* | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ΣI | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | ΣI | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | DateTime this version was made
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | I | 0..* | Reference(Specimen) | Specimens this report is based on
result | I | 0..* | Reference(Observation) | Observations
imagingStudy | I | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | ΣI | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
comment | I | 0..1 | string | Comment about the image (e.g. explanation)
link | ΣI | 1..1 | Reference(Media) | Reference to the image source
conclusion | I | 0..1 | string | Clinical conclusion (interpretation) of test results
presentedForm | I | 0..* | Attachment | Entire report as issued
Documentation for this format

XML Template

```
<!-- DiagnosticReport-Genetics -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) What was requested --></basedOn>
 <status value="[code]"/><!--  1..1 registered | partial | preliminary | final +   -->
 <category><!--  0..* CodeableConcept Service category   --></category>
 <code><!--  1..1 CodeableConcept Name/Code for this diagnostic report   --></code>
 <subject><!--  0..1 Reference(Patient)
     The subject of the report - usually, but not always, the patient --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Health care event when test ordered --></encounter>
 <effective[x]><!--  0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!--  0..1 DateTime this version was made -->
 <performer><!--  0..* Reference(Practitioner)
     Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!--  0..* Reference(Practitioner)
     Primary result interpreter --></resultsInterpreter>
 <specimen><!--  0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
```

```
 <result><!--  0..* Reference(Observation) Observations --></result>
 <imagingStudy><!--  0..* Reference(ImagingStudy)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <media>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <comment value="[string]"/><!--  0..1 Comment about the image (e.g. explanation) -->
  <link><!--  1..1 Reference(Media)
      Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!--  0..1 Clinical conclusion (interpretation) of test results -->
 <conclusionCode><!--  0..0 CodeableConcept Codes for the clinical conclusion of test results   --></conclusionCode>
 <presentedForm><!--  0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{ // DiagnosticReport-Genetics
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // AssessedCondition // C?
        // from Element: extension
      },
      { // FamilyHistory // C?
        // from Element: extension
      },
      { // Analysis // C?
        // from Element: extension
      },
      { // References // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business identifier for report
    "basedOn" : [{ Reference(CarePlan) }], // C?What was requested
    "status" : "<code>", // C? R! registered | partial | preliminary | final +
    "category" : [{ CodeableConcept }], // C?Service category
    "code" : { CodeableConcept }, // C? R! Name/Code for this diagnostic report
    "subject" : { Reference(Patient) }, // C?The subject of the report - usually, but not always, the patient
    "encounter" : { Reference(Encounter) }, // C?Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for report
    "issued" : "<instant>", // C?DateTime this version was made
    "performer" : [{ Reference(Practitioner) }], // C?Responsible Diagnostic Service
    "resultsInterpreter" : [{ Reference(Practitioner) }], // C?Primary result interpreter
    "specimen" : [{ Reference(Specimen) }], // C?Specimens this report is based on
    "result" : [{ Reference(Observation) }], // C?Observations
    "imagingStudy" : [{ Reference(ImagingStudy) }], // C?Reference to full details of imaging associated with the
    diagnostic report
    "media" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", // C?Clinical conclusion (interpretation) of test results
    "presentedForm" : [{ Attachment }], // C?Entire report as issued
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.14.3.2 Terminology Bindings

Path	Name	Conformance	ValueSet
DiagnosticReport.language	Common Languages	preferred	Common Languages
DiagnosticReport.status	DiagnosticReportStatus	required	DiagnosticReportStatus
DiagnosticReport.category	Diagnostic Service Section Codes	example	Diagnostic Service Section Codes
DiagnosticReport.code	LOINC Diagnostic Report Codes	preferred	LOINC Diagnostic Report Codes

10.3.14.8 Diagnosticreport-genetic-DiagnosticReport-assessed-condition

URL: https://hl7.org/fhir/R4/diagnosticreport-genetic-DiagnosticReport-assessed-condition.html

10.3.14.8 Diagnosticreport-genetic-DiagnosticReport-assessed-condition

Search Parameter assessed-condition (type: reference) defined as part of the diagnosticreport-genetic Profile.

Orders and Observations | Maturity Level: 0 | Informative

Condition assessed by genetic test

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/diagnosticreport-genetic-DiagnosticReport-assessed-condition

• Paths :
• Targets :

10.3.15 Profile for HLA Genotyping Results

URL: https://hl7.org/fhir/R4/diagnosticreport-hla-results.html

10.3.15 Profile for HLA Genotyping Results

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

Trial-Use Note:

The material on this page is currently undergoing work to be refactored or potentially removed in a future release as further analysis is done as part of the Genomics Reporting Implementation Guide .

10.3.15.1 Content

Profiles:
HLAResult | Describes how the HLA genotyping results
Extensions:
hla-genotyping-results-allele-database | allele-database : Allele Database.
hla-genotyping-results-glstring | glstring : glstring.
hla-genotyping-results-haploid | haploid : haploid.
hla-genotyping-results-method | method : The platform, methodology and software applied at the time of the genotyping.
Examples:
HLA genotyping | An example of a HLA genotyping results report

10.3.15.2 StructureDefinition: HLAResult

URL: https://hl7.org/fhir/R4/hlaresult.html

10.3.15.2 StructureDefinition: HLAResult

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/hlaresult`

Describes how the HLA genotyping results

This profile was published on Fri, Oct 9, 2015 00:00+1100 as a draft by Health Level Seven International (Clinical Genomics).

10.3.15.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from DiagnosticReport.

Summary

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
• http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
• http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
• http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method

This structure is derived from DiagnosticReport.

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*
hla-genotyping-results-allele-database | | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring | | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
hla-genotyping-results-haploid | | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method

Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
hla-genotyping-results-allele-database | I | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring | I | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
hla-genotyping-results-haploid | I | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | I | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business identifier for report
basedOn | I | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | What was requested
status | ?!ΣI | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ΣI | 0..* | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ΣI | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | ΣI | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | DateTime this version was made
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | I | 0..* | Reference(Specimen) | Specimens this report is based on
result | I | 0..* | Reference(Observation) | Observations
imagingStudy | I | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | ΣI | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
comment | I | 0..1 | string | Comment about the image (e.g. explanation)
link | ΣI | 1..1 | Reference(Media) | Reference to the image source
conclusion | I | 0..1 | string | Clinical conclusion (interpretation) of test results
conclusionCode | I | 0..* | CodeableConcept | Codes for the clinical conclusion of test results Binding: SNOMED CT Clinical Findings (example)
presentedForm | I | 0..* | Attachment | Entire report as issued

Documentation for this format

```
<!-- Profile for HLA Genotyping Results -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) What was requested --></basedOn>
 <status value="[code]"/><!--  1..1 registered | partial | preliminary | final +   -->
 <category><!--  0..* CodeableConcept Service category   --></category>
 <code><!--  1..1 CodeableConcept Name/Code for this diagnostic report   --></code>
 <subject><!--  0..1 Reference(Patient)
     The subject of the report - usually, but not always, the patient --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Health care event when test ordered --></encounter>
 <effective[x]><!--  0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!--  0..1 DateTime this version was made -->
 <performer><!--  0..* Reference(Practitioner)
     Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!--  0..* Reference(Practitioner)
```

```
    Primary result interpreter --></resultsInterpreter>
 <specimen><!--  0..* Reference(Specimen)
    Specimens this report is based on --></specimen>
 <result><!--  0..* Reference(Observation) Observations --></result>
 <imagingStudy><!--  0..* Reference(ImagingStudy)
    Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <media>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <comment value="[string]"/><!--  0..1 Comment about the image (e.g. explanation) -->
  <link><!--  1..1 Reference(Media)
     Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!--  0..1 Clinical conclusion (interpretation) of test results -->
 <conclusionCode><!--  0..* CodeableConcept Codes for the clinical conclusion of test results   --></conclusionCode>
 <presentedForm><!--  0..* Attachment
    Entire report as issued --></presentedForm>
</DiagnosticReport>
```

```
{ // Profile for HLA Genotyping Results
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Allele Database // C?
        // from Element: extension
      },
      { // Glstring // C?
        // from Element: extension
      },
      { // Haploid // C?
        // from Element: extension
      },
      { // The platform, methodology and software applied at the time of the
genotyping // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business identifier for report
    "basedOn" : [{ Reference(CarePlan) }], // C?What was requested
    "status" : "<code>", // C? R! registered | partial | preliminary | final +
    "category" : [{ CodeableConcept }], // C?Service category
    "code" : { CodeableConcept }, // C? R! Name/Code for this diagnostic report
    "subject" : { Reference(Patient) }, // C?The subject of the report - usually, but not always, the patient
    "encounter" : { Reference(Encounter) }, // C?Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for report
    "issued" : "<instant>", // C?DateTime this version was made
    "performer" : [{ Reference(Practitioner) }], // C?Responsible Diagnostic Service
    "resultsInterpreter" : [{ Reference(Practitioner) }], // C?Primary result interpreter
    "specimen" : [{ Reference(Specimen) }], // C?Specimens this report is based on
    "result" : [{ Reference(Observation) }], // C?Observations
    "imagingStudy" : [{ Reference(ImagingStudy) }], // C?Reference to full details of imaging associated with the
    diagnostic report
    "media" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", // C?Clinical conclusion (interpretation) of test results
    "conclusionCode" : [{ CodeableConcept }], // C?Codes for the clinical conclusion of test results
    "presentedForm" : [{ Attachment }], // C?Entire report as issued
  }
```

This structure is derived from DiagnosticReport.

Summary

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
- http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
- http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
- http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method

Differential View

This structure is derived from DiagnosticReport.

Name	Flags	Card.	Type	Description & Constraints
DiagnosticReport		0..*		
hla-genotyping-results-allele-database		0..1	CodeableConcept	Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring		0..1	(Complex)	Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring

hla-genotyping-results-haploid | | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | I | 0..* | | A Diagnostic report - a combination of request information, atomic results, images, interpretation, as well as formatted reports
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
hla-genotyping-results-allele-database | I | 0..1 | CodeableConcept | Allele Database URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database
hla-genotyping-results-glstring | I | 0..1 | (Complex) | Glstring URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring
hla-genotyping-results-haploid | I | 0..* | (Complex) | Haploid URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid
hla-genotyping-results-method | I | 0..1 | CodeableConcept | The platform, methodology and software applied at the time of the genotyping URL: http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | Business identifier for report
basedOn | I | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | What was requested
status | ?!ΣI | 1..1 | code | registered | partial | preliminary | final + Binding: DiagnosticReportStatus (required)
category | ΣI | 0..* | CodeableConcept | Service category Binding: Diagnostic Service Section Codes (example)
code | ΣI | 1..1 | CodeableConcept | Name/Code for this diagnostic report Binding: LOINC Diagnostic Report Codes (preferred)
subject | ΣI | 0..1 | Reference(Patient | Group | Device | Location) | The subject of the report - usually, but not always, the patient
encounter | ΣI | 0..1 | Reference(Encounter) | Health care event when test ordered
effective[x] | ΣI | 0..1 | | Clinically relevant time/time-period for report
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
issued | ΣI | 0..1 | instant | DateTime this version was made
performer | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Responsible Diagnostic Service
resultsInterpreter | ΣI | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Primary result interpreter
specimen | I | 0..* | Reference(Specimen) | Specimens this report is based on
result | I | 0..* | Reference(Observation) | Observations
imagingStudy | I | 0..* | Reference(ImagingStudy) | Reference to full details of imaging associated with the diagnostic report
media | ΣI | 0..* | BackboneElement | Key images associated with this report
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
comment | I | 0..1 | string | Comment about the image (e.g. explanation)
link | ΣI | 1..1 | Reference(Media) | Reference to the image source
conclusion | I | 0..1 | string | Clinical conclusion (interpretation) of test results
conclusionCode | I | 0..* | CodeableConcept | Codes for the clinical conclusion of test results Binding: SNOMED CT Clinical Findings (example)
presentedForm | I | 0..* | Attachment | Entire report as issued
Documentation for this format

XML Template

```
<!-- Profile for HLA Genotyping Results -->

<DiagnosticReport xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     Business identifier for report --></identifier>
 <basedOn><!--  0..* Reference(CarePlan) What was requested --></basedOn>
 <status value="[code]"/><!--  1..1 registered | partial | preliminary | final +   -->
 <category><!--  0..* CodeableConcept Service category   --></category>
 <code><!--  1..1 CodeableConcept Name/Code for this diagnostic report   --></code>
 <subject><!--  0..1 Reference(Patient)
     The subject of the report - usually, but not always, the patient --></subject>
```

```
 <encounter><!--  0..1 Reference(Encounter)
     Health care event when test ordered --></encounter>
 <effective[x]><!--  0..1 dateTime|Period
     Clinically relevant time/time-period for report --></effective[x]>
 <issued value="[instant]"/><!--  0..1 DateTime this version was made -->
 <performer><!--  0..* Reference(Practitioner)
     Responsible Diagnostic Service --></performer>
 <resultsInterpreter><!--  0..* Reference(Practitioner)
     Primary result interpreter --></resultsInterpreter>
 <specimen><!--  0..* Reference(Specimen)
     Specimens this report is based on --></specimen>
 <result><!--  0..* Reference(Observation) Observations --></result>
 <imagingStudy><!--  0..* Reference(ImagingStudy)
     Reference to full details of imaging associated with the diagnostic report --></imagingStudy>
 <media>  0..* BackboneElement  <!--  0..* Key images associated with this report -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <comment value="[string]"/><!--  0..1 Comment about the image (e.g. explanation) -->
  <link><!--  1..1 Reference(Media)
      Reference to the image source --></link>
 </media>
 <conclusion value="[string]"/><!--  0..1 Clinical conclusion (interpretation) of test results -->
 <conclusionCode><!--  0..* CodeableConcept Codes for the clinical conclusion of test results   --></conclusionCode>
 <presentedForm><!--  0..* Attachment
     Entire report as issued --></presentedForm>
</DiagnosticReport>
```

JSON Template

```
{ // Profile for HLA Genotyping Results
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Allele Database // C?
        // from Element: extension
      },
      { // Glstring // C?
        // from Element: extension
      },
      { // Haploid // C?
        // from Element: extension
      },
      { // The platform, methodology and software applied at the time of the
genotyping // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?Business identifier for report
    "basedOn" : [{ Reference(CarePlan) }], // C?What was requested
    "status" : "<code>", // C? R! registered | partial | preliminary | final +
    "category" : [{ CodeableConcept }], // C?Service category
    "code" : { CodeableConcept }, // C? R! Name/Code for this diagnostic report
    "subject" : { Reference(Patient) }, // C?The subject of the report - usually, but not always, the patient
    "encounter" : { Reference(Encounter) }, // C?Health care event when test ordered
// value[x]: Clinically relevant time/time-period for report. One of these 2:
    "effectiveDateTime" : "<dateTime>", // C?Clinically relevant time/time-period for report
    "effectivePeriod" : { Period }, // C?Clinically relevant time/time-period for report
    "issued" : "<instant>", // C?DateTime this version was made
    "performer" : [{ Reference(Practitioner) }], // C?Responsible Diagnostic Service
    "resultsInterpreter" : [{ Reference(Practitioner) }], // C?Primary result interpreter
    "specimen" : [{ Reference(Specimen) }], // C?Specimens this report is based on
    "result" : [{ Reference(Observation) }], // C?Observations
    "imagingStudy" : [{ Reference(ImagingStudy) }], // C?Reference to full details of imaging associated with the
    diagnostic report
    "media" : [{ BackboneElement }], // C?Key images associated with this report
    "conclusion" : "<string>", // C?Clinical conclusion (interpretation) of test results
    "conclusionCode" : [{ CodeableConcept }], // C?Codes for the clinical conclusion of test results
    "presentedForm" : [{ Attachment }], // C?Entire report as issued
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

10.3.15.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
DiagnosticReport.language	Common Languages	preferred	Common Languages
DiagnosticReport.status	DiagnosticReportStatus	required	DiagnosticReportStatus
DiagnosticReport.category	Diagnostic Service Section Codes	example	Diagnostic Service Section Codes
DiagnosticReport.code	LOINC Diagnostic Report Codes	preferred	LOINC Diagnostic Report Codes
DiagnosticReport.conclusionCode	SNOMED CT Clinical Findings	example	SNOMED CT Clinical Findings

10.3.16 JSON Schema for DiagnosticReport

URL: https://hl7.org/fhir/R4/diagnosticreport.schema.json.html

10.3.16 JSON Schema for DiagnosticReport

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

This is a representation of the json schema for DiagnosticReport, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DiagnosticReport",
  "$ref": "#/definitions/DiagnosticReport",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DiagnosticReport": {
      "description": "The findings and interpretation of diagnostic  tests performed on patients, groups of patients,
    devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting
     and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted
    representation of diagnostic reports.",
      "properties": {
        "resourceType": {
          "description": "This is a DiagnosticReport resource",
          "const": "DiagnosticReport"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
```

```
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers assigned to this report by the performer or other systems.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "Details concerning a service requested.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of the diagnostic report.",
      "enum": [
        "registered",
        "partial",
        "preliminary",
        "final",
        "amended",
        "corrected",
        "appended",
        "cancelled",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "A code that classifies the clinical discipline, department or diagnostic service that created
the report (e.g. cardiology, biochemistry, hematology, MRI). This is used for searching, sorting and display
purposes.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "code": {
      "description": "A code or name that describes this diagnostic report.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The subject of the report. Usually, but not always, this is a patient. However, diagnostic
services also perform analyses on specimens collected from a variety of other sources.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The healthcare event  (e.g. a patient and healthcare provider interaction) which this
DiagnosticReport is about.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "effectiveDateTime": {
      "description": "The time or time-period the observed values are related to. When the subject of the report is
a patient, this is usually either the time of the procedure or of specimen collection(s), but very often the source
of the date/time is not known, only the date/time itself.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_effectiveDateTime": {
      "description": "Extensions for effectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The time or time-period the observed values are related to. When the subject of the report is
a patient, this is usually either the time of the procedure or of specimen collection(s), but very often the source
of the date/time is not known, only the date/time itself.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "issued": {
      "description": "The date and time that this version of the report was made available to providers, typically
after the report was reviewed and verified.",
      "$ref": "#/definitions/instant"
    },
    "_issued": {
      "description": "Extensions for issued",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "performer": {
      "description": "The diagnostic service that is responsible for issuing the report.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
```

```
      },
      "type": "array"
    },
    "resultsInterpreter": {
      "description": "The practitioner or organization that is responsible for the report\u0027s conclusions and
interpretations.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "specimen": {
      "description": "Details about the specimens on which this diagnostic report is based.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "result": {
      "description": "[Observations](observation.html)  that are part of this diagnostic report.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "imagingStudy": {
      "description": "One or more links to full details of any imaging performed during the diagnostic
investigation. Typically, this is imaging performed by DICOM enabled modalities, but this is not required. A fully
enabled PACS viewer can use this information to provide views of the source images.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "media": {
      "description": "A list of key images associated with this report. The images are generally created during the
diagnostic process, and may be directly of the patient, or of treated specimens (i.e. slides of interest).",
      "items": {
        "$ref": "#/definitions/DiagnosticReport_Media"
      },
      "type": "array"
    },
    "conclusion": {
      "description": "Concise and clinically contextualized summary conclusion (interpretation/impression) of the
diagnostic report.",
      "$ref": "#/definitions/string"
    },
    "_conclusion": {
      "description": "Extensions for conclusion",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "conclusionCode": {
      "description": "One or more codes that represent the summary conclusion (interpretation/impression) of the
diagnostic report.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "presentedForm": {
      "description": "Rich text representation of the entire result as issued by the diagnostic service. Multiple
formats are allowed but they SHALL be semantically equivalent.",
      "items": {
        "$ref": "Attachment.schema.json#/definitions/Attachment"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "code",
    "resourceType"
  ]
},
"DiagnosticReport_Media": {
  "description": "The findings and interpretation of diagnostic  tests performed on patients, groups of patients,
devices, and locations, and/or specimens derived from these. The report includes clinical context such as requesting
 and provider information, and some mix of atomic results, images, textual and coded interpretations, and formatted
representation of diagnostic reports.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "comment": {
          "description": "A comment about the image. Typically, this is used to provide an explanation for why the image
     is included, or to draw the viewer\u0027s attention to important features.",
          "$ref": "#/definitions/string"
        },
        "_comment": {
          "description": "Extensions for comment",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "link": {
          "description": "Reference to the image source.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "link"
      ]
    }
  }
}
```


10.3.17 Resource DiagnosticReport - Maturity Dependency View

URL: https://hl7.org/fhir/R4/diagnosticreport-dependencies.html

10.3.17 Resource DiagnosticReport - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner

Dependency Graph for DiagnosticReport FMM level 3

Name | Card. | Type | Dependency Analysis
DiagnosticReport | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(CarePlan | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ImmunizationRecommendation = FMM1-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (NutritionOrder = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use)
status | 1..1 | code
category | 0..* | CodeableConcept
code | 1..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group | Device | Location) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
effective[x] | 0..1
effectiveDateTime | | dateTime
effectivePeriod | | Period
issued | 0..1 | instant
performer | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (CareTeam = FMM2-Trial Use vs. Element = FMM3-Trial Use)
resultsInterpreter | 0..* | Reference(Practitioner | PractitionerRole | Organization | CareTeam) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (CareTeam = FMM2-Trial Use vs. Element = FMM3-Trial Use)
specimen | 0..* | Reference(Specimen) | Type Warning: (Specimen = FMM2-Trial Use vs. Element = FMM3-Trial Use)
result | 0..* | Reference(Observation)
imagingStudy | 0..* | Reference(ImagingStudy)
media | 0..* | BackboneElement
comment | 0..1 | string
link | 1..1 | Reference(Media) | Type Warning: (Media = FMM1-Trial Use vs. Element = FMM3-Trial Use)
conclusion | 0..1 | string
conclusionCode | 0..* | CodeableConcept

presentedForm | 0..* | Attachment

10.3.18 Value Set Sc-valueset-diagnostic-report-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-diagnostic-report-status.html

10.3.18 Value Set Sc-valueset-diagnostic-report-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

10.3.19 DiagnosticReportStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-diagnostic-report-status)

Mapping from http://hl7.org/fhir/ValueSet/diagnostic-report-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the diagnostic report."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
partial,preliminary	equivalent	draft
registered	equivalent	received
corrected, amended, appended	equivalent	replaced
final	equivalent	complete
cancelled	equivalent	abandoned
unknown	equivalent	unknown

10.4 Resource ImagingStudy - Content

URL: https://hl7.org/fhir/R4/imagingstudy.html

we can construct the WADO-RS URL to issue an HTTP GET for a native DICOM PS3.10 instance file (if consistent with the Accept header): we can construct the WADO-URI URL to issue an HTTP GET for a native DICOM PS3.10 instance file (if consistent with the Accept header): we can construct the IID URL to invoke a diagnostic quality viewer on the indicated study:

Imaging Integration Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient

Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data) acquired or produced in a common context. A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a study may have multiple series of different modalities.

10.4.1 Scope and Usage

ImagingStudy provides information on a DICOM imaging study, and the series and imaging objects in that study. It also provides information on how to retrieve that information (in a native DICOM format, or in a rendered format, such as JPEG). ImagingStudy is used to make available information about all parts of a single DICOM study.

This resource provides mappings of its elements to DICOM attributes. DICOM attributes are identified by a 32-bit tag, presented in canonical form as two four-digit hexadecimal values within parentheses and separated by a comma, e.g. (0008,103E). The name and value representation (data type) of each attribute can be found in DICOM Part 6 Data Dictionary . The use of the attributes in the context of information objects, including detailed description of use, can be found in DICOM Part 3 Information Object Definitions . Attributes used in the DICOM query information models, such as "Number of Instances in Study", can be found in DICOM Part 4 Annex C .

ImagingStudy provides access to significant DICOM information but will only eliminate the need for DICOM query (e.g., QIDO-RS) in the simplest cases. The DICOM instances are not stored in the ImagingStudy resource; use of a DICOM WADO-RS server or other storage mechanism is needed.

An ImagingStudy SHALL reference one DICOM Study, and MAY reference a subset of that Study. More than one ImagingStudy MAY reference the same DICOM Study or different subsets of the same DICOM Study.

10.4.2 Boundaries and Relationships

ImagingStudy is used for DICOM imaging and associated information. Use Media to track non-DICOM images, video, or audio. Binary can be used to store arbitrary content. DocumentReference allow indexing and retrieval of clinical “documents” with relevant metadata.

This resource is referenced by ChargeItem, ClinicalImpression, DiagnosticReport and Observation

10.4.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ImagingStudy | TU | | DomainResource | A set of images produced in single study (one or more series of references images)
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifiers for the whole study
status | ?!Σ | 1..1 | code | registered | available | cancelled | entered-in-error | unknown ImagingStudyStatus (Required)
modality | Σ | 0..* | Coding | All series modality if actual acquisition modalities AcquisitionModality (Extensible)
subject | Σ | 1..1 | Reference(Patient | Device | Group) | Who or what is the subject of the study
encounter | Σ | 0..1 | Reference(Encounter) | Encounter with which this imaging study is associated
started | Σ | 0..1 | dateTime | When the study was started
basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest | Appointment | AppointmentResponse | Task) | Request fulfilled
referrer | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Referring physician
interpreter | Σ | 0..* | Reference(Practitioner | PractitionerRole) | Who interpreted images
endpoint | Σ | 0..* | Reference(Endpoint) | Study access endpoint
numberOfSeries | Σ | 0..1 | unsignedInt | Number of Study Related Series
numberOfInstances | Σ | 0..1 | unsignedInt | Number of Study Related Instances
procedureReference | Σ | 0..1 | Reference(Procedure) | The performed Procedure reference
procedureCode | Σ | 0..* | CodeableConcept | The performed procedure code ImagingProcedureCode (Extensible)
location | Σ | 0..1 | Reference(Location) | Where ImagingStudy occurred
reasonCode | Σ | 0..* | CodeableConcept | Why the study was requested Procedure Reason Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | Media | DiagnosticReport | DocumentReference) | Why was study performed
note | Σ | 0..* | Annotation | User-defined comments
description | Σ | 0..1 | string | Institution-generated description
series | Σ | 0..* | BackboneElement | Each study has one or more series of instances
uid | Σ | 1..1 | id | DICOM Series Instance UID for the series
number | Σ | 0..1 | unsignedInt | Numeric identifier of this series
modality | Σ | 1..1 | Coding | The modality of the instances in the series AcquisitionModality (Extensible)
description | Σ | 0..1 | string | A short human readable summary of the series
numberOfInstances | Σ | 0..1 | unsignedInt | Number of Series Related Instances
endpoint | Σ | 0..* | Reference(Endpoint) | Series access endpoint
bodySite | Σ | 0..1 | Coding | Body part examined SNOMED CT Body Structures (Example)
laterality | Σ | 0..1 | Coding | Body part laterality Laterality (Example)
specimen | Σ | 0..* | Reference(Specimen) | Specimen imaged
started | Σ | 0..1 | dateTime | When the series started
performer | Σ | 0..* | BackboneElement | Who performed the series
function | Σ | 0..1 | CodeableConcept | Type of performance ImagingStudy series performer function (Extensible)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | Who performed the series
instance | | 0..* | BackboneElement | A single SOP instance from the series
uid | | 1..1 | id | DICOM SOP Instance UID
sopClass | | 1..1 | Coding | DICOM class type sopClass (Extensible)
number | | 0..1 | unsignedInt | The number of this instance in the series
title | | 0..1 | string | Description of instance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImagingStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers for the whole study --></identifier>
 <status value="[code]"/><!-- 1..1 registered | available | cancelled | entered-in-error | unknown -->
 <modality><!-- 0..* Coding All series modality if actual acquisition modalities  --></modality>
 <subject><!-- 1..1 Reference(Patient|Device|Group) Who or what is the subject of the study --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter with which this imaging study is associated --></encounter>
 <started value="[dateTime]"/><!-- 0..1 When the study was started -->
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest|Appointment|
   AppointmentResponse|Task) Request fulfilled --></basedOn>
 <referrer><!-- 0..1 Reference(Practitioner|PractitionerRole) Referring physician --></referrer>
 <interpreter><!-- 0..* Reference(Practitioner|PractitionerRole) Who interpreted images --></interpreter>
```

```
 <endpoint><!-- 0..* Reference(Endpoint) Study access endpoint --></endpoint>
 <numberOfSeries value="[unsignedInt]"/><!-- 0..1 Number of Study Related Series -->
 <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Study Related Instances -->
 <procedureReference><!-- 0..1 Reference(Procedure) The performed Procedure reference --></procedureReference>
 <procedureCode><!-- 0..* CodeableConcept The performed procedure code  --></procedureCode>
 <location><!-- 0..1 Reference(Location) Where ImagingStudy occurred --></location>
 <reasonCode><!-- 0..* CodeableConcept Why the study was requested --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|Media|
   DiagnosticReport|DocumentReference) Why was study performed --></reasonReference>
 <note><!-- 0..* Annotation User-defined comments --></note>
 <description value="[string]"/><!-- 0..1 Institution-generated description -->
 <series>  <!-- 0..* Each study has one or more series of instances -->
  <uid value="[id]"/><!-- 1..1 DICOM Series Instance UID for the series -->
  <number value="[unsignedInt]"/><!-- 0..1 Numeric identifier of this series -->
  <modality><!-- 1..1 Coding The modality of the instances in the series  --></modality>
  <description value="[string]"/><!-- 0..1 A short human readable summary of the series -->
  <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Series Related Instances -->
  <endpoint><!-- 0..* Reference(Endpoint) Series access endpoint --></endpoint>
  <bodySite><!-- 0..1 Coding Body part examined --></bodySite>
  <laterality><!-- 0..1 Coding Body part laterality --></laterality>
  <specimen><!-- 0..* Reference(Specimen) Specimen imaged --></specimen>
  <started value="[dateTime]"/><!-- 0..1 When the series started -->
  <performer>  <!-- 0..* Who performed the series -->
   <function><!-- 0..1 CodeableConcept Type of performance --></function>
   <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|CareTeam|
     Patient|Device|RelatedPerson) Who performed the series --></actor>
  </performer>
  <instance>  <!-- 0..* A single SOP instance from the series -->
   <uid value="[id]"/><!-- 1..1 DICOM SOP Instance UID -->
   <sopClass><!-- 1..1 Coding DICOM class type  --></sopClass>
   <number value="[unsignedInt]"/><!-- 0..1 The number of this instance in the series -->
   <title value="[string]"/><!-- 0..1 Description of instance -->
  </instance>
 </series>
</ImagingStudy>
```

JSON Template

```
{
  "resourceType" : "ImagingStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers for the whole study
  "status" : "<code>", // R!  registered | available | cancelled | entered-in-error | unknown
  "modality" : [{ Coding }], // All series modality if actual acquisition modalities
  "subject" : { Reference(Patient|Device|Group) }, // R!  Who or what is the subject of the study
  "encounter" : { Reference(Encounter) }, // Encounter with which this imaging study is associated
  "started" : "<dateTime>", // When the study was started
  "basedOn" : [{ Reference(CarePlan|ServiceRequest|Appointment|
   AppointmentResponse|Task) }], // Request fulfilled
  "referrer" : { Reference(Practitioner|PractitionerRole) }, // Referring physician
  "interpreter" : [{ Reference(Practitioner|PractitionerRole) }], // Who interpreted images
  "endpoint" : [{ Reference(Endpoint) }], // Study access endpoint
  "numberOfSeries" : "<unsignedInt>", // Number of Study Related Series
  "numberOfInstances" : "<unsignedInt>", // Number of Study Related Instances
  "procedureReference" : { Reference(Procedure) }, // The performed Procedure reference
  "procedureCode" : [{ CodeableConcept }], // The performed procedure code
  "location" : { Reference(Location) }, // Where ImagingStudy occurred
  "reasonCode" : [{ CodeableConcept }], // Why the study was requested
  "reasonReference" : [{ Reference(Condition|Observation|Media|
   DiagnosticReport|DocumentReference) }], // Why was study performed
  "note" : [{ Annotation }], // User-defined comments
  "description" : "<string>", // Institution-generated description
  "series" : [{ // Each study has one or more series of instances
    "uid" : "<id>", // R!  DICOM Series Instance UID for the series
    "number" : "<unsignedInt>", // Numeric identifier of this series
    "modality" : { Coding }, // R!  The modality of the instances in the series
    "description" : "<string>", // A short human readable summary of the series
    "numberOfInstances" : "<unsignedInt>", // Number of Series Related Instances
    "endpoint" : [{ Reference(Endpoint) }], // Series access endpoint
    "bodySite" : { Coding }, // Body part examined
    "laterality" : { Coding }, // Body part laterality
    "specimen" : [{ Reference(Specimen) }], // Specimen imaged
    "started" : "<dateTime>", // When the series started
    "performer" : [{ // Who performed the series
      "function" : { CodeableConcept }, // Type of performance
      "actor" : { Reference(Practitioner|PractitionerRole|Organization|CareTeam|
     Patient|Device|RelatedPerson) } // R!  Who performed the series
    }],
    "instance" : [{ // A single SOP instance from the series
      "uid" : "<id>", // R!  DICOM SOP Instance UID
      "sopClass" : { Coding }, // R!  DICOM class type
      "number" : "<unsignedInt>", // The number of this instance in the series
      "title" : "<string>" // Description of instance
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:ImagingStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImagingStudy.identifier [ Identifier ], ... ; # 0..* Identifiers for the whole study
  fhir:ImagingStudy.status [ code ]; # 1..1 registered | available | cancelled | entered-in-error | unknown
  fhir:ImagingStudy.modality [ Coding ], ... ; # 0..* All series modality if actual acquisition modalities
  fhir:ImagingStudy.subject [ Reference(Patient|Device|Group) ]; # 1..1 Who or what is the subject of the study
  fhir:ImagingStudy.encounter [ Reference(Encounter) ]; # 0..1 Encounter with which this imaging study is associated
  fhir:ImagingStudy.started [ dateTime ]; # 0..1 When the study was started
  fhir:ImagingStudy.basedOn [ Reference(CarePlan|ServiceRequest|Appointment|AppointmentResponse|Task) ], ... ; # 0..*
    Request fulfilled
  fhir:ImagingStudy.referrer [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Referring physician
  fhir:ImagingStudy.interpreter [ Reference(Practitioner|PractitionerRole) ], ... ; # 0..* Who interpreted images
  fhir:ImagingStudy.endpoint [ Reference(Endpoint) ], ... ; # 0..* Study access endpoint
  fhir:ImagingStudy.numberOfSeries [ unsignedInt ]; # 0..1 Number of Study Related Series
  fhir:ImagingStudy.numberOfInstances [ unsignedInt ]; # 0..1 Number of Study Related Instances
  fhir:ImagingStudy.procedureReference [ Reference(Procedure) ]; # 0..1 The performed Procedure reference
  fhir:ImagingStudy.procedureCode [ CodeableConcept ], ... ; # 0..* The performed procedure code
  fhir:ImagingStudy.location [ Reference(Location) ]; # 0..1 Where ImagingStudy occurred
  fhir:ImagingStudy.reasonCode [ CodeableConcept ], ... ; # 0..* Why the study was requested
  fhir:ImagingStudy.reasonReference [ Reference(Condition|Observation|Media|DiagnosticReport|DocumentReference) ], ... ;
     # 0..* Why was study performed
  fhir:ImagingStudy.note [ Annotation ], ... ; # 0..* User-defined comments
  fhir:ImagingStudy.description [ string ]; # 0..1 Institution-generated description
  fhir:ImagingStudy.series [ # 0..* Each study has one or more series of instances
    fhir:ImagingStudy.series.uid [ id ]; # 1..1 DICOM Series Instance UID for the series
    fhir:ImagingStudy.series.number [ unsignedInt ]; # 0..1 Numeric identifier of this series
    fhir:ImagingStudy.series.modality [ Coding ]; # 1..1 The modality of the instances in the series
    fhir:ImagingStudy.series.description [ string ]; # 0..1 A short human readable summary of the series
    fhir:ImagingStudy.series.numberOfInstances [ unsignedInt ]; # 0..1 Number of Series Related Instances
    fhir:ImagingStudy.series.endpoint [ Reference(Endpoint) ], ... ; # 0..* Series access endpoint
    fhir:ImagingStudy.series.bodySite [ Coding ]; # 0..1 Body part examined
    fhir:ImagingStudy.series.laterality [ Coding ]; # 0..1 Body part laterality
    fhir:ImagingStudy.series.specimen [ Reference(Specimen) ], ... ; # 0..* Specimen imaged
    fhir:ImagingStudy.series.started [ dateTime ]; # 0..1 When the series started
    fhir:ImagingStudy.series.performer [ # 0..* Who performed the series
      fhir:ImagingStudy.series.performer.function [ CodeableConcept ]; # 0..1 Type of performance
      fhir:ImagingStudy.series.performer.actor [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson) ]; # 1..1 Who performed
    the series
    ], ...;
    fhir:ImagingStudy.series.instance [ # 0..* A single SOP instance from the series
      fhir:ImagingStudy.series.instance.uid [ id ]; # 1..1 DICOM SOP Instance UID
      fhir:ImagingStudy.series.instance.sopClass [ Coding ]; # 1..1 DICOM class type
      fhir:ImagingStudy.series.instance.number [ unsignedInt ]; # 0..1 The number of this instance in the series
      fhir:ImagingStudy.series.instance.title [ string ]; # 0..1 Description of instance
    ], ...;
  ], ...;
]
```

Changes since R3

ImagingStudy

ImagingStudy.status | Added Mandatory Element

ImagingStudy.modality | Renamed from modalityList to modality Change value set from http://hl7.org/fhir/ValueSet/dicom-cid29 to http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html

ImagingStudy.subject | Renamed from patient to subject Type Reference: Added Target Types Device, Group

ImagingStudy.encounter | Added Element

ImagingStudy.basedOn | Type Reference: Added Target Types ServiceRequest, Appointment, AppointmentResponse, Task Type Reference: Removed Target Types ReferralRequest, ProcedureRequest

ImagingStudy.referrer | Type Reference: Added Target Type PractitionerRole

ImagingStudy.interpreter | Type Reference: Added Target Type PractitionerRole

ImagingStudy.procedureReference | Max Cardinality changed from * to 1

ImagingStudy.procedureCode | Add Binding http://www.rsna.org/RadLex_Playbook.aspx (extensible)

ImagingStudy.location | Added Element

ImagingStudy.reasonCode | Added Element

ImagingStudy.reasonReference | Added Element

ImagingStudy.note | Added Element

ImagingStudy.series.uid | Type changed from oid to id

ImagingStudy.series.modality | Change value set from http://hl7.org/fhir/ValueSet/dicom-cid29 to http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html

ImagingStudy.series.specimen | Added Element

ImagingStudy.series.performer | Type changed from Reference(Practitioner) to BackboneElement

ImagingStudy.series.performer.function | Added Element

ImagingStudy.series.performer.actor | Added Mandatory Element

ImagingStudy.series.instance.uid | Type changed from oid to id

ImagingStudy.series.instance.sopClass | Type changed from oid to Coding Add Binding http://dicom.nema.org/medical/dicom/current/output/chtml/part04/sect_B.5.html#table_B.5-1 (extensible)

ImagingStudy.uid | deleted

ImagingStudy.accession | deleted

ImagingStudy.availability | deleted

ImagingStudy.context | deleted
ImagingStudy.reason | deleted
ImagingStudy.series.availability | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
ImagingStudy | TU | | DomainResource | A set of images produced in single study (one or more series of references images)
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Identifiers for the whole study
status | ?!Σ | 1..1 | code | registered | available | cancelled | entered-in-error | unknown ImagingStudyStatus (Required)
modality | Σ | 0..* | Coding | All series modality if actual acquisition modalities AcquisitionModality (Extensible)
subject | Σ | 1..1 | Reference(Patient | Device | Group) | Who or what is the subject of the study
encounter | Σ | 0..1 | Reference(Encounter) | Encounter with which this imaging study is associated
started | Σ | 0..1 | dateTime | When the study was started
basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest | Appointment | AppointmentResponse | Task) | Request fulfilled
referrer | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Referring physician
interpreter | Σ | 0..* | Reference(Practitioner | PractitionerRole) | Who interpreted images
endpoint | Σ | 0..* | Reference(Endpoint) | Study access endpoint
numberOfSeries | Σ | 0..1 | unsignedInt | Number of Study Related Series
numberOfInstances | Σ | 0..1 | unsignedInt | Number of Study Related Instances
procedureReference | Σ | 0..1 | Reference(Procedure) | The performed Procedure reference
procedureCode | Σ | 0..* | CodeableConcept | The performed procedure code ImagingProcedureCode (Extensible)
location | Σ | 0..1 | Reference(Location) | Where ImagingStudy occurred
reasonCode | Σ | 0..* | CodeableConcept | Why the study was requested Procedure Reason Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | Media | DiagnosticReport | DocumentReference) | Why was study performed
note | Σ | 0..* | Annotation | User-defined comments
description | Σ | 0..1 | string | Institution-generated description
series | Σ | 0..* | BackboneElement | Each study has one or more series of instances
uid | Σ | 1..1 | id | DICOM Series Instance UID for the series
number | Σ | 0..1 | unsignedInt | Numeric identifier of this series
modality | Σ | 1..1 | Coding | The modality of the instances in the series AcquisitionModality (Extensible)
description | Σ | 0..1 | string | A short human readable summary of the series
numberOfInstances | Σ | 0..1 | unsignedInt | Number of Series Related Instances
endpoint | Σ | 0..* | Reference(Endpoint) | Series access endpoint
bodySite | Σ | 0..1 | Coding | Body part examined SNOMED CT Body Structures (Example)
laterality | Σ | 0..1 | Coding | Body part laterality Laterality (Example)
specimen | Σ | 0..* | Reference(Specimen) | Specimen imaged
started | Σ | 0..1 | dateTime | When the series started
performer | Σ | 0..* | BackboneElement | Who performed the series
function | Σ | 0..1 | CodeableConcept | Type of performance ImagingStudy series performer function (Extensible)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | Who performed the series
instance | | 0..* | BackboneElement | A single SOP instance from the series
uid | | 1..1 | id | DICOM SOP Instance UID
sopClass | | 1..1 | Coding | DICOM class type sopClass (Extensible)
number | | 0..1 | unsignedInt | The number of this instance in the series
title | | 0..1 | string | Description of instance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImagingStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifiers for the whole study --></identifier>
 <status value="[code]"/><!-- 1..1 registered | available | cancelled | entered-in-error | unknown -->
 <modality><!-- 0..* Coding All series modality if actual acquisition modalities  --></modality>
 <subject><!-- 1..1 Reference(Patient|Device|Group) Who or what is the subject of the study --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter with which this imaging study is associated --></encounter>
 <started value="[dateTime]"/><!-- 0..1 When the study was started -->
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest|Appointment|
   AppointmentResponse|Task) Request fulfilled --></basedOn>
 <referrer><!-- 0..1 Reference(Practitioner|PractitionerRole) Referring physician --></referrer>
 <interpreter><!-- 0..* Reference(Practitioner|PractitionerRole) Who interpreted images --></interpreter>
 <endpoint><!-- 0..* Reference(Endpoint) Study access endpoint --></endpoint>
 <numberOfSeries value="[unsignedInt]"/><!-- 0..1 Number of Study Related Series -->
 <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Study Related Instances -->
 <procedureReference><!-- 0..1 Reference(Procedure) The performed Procedure reference --></procedureReference>
 <procedureCode><!-- 0..* CodeableConcept The performed procedure code  --></procedureCode>
 <location><!-- 0..1 Reference(Location) Where ImagingStudy occurred --></location>
```

```
 <reasonCode><!-- 0..* CodeableConcept Why the study was requested --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|Media|
   DiagnosticReport|DocumentReference) Why was study performed --></reasonReference>
 <note><!-- 0..* Annotation User-defined comments --></note>
 <description value="[string]"/><!-- 0..1 Institution-generated description -->
 <series>  <!-- 0..* Each study has one or more series of instances -->
  <uid value="[id]"/><!-- 1..1 DICOM Series Instance UID for the series -->
  <number value="[unsignedInt]"/><!-- 0..1 Numeric identifier of this series -->
  <modality><!-- 1..1 Coding The modality of the instances in the series  --></modality>
  <description value="[string]"/><!-- 0..1 A short human readable summary of the series -->
  <numberOfInstances value="[unsignedInt]"/><!-- 0..1 Number of Series Related Instances -->
  <endpoint><!-- 0..* Reference(Endpoint) Series access endpoint --></endpoint>
  <bodySite><!-- 0..1 Coding Body part examined --></bodySite>
  <laterality><!-- 0..1 Coding Body part laterality --></laterality>
  <specimen><!-- 0..* Reference(Specimen) Specimen imaged --></specimen>
  <started value="[dateTime]"/><!-- 0..1 When the series started -->
  <performer>  <!-- 0..* Who performed the series -->
   <function><!-- 0..1 CodeableConcept Type of performance --></function>
   <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|CareTeam|
     Patient|Device|RelatedPerson) Who performed the series --></actor>
  </performer>
  <instance>  <!-- 0..* A single SOP instance from the series -->
   <uid value="[id]"/><!-- 1..1 DICOM SOP Instance UID -->
   <sopClass><!-- 1..1 Coding DICOM class type  --></sopClass>
   <number value="[unsignedInt]"/><!-- 0..1 The number of this instance in the series -->
   <title value="[string]"/><!-- 0..1 Description of instance -->
  </instance>
 </series>
</ImagingStudy>
```

JSON Template

```
{
  "resourceType" : "ImagingStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifiers for the whole study
  "status" : "<code>", // R!  registered | available | cancelled | entered-in-error | unknown
  "modality" : [{ Coding }], // All series modality if actual acquisition modalities
  "subject" : { Reference(Patient|Device|Group) }, // R!  Who or what is the subject of the study
  "encounter" : { Reference(Encounter) }, // Encounter with which this imaging study is associated
  "started" : "<dateTime>", // When the study was started
  "basedOn" : [{ Reference(CarePlan|ServiceRequest|Appointment|
   AppointmentResponse|Task) }], // Request fulfilled
  "referrer" : { Reference(Practitioner|PractitionerRole) }, // Referring physician
  "interpreter" : [{ Reference(Practitioner|PractitionerRole) }], // Who interpreted images
  "endpoint" : [{ Reference(Endpoint) }], // Study access endpoint
  "numberOfSeries" : "<unsignedInt>", // Number of Study Related Series
  "numberOfInstances" : "<unsignedInt>", // Number of Study Related Instances
  "procedureReference" : { Reference(Procedure) }, // The performed Procedure reference
  "procedureCode" : [{ CodeableConcept }], // The performed procedure code
  "location" : { Reference(Location) }, // Where ImagingStudy occurred
  "reasonCode" : [{ CodeableConcept }], // Why the study was requested
  "reasonReference" : [{ Reference(Condition|Observation|Media|
   DiagnosticReport|DocumentReference) }], // Why was study performed
  "note" : [{ Annotation }], // User-defined comments
  "description" : "<string>", // Institution-generated description
  "series" : [{ // Each study has one or more series of instances
    "uid" : "<id>", // R!  DICOM Series Instance UID for the series
    "number" : "<unsignedInt>", // Numeric identifier of this series
    "modality" : { Coding }, // R!  The modality of the instances in the series
    "description" : "<string>", // A short human readable summary of the series
    "numberOfInstances" : "<unsignedInt>", // Number of Series Related Instances
    "endpoint" : [{ Reference(Endpoint) }], // Series access endpoint
    "bodySite" : { Coding }, // Body part examined
    "laterality" : { Coding }, // Body part laterality
    "specimen" : [{ Reference(Specimen) }], // Specimen imaged
    "started" : "<dateTime>", // When the series started
    "performer" : [{ // Who performed the series
      "function" : { CodeableConcept }, // Type of performance
      "actor" : { Reference(Practitioner|PractitionerRole|Organization|CareTeam|
     Patient|Device|RelatedPerson) } // R!  Who performed the series
    }],
    "instance" : [{ // A single SOP instance from the series
      "uid" : "<id>", // R!  DICOM SOP Instance UID
      "sopClass" : { Coding }, // R!  DICOM class type
      "number" : "<unsignedInt>", // The number of this instance in the series
      "title" : "<string>" // Description of instance
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImagingStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
```

```
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImagingStudy.identifier [ Identifier ], ... ; # 0..* Identifiers for the whole study
  fhir:ImagingStudy.status [ code ]; # 1..1 registered | available | cancelled | entered-in-error | unknown
  fhir:ImagingStudy.modality [ Coding ], ... ; # 0..* All series modality if actual acquisition modalities
  fhir:ImagingStudy.subject [ Reference(Patient|Device|Group) ]; # 1..1 Who or what is the subject of the study
  fhir:ImagingStudy.encounter [ Reference(Encounter) ]; # 0..1 Encounter with which this imaging study is associated
  fhir:ImagingStudy.started [ dateTime ]; # 0..1 When the study was started
  fhir:ImagingStudy.basedOn [ Reference(CarePlan|ServiceRequest|Appointment|AppointmentResponse|Task) ], ... ; # 0..*
    Request fulfilled
  fhir:ImagingStudy.referrer [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Referring physician
  fhir:ImagingStudy.interpreter [ Reference(Practitioner|PractitionerRole) ], ... ; # 0..* Who interpreted images
  fhir:ImagingStudy.endpoint [ Reference(Endpoint) ], ... ; # 0..* Study access endpoint
  fhir:ImagingStudy.numberOfSeries [ unsignedInt ]; # 0..1 Number of Study Related Series
  fhir:ImagingStudy.numberOfInstances [ unsignedInt ]; # 0..1 Number of Study Related Instances
  fhir:ImagingStudy.procedureReference [ Reference(Procedure) ]; # 0..1 The performed Procedure reference
  fhir:ImagingStudy.procedureCode [ CodeableConcept ], ... ; # 0..* The performed procedure code
  fhir:ImagingStudy.location [ Reference(Location) ]; # 0..1 Where ImagingStudy occurred
  fhir:ImagingStudy.reasonCode [ CodeableConcept ], ... ; # 0..* Why the study was requested
  fhir:ImagingStudy.reasonReference [ Reference(Condition|Observation|Media|DiagnosticReport|DocumentReference) ], ... ;
     # 0..* Why was study performed
  fhir:ImagingStudy.note [ Annotation ], ... ; # 0..* User-defined comments
  fhir:ImagingStudy.description [ string ]; # 0..1 Institution-generated description
  fhir:ImagingStudy.series [ # 0..* Each study has one or more series of instances
    fhir:ImagingStudy.series.uid [ id ]; # 1..1 DICOM Series Instance UID for the series
    fhir:ImagingStudy.series.number [ unsignedInt ]; # 0..1 Numeric identifier of this series
    fhir:ImagingStudy.series.modality [ Coding ]; # 1..1 The modality of the instances in the series
    fhir:ImagingStudy.series.description [ string ]; # 0..1 A short human readable summary of the series
    fhir:ImagingStudy.series.numberOfInstances [ unsignedInt ]; # 0..1 Number of Series Related Instances
    fhir:ImagingStudy.series.endpoint [ Reference(Endpoint) ], ... ; # 0..* Series access endpoint
    fhir:ImagingStudy.series.bodySite [ Coding ]; # 0..1 Body part examined
    fhir:ImagingStudy.series.laterality [ Coding ]; # 0..1 Body part laterality
    fhir:ImagingStudy.series.specimen [ Reference(Specimen) ], ... ; # 0..* Specimen imaged
    fhir:ImagingStudy.series.started [ dateTime ]; # 0..1 When the series started
    fhir:ImagingStudy.series.performer [ # 0..* Who performed the series
      fhir:ImagingStudy.series.performer.function [ CodeableConcept ]; # 0..1 Type of performance
      fhir:ImagingStudy.series.performer.actor [
    Reference(Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson) ]; # 1..1 Who performed
    the series
    ], ...;
    fhir:ImagingStudy.series.instance [ # 0..* A single SOP instance from the series
      fhir:ImagingStudy.series.instance.uid [ id ]; # 1..1 DICOM SOP Instance UID
      fhir:ImagingStudy.series.instance.sopClass [ Coding ]; # 1..1 DICOM class type
      fhir:ImagingStudy.series.instance.number [ unsignedInt ]; # 0..1 The number of this instance in the series
      fhir:ImagingStudy.series.instance.title [ string ]; # 0..1 Description of instance
    ], ...;
  ], ...;
]
```

Changes since Release 3

ImagingStudy

ImagingStudy.status | Added Mandatory Element

ImagingStudy.modality | Renamed from modalityList to modality Change value set from http://hl7.org/fhir/ValueSet/dicom-cid29 to http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html

ImagingStudy.subject | Renamed from patient to subject Type Reference: Added Target Types Device, Group

ImagingStudy.encounter | Added Element

ImagingStudy.basedOn | Type Reference: Added Target Types ServiceRequest, Appointment, AppointmentResponse, Task Type Reference: Removed Target Types ReferralRequest, ProcedureRequest

ImagingStudy.referrer | Type Reference: Added Target Type PractitionerRole

ImagingStudy.interpreter | Type Reference: Added Target Type PractitionerRole

ImagingStudy.procedureReference | Max Cardinality changed from * to 1

ImagingStudy.procedureCode | Add Binding http://www.rsna.org/RadLex_Playbook.aspx (extensible)

ImagingStudy.location | Added Element

ImagingStudy.reasonCode | Added Element

ImagingStudy.reasonReference | Added Element

ImagingStudy.note | Added Element

ImagingStudy.series.uid | Type changed from oid to id

ImagingStudy.series.modality | Change value set from http://hl7.org/fhir/ValueSet/dicom-cid29 to http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html

ImagingStudy.series.specimen | Added Element

ImagingStudy.series.performer | Type changed from Reference(Practitioner) to BackboneElement

ImagingStudy.series.performer.function | Added Element

ImagingStudy.series.performer.actor | Added Mandatory Element

ImagingStudy.series.instance.uid | Type changed from oid to id

ImagingStudy.series.instance.sopClass | Type changed from oid to Coding Add Binding http://dicom.nema.org/medical/dicom/current/output/chtml/part04/sect_B.5.html#table_B.5-1 (extensible)

ImagingStudy.uid | deleted

ImagingStudy.accession | deleted

ImagingStudy.availability | deleted

ImagingStudy.context | deleted

ImagingStudy.reason | deleted

ImagingStudy.series.availability | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.4.3.1 Terminology Bindings

Path | Definition | Type | Reference
ImagingStudy.status | The status of the ImagingStudy. | Required | ImagingStudyStatus
ImagingStudy.modality ImagingStudy.series.modality | Type of acquired data in the instance. | Extensible | AcquisitionModality
ImagingStudy.procedureCode | The performed procedure type. | Extensible | http://www.rsna.org/RadLex_Playbook.aspx
ImagingStudy.reasonCode | The reason for the study. | Example | ProcedureReasonCodes
ImagingStudy.series.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures
ImagingStudy.series.laterality | Codes describing body site laterality (left, right, etc.). | Example | Laterality
ImagingStudy.series.performer.function | The type of involvement of the performer. | Extensible | ImagingStudySeriesPerformerFunction
ImagingStudy.series.instance.sopClass | The sopClass for the instance. | Extensible | http://dicom.nema.org/medical/dicom/current/output/chtml/part04/sect_B.5.html#table_B.5-1

10.4.3.2 Implementation Notes

A referenced DICOM SOP instance could be:

• A single- or multi-frame, still or video image captured by a variety of imaging modalities, such as X-ray, MR, and ultrasound;

• A set of various presentation parameters, including annotation and markup;

• A set of measurements or a report, including radiation dose report and CAD analysis;

• An encapsulated PDF or CDA document;

• A list of instances, such as key “of interest” images, or instances to be “deleted”; or

• Other DICOM content.

DICOM Series Instance UID and SOP Instance UID use the id datatype, and are encoded directly. For example, an image with SOP Instance UID of 2.16.124.113543.1154777499.30246.19789.3503430045.1.1 is encoded in ImagingStudy.series.instance.uid as “2.16.124.113543.1154777499.30246.19789.3503430045.1.1”.

The ImagingStudy’s DICOM Study Instance UID is encoded in the ImagingStudy.identifier element, which is of the Identifier datatype. When encoding a DICOM UID in an Identifier datatype, use the Identifier system of “urn:dicom:uid”, and prefix the UID value with “urn:oid:”. Therefore, an ImagingStudy with DICOM Study Instance UID of 2.16.124.113543.1154777499.30246.19789.3503430046 is encoded as:

```
"identifier":{
  "system":"urn:dicom:uid",
  "value":"urn:oid:2.16.124.113543.1154777499.30246.19789.3503430046"
}
```

The study accession number can also be encoded as an Identifier using the “ACSN” identifier type, as follows:

```
"identifier":{
  "type" : {
    "coding" : [
      {
        "system" : "http://terminology.hl7.org/CodeSystem/v2-0203",
        "code" : "ACSN"
      }
    ]
  },
  "system":"http://ginormoushospital.org/accession",
  "value":"GH334103"
}
```

The ImagingStudy.endpoint elements and ImagingStudy.series.endpoint elements indicate network services that can be used to access the studies, series, or instances; for example, a DICOM WADO-RS server. An ImagingStudy.series.endpoint of a particular Endpoint.connectionType provides that service for that series, and all contained instances. An ImagingStudy.endpoint of a particular connection type provides that service for all series in that study that do not have a specified Endpoint of that type, and their contained instances. That is, an ImagingStudy.series.endpoint overrides an ImagingStudy.endpoint of the same connection type. Systems can determine if a particular study, series, or instance is available or offline by interacting with the endpoint. Since each study, or individual series of a study can be stored on different imaging archive servers, per-series endpoints are required. For the identified services and use cases, all instances within a series would be stored together, and thus instance-level endpoints are not defined.

Different Endpoint connection types may have different capabilities, protocols or requirements; and the specified Endpoint.address may require manipulation. See below for the details on use of imaging-related Endpoint connection types.

10.4.3.2.1 WADO-RS

An Endpoint.connectionType of code dicom-wado-rs, system http://terminology.hl7.org/CodeSystem/endpoint-connection-type, identifies a DICOM WADO-RS service. The Endpoint.address identifies the HTTP(S) service base url. That is, only the scheme, authority and path are included. Sub-services, such as study, shall not be specified. The path shall not contain a trailing slash.

The DICOM WADO-RS (Web Access to DICOM Objects, RESTful mode) service uses a RESTful approach to instance retrieval. This service allows for retrieval of native DICOM SOP instances, or instances “rendered” into other formats, including JPEG and MPEG. The media type of a response is specified by the request Accept header (preferred); or, by the accept query parameters. Supported media types depend on the capabilities of the WADO-RS server and the classification of the instance as “single frame,” “multi-frame,” “video,” “text,” or “other.” The WADO-RS service also allows retrieval of study or series level information.

The path to retrieve a DICOM instance is constructed by appending the appropriate sub-resource paths to the Endpoint.address value.

For example, using the following information in a fictional ImagingStudy resource:

• the WADO-RS service base url of “https://pacs.hospital.org/wado-rs” found in an ImagingStudy.endpoint.address,

• the DICOM Study Instance UID of “1.2.250.1.59.40211.12345678.678910” found in an ImagingStudy.identifier having Identifier.system of “urn:dicom:uid”,

• the DICOM Series Instance UID of “1.2.250.1.59.40211.789001276.14556172.67789” found in ImagingStudy.series.uid, and

• the DICOM SOP Instance UID of “1.2.250.1.59.40211.2678810.87991027.899772.2” found in ImagingStudy.series.instance.uid

```
https://pacs.hospital.org/wado-rs/studies/1.2.250.1.59.40211.12345678.678910/series/1.2.250.1.59.40211.789001276.1455617
    2.67789/instances/1.2.250.1.59.40211.2678810.87991027.899772.2
```

Query parameters on the "rendered" sub-resource can control other aspects of the rendering including: the rendered dimensions, the quality (compression ratio), the region of interest to render, the brightness/contrast (window center/width) adjustments, and whether to “burn” patient or study demographics into the rendered result. Specific frames of a multi-frame instance may be retrieved using the frames sub-resource.

For example, provided the Accept header indicates a preference for image/jpeg, the example above can be extended with parameters that cause a JPEG thumbnail (100 columns by 100 rows) of a region extending from the top-left corner of the original image, across 1000 and down 3000 pixels, to be retrieved (additional sub-resource and parameters emphasized):

```
https://pacs.hospital.org/wado-rs/studies/1.2.250.1.59.40211.12345678.678910/series/1.2.250.1.59.40211.789001276.1455617
    2.67789/instances/1.2.250.1.59.40211.2678810.87991027.899772.2/rendered?viewport=100,100,0,0,1000,3000
```

For further details on DICOM WADO-RS capabilities including additional rendering parameters, see DICOM PS 3.18 .

10.4.3.2.2 WADO-URI

An Endpoint.connectionType of code dicom-wado-uri, system http://terminology.hl7.org/CodeSystem/endpoint-connection-type, identifies a DICOM WADO-URI service. The Endpoint.address identifies the HTTP(S) service base url. That is, only the scheme, authority and path are included. Neither a question mark (“?”) nor any query parameters shall be included.

The DICOM WADO-URI (Web Access to DICOM Objects, URI mode) service uses HTTP query parameter syntax. This service allows for retrieval of native DICOM instances, or instances “rendered” into other formats, including JPEG and MPEG. The media type of a response is specified by the request Accept header (preferred); or, by the contentType query parameter. Supported media types depend on the classification of the instance as “single frame,” “multi-frame,” “video,” “text,” or “other.”

The query to retrieve a DICOM instance is constructed by appending the appropriate query parameters to the Endpoint.address value.

For example, using the following information in a fictional ImagingStudy resource:

• the WADO-URI service base url of “https://pacs.hospital.org/wado-uri” found in an ImagingStudy.endpoint.address,

• the DICOM Study Instance UID of 1.2.250.1.59.40211.12345678.678910” found in an ImagingStudy.identifier having Identifier.system of “urn:dicom:uid”,

• the DICOM Series Instance UID of “1.2.250.1.59.40211.789001276.14556172.67789” found in ImagingStudy.series.uid, and

• the DICOM SOP Instance UID of “1.2.250.1.59.40211.2678810.87991027.899772.2” found in ImagingStudy.series.instance.uid

```
https://pacs.hospital.org/wado-uri?requestType=WADO&studyUID=1.2.250.1.59.40211.12345678.678910&seriesUID=1.2.250.1.59.4
    0211.789001276.14556172.67789&objectUID=1.2.250.1.59.40211.2678810.87991027.899772.2
```

Additional query parameters can control other aspects of the rendering including rendered dimensions, quality (compression ratio), the region of interest within the image to render, brightness/contrast (window center/width) adjustments, whether to “burn” patient or study demographics into the rendered result, and which frame of a multi-frame instance to retrieve.

For example, provided the Accept header indicates a preference for image/jpeg, the example above can be extended with parameters that cause a JPEG thumbnail (100 columns by 100 rows) of the left half of the image to be retrieved (additional parameters emphasized):

```
https://pacs.hospital.org/wado-uri?requestType=WADO&studyUID=1.2.250.1.59.40211.12345678.678910&seriesUID=1.2.250.1.59.40211.789001276.14556172.67789&objectUID=1.2.250.1.59.40211.2678810.87991027.899772.2&rows=100&columns=100&region=0,0,0.5,1
```

For further details on DICOM WADO-URI capabilities including additional rendering parameters, see DICOM PS 3.18 .

10.4.3.2.3 IID

An Endpoint.connectionType of code ihe-iid, system http://terminology.hl7.org/CodeSystem/endpoint-connection-type, identifies an IHE Invoke Image Display (IID) service. The Endpoint.address identifies the HTTP(S) service base url. That is, only the scheme, authority and path are included. Neither a question mark (“?”) nor any query parameters shall be included.

The IHE Invoke Image Display (IID) service provides a standardized mechanism to launch a viewer in a particular study context. (IID also supports invoking a particular patient context, but that is not profiled here.) An IID-type Endpoint should be used only at the study level. As well as invoking the viewer on a particular study, query parameters can request particular viewer capabilities, image quality, and more.

To launch a viewer, append the appropriate query parameters to Endpoint.address value.

For example, using the following information in a fictional ImagingStudy resource:

• the WADO-URI service base url of “https://pacs.hospital.org/IHEInvokeImageDisplay” found in an ImagingStudy.endpoint.address,

• the DICOM Study Instance UID of “urn:oid:1.2.250.1.59.40211.12345678.678910” found in an ImagingStudy.identifier having Identifier.system of “urn:dicom:uid”,

```
https://pacs.hospital.org/IHEInvokeImageDisplay?requestType=STUDY&studyUID=1.2.250.1.59.40211.12345678.678910&diagnosticQuality=true
```

For further details on IHE Invoke Image Display capabilities including additional parameters, see the IHE Technical Frameworks , or the introduction on the IHE IID Profile Wiki .

10.4.3.3 Use Case

10.4.3.3.1 EHR access to imaging studies

Amy, a family physician, would like to see a list of available studies for her patient, Alex. Her EHR client makes a FHIR call for all ImagingStudy objects available for Alex. In the response, she is able to see the study date, procedure, modality, and accession number, for each study returned. There is enough information provided in the response to obtain a thumbnail via a WADO-RS call, or to launch a viewer using an IHE Radiology - Invoke Image Display (IID) profile call using the url elements found in the ImagingStudy.

10.4.3.3.2 Comprehensive Imaging Scheduled Workflow

Joe Angina complains of shortness of breath and occasional chest pain to his primary care physician, Dr. Pat Down at Local MultiClinic, who orders a stress echocardiogram; the order is created as a FHIR Task resource to manage the workflow, with a link to a ServiceRequest resource with the details of the request. The order is scheduled and assigned to cardiologist Dr. Art Skann, also at Local MultiClinic.

On the scheduled day of the exam, Joe arrives at the echo lab to meet with Dr. Skann and have the study done. Dr. Skann’s workstation shows the daily list of Task, and he follows the link to retrieve the ServiceRequest. (He may follow the links through the referenced Patient resource to access Joe’s electronic medical record, but that is not the concern of this storyboard.)

The Task and ServiceRequest has been transcoded to a DICOM Modality Worklist Scheduled Procedure Step, and in the echo lab the equipment has downloaded the Modality Worklist. The study is performed, and the acquired images and sonographer’s preliminary measurements are stored in the Local MultiClinic Picture Archiving and Communication System (PACS). The PACS creates an ImagingStudy resource for each study it manages.

Dr. Skann interprets the study on a PACS workstation, and he selects two key image frames to be included in the diagnostic report; this selection is stored back to the PACS as a DICOM Key Object Selection with the title "For Report Attachment", and the PACS makes it available (transcodes it) as a FHIR ImagingStudy resource. Dr. Skann dictates the report using a structured data entry report writing program, including a recommendation for a cardiac catheterization procedure, and signs it. The report writing program formats the report as a CDA document, retrieves the ImagingStudy resource, and inserts the referenced key images into the report.

Dr. Down meets again with Joe, and they review the results of the stress test. Joe has a question about the findings that the key images in the report do not show, so Dr. Down uses the Local MultiClinic EMR to query the PACS for the full ImagingStudy resource and uses the references there to open an image display for the full study. Joe agrees to proceed to catheterization, and Dr. Down sends a referral to the Ginormous University Hospital cath department and triggers the PACS to share the echo study through the Metropolitan Health Information Exchange.

The PACS creates an imaging study as an ImagingStudy resource, which includes all the images but excludes the sonographer's preliminary measurements (which as a matter of policy are not shared outside the Local MultiClinic). The imaging study is published to the Metro HIE. (In accordance with IHE XDS-I , the images themselves are not directly published to the HIE, but available for on-demand retrieval from the PACS.)

At Ginormous Hospital, Dr. Cora Plummer receives the cath referral, and looks up the study in the Metro HIE registry. She retrieves the study ImagingStudy, and uses it to access the shared images, which she uses to prepare for the cath procedure.

10.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
basedon | reference | The order for the image | ImagingStudy.basedOn (Appointment, AppointmentResponse, CarePlan, Task, ServiceRequest)
bodysite | token | The body site studied | ImagingStudy.series.bodySite
dicom-class | token | The type of the instance | ImagingStudy.series.instance.sopClass
encounter | reference | The context of the study | ImagingStudy.encounter (Encounter)
endpoint | reference | The endpoint for the study or series | ImagingStudy.endpoint | ImagingStudy.series.endpoint (Endpoint)
identifier | token | Identifiers for the Study, such as DICOM Study Instance UID and Accession number | ImagingStudy.identifier | 30 Resources
instance | token | SOP Instance UID for an instance | ImagingStudy.series.instance.uid
interpreter | reference | Who interpreted the images | ImagingStudy.interpreter (Practitioner, PractitionerRole)
modality | token | The modality of the series | ImagingStudy.series.modality
patient | reference | Who the study is about | ImagingStudy.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | The person who performed the study | ImagingStudy.series.performer.actor (Practitioner, Organization, CareTeam, Device, Patient, PractitionerRole, RelatedPerson)
reason | token | The reason for the study | ImagingStudy.reasonCode
referrer | reference | The referring physician | ImagingStudy.referrer (Practitioner, PractitionerRole)
series | token | DICOM Series Instance UID for a series | ImagingStudy.series.uid
started | date | When the study was started | ImagingStudy.started
status | token | The status of the study | ImagingStudy.status
subject | reference | Who the study is about | ImagingStudy.subject (Group, Device, Patient)

10.4.5 Resource ImagingStudy - Examples

URL: https://hl7.org/fhir/R4/imagingstudy-examples.html

10.4.5 Resource ImagingStudy - Examples

Imaging Integration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
Example of imagingstudy | example | XML | JSON | Turtle
Example XR imaging study with full elements | example-xr | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.4.6 Resource ImagingStudy - Detailed Descriptions

URL: https://hl7.org/fhir/R4/imagingstudy-definitions.html

10.4.6 Resource ImagingStudy - Detailed Descriptions

Imaging Integration Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the ImagingStudy resource.

ImagingStudy
Element Id | ImagingStudy
Definition | Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data) acquired or produced in a common context. A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a study may have multiple series of different modalities.
Cardinality | 0..*
Type | DomainResource
ImagingStudy.identifier

Element Id | ImagingStudy.identifier
Definition | Identifiers for the ImagingStudy such as DICOM Study Instance UID, and Accession Number.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | If one or more series elements are present in the ImagingStudy, then there shall be one DICOM Study UID identifier (see DICOM PS 3.3 C.7.2 .
Alternate Names | StudyInstanceUID; AccessionNumber
Summary | true
Comments | See discussion under Imaging Study Implementation Notes for encoding of DICOM Study Instance UID. Accession Number should use ACSN Identifier type.
ImagingStudy.status
Element Id | ImagingStudy.status
Definition | The current state of the ImagingStudy.
Cardinality | 1..1
Terminology Binding | ImagingStudyStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | Unknown does not represent "other" - one of the defined statuses must apply. Unknown is used when the authoring system is not sure what the current status is.
ImagingStudy.modality
Element Id | ImagingStudy.modality
Definition | A list of all the series.modality values that are actual acquisition modalities, i.e. those in the DICOM Context Group 29 (value set OID 1.2.840.10008.6.1.19).
Cardinality | 0..*
Terminology Binding | AcquisitionModality (Extensible)
Type | Coding
Alternate Names | ModalitiesInStudy
Summary | true
ImagingStudy.subject
Element Id | ImagingStudy.subject
Definition | The subject, typically a patient, of the imaging study.
Cardinality | 1..1
Type | Reference(Patient | Device | Group)
Summary | true
Comments | QA phantoms can be recorded with a Device; multiple subjects (such as mice) can be recorded with a Group.
ImagingStudy.encounter
Element Id | ImagingStudy.encounter
Definition | The healthcare event (e.g. a patient and healthcare provider interaction) during which this ImagingStudy is made.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some events may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter (e.g. pre-admission test).
ImagingStudy.started
Element Id | ImagingStudy.started
Definition | Date and time the study started.
Cardinality | 0..1
Type | dateTime
Alternate Names | StudyDate; StudyTime
Summary | true
ImagingStudy.basedOn
Element Id | ImagingStudy.basedOn
Definition | A list of the diagnostic requests that resulted in this imaging study being performed.
Cardinality | 0..*
Type | Reference(CarePlan | ServiceRequest | Appointment | AppointmentResponse | Task)
Requirements | To support grouped procedures (one imaging study supporting multiple ordered procedures, e.g. chest/abdomen/pelvis CT).
Summary | true
ImagingStudy.referrer
Element Id | ImagingStudy.referrer
Definition | The requesting/referring physician.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Alternate Names | ReferringPhysiciansName
Summary | true
ImagingStudy.interpreter
Element Id | ImagingStudy.interpreter
Definition | Who read the study and interpreted the images or other content.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole)

Alternate Names | Name of Physician(s) Reading Study
Summary | true
ImagingStudy.endpoint
Element Id | ImagingStudy.endpoint
Definition | The network service providing access (e.g., query, view, or retrieval) for the study. See implementation notes for information about using DICOM endpoints. A study-level endpoint applies to each series in the study, unless overridden by a series-level endpoint with the same Endpoint.connectionType.
Cardinality | 0..*
Type | Reference(Endpoint)
Requirements | Access methods for viewing (e.g., IHE's IID profile) or retrieving (e.g., DICOM's WADO-URI and WADO-RS) the study or the study's series or instances. The study-level baseLocation applies to each series in the study, unless overridden in series by a series-level baseLocation of the same type.
Summary | true
Comments | Typical endpoint types include DICOM WADO-RS, which is used to retrieve DICOM instances in native or rendered (e.g., JPG, PNG), formats using a RESTful API; DICOM WADO-URI, which can similarly retrieve native or rendered instances, except using an HTTP query-based approach; DICOM QIDO-RS, which allows RESTful query for DICOM information without retrieving the actual instances; or IHE Invoke Image Display (IID), which provides standard invocation of an imaging web viewer.
ImagingStudy.numberOfSeries
Element Id | ImagingStudy.numberOfSeries
Definition | Number of Series in the Study. This value given may be larger than the number of series elements this Resource contains due to resource availability, security, or other factors. This element should be present if any series elements are present.
Cardinality | 0..1
Type | unsignedInt
Alternate Names | NumberOfStudyRelatedSeries
Summary | true
ImagingStudy.numberOfInstances
Element Id | ImagingStudy.numberOfInstances
Definition | Number of SOP Instances in Study. This value given may be larger than the number of instance elements this resource contains due to resource availability, security, or other factors. This element should be present if any instance elements are present.
Cardinality | 0..1
Type | unsignedInt
Alternate Names | NumberOfStudyRelatedInstances
Summary | true
ImagingStudy.procedureReference
Element Id | ImagingStudy.procedureReference
Definition | The procedure which this ImagingStudy was part of.
Cardinality | 0..1
Type | Reference(Procedure)
Summary | true
ImagingStudy.procedureCode
Element Id | ImagingStudy.procedureCode
Definition | The code for the performed procedure type.
Cardinality | 0..*
Terminology Binding | ImagingProcedureCode :
Type | CodeableConcept
Alternate Names | ProcedureCodeSequence
Summary | true
ImagingStudy.location
Element Id | ImagingStudy.location
Definition | The principal physical location where the ImagingStudy was performed.
Cardinality | 0..1
Type | Reference(Location)
Requirements | Ties the event to where the records are likely kept and provides context around the event occurrence (e.g. if it occurred inside or outside a dedicated healthcare setting).
Summary | true
ImagingStudy.reasonCode
Element Id | ImagingStudy.reasonCode
Definition | Description of clinical condition indicating why the ImagingStudy was requested.
Cardinality | 0..*
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept
Summary | true
ImagingStudy.reasonReference
Element Id | ImagingStudy.reasonReference
Definition | Indicates another resource whose existence justifies this Study.
Cardinality | 0..*
Type | Reference(Condition | Observation | Media | DiagnosticReport | DocumentReference)
Summary | true
ImagingStudy.note
Element Id | ImagingStudy.note
Definition | Per the recommended DICOM mapping, this element is derived from the Study Description attribute (0008,1030). Observations or findings about the imaging study should be recorded in another resource, e.g. Observation, and not in this

element.
Cardinality | 0..*
Type | Annotation
Summary | true
ImagingStudy.description
Element Id | ImagingStudy.description
Definition | The Imaging Manager description of the study. Institution-generated description or classification of the Study (component) performed.
Cardinality | 0..1
Type | string
Alternate Names | StudyDescription
Summary | true
ImagingStudy.series
Element Id | ImagingStudy.series
Definition | Each study has one or more series of images or other content.
Cardinality | 0..*
Summary | true
ImagingStudy.series.uid
Element Id | ImagingStudy.series.uid
Definition | The DICOM Series Instance UID for the series.
Cardinality | 1..1
Type | id
Requirements | DICOM Series Instance UID.
Alternate Names | SeriesInstanceUID
Summary | true
Comments | See DICOM PS3.3 C.7.3 .
ImagingStudy.series.number
Element Id | ImagingStudy.series.number
Definition | The numeric identifier of this series in the study.
Cardinality | 0..1
Type | unsignedInt
Alternate Names | SeriesNumber
Summary | true
ImagingStudy.series.modality
Element Id | ImagingStudy.series.modality
Definition | The modality of this series sequence.
Cardinality | 1..1
Terminology Binding | AcquisitionModality (Extensible)
Type | Coding
Alternate Names | Modality
Summary | true
ImagingStudy.series.description
Element Id | ImagingStudy.series.description
Definition | A description of the series.
Cardinality | 0..1
Type | string
Alternate Names | SeriesDescription
Summary | true
ImagingStudy.series.numberOfInstances
Element Id | ImagingStudy.series.numberOfInstances
Definition | Number of SOP Instances in the Study. The value given may be larger than the number of instance elements this resource contains due to resource availability, security, or other factors. This element should be present if any instance elements are present.
Cardinality | 0..1
Type | unsignedInt
Alternate Names | NumberOfSeriesRelatedInstances
Summary | true
ImagingStudy.series.endpoint
Element Id | ImagingStudy.series.endpoint
Definition | The network service providing access (e.g., query, view, or retrieval) for this series. See implementation notes for information about using DICOM endpoints. A series-level endpoint, if present, has precedence over a study-level endpoint with the same Endpoint.connectionType.
Cardinality | 0..*
Type | Reference(Endpoint)
Requirements | Access methods for retrieving (e.g., DICOM’s WADO-URI and WADO-RS) the series or the series’ instances. A baseLocation specified at the series level has precedence over a baseLocation of the same type specified at the study level.
Summary | true
Comments | Typical endpoint types include DICOM WADO-RS, which is used to retrieve DICOM instances in native or rendered (e.g., JPG, PNG) formats using a RESTful API; DICOM WADO-URI, which can similarly retrieve native or rendered instances, except using an HTTP query-based approach; and DICOM QIDO-RS, which allows RESTful query for DICOM information without retrieving the actual instances.
ImagingStudy.series.bodySite
Element Id | ImagingStudy.series.bodySite

Definition | The anatomic structures examined. See DICOM Part 16 Annex L (http://dicom.nema.org/medical/dicom/current/output/chtml/part16/chapter_L.html) for DICOM to SNOMED-CT mappings. The bodySite may indicate the laterality of body part imaged; if so, it shall be consistent with any content of ImagingStudy.series.laterality.
Cardinality | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | Coding
Alternate Names | BodyPartExamined
Summary | true
ImagingStudy.series.laterality
Element Id | ImagingStudy.series.laterality
Definition | The laterality of the (possibly paired) anatomic structures examined. E.g., the left knee, both lungs, or unpaired abdomen. If present, shall be consistent with any laterality information indicated in ImagingStudy.series.bodySite.
Cardinality | 0..1
Terminology Binding | Laterality (Example)
Type | Coding
Summary | true
ImagingStudy.series.specimen
Element Id | ImagingStudy.series.specimen
Definition | The specimen imaged, e.g., for whole slide imaging of a biopsy.
Cardinality | 0..*
Type | Reference(Specimen)
Summary | true
ImagingStudy.series.started
Element Id | ImagingStudy.series.started
Definition | The date and time the series was started.
Cardinality | 0..1
Type | dateTime
Summary | true
ImagingStudy.series.performer
Element Id | ImagingStudy.series.performer
Definition | Indicates who or what performed the series and how they were involved.
Cardinality | 0..*
Requirements | The performer is recorded at the series level, since each series in a study may be performed by a different performer, at different times, and using different devices. A series may be performed by multiple performers.
Alternate Names | PerformingPhysicianName; OperatorName
Summary | true
Comments | If the person who performed the series is not known, their Organization may be recorded. A patient, or related person, may be the performer, e.g. for patient-captured images.
ImagingStudy.series.performer.function
Element Id | ImagingStudy.series.performer.function
Definition | Distinguishes the type of involvement of the performer in the series.
Cardinality | 0..1
Terminology Binding | ImagingStudy series performer function (Extensible)
Type | CodeableConcept
Requirements | Allows disambiguation of the types of involvement of different performers.
Summary | true
ImagingStudy.series.performer.actor
Element Id | ImagingStudy.series.performer.actor
Definition | Indicates who or what performed the series.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson)
Summary | true
ImagingStudy.series.instance
Element Id | ImagingStudy.series.instance
Definition | A single SOP instance within the series, e.g. an image, or presentation state.
Cardinality | 0..*
ImagingStudy.series.instance.uid
Element Id | ImagingStudy.series.instance.uid
Definition | The DICOM SOP Instance UID for this image or other DICOM content.
Cardinality | 1..1
Type | id
Requirements | DICOM SOP Instance UID.
Alternate Names | SOPInstanceUID
Comments | See DICOM PS3.3 C.12.1 .
ImagingStudy.series.instance.sopClass
Element Id | ImagingStudy.series.instance.sopClass
Definition | DICOM instance type.
Cardinality | 1..1
Terminology Binding | sopClass :
Type | Coding
Alternate Names | SOPClassUID
ImagingStudy.series.instance.number

Element Id | ImagingStudy.series.instance.number
Definition | The number of instance in the series.
Cardinality | 0..1
Type | unsignedInt
Alternate Names | InstanceNumber
ImagingStudy.series.instance.title
Element Id | ImagingStudy.series.instance.title
Definition | The description of the instance.
Cardinality | 0..1
Type | string
Comments | Particularly for post-acquisition analytic objects, such as SR, presentation states, value mapping, etc.

10.4.7 Resource ImagingStudy - Mappings

URL: https://hl7.org/fhir/R4/imagingstudy-mappings.html

10.4.7 Resource ImagingStudy - Mappings

Imaging Integration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

DICOM Tag Mapping

Mappings for the imagingstudy resource (see Mappings to Other Standards for further information & status).

10.4.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ImagingStudy
identifier | FiveWs.identifier
status | FiveWs.status
modality | FiveWs.class
subject | FiveWs.subject[x]
encounter | FiveWs.context
started | FiveWs.init
basedOn | FiveWs.cause
referrer | FiveWs.cause
interpreter | FiveWs.witness
location | FiveWs.where[x]
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]
actor | FiveWs.actor

10.4.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ImagingStudy | Event
identifier | Event.identifier
status | Event.status
subject | Event.subject
encounter | Event.encounter
started | Event.occurrence[x]
basedOn | Event.basedOn
procedureReference | Event.partOf
procedureCode | Event.code
location | Event.location
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note
performer | Event.performer

10.4.7.3 HL7 v2 Mapping (http://hl7.org/v2)

ImagingStudy
identifier
status | Varies by domain
modality
subject
encounter
started
basedOn

referrer
interpreter
endpoint
numberOfSeries
numberOfInstances
procedureReference
procedureCode
location | EVN.7
reasonCode
reasonReference | EVN.4 or by domain
note
description
series
uid
number
modality
description
numberOfInstances
endpoint
bodySite
laterality
specimen
started
performer
function
actor | EVN.5 or by domain
instance
uid
sopClass
number
title

10.4.7.4 RIM Mapping (http://hl7.org/v3)

ImagingStudy | Act[classCode=COMPOSITION, moodCode=EVN]
identifier | .id and .inboundRelationship[typeCode=COMP].source[classCode=ACSN, moodCode=EVN].id
status | .status
modality | .code
subject | .participation[typeCode=SBJ].role
encounter | inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
started | .effectiveTime
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=DGIMG, moodCode=RQO]
referrer | .outboundRelationship[typeCode=FLFS].target[classCode=DGIMG, moodCode=RQO].participation[typeCode=AUT].role
interpreter | .participation[typeCode=PRF].role
endpoint | Not supported
numberOfSeries | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target[classCode=OBSSER, moodCode=EVN].repeatNumber
numberOfInstances | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target[classCode=DGIMG, moodCode=EVN].repeatNumber
procedureReference | .outboundRelationship[typeCode=COMP].target[classCode=PROC, moodCode=EVN]
procedureCode | .outboundRelationship[typeCode=COMP].target[classCode=PROC, moodCode=EVN]
location | .participation[typeCode=LOC].role
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
note | unknown
description | .text
series | .outboundRelationship[typeCode=COMP].target[classCode=OBSSER, moodCode=EVN]
uid | .id
number | .inboundRelationship[type=COMP, source[classCode=COMPOSITION, moodCode=EVN]].sequenceNumber
modality | .code
description | .text
numberOfInstances | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target[classCode=DGIMG, moodCode=EVN].repeatNumber
endpoint | Not supported
bodySite | .targetSiteCode
laterality | .targetSiteCode
specimen | Role[classCode=SPEC]
started | .effectiveTime
performer | .participation[typeCode=PRF].role
function | .participation[typeCode=PRF].role
actor | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
instance | .outboundRelationship[typeCode=COMP].target[classCode=DGIMG, moodCode=EVN]
uid | .id

sopClass | .inboundRelationship[typeCode=COMP]. source[classCode=OBS, moodCode=EVN, code="sop class"].value
number | .inboundRelationship[typeCode=COMP, source[classCode=OBSSER, moodCode=EVN]].sequenceNumber
title | .title

10.4.7.5 DICOM Tag Mapping (http://nema.org/dicom)

ImagingStudy | Reference IHE radiology TF vol 2 table 4.14-1
identifier | StudyInstanceUID (0020,000D) | Accession Number and Issuer (0080,0050)+(0080,0051) | Study ID (0020,0010)
status
modality | (0008,0061)
subject | (0010/*)
encounter
started | (0008,0020)+(0008,0030)
basedOn | (0032,1064)
referrer | (0008,0090)+(0008,0096)
interpreter | (0008,1060)
endpoint
numberOfSeries | (0020,1206)
numberOfInstances | (0020,1208)
procedureReference | (0008,1032)
procedureCode | (0008,1032)
location | (0008,1040) | (0040,0243)
reasonCode | (0040,1002)
reasonReference
note
description | (0008,1030)
series
uid | (0020,000E)
number | (0020,0011)
modality | (0008,0060)
description | (0008,103E)
numberOfInstances | (0020,1209)
endpoint
bodySite | (0018,0015)
laterality | (0020,0060)
specimen | (0040,0551) + (0040,0562)
started | (0008,0021) + (0008,0031)
performer | (0008, 1050) | (0008, 1052) | (0008, 1070) | (0008, 1072)
function
actor
instance
uid | (0008,0018)
sopClass | (0008,0016)
number | (0020,0013)
title | (0008,0008) | (0007,0080) | (0040,A043) + (0008,0104) | (0042,0010)

10.4.8 Resource ImagingStudy - Extensions & Profiles

URL: https://hl7.org/fhir/R4/imagingstudy-profiles.html

10.4.8 Resource ImagingStudy - Extensions & Profiles

Imaging Integration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the ImagingStudy resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.4.8.1 Profiles

No Profiles defined for this resource

10.4.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

auditevent-SOPClass | SOPClass | for AuditEvent HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions

flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.4.8.3 Search Extensions

No Search Extensions defined for this resource

10.4.9 Resource ImagingStudy - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/imagingstudy-version-maps.html

No validation errors - all conversions are clean

10.4.9 Resource ImagingStudy - R3/R4 Conversions

Imaging Integration Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for ImagingStudy.

Functional status for this map: 2 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

10.4.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ImagingStudy3to4" = "R3 to R4 Conversions for ImagingStudy"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ImagingStudy" alias ImagingStudyR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ImagingStudy" alias ImagingStudy as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ImagingStudy(source src : ImagingStudyR3, target tgt : ImagingStudy) extends DomainResource <<type+>> {
  src.uid as s ->  tgt.identifier as i,  i.system = 'urn:dicom:uid',  i.value = s;
  src.identifier -> tgt.identifier;
  src.accession -> tgt.identifier;
  src.availability where (value = 'OFFLINE') or (value = 'UNAVAILABLE') -> tgt.status = 'registered';
  src.availability where (value = 'ONLINE') or (value = 'NEARLINE') -> tgt.status = 'available';
  src.availability where value.empty() -> tgt.status = 'unknown';
  src.modalityList -> tgt.modality;
  src.patient -> tgt.subject;
  src.context -> tgt.encounter;
  src.started -> tgt.started;
  src.basedOn -> tgt.basedOn;
  src.referrer -> tgt.referrer;
  src.interpreter -> tgt.interpreter;
  src.endpoint -> tgt.endpoint;
  src.numberOfSeries -> tgt.numberOfSeries;
  src.numberOfInstances -> tgt.numberOfInstances;
  src.procedureReference -> tgt.procedureReference;
  src.procedureCode -> tgt.procedureCode;
  src.reason -> tgt.reasonCode;
  src.description -> tgt.description;
  src.series as vs -> tgt.series as vt then series(vs, vt);
}

group series(source src, target tgt) extends BackboneElement {
  src.uid -> tgt.uid;
  src.number -> tgt.number;
  src.modality -> tgt.modality;
  src.description -> tgt.description;
  src.numberOfInstances -> tgt.numberOfInstances;
  src.availability as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImagingStudy.series.availability',  vt.value = v;
  src.endpoint -> tgt.endpoint;
  src.bodySite -> tgt.bodySite;
  src.laterality -> tgt.laterality;
  src.started -> tgt.started;
  src.performer as vs -> tgt.performer as vt then performer(vs, vt);
  src.instance as vs -> tgt.instance as vt then instance(vs, vt);
}

group performer(source src, target tgt) extends BackboneElement {
  src ->  tgt.function as t,  t.coding as c,  c.system = 'http://terminology.hl7.org/CodeSystem/v3-ParticipationType',
    c.code = 'PRF' "function";
  src -> tgt.actor as t then Reference(src, t) "actor";
}

group instance(source src, target tgt) extends BackboneElement {
  src.uid -> tgt.uid;
  src.number -> tgt.number;
  src.sopClass as v ->  tgt.sopClass as t,  t.system = 'urn:ietf:rfc:3986',  t.code = v;
  src.title -> tgt.title;
}
```

10.4.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ImagingStudy4to3" = "R4 to R3 Conversion for ImagingStudy"

uses "http://hl7.org/fhir/StructureDefinition/ImagingStudy" alias ImagingStudy as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ImagingStudy" alias ImagingStudyR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ImagingStudy(source src : ImagingStudyR3, target tgt : ImagingStudy) extends DomainResource <<type+>> {
  src.identifier as vs where system != 'urn:dicom:uid' -> tgt.identifier as vt then Identifier(vs, vt);
  src.identifier as vs where system = 'urn:dicom:uid' then {
    vs.value -> tgt.uid;
  };
  src.status where value = 'registered' -> tgt.availability = 'OFFLINE';
  src.status where value = 'available' -> tgt.availability = 'ONLINE';
  src.modality -> tgt.modalityList;
  src.subject -> tgt.patient;
  src.encounter -> tgt.context;
  src.started -> tgt.started;
  src.basedOn -> tgt.basedOn;
  src.referrer -> tgt.referrer;
  src.interpreter -> tgt.interpreter;
  src.endpoint -> tgt.endpoint;
  src.numberOfSeries -> tgt.numberOfSeries;
  src.numberOfInstances -> tgt.numberOfInstances;
  src.procedureReference -> tgt.procedureReference;
  src.procedureCode -> tgt.procedureCode;
  src.reasonCode -> tgt.reason;
  src.description -> tgt.description;
  src.series as vs -> tgt.series as vt then series(vs, vt);
}

group series(source src, target tgt) extends BackboneElement {
  src.uid -> tgt.uid;
  src.number -> tgt.number;
  src.modality -> tgt.modality;
  src.description -> tgt.description;
  src.numberOfInstances -> tgt.numberOfInstances;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImagingStudy.series.availability' then {
    e.value -> tgt.availability;
  };
  src.endpoint -> tgt.endpoint;
  src.bodySite -> tgt.bodySite;
  src.laterality -> tgt.laterality;
  src.started -> tgt.started;
  src.performer as vs where function.coding.where((system =
    'http://terminology.hl7.org/CodeSystem/v3-ParticipationType') and (code = 'PRF')) then {
    vs.actor -> tgt.performer;
  };
  src.instance as vs -> tgt.instance as vt then instance(vs, vt);
}

group instance(source src, target tgt) extends BackboneElement {
  src.uid -> tgt.uid;
  src.number -> tgt.number;
  src.sopClass as s where system = 'urn:ietf:rfc:3986' then {
    s.code -> tgt.sopClass;
  };
  src.title -> tgt.title;
}
```

10.4.9.3 R4 Validation Errors

10.4.10 JSON Schema for ImagingStudy

URL: https://hl7.org/fhir/R4/imagingstudy.schema.json.html

10.4.10 JSON Schema for ImagingStudy

Imaging Integration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for ImagingStudy, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ImagingStudy",
  "$ref": "#/definitions/ImagingStudy",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ImagingStudy": {
      "description": "Representation of the content produced in a DICOM imaging study. A study comprises a set of
    series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data)
    acquired or produced in a common context.  A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a
    study may have multiple series of different modalities.",
      "properties": {
        "resourceType": {
          "description": "This is a ImagingStudy resource",
```

```
      "const": "ImagingStudy"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers for the ImagingStudy such as DICOM Study Instance UID, and Accession Number.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The current state of the ImagingStudy.",
      "enum": [
        "registered",
        "available",
        "cancelled",
        "entered-in-error",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "modality": {
      "description": "A list of all the series.modality values that are actual acquisition modalities, i.e. those in
 the DICOM Context Group 29 (value set OID 1.2.840.10008.6.1.19).",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "subject": {
      "description": "The subject, typically a patient, of the imaging study.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The healthcare event (e.g. a patient and healthcare provider interaction) during which this
ImagingStudy is made.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "started": {
      "description": "Date and time the study started.",
      "$ref": "#/definitions/dateTime"
    },
    "_started": {
      "description": "Extensions for started",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "basedOn": {
      "description": "A list of the diagnostic requests that resulted in this imaging study being performed.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "referrer": {
      "description": "The requesting/referring physician.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "interpreter": {
      "description": "Who read the study and interpreted the images or other content.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "endpoint": {
      "description": "The network service providing access (e.g., query, view, or retrieval) for the study. See
implementation notes for information about using DICOM endpoints. A study-level endpoint applies to each series in
the study, unless overridden by a series-level endpoint with the same Endpoint.connectionType.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "numberOfSeries": {
      "description": "Number of Series in the Study. This value given may be larger than the number of series
elements this Resource contains due to resource availability, security, or other factors. This element should be
present if any series elements are present.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_numberOfSeries": {
      "description": "Extensions for numberOfSeries",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numberOfInstances": {
      "description": "Number of SOP Instances in Study. This value given may be larger than the number of instance
elements this resource contains due to resource availability, security, or other factors. This element should be
present if any instance elements are present.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_numberOfInstances": {
      "description": "Extensions for numberOfInstances",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "procedureReference": {
      "description": "The procedure which this ImagingStudy was part of.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "procedureCode": {
      "description": "The code for the performed procedure type.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "location": {
      "description": "The principal physical location where the ImagingStudy was performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
```

```
      "description": "Description of clinical condition indicating why the ImagingStudy was requested.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates another resource whose existence justifies this Study.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Per the recommended DICOM mapping, this element is derived from the Study Description
attribute (0008,1030). Observations or findings about the imaging study should be recorded in another resource, e.g.
 Observation, and not in this element.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "description": {
      "description": "The Imaging Manager description of the study. Institution-generated description or
classification of the Study (component) performed.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "series": {
      "description": "Each study has one or more series of images or other content.",
      "items": {
        "$ref": "#/definitions/ImagingStudy_Series"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"ImagingStudy_Series": {
  "description": "Representation of the content produced in a DICOM imaging study. A study comprises a set of
series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data)
acquired or produced in a common context.  A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a
study may have multiple series of different modalities.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "uid": {
      "description": "The DICOM Series Instance UID for the series.",
      "$ref": "#/definitions/id"
    },
    "_uid": {
      "description": "Extensions for uid",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "number": {
```

```
      "description": "The numeric identifier of this series in the study.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_number": {
      "description": "Extensions for number",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "modality": {
      "description": "The modality of this series sequence.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "description": {
      "description": "A description of the series.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numberOfInstances": {
      "description": "Number of SOP Instances in the Study. The value given may be larger than the number of instance elements this resource contains due to resource availability, security, or other factors. This element should be present if any instance elements are present.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_numberOfInstances": {
      "description": "Extensions for numberOfInstances",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "endpoint": {
      "description": "The network service providing access (e.g., query, view, or retrieval) for this series. See implementation notes for information about using DICOM endpoints. A series-level endpoint, if present, has precedence over a study-level endpoint with the same Endpoint.connectionType.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "bodySite": {
      "description": "The anatomic structures examined. See DICOM Part 16 Annex L (http://dicom.nema.org/medical/dicom/current/output/chtml/part16/chapter_L.html) for DICOM to SNOMED-CT mappings. The bodySite may indicate the laterality of body part imaged; if so, it shall be consistent with any content of ImagingStudy.series.laterality.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "laterality": {
      "description": "The laterality of the (possibly paired) anatomic structures examined. E.g., the left knee, both lungs, or unpaired abdomen. If present, shall be consistent with any laterality information indicated in ImagingStudy.series.bodySite.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "specimen": {
      "description": "The specimen imaged, e.g., for whole slide imaging of a biopsy.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "started": {
      "description": "The date and time the series was started.",
      "$ref": "#/definitions/dateTime"
    },
    "_started": {
      "description": "Extensions for started",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "performer": {
      "description": "Indicates who or what performed the series and how they were involved.",
      "items": {
        "$ref": "#/definitions/ImagingStudy_Performer"
      },
      "type": "array"
    },
    "instance": {
      "description": "A single SOP instance within the series, e.g. an image, or presentation state.",
      "items": {
        "$ref": "#/definitions/ImagingStudy_Instance"
      },
      "type": "array"

    }
  },
  "additionalProperties": false,
  "required": [
    "modality"
  ]
},
"ImagingStudy_Performer": {
  "description": "Representation of the content produced in a DICOM imaging study. A study comprises a set of series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data)
```

```
acquired or produced in a common context.  A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a
study may have multiple series of different modalities.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "function": {
      "description": "Distinguishes the type of involvement of the performer in the series.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "actor": {
      "description": "Indicates who or what performed the series.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "actor"
  ]
},
"ImagingStudy_Instance": {
  "description": "Representation of the content produced in a DICOM imaging study. A study comprises a set of
series, each of which includes a set of Service-Object Pair Instances (SOP Instances - images or other data)
acquired or produced in a common context.  A series is of only one modality (e.g. X-ray, CT, MR, ultrasound), but a
study may have multiple series of different modalities.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "uid": {
      "description": "The DICOM SOP Instance UID for this image or other DICOM content.",
      "$ref": "#/definitions/id"
    },
    "_uid": {
      "description": "Extensions for uid",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
        "sopClass": {
          "description": "DICOM instance  type.",
          "$ref": "Coding.schema.json#/definitions/Coding"
        },
        "number": {
          "description": "The number of instance in the series.",
          "$ref": "#/definitions/unsignedInt"
        },
        "_number": {
          "description": "Extensions for number",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "title": {
          "description": "The description of the instance.",
          "$ref": "#/definitions/string"
        },
        "_title": {
          "description": "Extensions for title",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false,
      "required": [
        "sopClass"
      ]
    }
  }
}
```


10.4.11 Resource ImagingStudy - Maturity Dependency View

URL: https://hl7.org/fhir/R4/imagingstudy-dependencies.html

10.4.11 Resource ImagingStudy - Maturity Dependency View

Imaging Integration Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient

Dependency Graph for ImagingStudy FMM level 3

Name | Card. | Type | Dependency Analysis
ImagingStudy | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
modality | 0..* | Coding | Binding Error: Unable to resolve vs 'http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html' to check dependencies
subject | 1..1 | Reference(Patient | Device | Group) | Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
started | 0..1 | dateTime
basedOn | 0..* | Reference(CarePlan | ServiceRequest | Appointment | AppointmentResponse | Task) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Task = FMM2-Trial Use vs. Element = FMM3-Trial Use)
referrer | 0..1 | Reference(Practitioner | PractitionerRole) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
interpreter | 0..* | Reference(Practitioner | PractitionerRole) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
endpoint | 0..* | Reference(Endpoint) | Type Warning: (Endpoint = FMM2-Trial Use vs. Element = FMM3-Trial Use)
numberOfSeries | 0..1 | unsignedInt
numberOfInstances | 0..1 | unsignedInt
procedureReference | 0..1 | Reference(Procedure)
procedureCode | 0..* | CodeableConcept | Binding Error: Unable to resolve vs 'http://www.rsna.org/RadLex_Playbook.aspx' to check dependencies
location | 0..1 | Reference(Location)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | Media | DiagnosticReport | DocumentReference) | Type Warning: (Media = FMM1-Trial Use vs. Element = FMM3-Trial Use)
note | 0..* | Annotation
description | 0..1 | string
series | 0..* | BackboneElement
uid | 1..1 | id
number | 0..1 | unsignedInt
modality | 1..1 | Coding | Binding Error: Unable to resolve vs 'http://dicom.nema.org/medical/dicom/current/output/chtml/part16/sect_CID_29.html' to check dependencies
description | 0..1 | string
numberOfInstances | 0..1 | unsignedInt
endpoint | 0..* | Reference(Endpoint) | Type Warning: (Endpoint = FMM2-Trial Use vs. Element = FMM3-Trial Use)
bodySite | 0..1 | Coding
laterality | 0..1 | Coding
specimen | 0..* | Reference(Specimen) | Type Warning: (Specimen = FMM2-Trial Use vs. Element = FMM3-Trial Use)
started | 0..1 | dateTime

performer | 0..* | BackboneElement
function | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | Device | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (CareTeam = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
instance | 0..* | BackboneElement
uid | 1..1 | id
sopClass | 1..1 | Coding | Binding Error: Unable to resolve vs 'http://dicom.nema.org/medical/dicom/current/output/chtml/part04/sect_B.5.html#table_B.5-1' to check dependencies
number | 0..1 | unsignedInt
title | 0..1 | string

10.4.12 Value Set Sc-valueset-imagingstudy-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-imagingstudy-status.html

10.4.12 Value Set Sc-valueset-imagingstudy-status - Status Code Mapping

Imaging Integration Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

10.4.13 ImagingStudyStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-imagingstudy-status)

Mapping from http://hl7.org/fhir/ValueSet/imagingstudy-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the ImagingStudy."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
registered | equivalent | received
available | equivalent | active
cancelled | equivalent | abandoned
unknown | equivalent | unknown

10.6 Resource MolecularSequence - Content

URL: https://hl7.org/fhir/R4/molecularsequence.html

while the amino acid string should be constrained within the range:

Clinical Genomics Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

Raw data describing a biological sequence.

The Clinical Genomics committee has identified overlaps and redundancies between content in the MolecularSequence resource and content in Observation profiles in the evolving Implementation Guide for Clinical Genomics Reporting found here . The committee is considering options for modifying the resource and anticipates potential changes being brought forward in an upcoming ballot.

10.6.1 Scope and Usage

The MolecularSequence resource is designed to describe an atomic sequence which contains the alignment sequencing test result and multiple variations. Atomic sequences can be connected by link element and they will lead to sequence graph. By this method, a sequence can be reported. Complete genetic sequence information, of which specific genetic variations are a part, is reported by reference to the GA4GH repository. Thus, the FHIR MolecularSequence resource avoids large genomic payloads in a manner analogous to how the FHIR ImagingStudy resource references large images maintained in other systems. For use cases, details on how this resource interact with other Clinical Genomics resources or profiles, please refer to implementation guidance document here.

10.6.1.1 Genetic Standards and Resources include:

• Variant Databases: dbSNP , ClinVar , and COSMIC

• Reference Sequences: RefSeq and ENSEMBL

This resource is designed to describe sequence variations with clinical significance with information such as:

• Name of the variation represented

• Type of the variation

• Gene region occupied by the variation

• Tissue source used to determine genotype of the variation

• Quality of the result

It is strongly encouraged to provide all available information in this resource for any reported variants, because receiving systems (e.g. discovery research, outcomes analysis, and public health reporting) may use this information to normalize variants over time or across sources. However, these data should not be used to dynamically correct/change variant representations for clinical use outside of the laboratory, due to insufficient information.

Implementers should be aware that semantic equivalency of results of genetic variants cannot be guaranteed unless there is an agreed upon standard between sending and receiving systems.

10.6.2 Boundaries and Relationships

Focus of the resource is to provide sequencing alignment data immediately relevant to what the interpretation on clinical decision-making originates from. Hence data such as precise read of DNA sequences and sequence alignment are not included; such data are nonetheless accessible through references to GA4GH (Global Alliance for Genomics and Health) API. The MolecularSequence resource will be referenced by Observation to provide variant information. As clinical assessments/diagnosis of a patient are typically captured in the Condition resource or the ClinicalImpression resource, the MolecularSequence resource can be referenced by the Condition resource to provide specific genetic data to support assertions. This is analogous to how Condition references other resources, such as AllergyIntolerance, Procedure, and Questionnaire resources.

This resource is referenced by itself and Observation

10.6.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
MolecularSequence | ΣITU | | DomainResource | Information about a biological sequence + Rule: Only 0 and 1 are valid for coordinateSystem Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique ID for this particular sequence. This is a FHIR-defined id
type | Σ | 0..1 | code | aa | dna | rna sequenceType (Required)
coordinateSystem | Σ | 1..1 | integer | Base number of coordinate system (0 for 0-based numbering or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
patient | Σ | 0..1 | Reference(Patient) | Who and/or what this is about
specimen | Σ | 0..1 | Reference(Specimen) | Specimen used for sequencing
device | Σ | 0..1 | Reference(Device) | The method for sequencing
performer | Σ | 0..1 | Reference(Organization) | Who should be responsible for test result
quantity | Σ | 0..1 | Quantity | The number of copies of the sequence of interest. (RNASeq)
referenceSeq | ΣI | 0..1 | BackboneElement | A sequence used as reference + Rule: GenomeBuild and chromosome must be both contained if either one of them is contained + Rule: Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString;
chromosome | Σ | 0..1 | CodeableConcept | Chromosome containing genetic finding chromosome-human (Example)
genomeBuild | Σ | 0..1 | string | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
orientation | Σ | 0..1 | code | sense | antisense orientationType (Required)
referenceSeqId | Σ | 0..1 | CodeableConcept | Reference identifier E n s e m b l (Example)
referenceSeqPointer | Σ | 0..1 | Reference(MolecularSequence) | A pointer to another MolecularSequence entity as reference sequence
referenceSeqString | Σ | 0..1 | string | A string to represent reference sequence
strand | Σ | 0..1 | code | watson | crick strandType (Required)
windowStart | Σ | 0..1 | integer | Start position of the window on the reference sequence
windowEnd | Σ | 0..1 | integer | End position of the window on the reference sequence
variant | Σ | 0..* | BackboneElement | Variant in sequence
start | Σ | 0..1 | integer | Start position of the variant on the reference sequence
end | Σ | 0..1 | integer | End position of the variant on the reference sequence
observedAllele | Σ | 0..1 | string | Allele that was observed
referenceAllele | Σ | 0..1 | string | Allele in the reference sequence
cigar | Σ | 0..1 | string | Extended CIGAR string for aligning the sequence with reference bases
variantPointer | Σ | 0..1 | Reference(Observation) | Pointer to observed variant information
observedSeq | Σ | 0..1 | string | Sequence that was observed
quality | Σ | 0..* | BackboneElement | An set of value as quality of sequence
type | Σ | 1..1 | code | indel | snp | unknown qualityType (Required)
standardSequence | Σ | 0..1 | CodeableConcept | Standard sequence for comparison F d a- standard sequence (Example)

start | Σ | 0..1 | integer | Start position of the sequence
end | Σ | 0..1 | integer | End position of the sequence
score | Σ | 0..1 | Quantity | Quality score for the comparison
method | Σ | 0..1 | CodeableConcept | Method to get quality F d a- method (Example)
truthTP | Σ | 0..1 | decimal | True positives from the perspective of the truth data
queryTP | Σ | 0..1 | decimal | True positives from the perspective of the query data
truthFN | Σ | 0..1 | decimal | False negatives
queryFP | Σ | 0..1 | decimal | False positives
gtFP | Σ | 0..1 | decimal | False positives where the non-REF alleles in the Truth and Query Call Sets match
precision | Σ | 0..1 | decimal | Precision of comparison
recall | Σ | 0..1 | decimal | Recall of comparison
fScore | Σ | 0..1 | decimal | F-score
roc | Σ | 0..1 | BackboneElement | Receiver Operator Characteristic (ROC) Curve
score | Σ | 0..* | integer | Genotype quality score
numTP | Σ | 0..* | integer | Roc score true positive numbers
numFP | Σ | 0..* | integer | Roc score false positive numbers
numFN | Σ | 0..* | integer | Roc score false negative numbers
precision | Σ | 0..* | decimal | Precision of the GQ score
sensitivity | Σ | 0..* | decimal | Sensitivity of the GQ score
fMeasure | Σ | 0..* | decimal | FScore of the GQ score
readCoverage | Σ | 0..1 | integer | Average number of reads representing a given nucleotide in the reconstructed sequence
repository | Σ | 0..* | BackboneElement | External repository which contains detailed report related with observedSeq in this resource
type | Σ | 1..1 | code | directlink | openapi | login | oauth | other repositoryType (Required)
url | Σ | 0..1 | uri | URI of the repository
name | Σ | 0..1 | string | Repository's name
datasetId | Σ | 0..1 | string | Id of the dataset that used to call for dataset in repository
variantsetId | Σ | 0..1 | string | Id of the variantset that used to call for variantset in repository
readsetId | Σ | 0..1 | string | Id of the read
pointer | Σ | 0..* | Reference(MolecularSequence) | Pointer to next atomic sequence
structureVariant | Σ | 0..* | BackboneElement | Structural variant
variantType | Σ | 0..1 | CodeableConcept | Structural variant change type LOINC LL379-9 answerlist (Required)
exact | Σ | 0..1 | boolean | Does the structural variant have base pair resolution breakpoints?
length | Σ | 0..1 | integer | Structural variant length
outer | Σ | 0..1 | BackboneElement | Structural variant outer
start | Σ | 0..1 | integer | Structural variant outer start
end | Σ | 0..1 | integer | Structural variant outer end
inner | Σ | 0..1 | BackboneElement | Structural variant inner
start | Σ | 0..1 | integer | Structural variant inner start
end | Σ | 0..1 | integer | Structural variant inner end
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MolecularSequence xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique ID for this particular sequence. This is a FHIR-defined id --></identifier>
 <type value="[code]"/><!-- 0..1 aa | dna | rna -->
 <coordinateSystem value="[integer]"/><!-- 1..1 Base number of coordinate system (0 for 0-based numbering or
    coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end) -->
 <patient><!-- 0..1 Reference(Patient) Who and/or what this is about --></patient>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for sequencing --></specimen>
 <device><!-- 0..1 Reference(Device) The method for sequencing --></device>
 <performer><!-- 0..1 Reference(Organization) Who should be responsible for test result --></performer>
 <quantity><!-- 0..1 Quantity The number of copies of the sequence of interest.  (RNASeq) --></quantity>
 <referenceSeq>  <!-- 0..1 A sequence used as reference -->
  <chromosome><!-- 0..1 CodeableConcept Chromosome containing genetic finding --></chromosome>
  <genomeBuild value="[string]"/><!-- 0..1 The Genome Build used for reference, following GRCh build versions e.g. 'GRCh
     37' -->
  <orientation value="[code]"/><!-- 0..1 sense | antisense -->
  <referenceSeqId><!-- 0..1 CodeableConcept Reference identifier --></referenceSeqId>
  <referenceSeqPointer><!-- 0..1 Reference(MolecularSequence) A pointer to another MolecularSequence entity as reference
     sequence --></referenceSeqPointer>
  <referenceSeqString value="[string]"/><!-- 0..1 A string to represent reference sequence -->
  <strand value="[code]"/><!-- 0..1 watson | crick -->
  <windowStart value="[integer]"/><!-- 0..1 Start position of the window on the  reference sequence -->
  <windowEnd value="[integer]"/><!-- 0..1 End position of the window on the reference sequence -->
 </referenceSeq>
 <variant>  <!-- 0..* Variant in sequence -->
  <start value="[integer]"/><!-- 0..1 Start position of the variant on the  reference sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the variant on the reference sequence -->
  <observedAllele value="[string]"/><!-- 0..1 Allele that was observed -->
  <referenceAllele value="[string]"/><!-- 0..1 Allele in the reference sequence -->
  <cigar value="[string]"/><!-- 0..1 Extended CIGAR string for aligning the sequence with reference bases -->
  <variantPointer><!-- 0..1 Reference(Observation) Pointer to observed variant information --></variantPointer>
 </variant>
 <observedSeq value="[string]"/><!-- 0..1 Sequence that was observed -->
 <quality>  <!-- 0..* An set of value as quality of sequence -->
```

```
  <type value="[code]"/><!-- 1..1 indel | snp | unknown -->
  <standardSequence><!-- 0..1 CodeableConcept Standard sequence for comparison --></standardSequence>
  <start value="[integer]"/><!-- 0..1 Start position of the sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the sequence -->
  <score><!-- 0..1 Quantity Quality score for the comparison --></score>
  <method><!-- 0..1 CodeableConcept Method to get quality --></method>
  <truthTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the truth data -->
  <queryTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the query data -->
  <truthFN value="[decimal]"/><!-- 0..1 False negatives -->
  <queryFP value="[decimal]"/><!-- 0..1 False positives -->
  <gtFP value="[decimal]"/><!-- 0..1 False positives where the non-REF alleles in the Truth and Query Call Sets match
    -->
  <precision value="[decimal]"/><!-- 0..1 Precision of comparison -->
  <recall value="[decimal]"/><!-- 0..1 Recall of comparison -->
  <fScore value="[decimal]"/><!-- 0..1 F-score -->
  <roc>  <!-- 0..1 Receiver Operator Characteristic (ROC) Curve -->
   <score value="[integer]"/><!-- 0..* Genotype quality score -->
   <numTP value="[integer]"/><!-- 0..* Roc score true positive numbers -->
   <numFP value="[integer]"/><!-- 0..* Roc score false positive numbers -->
   <numFN value="[integer]"/><!-- 0..* Roc score false negative numbers -->
   <precision value="[decimal]"/><!-- 0..* Precision of the GQ score -->
   <sensitivity value="[decimal]"/><!-- 0..* Sensitivity of the GQ score -->
   <fMeasure value="[decimal]"/><!-- 0..* FScore of the GQ score -->
  </roc>
 </quality>
 <readCoverage value="[integer]"/><!-- 0..1 Average number of reads representing a given nucleotide in the reconstructed
     sequence -->
 <repository>  <!-- 0..* External repository which contains detailed report related with observedSeq in this resource
    -->
  <type value="[code]"/><!-- 1..1 directlink | openapi | login | oauth | other -->
  <url value="[uri]"/><!-- 0..1 URI of the repository -->
  <name value="[string]"/><!-- 0..1 Repository's name -->
  <datasetId value="[string]"/><!-- 0..1 Id of the dataset that used to call for dataset in repository -->
  <variantsetId value="[string]"/><!-- 0..1 Id of the variantset that used to call for variantset in repository -->
  <readsetId value="[string]"/><!-- 0..1 Id of the read -->
 </repository>
 <pointer><!-- 0..* Reference(MolecularSequence) Pointer to next atomic sequence --></pointer>
 <structureVariant>  <!-- 0..* Structural variant -->
  <variantType><!-- 0..1 CodeableConcept Structural variant change type  --></variantType>
  <exact value="[boolean]"/><!-- 0..1 Does the structural variant have base pair resolution breakpoints? -->
  <length value="[integer]"/><!-- 0..1 Structural variant length -->
  <outer>  <!-- 0..1 Structural variant outer -->
   <start value="[integer]"/><!-- 0..1 Structural variant outer start -->
   <end value="[integer]"/><!-- 0..1 Structural variant outer end -->
  </outer>
  <inner>  <!-- 0..1 Structural variant inner -->
   <start value="[integer]"/><!-- 0..1 Structural variant inner start -->
   <end value="[integer]"/><!-- 0..1 Structural variant inner end -->
  </inner>
 </structureVariant>
</MolecularSequence>
```

JSON Template

```
{
  "resourceType" : "MolecularSequence",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique ID for this particular sequence. This is a FHIR-defined id
  "type" : "<code>", // aa | dna | rna
  "coordinateSystem" : <integer>, // R!  Base number of coordinate system (0 for 0-based numbering or coordinates,
    inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  "patient" : { Reference(Patient) }, // Who and/or what this is about
  "specimen" : { Reference(Specimen) }, // Specimen used for sequencing
  "device" : { Reference(Device) }, // The method for sequencing
  "performer" : { Reference(Organization) }, // Who should be responsible for test result
  "quantity" : { Quantity }, // The number of copies of the sequence of interest.  (RNASeq)
  "referenceSeq" : { // A sequence used as reference
    "chromosome" : { CodeableConcept }, // Chromosome containing genetic finding
    "genomeBuild" : "<string>", // The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
    "orientation" : "<code>", // sense | antisense
    "referenceSeqId" : { CodeableConcept }, // Reference identifier
    "referenceSeqPointer" : { Reference(MolecularSequence) }, // A pointer to another MolecularSequence entity as
    reference sequence
    "referenceSeqString" : "<string>", // A string to represent reference sequence
    "strand" : "<code>", // watson | crick
    "windowStart" : <integer>, // Start position of the window on the  reference sequence
    "windowEnd" : <integer> // End position of the window on the reference sequence
  },
  "variant" : [{ // Variant in sequence
    "start" : <integer>, // Start position of the variant on the  reference sequence
    "end" : <integer>, // End position of the variant on the reference sequence
    "observedAllele" : "<string>", // Allele that was observed
    "referenceAllele" : "<string>", // Allele in the reference sequence
    "cigar" : "<string>", // Extended CIGAR string for aligning the sequence with reference bases
    "variantPointer" : { Reference(Observation) } // Pointer to observed variant information
  }],
  "observedSeq" : "<string>", // Sequence that was observed
  "quality" : [{ // An set of value as quality of sequence
    "type" : "<code>", // R!  indel | snp | unknown
```

```
    "standardSequence" : { CodeableConcept }, // Standard sequence for comparison
    "start" : <integer>, // Start position of the sequence
    "end" : <integer>, // End position of the sequence
    "score" : { Quantity }, // Quality score for the comparison
    "method" : { CodeableConcept }, // Method to get quality
    "truthTP" : <decimal>, // True positives from the perspective of the truth data
    "queryTP" : <decimal>, // True positives from the perspective of the query data
    "truthFN" : <decimal>, // False negatives
    "queryFP" : <decimal>, // False positives
    "gtFP" : <decimal>, // False positives where the non-REF alleles in the Truth and Query Call Sets match
    "precision" : <decimal>, // Precision of comparison
    "recall" : <decimal>, // Recall of comparison
    "fScore" : <decimal>, // F-score
    "roc" : { // Receiver Operator Characteristic (ROC) Curve
      "score" : [<integer>], // Genotype quality score
      "numTP" : [<integer>], // Roc score true positive numbers
      "numFP" : [<integer>], // Roc score false positive numbers
      "numFN" : [<integer>], // Roc score false negative numbers
      "precision" : [<decimal>], // Precision of the GQ score
      "sensitivity" : [<decimal>], // Sensitivity of the GQ score
      "fMeasure" : [<decimal>] // FScore of the GQ score
    }
  }],
  "readCoverage" : <integer>, // Average number of reads representing a given nucleotide in the reconstructed sequence
  "repository" : [{ // External repository which contains detailed report related with observedSeq in this resource
    "type" : "<code>", // R!  directlink | openapi | login | oauth | other
    "url" : "<uri>", // URI of the repository
    "name" : "<string>", // Repository's name
    "datasetId" : "<string>", // Id of the dataset that used to call for dataset in repository
    "variantsetId" : "<string>", // Id of the variantset that used to call for variantset in repository
    "readsetId" : "<string>" // Id of the read
  }],
  "pointer" : [{ Reference(MolecularSequence) }], // Pointer to next atomic sequence
  "structureVariant" : [{ // Structural variant
    "variantType" : { CodeableConcept }, // Structural variant change type
    "exact" : <boolean>, // Does the structural variant have base pair resolution breakpoints?
    "length" : <integer>, // Structural variant length
    "outer" : { // Structural variant outer
      "start" : <integer>, // Structural variant outer start
      "end" : <integer> // Structural variant outer end
    },
    "inner" : { // Structural variant inner
      "start" : <integer>, // Structural variant inner start
      "end" : <integer> // Structural variant inner end
    }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MolecularSequence;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MolecularSequence.identifier [ Identifier ], ... ; # 0..* Unique ID for this particular sequence. This is a FHIR-
    defined id
  fhir:MolecularSequence.type [ code ]; # 0..1 aa | dna | rna
  fhir:MolecularSequence.coordinateSystem [ integer ]; # 1..1 Base number of coordinate system (0 for 0-based numbering
    or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  fhir:MolecularSequence.patient [ Reference(Patient) ]; # 0..1 Who and/or what this is about
  fhir:MolecularSequence.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for sequencing
  fhir:MolecularSequence.device [ Reference(Device) ]; # 0..1 The method for sequencing
  fhir:MolecularSequence.performer [ Reference(Organization) ]; # 0..1 Who should be responsible for test result
  fhir:MolecularSequence.quantity [ Quantity ]; # 0..1 The number of copies of the sequence of interest.  (RNASeq)
  fhir:MolecularSequence.referenceSeq [ # 0..1 A sequence used as reference
    fhir:MolecularSequence.referenceSeq.chromosome [ CodeableConcept ]; # 0..1 Chromosome containing genetic finding
    fhir:MolecularSequence.referenceSeq.genomeBuild [ string ]; # 0..1 The Genome Build used for reference, following
    GRCh build versions e.g. 'GRCh 37'
    fhir:MolecularSequence.referenceSeq.orientation [ code ]; # 0..1 sense | antisense
    fhir:MolecularSequence.referenceSeq.referenceSeqId [ CodeableConcept ]; # 0..1 Reference identifier
    fhir:MolecularSequence.referenceSeq.referenceSeqPointer [ Reference(MolecularSequence) ]; # 0..1 A pointer to
    another MolecularSequence entity as reference sequence
    fhir:MolecularSequence.referenceSeq.referenceSeqString [ string ]; # 0..1 A string to represent reference sequence
    fhir:MolecularSequence.referenceSeq.strand [ code ]; # 0..1 watson | crick
    fhir:MolecularSequence.referenceSeq.windowStart [ integer ]; # 0..1 Start position of the window on the  reference
    sequence
    fhir:MolecularSequence.referenceSeq.windowEnd [ integer ]; # 0..1 End position of the window on the reference
    sequence
  ];
  fhir:MolecularSequence.variant [ # 0..* Variant in sequence
    fhir:MolecularSequence.variant.start [ integer ]; # 0..1 Start position of the variant on the  reference sequence
    fhir:MolecularSequence.variant.end [ integer ]; # 0..1 End position of the variant on the reference sequence
    fhir:MolecularSequence.variant.observedAllele [ string ]; # 0..1 Allele that was observed
    fhir:MolecularSequence.variant.referenceAllele [ string ]; # 0..1 Allele in the reference sequence
    fhir:MolecularSequence.variant.cigar [ string ]; # 0..1 Extended CIGAR string for aligning the sequence with
    reference bases
```

```
    fhir:MolecularSequence.variant.variantPointer [ Reference(Observation) ]; # 0..1 Pointer to observed variant
    information
  ], ...;
  fhir:MolecularSequence.observedSeq [ string ]; # 0..1 Sequence that was observed
  fhir:MolecularSequence.quality [ # 0..* An set of value as quality of sequence
    fhir:MolecularSequence.quality.type [ code ]; # 1..1 indel | snp | unknown
    fhir:MolecularSequence.quality.standardSequence [ CodeableConcept ]; # 0..1 Standard sequence for comparison
    fhir:MolecularSequence.quality.start [ integer ]; # 0..1 Start position of the sequence
    fhir:MolecularSequence.quality.end [ integer ]; # 0..1 End position of the sequence
    fhir:MolecularSequence.quality.score [ Quantity ]; # 0..1 Quality score for the comparison
    fhir:MolecularSequence.quality.method [ CodeableConcept ]; # 0..1 Method to get quality
    fhir:MolecularSequence.quality.truthTP [ decimal ]; # 0..1 True positives from the perspective of the truth data
    fhir:MolecularSequence.quality.queryTP [ decimal ]; # 0..1 True positives from the perspective of the query data
    fhir:MolecularSequence.quality.truthFN [ decimal ]; # 0..1 False negatives
    fhir:MolecularSequence.quality.queryFP [ decimal ]; # 0..1 False positives
    fhir:MolecularSequence.quality.gtFP [ decimal ]; # 0..1 False positives where the non-REF alleles in the Truth and
    Query Call Sets match
    fhir:MolecularSequence.quality.precision [ decimal ]; # 0..1 Precision of comparison
    fhir:MolecularSequence.quality.recall [ decimal ]; # 0..1 Recall of comparison
    fhir:MolecularSequence.quality.fScore [ decimal ]; # 0..1 F-score
    fhir:MolecularSequence.quality.roc [ # 0..1 Receiver Operator Characteristic (ROC) Curve
      fhir:MolecularSequence.quality.roc.score [ integer ], ... ; # 0..* Genotype quality score
      fhir:MolecularSequence.quality.roc.numTP [ integer ], ... ; # 0..* Roc score true positive numbers
      fhir:MolecularSequence.quality.roc.numFP [ integer ], ... ; # 0..* Roc score false positive numbers
      fhir:MolecularSequence.quality.roc.numFN [ integer ], ... ; # 0..* Roc score false negative numbers
      fhir:MolecularSequence.quality.roc.precision [ decimal ], ... ; # 0..* Precision of the GQ score
      fhir:MolecularSequence.quality.roc.sensitivity [ decimal ], ... ; # 0..* Sensitivity of the GQ score
      fhir:MolecularSequence.quality.roc.fMeasure [ decimal ], ... ; # 0..* FScore of the GQ score
    ];
  ], ...;
  fhir:MolecularSequence.readCoverage [ integer ]; # 0..1 Average number of reads representing a given nucleotide in the
     reconstructed sequence
  fhir:MolecularSequence.repository [ # 0..* External repository which contains detailed report related with observedSeq
     in this resource
    fhir:MolecularSequence.repository.type [ code ]; # 1..1 directlink | openapi | login | oauth | other
    fhir:MolecularSequence.repository.url [ uri ]; # 0..1 URI of the repository
    fhir:MolecularSequence.repository.name [ string ]; # 0..1 Repository's name
    fhir:MolecularSequence.repository.datasetId [ string ]; # 0..1 Id of the dataset that used to call for dataset in
    repository
    fhir:MolecularSequence.repository.variantsetId [ string ]; # 0..1 Id of the variantset that used to call for
    variantset in repository
    fhir:MolecularSequence.repository.readsetId [ string ]; # 0..1 Id of the read
  ], ...;
  fhir:MolecularSequence.pointer [ Reference(MolecularSequence) ], ... ; # 0..* Pointer to next atomic sequence
  fhir:MolecularSequence.structureVariant [ # 0..* Structural variant
    fhir:MolecularSequence.structureVariant.variantType [ CodeableConcept ]; # 0..1 Structural variant change type
    fhir:MolecularSequence.structureVariant.exact [ boolean ]; # 0..1 Does the structural variant have base pair
    resolution breakpoints?
    fhir:MolecularSequence.structureVariant.length [ integer ]; # 0..1 Structural variant length
    fhir:MolecularSequence.structureVariant.outer [ # 0..1 Structural variant outer
      fhir:MolecularSequence.structureVariant.outer.start [ integer ]; # 0..1 Structural variant outer start
      fhir:MolecularSequence.structureVariant.outer.end [ integer ]; # 0..1 Structural variant outer end
    ];
    fhir:MolecularSequence.structureVariant.inner [ # 0..1 Structural variant inner
      fhir:MolecularSequence.structureVariant.inner.start [ integer ]; # 0..1 Structural variant inner start
      fhir:MolecularSequence.structureVariant.inner.end [ integer ]; # 0..1 Structural variant inner end
    ];
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

MolecularSequence | ΣITU | | DomainResource | Information about a biological sequence + Rule: Only 0 and 1 are valid for coordinateSystem Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Unique ID for this particular sequence. This is a FHIR-defined id

type | Σ | 0..1 | code | aa | dna | rna sequenceType (Required)

coordinateSystem | Σ | 1..1 | integer | Base number of coordinate system (0 for 0-based numbering or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)

patient | Σ | 0..1 | Reference(Patient) | Who and/or what this is about

specimen | Σ | 0..1 | Reference(Specimen) | Specimen used for sequencing

device | Σ | 0..1 | Reference(Device) | The method for sequencing

performer | Σ | 0..1 | Reference(Organization) | Who should be responsible for test result

quantity | Σ | 0..1 | Quantity | The number of copies of the sequence of interest. (RNASeq)

referenceSeq | ΣI | 0..1 | BackboneElement | A sequence used as reference + Rule: GenomeBuild and chromosome must be both contained if either one of them is contained + Rule: Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString;

chromosome | Σ | 0..1 | CodeableConcept | Chromosome containing genetic finding chromosome-human (Example)

genomeBuild | Σ | 0..1 | string | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
orientation | Σ | 0..1 | code | sense | antisense orientationType (Required)
referenceSeqId | Σ | 0..1 | CodeableConcept | Reference identifier E n s e m b l (Example)
referenceSeqPointer | Σ | 0..1 | Reference(MolecularSequence) | A pointer to another MolecularSequence entity as reference sequence
referenceSeqString | Σ | 0..1 | string | A string to represent reference sequence
strand | Σ | 0..1 | code | watson | crick strandType (Required)
windowStart | Σ | 0..1 | integer | Start position of the window on the reference sequence
windowEnd | Σ | 0..1 | integer | End position of the window on the reference sequence
variant | Σ | 0..* | BackboneElement | Variant in sequence
start | Σ | 0..1 | integer | Start position of the variant on the reference sequence
end | Σ | 0..1 | integer | End position of the variant on the reference sequence
observedAllele | Σ | 0..1 | string | Allele that was observed
referenceAllele | Σ | 0..1 | string | Allele in the reference sequence
cigar | Σ | 0..1 | string | Extended CIGAR string for aligning the sequence with reference bases
variantPointer | Σ | 0..1 | Reference(Observation) | Pointer to observed variant information
observedSeq | Σ | 0..1 | string | Sequence that was observed
quality | Σ | 0..* | BackboneElement | An set of value as quality of sequence
type | Σ | 1..1 | code | indel | snp | unknown qualityType (Required)
standardSequence | Σ | 0..1 | CodeableConcept | Standard sequence for comparison F d a- standard sequence (Example)
start | Σ | 0..1 | integer | Start position of the sequence
end | Σ | 0..1 | integer | End position of the sequence
score | Σ | 0..1 | Quantity | Quality score for the comparison
method | Σ | 0..1 | CodeableConcept | Method to get quality F d a- method (Example)
truthTP | Σ | 0..1 | decimal | True positives from the perspective of the truth data
queryTP | Σ | 0..1 | decimal | True positives from the perspective of the query data
truthFN | Σ | 0..1 | decimal | False negatives
queryFP | Σ | 0..1 | decimal | False positives
gtFP | Σ | 0..1 | decimal | False positives where the non-REF alleles in the Truth and Query Call Sets match
precision | Σ | 0..1 | decimal | Precision of comparison
recall | Σ | 0..1 | decimal | Recall of comparison
fScore | Σ | 0..1 | decimal | F-score
roc | Σ | 0..1 | BackboneElement | Receiver Operator Characteristic (ROC) Curve
score | Σ | 0..* | integer | Genotype quality score
numTP | Σ | 0..* | integer | Roc score true positive numbers
numFP | Σ | 0..* | integer | Roc score false positive numbers
numFN | Σ | 0..* | integer | Roc score false negative numbers
precision | Σ | 0..* | decimal | Precision of the GQ score
sensitivity | Σ | 0..* | decimal | Sensitivity of the GQ score
fMeasure | Σ | 0..* | decimal | FScore of the GQ score
readCoverage | Σ | 0..1 | integer | Average number of reads representing a given nucleotide in the reconstructed sequence
repository | Σ | 0..* | BackboneElement | External repository which contains detailed report related with observedSeq in this resource
type | Σ | 1..1 | code | directlink | openapi | login | oauth | other repositoryType (Required)
url | Σ | 0..1 | uri | URI of the repository
name | Σ | 0..1 | string | Repository's name
datasetId | Σ | 0..1 | string | Id of the dataset that used to call for dataset in repository
variantsetId | Σ | 0..1 | string | Id of the variantset that used to call for variantset in repository
readsetId | Σ | 0..1 | string | Id of the read
pointer | Σ | 0..* | Reference(MolecularSequence) | Pointer to next atomic sequence
structureVariant | Σ | 0..* | BackboneElement | Structural variant
variantType | Σ | 0..1 | CodeableConcept | Structural variant change type LOINC LL379-9 answerlist (Required)
exact | Σ | 0..1 | boolean | Does the structural variant have base pair resolution breakpoints?
length | Σ | 0..1 | integer | Structural variant length
outer | Σ | 0..1 | BackboneElement | Structural variant outer
start | Σ | 0..1 | integer | Structural variant outer start
end | Σ | 0..1 | integer | Structural variant outer end
inner | Σ | 0..1 | BackboneElement | Structural variant inner
start | Σ | 0..1 | integer | Structural variant inner start
end | Σ | 0..1 | integer | Structural variant inner end
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MolecularSequence xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique ID for this particular sequence. This is a FHIR-defined id --></identifier>
 <type value="[code]"/><!-- 0..1 aa | dna | rna -->
 <coordinateSystem value="[integer]"/><!-- 1..1 Base number of coordinate system (0 for 0-based numbering or
    coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end) -->
 <patient><!-- 0..1 Reference(Patient) Who and/or what this is about --></patient>
 <specimen><!-- 0..1 Reference(Specimen) Specimen used for sequencing --></specimen>
 <device><!-- 0..1 Reference(Device) The method for sequencing --></device>
```

```
 <performer><!-- 0..1 Reference(Organization) Who should be responsible for test result --></performer>
 <quantity><!-- 0..1 Quantity The number of copies of the sequence of interest.  (RNASeq) --></quantity>
 <referenceSeq>  <!-- 0..1 A sequence used as reference -->
  <chromosome><!-- 0..1 CodeableConcept Chromosome containing genetic finding --></chromosome>
  <genomeBuild value="[string]"/><!-- 0..1 The Genome Build used for reference, following GRCh build versions e.g. 'GRCh
     37' -->
  <orientation value="[code]"/><!-- 0..1 sense | antisense -->
  <referenceSeqId><!-- 0..1 CodeableConcept Reference identifier --></referenceSeqId>
  <referenceSeqPointer><!-- 0..1 Reference(MolecularSequence) A pointer to another MolecularSequence entity as reference
     sequence --></referenceSeqPointer>
  <referenceSeqString value="[string]"/><!-- 0..1 A string to represent reference sequence -->
  <strand value="[code]"/><!-- 0..1 watson | crick -->
  <windowStart value="[integer]"/><!-- 0..1 Start position of the window on the  reference sequence -->
  <windowEnd value="[integer]"/><!-- 0..1 End position of the window on the reference sequence -->
 </referenceSeq>
 <variant>  <!-- 0..* Variant in sequence -->
  <start value="[integer]"/><!-- 0..1 Start position of the variant on the  reference sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the variant on the reference sequence -->
  <observedAllele value="[string]"/><!-- 0..1 Allele that was observed -->
  <referenceAllele value="[string]"/><!-- 0..1 Allele in the reference sequence -->
  <cigar value="[string]"/><!-- 0..1 Extended CIGAR string for aligning the sequence with reference bases -->
  <variantPointer><!-- 0..1 Reference(Observation) Pointer to observed variant information --></variantPointer>
 </variant>
 <observedSeq value="[string]"/><!-- 0..1 Sequence that was observed -->
 <quality>  <!-- 0..* An set of value as quality of sequence -->
  <type value="[code]"/><!-- 1..1 indel | snp | unknown -->
  <standardSequence><!-- 0..1 CodeableConcept Standard sequence for comparison --></standardSequence>
  <start value="[integer]"/><!-- 0..1 Start position of the sequence -->
  <end value="[integer]"/><!-- 0..1 End position of the sequence -->
  <score><!-- 0..1 Quantity Quality score for the comparison --></score>
  <method><!-- 0..1 CodeableConcept Method to get quality --></method>
  <truthTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the truth data -->
  <queryTP value="[decimal]"/><!-- 0..1 True positives from the perspective of the query data -->
  <truthFN value="[decimal]"/><!-- 0..1 False negatives -->
  <queryFP value="[decimal]"/><!-- 0..1 False positives -->
  <gtFP value="[decimal]"/><!-- 0..1 False positives where the non-REF alleles in the Truth and Query Call Sets match
     -->
  <precision value="[decimal]"/><!-- 0..1 Precision of comparison -->
  <recall value="[decimal]"/><!-- 0..1 Recall of comparison -->
  <fScore value="[decimal]"/><!-- 0..1 F-score -->
  <roc>  <!-- 0..1 Receiver Operator Characteristic (ROC) Curve -->
   <score value="[integer]"/><!-- 0..* Genotype quality score -->
   <numTP value="[integer]"/><!-- 0..* Roc score true positive numbers -->
   <numFP value="[integer]"/><!-- 0..* Roc score false positive numbers -->
   <numFN value="[integer]"/><!-- 0..* Roc score false negative numbers -->
   <precision value="[decimal]"/><!-- 0..* Precision of the GQ score -->
   <sensitivity value="[decimal]"/><!-- 0..* Sensitivity of the GQ score -->
   <fMeasure value="[decimal]"/><!-- 0..* FScore of the GQ score -->
  </roc>
 </quality>
 <readCoverage value="[integer]"/><!-- 0..1 Average number of reads representing a given nucleotide in the reconstructed
     sequence -->
 <repository>  <!-- 0..* External repository which contains detailed report related with observedSeq in this resource
    -->
  <type value="[code]"/><!-- 1..1 directlink | openapi | login | oauth | other -->
  <url value="[uri]"/><!-- 0..1 URI of the repository -->
  <name value="[string]"/><!-- 0..1 Repository's name -->
  <datasetId value="[string]"/><!-- 0..1 Id of the dataset that used to call for dataset in repository -->
  <variantsetId value="[string]"/><!-- 0..1 Id of the variantset that used to call for variantset in repository -->
  <readsetId value="[string]"/><!-- 0..1 Id of the read -->
 </repository>
 <pointer><!-- 0..* Reference(MolecularSequence) Pointer to next atomic sequence --></pointer>
 <structureVariant>  <!-- 0..* Structural variant -->
  <variantType><!-- 0..1 CodeableConcept Structural variant change type  --></variantType>
  <exact value="[boolean]"/><!-- 0..1 Does the structural variant have base pair resolution breakpoints? -->
  <length value="[integer]"/><!-- 0..1 Structural variant length -->
  <outer>  <!-- 0..1 Structural variant outer -->
   <start value="[integer]"/><!-- 0..1 Structural variant outer start -->
   <end value="[integer]"/><!-- 0..1 Structural variant outer end -->
  </outer>
  <inner>  <!-- 0..1 Structural variant inner -->
   <start value="[integer]"/><!-- 0..1 Structural variant inner start -->
   <end value="[integer]"/><!-- 0..1 Structural variant inner end -->
  </inner>
 </structureVariant>
</MolecularSequence>
```

JSON Template

```
{
  "resourceType" : "MolecularSequence",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique ID for this particular sequence. This is a FHIR-defined id
  "type" : "<code>", // aa | dna | rna
  "coordinateSystem" : <integer>, // R!  Base number of coordinate system (0 for 0-based numbering or coordinates,
    inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  "patient" : { Reference(Patient) }, // Who and/or what this is about
  "specimen" : { Reference(Specimen) }, // Specimen used for sequencing
  "device" : { Reference(Device) }, // The method for sequencing
```

```
  "performer" : { Reference(Organization) }, // Who should be responsible for test result
  "quantity" : { Quantity }, // The number of copies of the sequence of interest.  (RNASeq)
  "referenceSeq" : { // A sequence used as reference
    "chromosome" : { CodeableConcept }, // Chromosome containing genetic finding
    "genomeBuild" : "<string>", // The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
    "orientation" : "<code>", // sense | antisense
    "referenceSeqId" : { CodeableConcept }, // Reference identifier
    "referenceSeqPointer" : { Reference(MolecularSequence) }, // A pointer to another MolecularSequence entity as
    reference sequence
    "referenceSeqString" : "<string>", // A string to represent reference sequence
    "strand" : "<code>", // watson | crick
    "windowStart" : <integer>, // Start position of the window on the  reference sequence
    "windowEnd" : <integer> // End position of the window on the reference sequence
  },
  "variant" : [{ // Variant in sequence
    "start" : <integer>, // Start position of the variant on the  reference sequence
    "end" : <integer>, // End position of the variant on the reference sequence
    "observedAllele" : "<string>", // Allele that was observed
    "referenceAllele" : "<string>", // Allele in the reference sequence
    "cigar" : "<string>", // Extended CIGAR string for aligning the sequence with reference bases
    "variantPointer" : { Reference(Observation) } // Pointer to observed variant information
  }],
  "observedSeq" : "<string>", // Sequence that was observed
  "quality" : [{ // An set of value as quality of sequence
    "type" : "<code>", // R!  indel | snp | unknown
    "standardSequence" : { CodeableConcept }, // Standard sequence for comparison
    "start" : <integer>, // Start position of the sequence
    "end" : <integer>, // End position of the sequence
    "score" : { Quantity }, // Quality score for the comparison
    "method" : { CodeableConcept }, // Method to get quality
    "truthTP" : <decimal>, // True positives from the perspective of the truth data
    "queryTP" : <decimal>, // True positives from the perspective of the query data
    "truthFN" : <decimal>, // False negatives
    "queryFP" : <decimal>, // False positives
    "gtFP" : <decimal>, // False positives where the non-REF alleles in the Truth and Query Call Sets match
    "precision" : <decimal>, // Precision of comparison
    "recall" : <decimal>, // Recall of comparison
    "fScore" : <decimal>, // F-score
    "roc" : { // Receiver Operator Characteristic (ROC) Curve
      "score" : [<integer>], // Genotype quality score
      "numTP" : [<integer>], // Roc score true positive numbers
      "numFP" : [<integer>], // Roc score false positive numbers
      "numFN" : [<integer>], // Roc score false negative numbers
      "precision" : [<decimal>], // Precision of the GQ score
      "sensitivity" : [<decimal>], // Sensitivity of the GQ score
      "fMeasure" : [<decimal>] // FScore of the GQ score
    }
  }],
  "readCoverage" : <integer>, // Average number of reads representing a given nucleotide in the reconstructed sequence
  "repository" : [{ // External repository which contains detailed report related with observedSeq in this resource
    "type" : "<code>", // R!  directlink | openapi | login | oauth | other
    "url" : "<uri>", // URI of the repository
    "name" : "<string>", // Repository's name
    "datasetId" : "<string>", // Id of the dataset that used to call for dataset in repository
    "variantsetId" : "<string>", // Id of the variantset that used to call for variantset in repository
    "readsetId" : "<string>" // Id of the read
  }],
  "pointer" : [{ Reference(MolecularSequence) }], // Pointer to next atomic sequence
  "structureVariant" : [{ // Structural variant
    "variantType" : { CodeableConcept }, // Structural variant change type
    "exact" : <boolean>, // Does the structural variant have base pair resolution breakpoints?
    "length" : <integer>, // Structural variant length
    "outer" : { // Structural variant outer
      "start" : <integer>, // Structural variant outer start
      "end" : <integer> // Structural variant outer end
    },
    "inner" : { // Structural variant inner
      "start" : <integer>, // Structural variant inner start
      "end" : <integer> // Structural variant inner end
    }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MolecularSequence;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MolecularSequence.identifier [ Identifier ], ... ; # 0..* Unique ID for this particular sequence. This is a FHIR-
    defined id
  fhir:MolecularSequence.type [ code ]; # 0..1 aa | dna | rna
  fhir:MolecularSequence.coordinateSystem [ integer ]; # 1..1 Base number of coordinate system (0 for 0-based numbering
    or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
  fhir:MolecularSequence.patient [ Reference(Patient) ]; # 0..1 Who and/or what this is about
  fhir:MolecularSequence.specimen [ Reference(Specimen) ]; # 0..1 Specimen used for sequencing
```

```
  fhir:MolecularSequence.device [ Reference(Device) ]; # 0..1 The method for sequencing
  fhir:MolecularSequence.performer [ Reference(Organization) ]; # 0..1 Who should be responsible for test result
  fhir:MolecularSequence.quantity [ Quantity ]; # 0..1 The number of copies of the sequence of interest.  (RNASeq)
  fhir:MolecularSequence.referenceSeq [ # 0..1 A sequence used as reference
    fhir:MolecularSequence.referenceSeq.chromosome [ CodeableConcept ]; # 0..1 Chromosome containing genetic finding
    fhir:MolecularSequence.referenceSeq.genomeBuild [ string ]; # 0..1 The Genome Build used for reference, following
    GRCh build versions e.g. 'GRCh 37'
    fhir:MolecularSequence.referenceSeq.orientation [ code ]; # 0..1 sense | antisense
    fhir:MolecularSequence.referenceSeq.referenceSeqId [ CodeableConcept ]; # 0..1 Reference identifier
    fhir:MolecularSequence.referenceSeq.referenceSeqPointer [ Reference(MolecularSequence) ]; # 0..1 A pointer to
    another MolecularSequence entity as reference sequence
    fhir:MolecularSequence.referenceSeq.referenceSeqString [ string ]; # 0..1 A string to represent reference sequence
    fhir:MolecularSequence.referenceSeq.strand [ code ]; # 0..1 watson | crick
    fhir:MolecularSequence.referenceSeq.windowStart [ integer ]; # 0..1 Start position of the window on the  reference
    sequence
    fhir:MolecularSequence.referenceSeq.windowEnd [ integer ]; # 0..1 End position of the window on the reference
    sequence
  ];
  fhir:MolecularSequence.variant [ # 0..* Variant in sequence
    fhir:MolecularSequence.variant.start [ integer ]; # 0..1 Start position of the variant on the  reference sequence
    fhir:MolecularSequence.variant.end [ integer ]; # 0..1 End position of the variant on the reference sequence
    fhir:MolecularSequence.variant.observedAllele [ string ]; # 0..1 Allele that was observed
    fhir:MolecularSequence.variant.referenceAllele [ string ]; # 0..1 Allele in the reference sequence
    fhir:MolecularSequence.variant.cigar [ string ]; # 0..1 Extended CIGAR string for aligning the sequence with
    reference bases
    fhir:MolecularSequence.variant.variantPointer [ Reference(Observation) ]; # 0..1 Pointer to observed variant
    information
  ], ...;
  fhir:MolecularSequence.observedSeq [ string ]; # 0..1 Sequence that was observed
  fhir:MolecularSequence.quality [ # 0..* An set of value as quality of sequence
    fhir:MolecularSequence.quality.type [ code ]; # 1..1 indel | snp | unknown
    fhir:MolecularSequence.quality.standardSequence [ CodeableConcept ]; # 0..1 Standard sequence for comparison
    fhir:MolecularSequence.quality.start [ integer ]; # 0..1 Start position of the sequence
    fhir:MolecularSequence.quality.end [ integer ]; # 0..1 End position of the sequence
    fhir:MolecularSequence.quality.score [ Quantity ]; # 0..1 Quality score for the comparison
    fhir:MolecularSequence.quality.method [ CodeableConcept ]; # 0..1 Method to get quality
    fhir:MolecularSequence.quality.truthTP [ decimal ]; # 0..1 True positives from the perspective of the truth data
    fhir:MolecularSequence.quality.queryTP [ decimal ]; # 0..1 True positives from the perspective of the query data
    fhir:MolecularSequence.quality.truthFN [ decimal ]; # 0..1 False negatives
    fhir:MolecularSequence.quality.queryFP [ decimal ]; # 0..1 False positives
    fhir:MolecularSequence.quality.gtFP [ decimal ]; # 0..1 False positives where the non-REF alleles in the Truth and
    Query Call Sets match
    fhir:MolecularSequence.quality.precision [ decimal ]; # 0..1 Precision of comparison
    fhir:MolecularSequence.quality.recall [ decimal ]; # 0..1 Recall of comparison
    fhir:MolecularSequence.quality.fScore [ decimal ]; # 0..1 F-score
    fhir:MolecularSequence.quality.roc [ # 0..1 Receiver Operator Characteristic (ROC) Curve
      fhir:MolecularSequence.quality.roc.score [ integer ], ... ; # 0..* Genotype quality score
      fhir:MolecularSequence.quality.roc.numTP [ integer ], ... ; # 0..* Roc score true positive numbers
      fhir:MolecularSequence.quality.roc.numFP [ integer ], ... ; # 0..* Roc score false positive numbers
      fhir:MolecularSequence.quality.roc.numFN [ integer ], ... ; # 0..* Roc score false negative numbers
      fhir:MolecularSequence.quality.roc.precision [ decimal ], ... ; # 0..* Precision of the GQ score
      fhir:MolecularSequence.quality.roc.sensitivity [ decimal ], ... ; # 0..* Sensitivity of the GQ score
      fhir:MolecularSequence.quality.roc.fMeasure [ decimal ], ... ; # 0..* FScore of the GQ score
    ];
  ], ...;
  fhir:MolecularSequence.readCoverage [ integer ]; # 0..1 Average number of reads representing a given nucleotide in the
     reconstructed sequence
  fhir:MolecularSequence.repository [ # 0..* External repository which contains detailed report related with observedSeq
     in this resource
    fhir:MolecularSequence.repository.type [ code ]; # 1..1 directlink | openapi | login | oauth | other
    fhir:MolecularSequence.repository.url [ uri ]; # 0..1 URI of the repository
    fhir:MolecularSequence.repository.name [ string ]; # 0..1 Repository's name
    fhir:MolecularSequence.repository.datasetId [ string ]; # 0..1 Id of the dataset that used to call for dataset in
    repository
    fhir:MolecularSequence.repository.variantsetId [ string ]; # 0..1 Id of the variantset that used to call for
    variantset in repository
    fhir:MolecularSequence.repository.readsetId [ string ]; # 0..1 Id of the read
  ], ...;
  fhir:MolecularSequence.pointer [ Reference(MolecularSequence) ], ... ; # 0..* Pointer to next atomic sequence
  fhir:MolecularSequence.structureVariant [ # 0..* Structural variant
    fhir:MolecularSequence.structureVariant.variantType [ CodeableConcept ]; # 0..1 Structural variant change type
    fhir:MolecularSequence.structureVariant.exact [ boolean ]; # 0..1 Does the structural variant have base pair
    resolution breakpoints?
    fhir:MolecularSequence.structureVariant.length [ integer ]; # 0..1 Structural variant length
    fhir:MolecularSequence.structureVariant.outer [ # 0..1 Structural variant outer
      fhir:MolecularSequence.structureVariant.outer.start [ integer ]; # 0..1 Structural variant outer start
      fhir:MolecularSequence.structureVariant.outer.end [ integer ]; # 0..1 Structural variant outer end
    ];
    fhir:MolecularSequence.structureVariant.inner [ # 0..1 Structural variant inner
      fhir:MolecularSequence.structureVariant.inner.start [ integer ]; # 0..1 Structural variant inner start
      fhir:MolecularSequence.structureVariant.inner.end [ integer ]; # 0..1 Structural variant inner end
    ];
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.6.3.1 Terminology Bindings

Path | Definition | Type | Reference
MolecularSequence.type | Type if a sequence -- DNA, RNA, or amino acid sequence. | Required | sequenceType
MolecularSequence.referenceSeq.chromosome | Chromosome number for human. | Example | chromosome-human
MolecularSequence.referenceSeq.orientation | Type for orientation. | Required | orientationType
MolecularSequence.referenceSeq.referenceSeqId | Reference identifier. | Example | ENSEMBL
MolecularSequence.referenceSeq.strand | Type for strand. | Required | strandType
MolecularSequence.quality.type | Type for quality report. | Required | qualityType
MolecularSequence.quality.standardSequence | Reference identifier of the sequence that used to mark the quality of tested samples. | Example | FDA-StandardSequence
MolecularSequence.quality.method | The method used to evaluate the numerical quality of the observed sequence. | Example | FDA-Method
MolecularSequence.repository.type | Type for access of external URI. | Required | repositoryType
MolecularSequence.structureVariant.variantType | DNA change type. | Required | http://loinc.org/vs/LL379-9

10.6.3.2 Constraints

id | Level | Location | Description | Expression
msq-3 | Rule | (base) | Only 0 and 1 are valid for coordinateSystem | coordinateSystem = 1 or coordinateSystem = 0
msq-5 | Rule | MolecularSequence.referenceSeq | GenomeBuild and chromosome must be both contained if either one of them is contained | (chromosome.empty() and genomeBuild.empty()) or (chromosome.exists() and genomeBuild.exists())
msq-6 | Rule | MolecularSequence.referenceSeq | Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString; | (genomeBuild.count()+referenceSeqId.count()+ referenceSeqPointer.count()+ referenceSeqString.count()) = 1

10.6.4 Notes:

10.6.4.1 MolecularSequence Coordinate System

When saving the variant information, the nucleic acid will be numbered with order. Some files are using 0-based coordinates (e.g. BCD file format) while some files are using 1-based coordinates (e.g. VCF file format). The element coordinateSystem in MolecularSequence resource contains this information.

MolecularSequence.coordinateSystem constraints within two possible values: 0 for 0-based system, which will mark the sequence from number 0, while 1 for 1-based system, which will begin marking the first position with number 1. The significant difference between two system is the end position. In 0-based system, the end position is exclusive, which means the last position will not be contained in the sequence window while in 1-based system, the end position is inclusive , which means the last position is included in the sequence window. Note both systems has an inclusive start position.

For example, ACGTGCAT will be numbered from 1 to 8 in 1-based system and will be numbered from 0 to 8 in 0-based system to mark flanks (i.e. place between two Nucleotide). So the interval [3,5] in 1-based system is GTG while interval [2,5) in 0-based system is same segment GTG.

10.6.4.2 Choice of Strand

There are lots of definition concerning with the Directionality of DNA or RNA. Here we are using referenceSeq.orientation and referenceSeq.strand. orientation represents the sense of the sequence, which has different meanings depending on the MolecularSequence.type. strand represents the sequence writing order. Watson strand refers to 5' to 3' top strand (5' -> 3'), whereas Crick strand refers to 5' to 3' bottom strand (3' <- 5').

10.6.4.3 String usage for reference sequence and observed sequence

We hope that string of observedSeq can be constrained more than just any normal string but with notation tables. Here we present what the nucleotide acid string should be constrained within the range:

A --> adenosine | M --> A C (amino) | U --> uridine | H --> A C T | V --> G C A
C --> cytidine | S --> G C (strong) | D --> G A T | K --> G T (keto)
G --> guanine | W --> A T (weak) | R --> G A (purine) | N --> A G C T (any)
T --> thymidine | B --> G T C | Y --> T C (pyrimidine) | - --> gap of indeterminate length

A alanine | P proline | B aspartate or asparagine | Q glutamine
C cystine | R arginine | D aspartate | S serine
E glutamate | T threonine | F phenylalanine | U selenocysteine
G glycine | V valine | H histidine | W tryptophan
I isoleucine | Y tyrosine | K lysine | Z glutamate or glutamine
L leucine | X any | M methionine | * translation stop
N asparagine | - gap of indeterminate length

10.6.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

chromosome | token | Chromosome number of the reference sequence | MolecularSequence.referenceSeq.chromosome

chromosome-variant-coordinate | composite | Search parameter by chromosome and variant coordinate. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as `chromosome-variant-coordinate=1$lt345$gt123`, this means it will search for the MolecularSequence resource with variants on chromosome 1 and with position >123 and <345, where in 1-based system resource, all strings within region 1:124-344 will be revealed, while in 0-based system resource, all strings within region 1:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | On MolecularSequence.variant: chromosome: %resource.referenceSeq.chromosome variant-start: start variant-end: end

chromosome-window-coordinate | composite | Search parameter by chromosome and window. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as `chromosome-window-coordinate=1$lt345$gt123`, this means it will search for the MolecularSequence resource with a window on chromosome 1 and with position >123 and <345, where in 1-based system resource, all strings within region 1:124-344 will be revealed, while in 0-based system resource, all strings within region 1:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | On MolecularSequence.referenceSeq: chromosome: chromosome window-start: windowStart window-end: windowEnd

identifier | token | The unique identity for a particular sequence | MolecularSequence.identifier

patient | reference | The subject that the observation is about | MolecularSequence.patient (Patient)

referenceseqid | token | Reference Sequence of the sequence | MolecularSequence.referenceSeq.referenceSeqId

referenceseqid-variant-coordinate | composite | Search parameter by reference sequence and variant coordinate. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as `referenceSeqId-variant-coordinate=NC_000001.11$lt345$gt123`, this means it will search for the MolecularSequence resource with variants on NC_000001.11 and with position >123 and <345, where in 1-based system resource, all strings within region NC_000001.11:124-344 will be revealed, while in 0-based system resource, all strings within region NC_000001.11:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | On MolecularSequence.variant: referenceseqid: %resource.referenceSeq.referenceSeqId variant-start: start variant-end: end

referenceseqid-window-coordinate | composite | Search parameter by reference sequence and window. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as `referenceSeqId-window-coordinate=NC_000001.11$lt345$gt123`, this means it will search for the MolecularSequence resource with a window on NC_000001.11 and with position >123 and <345, where in 1-based system resource, all strings within region NC_000001.11:124-344 will be revealed, while in 0-based system resource, all strings within region NC_000001.11:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | On MolecularSequence.referenceSeq: referenceseqid: referenceSeqId window-start: windowStart window-end: windowEnd

type | token | Amino Acid Sequence/ DNA Sequence / RNA Sequence | MolecularSequence.type

variant-end | number | End position (0-based exclusive, which menas the acid at this position will not be included, 1-based inclusive, which means the acid at this position will be included) of the variant. | MolecularSequence.variant.end

variant-start | number | Start position (0-based inclusive, 1-based inclusive, that means the nucleic acid or amino acid at this position will be included) of the variant. | MolecularSequence.variant.start

window-end | number | End position (0-based exclusive, which menas the acid at this position will not be included, 1-based inclusive, which means the acid at this position will be included) of the reference sequence. | MolecularSequence.referenceSeq.windowEnd

window-start | number | Start position (0-based inclusive, 1-based inclusive, that means the nucleic acid or amino acid at this position will be included) of the reference sequence. | MolecularSequence.referenceSeq.windowStart

10.6.6 Resource MolecularSequence - Examples

URL: https://hl7.org/fhir/R4/molecularsequence-examples.html

10.6.6 Resource MolecularSequence - Examples

Clinical Genomics Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format

Simple sequence example | example | XML | JSON | Turtle

MolecularSequence example from precisionFDA | fda-example | XML | JSON | Turtle

Example of representing complex variants in MolecularSequence resource | sequence-complex-variant | XML | JSON | Turtle

Example of a single varaint on a reference MolecularSequence related to a haplotype observation in PGx example | example-pgx-1 | XML | JSON | Turtle

Example of another single varaint on a reference MolecularSequence related to a haplotype observation in PGx example, which will form with the other haplotype data to be a diplotype observation. | example-pgx-2 | XML | JSON | Turtle

Example of a TPMT SNP data that support a haplotype observation | example-TPMT-one | XML | JSON | Turtle

Example of another TPMT SNP data that support a haplotype observation | example-TPMT-two | XML | JSON | Turtle

MolecularSequence Graphic example-1 | graphic-example-1 | XML | JSON | Turtle

MolecularSequence Graphic example-2 | graphic-example-2 | XML | JSON | Turtle

MolecularSequence Graphic example-3 | graphic-example-3 | XML | JSON | Turtle

MolecularSequence Graphic example-4 | graphic-example-4 | XML | JSON | Turtle
MolecularSequence Graphic example-5 | graphic-example-5 | XML | JSON | Turtle
Example of a record with 0-based coordinatesystem | coord-0-base | XML | JSON | Turtle
Example of a record with 1-based coordinatesystem | coord-1-base | XML | JSON | Turtle
MolecularSequence example for vcf comparison on precisionFDA | fda-vcf-comparison | XML | JSON | Turtle
MolecularSequence example for vcf and Hap.py comparison on precisionFDA | fda-vcfeval-comparison | XML | JSON | Turtle
MolecularSequence example for patient with breast cancer gene variant | breastcancer | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.6.7 Resource MolecularSequence - Detailed Descriptions

URL: https://hl7.org/fhir/R4/molecularsequence-definitions.html

10.6.7 Resource MolecularSequence - Detailed Descriptions

Clinical Genomics Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the MolecularSequence resource.

MolecularSequence
Element Id | MolecularSequence
Definition | Raw data describing a biological sequence.
Cardinality | 0..*
Type | DomainResource
Summary | true
Invariants | Defined on this element msq-3 Rule Only 0 and 1 are valid for coordinateSystem coordinateSystem = 1 or coordinateSystem = 0
Defined on this element
msq-3 | Rule | Only 0 and 1 are valid for coordinateSystem | coordinateSystem = 1 or coordinateSystem = 0
MolecularSequence.identifier
Element Id | MolecularSequence.identifier
Definition | A unique identifier for this particular sequence instance. This is a FHIR-defined id.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows sequences to be distinguished and referenced.
Summary | true
MolecularSequence.type
Element Id | MolecularSequence.type
Definition | Amino Acid Sequence/ DNA Sequence / RNA Sequence.
Cardinality | 0..1
Terminology Binding | sequenceType (Required)
Type | code
Summary | true
MolecularSequence.coordinateSystem
Element Id | MolecularSequence.coordinateSystem
Definition | Whether the sequence is numbered starting at 0 (0-based numbering or coordinates, inclusive start, exclusive end) or starting at 1 (1-based numbering, inclusive start and inclusive end).
Cardinality | 1..1
Type | integer
Summary | true
MolecularSequence.patient
Element Id | MolecularSequence.patient
Definition | The patient whose sequencing results are described by this resource.
Cardinality | 0..1
Type | Reference(Patient)
Summary | true
MolecularSequence.specimen
Element Id | MolecularSequence.specimen
Definition | Specimen used for sequencing.
Cardinality | 0..1
Type | Reference(Specimen)
Summary | true
MolecularSequence.device
Element Id | MolecularSequence.device
Definition | The method for sequencing, for example, chip information.
Cardinality | 0..1
Type | Reference(Device)
Summary | true
MolecularSequence.performer
Element Id | MolecularSequence.performer
Definition | The organization or lab that should be responsible for this result.
Cardinality | 0..1

Type | Reference(Organization)
Summary | true
MolecularSequence.quantity
Element Id | MolecularSequence.quantity
Definition | The number of copies of the sequence of interest. (RNASeq).
Cardinality | 0..1
Type | Quantity
Summary | true
MolecularSequence.referenceSeq
Element Id | MolecularSequence.referenceSeq
Definition | A sequence that is used as a reference to describe variants that are present in a sequence analyzed.
Cardinality | 0..1
Summary | true
Invariants | Defined on this element msq-5 Rule GenomeBuild and chromosome must be both contained if either one of them is contained (chromosome.empty() and genomeBuild.empty()) or (chromosome.exists() and genomeBuild.exists()) msq-6 Rule Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString; (genomeBuild.count()+referenceSeqId.count()+ referenceSeqPointer.count()+ referenceSeqString.count()) = 1
Defined on this element
msq-5 | Rule | GenomeBuild and chromosome must be both contained if either one of them is contained | (chromosome.empty() and genomeBuild.empty()) or (chromosome.exists() and genomeBuild.exists())
msq-6 | Rule | Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString; | (genomeBuild.count()+referenceSeqId.count()+ referenceSeqPointer.count()+ referenceSeqString.count()) = 1
MolecularSequence.referenceSeq.chromosome
Element Id | MolecularSequence.referenceSeq.chromosome
Definition | Structural unit composed of a nucleic acid molecule which controls its own replication through the interaction of specific proteins at one or more origins of replication (SO:0000340).
Cardinality | 0..1
Terminology Binding | chromosome-human (Example)
Type | CodeableConcept
Summary | true
MolecularSequence.referenceSeq.genomeBuild
Element Id | MolecularSequence.referenceSeq.genomeBuild
Definition | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'. Version number must be included if a versioned release of a primary build was used.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.referenceSeq.orientation
Element Id | MolecularSequence.referenceSeq.orientation
Definition | A relative reference to a DNA strand based on gene orientation. The strand that contains the open reading frame of the gene is the "sense" strand, and the opposite complementary strand is the "antisense" strand.
Cardinality | 0..1
Terminology Binding | orientationType (Required)
Type | code
Summary | true
MolecularSequence.referenceSeq.referenceSeqId
Element Id | MolecularSequence.referenceSeq.referenceSeqId
Definition | Reference identifier of reference sequence submitted to NCBI. It must match the type in the MolecularSequence.type field. For example, the prefix, “NG_” identifies reference sequence for genes, “NM_” for messenger RNA transcripts, and “NP_” for amino acid sequences.
Cardinality | 0..1
Terminology Binding | E n s e m b l (Example)
Type | CodeableConcept
Summary | true
MolecularSequence.referenceSeq.referenceSeqPointer
Element Id | MolecularSequence.referenceSeq.referenceSeqPointer
Definition | A pointer to another MolecularSequence entity as reference sequence.
Cardinality | 0..1
Type | Reference(MolecularSequence)
Summary | true
MolecularSequence.referenceSeq.referenceSeqString
Element Id | MolecularSequence.referenceSeq.referenceSeqString
Definition | A string like "ACGT".
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.referenceSeq.strand
Element Id | MolecularSequence.referenceSeq.strand
Definition | An absolute reference to a strand. The Watson strand is the strand whose 5'-end is on the short arm of the chromosome, and the Crick strand as the one whose 5'-end is on the long arm.
Cardinality | 0..1

Terminology Binding | strandType (Required)
Type | code
Summary | true
MolecularSequence.referenceSeq.windowStart
Element Id | MolecularSequence.referenceSeq.windowStart
Definition | Start position of the window on the reference sequence. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.referenceSeq.windowEnd
Element Id | MolecularSequence.referenceSeq.windowEnd
Definition | End position of the window on the reference sequence. If the coordinate system is 0-based then end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.variant
Element Id | MolecularSequence.variant
Definition | The definition of variant here originates from Sequence ontology (variant_of). This element can represent amino acid or nucleic sequence change(including insertion,deletion,SNP,etc.) It can represent some complex mutation or segment variation with the assist of CIGAR string.
Cardinality | 0..*
Summary | true
MolecularSequence.variant.start
Element Id | MolecularSequence.variant.start
Definition | Start position of the variant on the reference sequence. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.variant.end
Element Id | MolecularSequence.variant.end
Definition | End position of the variant on the reference sequence. If the coordinate system is 0-based then end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.variant.observedAllele
Element Id | MolecularSequence.variant.observedAllele
Definition | An allele is one of a set of coexisting sequence variants of a gene (SO:0001023). Nucleotide(s)/amino acids from start position of sequence to stop position of sequence on the positive (+) strand of the observed sequence. When the sequence type is DNA, it should be the sequence on the positive (+) strand. This will lay in the range between variant.start and variant.end.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.variant.referenceAllele
Element Id | MolecularSequence.variant.referenceAllele
Definition | An allele is one of a set of coexisting sequence variants of a gene (SO:0001023). Nucleotide(s)/amino acids from start position of sequence to stop position of sequence on the positive (+) strand of the reference sequence. When the sequence type is DNA, it should be the sequence on the positive (+) strand. This will lay in the range between variant.start and variant.end.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.variant.cigar
Element Id | MolecularSequence.variant.cigar
Definition | Extended CIGAR string for aligning the sequence with reference bases. See detailed documentation here .
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.variant.variantPointer
Element Id | MolecularSequence.variant.variantPointer
Definition | A pointer to an Observation containing variant information.
Cardinality | 0..1
Type | Reference(Observation)
Summary | true
MolecularSequence.observedSeq
Element Id | MolecularSequence.observedSeq
Definition | Sequence that was observed. It is the result marked by referenceSeq along with variant records on referenceSeq. This shall start from referenceSeq.windowStart and end by referenceSeq.windowEnd.
Cardinality | 0..1
Type | string
Summary | true

MolecularSequence.quality
Element Id | MolecularSequence.quality
Definition | An experimental feature attribute that defines the quality of the feature in a quantitative way, such as a phred quality score (SO:0001686).
Cardinality | 0..*
Summary | true
MolecularSequence.quality.type
Element Id | MolecularSequence.quality.type
Definition | INDEL / SNP / Undefined variant.
Cardinality | 1..1
Terminology Binding | qualityType (Required)
Type | code
Summary | true
MolecularSequence.quality.standardSequence
Element Id | MolecularSequence.quality.standardSequence
Definition | Gold standard sequence used for comparing against.
Cardinality | 0..1
Terminology Binding | F d a- standard sequence (Example)
Type | CodeableConcept
Summary | true
MolecularSequence.quality.start
Element Id | MolecularSequence.quality.start
Definition | Start position of the sequence. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.quality.end
Element Id | MolecularSequence.quality.end
Definition | End position of the sequence. If the coordinate system is 0-based then end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.quality.score
Element Id | MolecularSequence.quality.score
Definition | The score of an experimentally derived feature such as a p-value (SO:0001685).
Cardinality | 0..1
Type | Quantity
Summary | true
MolecularSequence.quality.method
Element Id | MolecularSequence.quality.method
Definition | Which method is used to get sequence quality.
Cardinality | 0..1
Terminology Binding | F d a- method (Example)
Type | CodeableConcept
Summary | true
MolecularSequence.quality.truthTP
Element Id | MolecularSequence.quality.truthTP
Definition | True positives, from the perspective of the truth data, i.e. the number of sites in the Truth Call Set for which there are paths through the Query Call Set that are consistent with all of the alleles at this site, and for which there is an accurate genotype call for the event.
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.queryTP
Element Id | MolecularSequence.quality.queryTP
Definition | True positives, from the perspective of the query data, i.e. the number of sites in the Query Call Set for which there are paths through the Truth Call Set that are consistent with all of the alleles at this site, and for which there is an accurate genotype call for the event.
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.truthFN
Element Id | MolecularSequence.quality.truthFN
Definition | False negatives, i.e. the number of sites in the Truth Call Set for which there is no path through the Query Call Set that is consistent with all of the alleles at this site, or sites for which there is an inaccurate genotype call for the event. Sites with correct variant but incorrect genotype are counted here.
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.queryFP
Element Id | MolecularSequence.quality.queryFP

Definition | False positives, i.e. the number of sites in the Query Call Set for which there is no path through the Truth Call Set that is consistent with this site. Sites with correct variant but incorrect genotype are counted here.
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.gtFP
Element Id | MolecularSequence.quality.gtFP
Definition | The number of false positives where the non-REF alleles in the Truth and Query Call Sets match (i.e. cases where the truth is 1/1 and the query is 0/1 or similar).
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.precision
Element Id | MolecularSequence.quality.precision
Definition | QUERY.TP / (QUERY.TP + QUERY.FP).
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.recall
Element Id | MolecularSequence.quality.recall
Definition | TRUTH.TP / (TRUTH.TP + TRUTH.FN).
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.fScore
Element Id | MolecularSequence.quality.fScore
Definition | Harmonic mean of Recall and Precision, computed as: 2 * precision * recall / (precision + recall).
Cardinality | 0..1
Type | decimal
Summary | true
MolecularSequence.quality.roc
Element Id | MolecularSequence.quality.roc
Definition | Receiver Operator Characteristic (ROC) Curve to give sensitivity/specificity tradeoff.
Cardinality | 0..1
Summary | true
MolecularSequence.quality.roc.score
Element Id | MolecularSequence.quality.roc.score
Definition | Invidual data point representing the GQ (genotype quality) score threshold.
Cardinality | 0..*
Type | integer
Summary | true
MolecularSequence.quality.roc.numTP
Element Id | MolecularSequence.quality.roc.numTP
Definition | The number of true positives if the GQ score threshold was set to "score" field value.
Cardinality | 0..*
Type | integer
Summary | true
MolecularSequence.quality.roc.numFP
Element Id | MolecularSequence.quality.roc.numFP
Definition | The number of false positives if the GQ score threshold was set to "score" field value.
Cardinality | 0..*
Type | integer
Summary | true
MolecularSequence.quality.roc.numFN
Element Id | MolecularSequence.quality.roc.numFN
Definition | The number of false negatives if the GQ score threshold was set to "score" field value.
Cardinality | 0..*
Type | integer
Summary | true
MolecularSequence.quality.roc.precision
Element Id | MolecularSequence.quality.roc.precision
Definition | Calculated precision if the GQ score threshold was set to "score" field value.
Cardinality | 0..*
Type | decimal
Summary | true
MolecularSequence.quality.roc.sensitivity
Element Id | MolecularSequence.quality.roc.sensitivity
Definition | Calculated sensitivity if the GQ score threshold was set to "score" field value.
Cardinality | 0..*
Type | decimal
Summary | true
MolecularSequence.quality.roc.fMeasure

Element Id | MolecularSequence.quality.roc.fMeasure
Definition | Calculated fScore if the GQ score threshold was set to "score" field value.
Cardinality | 0..*
Type | decimal
Summary | true
MolecularSequence.readCoverage
Element Id | MolecularSequence.readCoverage
Definition | Coverage (read depth or depth) is the average number of reads representing a given nucleotide in the reconstructed sequence.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.repository
Element Id | MolecularSequence.repository
Definition | Configurations of the external repository. The repository shall store target's observedSeq or records related with target's observedSeq.
Cardinality | 0..*
Summary | true
MolecularSequence.repository.type
Element Id | MolecularSequence.repository.type
Definition | Click and see / RESTful API / Need login to see / RESTful API with authentication / Other ways to see resource.
Cardinality | 1..1
Terminology Binding | repositoryType (Required)
Type | code
Summary | true
MolecularSequence.repository.url
Element Id | MolecularSequence.repository.url
Definition | URI of an external repository which contains further details about the genetics data.
Cardinality | 0..1
Type | uri
Summary | true
MolecularSequence.repository.name
Element Id | MolecularSequence.repository.name
Definition | URI of an external repository which contains further details about the genetics data.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.repository.datasetId
Element Id | MolecularSequence.repository.datasetId
Definition | Id of the variant in this external repository. The server will understand how to use this id to call for more info about datasets in external repository.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.repository.variantsetId
Element Id | MolecularSequence.repository.variantsetId
Definition | Id of the variantset in this external repository. The server will understand how to use this id to call for more info about variantsets in external repository.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.repository.readsetId
Element Id | MolecularSequence.repository.readsetId
Definition | Id of the read in this external repository.
Cardinality | 0..1
Type | string
Summary | true
MolecularSequence.pointer
Element Id | MolecularSequence.pointer
Definition | Pointer to next atomic sequence which at most contains one variant.
Cardinality | 0..*
Type | Reference(MolecularSequence)
Summary | true
MolecularSequence.structureVariant
Element Id | MolecularSequence.structureVariant
Definition | Information about chromosome structure variation.
Cardinality | 0..*
Summary | true
MolecularSequence.structureVariant.variantType
Element Id | MolecularSequence.structureVariant.variantType
Definition | Information about chromosome structure variation DNA change type.
Cardinality | 0..1
Terminology Binding | LOINC LL379-9 answerlist :

Type | CodeableConcept
Summary | true
MolecularSequence.structureVariant.exact
Element Id | MolecularSequence.structureVariant.exact
Definition | Used to indicate if the outer and inner start-end values have the same meaning.
Cardinality | 0..1
Type | boolean
Summary | true
MolecularSequence.structureVariant.length
Element Id | MolecularSequence.structureVariant.length
Definition | Length of the variant chromosome.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.structureVariant.outer
Element Id | MolecularSequence.structureVariant.outer
Definition | Structural variant outer.
Cardinality | 0..1
Summary | true
MolecularSequence.structureVariant.outer.start
Element Id | MolecularSequence.structureVariant.outer.start
Definition | Structural variant outer start. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.structureVariant.outer.end
Element Id | MolecularSequence.structureVariant.outer.end
Definition | Structural variant outer end. If the coordinate system is 0-based then end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.structureVariant.inner
Element Id | MolecularSequence.structureVariant.inner
Definition | Structural variant inner.
Cardinality | 0..1
Summary | true
MolecularSequence.structureVariant.inner.start
Element Id | MolecularSequence.structureVariant.inner.start
Definition | Structural variant inner start. If the coordinate system is either 0-based or 1-based, then start position is inclusive.
Cardinality | 0..1
Type | integer
Summary | true
MolecularSequence.structureVariant.inner.end
Element Id | MolecularSequence.structureVariant.inner.end
Definition | Structural variant inner end. If the coordinate system is 0-based then end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last position.
Cardinality | 0..1
Type | integer
Summary | true

10.6.8 Resource MolecularSequence - Mappings

URL: https://hl7.org/fhir/R4/molecularsequence-mappings.html

10.6.8 Resource MolecularSequence - Mappings

Clinical Genomics Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Mappings for the molecularsequence resource (see Mappings to Other Standards for further information & status).

10.6.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MolecularSequence
identifier | FiveWs.identifier

10.6.9 Resource MolecularSequence - Extensions & Profiles

URL: https://hl7.org/fhir/R4/molecularsequence-profiles.html

10.6.9 Resource MolecularSequence - Extensions & Profiles

Clinical Genomics Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the MolecularSequence resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.6.9.1 Profiles

No Profiles defined for this resource

10.6.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

observation-geneticsPhaseSet | PhaseSet | for Observation-genetics
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.6.9.3 Search Extensions

No Search Extensions defined for this resource

10.6.10 Resource MolecularSequence - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/molecularsequence-version-maps.html

No validation errors - all conversions are clean

10.6.10 Resource MolecularSequence - R3/R4 Conversions

Clinical Genomics Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for MolecularSequence.

Functional status for this map: 14 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

10.6.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Sequence3to4" = "R3 to R4 Conversions for Sequence"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Sequence" alias SequenceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MolecularSequence" alias MolecularSequence as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Sequence(source src : SequenceR3, target tgt : MolecularSequence) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.type -> tgt.type;
  src.coordinateSystem -> tgt.coordinateSystem;
  src.patient -> tgt.patient;
  src.specimen -> tgt.specimen;
  src.device -> tgt.device;
  src.performer -> tgt.performer;
  src.quantity -> tgt.quantity;
  src.referenceSeq as vs0 -> tgt.referenceSeq as vt0 then referenceSeq(vs0, vt0);
  src.variant as vs0 -> tgt.variant as vt0 then variant(vs0, vt0);
  src.observedSeq -> tgt.observedSeq;
  src.quality as vs0 -> tgt.quality as vt0 then quality(vs0, vt0);
  src.readCoverage -> tgt.readCoverage;
  src.repository as vs0 -> tgt.repository as vt0 then repository(vs0, vt0);
  src.pointer -> tgt.pointer;
}

group referenceSeq(source src, target tgt) extends BackboneElement {
  src.chromosome -> tgt.chromosome;
  src.genomeBuild -> tgt.genomeBuild;
  src.referenceSeqId -> tgt.referenceSeqId;
  src.referenceSeqPointer -> tgt.referenceSeqPointer;
  src.referenceSeqString -> tgt.referenceSeqString;
  src.strand where value = 1 -> tgt.strand = 'watson';
```

```
  src.strand where value = -1 -> tgt.strand = 'crick';
  src.windowStart -> tgt.windowStart;
  src.windowEnd -> tgt.windowEnd;
}

group variant(source src, target tgt) extends BackboneElement {
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.observedAllele -> tgt.observedAllele;
  src.referenceAllele -> tgt.referenceAllele;
  src.cigar -> tgt.cigar;
  src.variantPointer -> tgt.variantPointer;
}

group quality(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.standardSequence -> tgt.standardSequence;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.score -> tgt.score;
  src.method -> tgt.method;
  src.truthTP -> tgt.truthTP;
  src.queryTP -> tgt.queryTP;
  src.truthFN -> tgt.truthFN;
  src.queryFP -> tgt.queryFP;
  src.gtFP -> tgt.gtFP;
  src.precision -> tgt.precision;
  src.recall -> tgt.recall;
  src.fScore -> tgt.fScore;
}

group repository(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.url -> tgt.url;
  src.name -> tgt.name;
  src.datasetId -> tgt.datasetId;
  src.variantsetId -> tgt.variantsetId;
  src.readsetId -> tgt.readsetId;
}
```

10.6.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Sequence4to3" = "R4 to R3 Conversion for Sequence"

uses "http://hl7.org/fhir/StructureDefinition/MolecularSequence" alias Sequence as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Sequence" alias SequenceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Sequence(source src : MolecularSequence, target tgt : SequenceR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.type -> tgt.type;
  src.coordinateSystem -> tgt.coordinateSystem;
  src.patient -> tgt.patient;
  src.specimen -> tgt.specimen;
  src.device -> tgt.device;
  src.performer -> tgt.performer;
  src.quantity -> tgt.quantity;
  src.referenceSeq as vs0 -> tgt.referenceSeq as vt0 then referenceSeq(vs0, vt0);
  src.variant as vs0 -> tgt.variant as vt0 then variant(vs0, vt0);
  src.observedSeq -> tgt.observedSeq;
  src.quality as vs0 -> tgt.quality as vt0 then quality(vs0, vt0);
  src.readCoverage -> tgt.readCoverage;
  src.repository as vs0 -> tgt.repository as vt0 then repository(vs0, vt0);
  src.pointer -> tgt.pointer;
}

group referenceSeq(source src, target tgt) extends BackboneElement {
  src.chromosome -> tgt.chromosome;
  src.genomeBuild -> tgt.genomeBuild;
  src.referenceSeqId -> tgt.referenceSeqId;
  src.referenceSeqPointer -> tgt.referenceSeqPointer;
  src.referenceSeqString -> tgt.referenceSeqString;
  src.strand where value = 'watson' -> tgt.strand = 1;
  src.strand as vvv where value = 'crick' -> tgt.strand = create('') as vvv "1";
  src.windowStart -> tgt.windowStart;
  src.windowEnd -> tgt.windowEnd;
}

group variant(source src, target tgt) extends BackboneElement {
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.observedAllele -> tgt.observedAllele;
  src.referenceAllele -> tgt.referenceAllele;
  src.cigar -> tgt.cigar;
  src.variantPointer -> tgt.variantPointer;
}

group quality(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
```

```
  src.standardSequence -> tgt.standardSequence;
  src.start -> tgt.start;
  src.end -> tgt.end;
  src.score -> tgt.score;
  src.method -> tgt.method;
  src.truthTP -> tgt.truthTP;
  src.queryTP -> tgt.queryTP;
  src.truthFN -> tgt.truthFN;
  src.queryFP -> tgt.queryFP;
  src.gtFP -> tgt.gtFP;
  src.precision -> tgt.precision;
  src.recall -> tgt.recall;
  src.fScore -> tgt.fScore;
}

group repository(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.url -> tgt.url;
  src.name -> tgt.name;
  src.datasetId -> tgt.datasetId;
  src.variantsetId -> tgt.variantsetId;
  src.readsetId -> tgt.readsetId;
}
```

10.6.10.3 R4 Validation Errors

10.6.11 JSON Schema for MolecularSequence

URL: https://hl7.org/fhir/R4/molecularsequence.schema.json.html

10.6.11 JSON Schema for MolecularSequence

Clinical Genomics Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for MolecularSequence, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MolecularSequence",
  "$ref": "#/definitions/MolecularSequence",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MolecularSequence": {
      "description": "Raw data describing a biological sequence.",
      "properties": {
        "resourceType": {
          "description": "This is a MolecularSequence resource",
          "const": "MolecularSequence"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
```

```
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A unique identifier for this particular sequence instance. This is a FHIR-defined id.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "Amino Acid Sequence/ DNA Sequence / RNA Sequence.",
      "enum": [
        "aa",
        "dna",
        "rna"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "coordinateSystem": {
      "description": "Whether the sequence is numbered starting at 0 (0-based numbering or coordinates, inclusive
start, exclusive end) or starting at 1 (1-based numbering, inclusive start and inclusive end).",
      "$ref": "#/definitions/integer"
    },
    "_coordinateSystem": {
      "description": "Extensions for coordinateSystem",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "The patient whose sequencing results are described by this resource.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "specimen": {
      "description": "Specimen used for sequencing.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "device": {
      "description": "The method for sequencing, for example, chip information.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performer": {
      "description": "The organization or lab that should be responsible for this result.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "quantity": {
      "description": "The number of copies of the sequence of interest. (RNASeq).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "referenceSeq": {
      "description": "A sequence that is used as a reference to describe variants that are present in a sequence
analyzed.",
      "$ref": "#/definitions/MolecularSequence_ReferenceSeq"
    },
    "variant": {
      "description": "The definition of variant here originates from Sequence ontology
([variant_of](http://www.sequenceontology.org/browser/current_svn/term/variant_of)). This element can represent
amino acid or nucleic sequence change(including insertion,deletion,SNP,etc.)  It can represent some complex mutation
 or segment variation with the assist of CIGAR string.",
      "items": {
        "$ref": "#/definitions/MolecularSequence_Variant"
```

```
      },
      "type": "array"
    },
    "observedSeq": {
      "description": "Sequence that was observed. It is the result marked by referenceSeq along with variant records
 on referenceSeq. This shall start from referenceSeq.windowStart and end by referenceSeq.windowEnd.",
      "$ref": "#/definitions/string"
    },
    "_observedSeq": {
      "description": "Extensions for observedSeq",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "quality": {
      "description": "An experimental feature attribute that defines the quality of the feature in a quantitative
way, such as a phred quality score
([SO:0001686](http://www.sequenceontology.org/browser/current_svn/term/SO:0001686)).",
      "items": {
        "$ref": "#/definitions/MolecularSequence_Quality"
      },
      "type": "array"
    },
    "readCoverage": {
      "description": "Coverage (read depth or depth) is the average number of reads representing a given nucleotide
in the reconstructed sequence.",
      "$ref": "#/definitions/integer"
    },
    "_readCoverage": {
      "description": "Extensions for readCoverage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "repository": {
      "description": "Configurations of the external repository. The repository shall store target\u0027s
observedSeq or records related with target\u0027s observedSeq.",
      "items": {
        "$ref": "#/definitions/MolecularSequence_Repository"
      },
      "type": "array"
    },
    "pointer": {
      "description": "Pointer to next atomic sequence which at most contains one variant.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "structureVariant": {
      "description": "Information about chromosome structure variation.",
      "items": {
        "$ref": "#/definitions/MolecularSequence_StructureVariant"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"MolecularSequence_ReferenceSeq": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
    },
    "chromosome": {
      "description": "Structural unit composed of a nucleic acid molecule which controls its own replication through
 the interaction of specific proteins at one or more origins of replication
([SO:0000340](http://www.sequenceontology.org/browser/current_svn/term/SO:0000340)).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "genomeBuild": {
      "description": "The Genome Build used for reference, following GRCh build versions e.g. \u0027GRCh 37\u0027.
Version number must be included if a versioned release of a primary build was used.",
      "$ref": "#/definitions/string"
    },
    "_genomeBuild": {
      "description": "Extensions for genomeBuild",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "orientation": {
      "description": "A relative reference to a DNA strand based on gene orientation. The strand that contains the
open reading frame of the gene is the \"sense\" strand, and the opposite complementary strand is the \"antisense\"
strand.",
      "enum": [
        "sense",
        "antisense"
      ]
    },
    "_orientation": {
      "description": "Extensions for orientation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "referenceSeqId": {
      "description": "Reference identifier of reference sequence submitted to NCBI. It must match the type in the
MolecularSequence.type field. For example, the prefix, “NG_” identifies reference sequence for genes, “NM_” for
messenger RNA transcripts, and “NP_” for amino acid sequences.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "referenceSeqPointer": {
      "description": "A pointer to another MolecularSequence entity as reference sequence.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "referenceSeqString": {
      "description": "A string like \"ACGT\".",
      "$ref": "#/definitions/string"
    },
    "_referenceSeqString": {
      "description": "Extensions for referenceSeqString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "strand": {
      "description": "An absolute reference to a strand. The Watson strand is the strand whose 5\u0027-end is on the
 short arm of the chromosome, and the Crick strand as the one whose 5\u0027-end is on the long arm.",
      "enum": [
        "watson",
        "crick"
      ]
    },
    "_strand": {
      "description": "Extensions for strand",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "windowStart": {
      "description": "Start position of the window on the reference sequence. If the coordinate system is either
0-based or 1-based, then start position is inclusive.",
      "$ref": "#/definitions/integer"
    },
    "_windowStart": {
      "description": "Extensions for windowStart",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "windowEnd": {
      "description": "End position of the window on the reference sequence. If the coordinate system is 0-based then
 end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is inclusive
and includes the last position.",
      "$ref": "#/definitions/integer"
    },
    "_windowEnd": {
      "description": "Extensions for windowEnd",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_Variant": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
```

```
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "start": {
      "description": "Start position of the variant on the  reference sequence. If the coordinate system is either
0-based or 1-based, then start position is inclusive.",
      "$ref": "#/definitions/integer"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "end": {
      "description": "End position of the variant on the reference sequence. If the coordinate system is 0-based
then end is exclusive and does not include the last position. If the coordinate system is 1-base, then end is
inclusive and includes the last position.",
      "$ref": "#/definitions/integer"
    },
    "_end": {
      "description": "Extensions for end",
      "$ref": "Element.schema.json#/definitions/Element"

    },
    "observedAllele": {
      "description": "An allele is one of a set of coexisting sequence variants of a gene
([SO:0001023](http://www.sequenceontology.org/browser/current_svn/term/SO:0001023)).  Nucleotide(s)/amino acids from
 start position of sequence to stop position of sequence on the positive (+) strand of the observed  sequence. When
the sequence  type is DNA, it should be the sequence on the positive (+) strand. This will lay in the range between
variant.start and variant.end.",
      "$ref": "#/definitions/string"
    },
    "_observedAllele": {
      "description": "Extensions for observedAllele",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "referenceAllele": {
      "description": "An allele is one of a set of coexisting sequence variants of a gene
([SO:0001023](http://www.sequenceontology.org/browser/current_svn/term/SO:0001023)). Nucleotide(s)/amino acids from
start position of sequence to stop position of sequence on the positive (+) strand of the reference sequence. When
the sequence  type is DNA, it should be the sequence on the positive (+) strand. This will lay in the range between
variant.start and variant.end.",
      "$ref": "#/definitions/string"
    },
    "_referenceAllele": {
      "description": "Extensions for referenceAllele",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cigar": {
      "description": "Extended CIGAR string for aligning the sequence with reference bases. See detailed
documentation [here](http://support.illumina.com/help/SequencingAnalysisWorkflow/Content/Vault/Informatics/Sequencin
g_Analysis/CASAVA/swSEQ_mCA_ExtendedCIGARFormat.htm).",
      "$ref": "#/definitions/string"
    },
    "_cigar": {
      "description": "Extensions for cigar",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "variantPointer": {
      "description": "A pointer to an Observation containing variant information.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_Quality": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
```

```
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "INDEL / SNP / Undefined variant.",
      "enum": [
        "indel",
        "snp",
        "unknown"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "standardSequence": {
      "description": "Gold standard sequence used for comparing against.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "start": {
      "description": "Start position of the sequence. If the coordinate system is either 0-based or 1-based, then
start position is inclusive.",
      "$ref": "#/definitions/integer"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "end": {
      "description": "End position of the sequence. If the coordinate system is 0-based then end is exclusive and
does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last
position.",
      "$ref": "#/definitions/integer"
    },
    "_end": {
      "description": "Extensions for end",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "score": {
      "description": "The score of an experimentally derived feature such as a p-value
([SO:0001685](http://www.sequenceontology.org/browser/current_svn/term/SO:0001685)).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "method": {
      "description": "Which method is used to get sequence quality.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "truthTP": {
      "description": "True positives, from the perspective of the truth data, i.e. the number of sites in the Truth
Call Set for which there are paths through the Query Call Set that are consistent with all of the alleles at this
site, and for which there is an accurate genotype call for the event.",
      "$ref": "#/definitions/decimal"
    },
    "_truthTP": {
      "description": "Extensions for truthTP",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "queryTP": {
      "description": "True positives, from the perspective of the query data, i.e. the number of sites in the Query
Call Set for which there are paths through the Truth Call Set that are consistent with all of the alleles at this
site, and for which there is an accurate genotype call for the event.",
      "$ref": "#/definitions/decimal"
    },
    "_queryTP": {
      "description": "Extensions for queryTP",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "truthFN": {
      "description": "False negatives, i.e. the number of sites in the Truth Call Set for which there is no path
through the Query Call Set that is consistent with all of the alleles at this site, or sites for which there is an
inaccurate genotype call for the event. Sites with correct variant but incorrect genotype are counted here.",
      "$ref": "#/definitions/decimal"
    },
    "_truthFN": {
      "description": "Extensions for truthFN",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "queryFP": {
      "description": "False positives, i.e. the number of sites in the Query Call Set for which there is no path
through the Truth Call Set that is consistent with this site. Sites with correct variant but incorrect genotype are
counted here.",
      "$ref": "#/definitions/decimal"
    },
    "_queryFP": {
      "description": "Extensions for queryFP",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "gtFP": {
      "description": "The number of false positives where the non-REF alleles in the Truth and Query Call Sets match
 (i.e. cases where the truth is 1/1 and the query is 0/1 or similar).",
      "$ref": "#/definitions/decimal"
    },
    "_gtFP": {
      "description": "Extensions for gtFP",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "precision": {
      "description": "QUERY.TP / (QUERY.TP + QUERY.FP).",
      "$ref": "#/definitions/decimal"
    },
    "_precision": {
      "description": "Extensions for precision",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recall": {
      "description": "TRUTH.TP / (TRUTH.TP + TRUTH.FN).",
      "$ref": "#/definitions/decimal"
    },
    "_recall": {
      "description": "Extensions for recall",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "fScore": {
      "description": "Harmonic mean of Recall and Precision, computed as: 2 * precision * recall / (precision +
recall).",
      "$ref": "#/definitions/decimal"
    },
    "_fScore": {
      "description": "Extensions for fScore",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "roc": {
      "description": "Receiver Operator Characteristic (ROC) Curve  to give sensitivity/specificity tradeoff.",
      "$ref": "#/definitions/MolecularSequence_Roc"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_Roc": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
```

```
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "score": {
      "description": "Invidual data point representing the GQ (genotype quality) score threshold.",
      "items": {
        "$ref": "#/definitions/integer"
      },
      "type": "array"
    },
    "_score": {
      "description": "Extensions for score",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "numTP": {
      "description": "The number of true positives if the GQ score threshold was set to \"score\" field value.",
      "items": {
        "$ref": "#/definitions/integer"
      },
      "type": "array"
    },
    "_numTP": {
      "description": "Extensions for numTP",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "numFP": {
      "description": "The number of false positives if the GQ score threshold was set to \"score\" field value.",
      "items": {
        "$ref": "#/definitions/integer"
      },
      "type": "array"
    },
    "_numFP": {
      "description": "Extensions for numFP",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "numFN": {
      "description": "The number of false negatives if the GQ score threshold was set to \"score\" field value.",
      "items": {
        "$ref": "#/definitions/integer"
      },
      "type": "array"
    },
    "_numFN": {
      "description": "Extensions for numFN",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "precision": {
      "description": "Calculated precision if the GQ score threshold was set to \"score\" field value.",
      "items": {
        "$ref": "#/definitions/decimal"
      },
      "type": "array"
    },
    "_precision": {
      "description": "Extensions for precision",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "sensitivity": {
      "description": "Calculated sensitivity if the GQ score threshold was set to \"score\" field value.",
      "items": {
        "$ref": "#/definitions/decimal"
      },
      "type": "array"
    },
    "_sensitivity": {
      "description": "Extensions for sensitivity",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
```

```
    },
    "fMeasure": {
      "description": "Calculated fScore if the GQ score threshold was set to \"score\" field value.",
      "items": {
        "$ref": "#/definitions/decimal"
      },
      "type": "array"
    },
    "_fMeasure": {
      "description": "Extensions for fMeasure",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_Repository": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Click and see / RESTful API / Need login to see / RESTful API with authentication / Other ways
 to see resource.",
      "enum": [
        "directlink",
        "openapi",
        "login",
        "oauth",
        "other"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "url": {
      "description": "URI of an external repository which contains further details about the genetics data.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "URI of an external repository which contains further details about the genetics data.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "datasetId": {
      "description": "Id of the variant in this external repository. The server will understand how to use this id
to call for more info about datasets in external repository.",
      "$ref": "#/definitions/string"
    },
    "_datasetId": {
      "description": "Extensions for datasetId",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "variantsetId": {
      "description": "Id of the variantset in this external repository. The server will understand how to use this
id to call for more info about variantsets in external repository.",
      "$ref": "#/definitions/string"
    },
    "_variantsetId": {
      "description": "Extensions for variantsetId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "readsetId": {
      "description": "Id of the read in this external repository.",
      "$ref": "#/definitions/string"
    },
    "_readsetId": {
      "description": "Extensions for readsetId",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_StructureVariant": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "variantType": {
      "description": "Information about chromosome structure variation DNA change type.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "exact": {
      "description": "Used to indicate if the outer and inner start-end values have the same meaning.",
      "$ref": "#/definitions/boolean"
    },
    "_exact": {
      "description": "Extensions for exact",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "length": {
      "description": "Length of the variant chromosome.",
      "$ref": "#/definitions/integer"
    },
    "_length": {
      "description": "Extensions for length",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "outer": {
      "description": "Structural variant outer.",
      "$ref": "#/definitions/MolecularSequence_Outer"
    },
    "inner": {
      "description": "Structural variant inner.",
      "$ref": "#/definitions/MolecularSequence_Inner"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_Outer": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
```

```
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "start": {
      "description": "Structural variant outer start. If the coordinate system is either 0-based or 1-based, then
start position is inclusive.",
      "$ref": "#/definitions/integer"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "end": {
      "description": "Structural variant outer end. If the coordinate system is 0-based then end is exclusive and
does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last
position.",
      "$ref": "#/definitions/integer"
    },
    "_end": {
      "description": "Extensions for end",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"MolecularSequence_Inner": {
  "description": "Raw data describing a biological sequence.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "start": {
      "description": "Structural variant inner start. If the coordinate system is either 0-based or 1-based, then
start position is inclusive.",
      "$ref": "#/definitions/integer"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
        },
        "end": {
          "description": "Structural variant inner end. If the coordinate system is 0-based then end is exclusive and
    does not include the last position. If the coordinate system is 1-base, then end is inclusive and includes the last
    position.",
          "$ref": "#/definitions/integer"
        },
        "_end": {
          "description": "Extensions for end",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

10.6.12 Resource MolecularSequence - Maturity Dependency View

URL: https://hl7.org/fhir/R4/molecularsequence-dependencies.html

10.6.12 Resource MolecularSequence - Maturity Dependency View

Clinical Genomics Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Dependency Graph for MolecularSequence FMM level 1

Name | Card. | Type | Dependency Analysis
MolecularSequence | | DomainResource
identifier | 0..* | Identifier
type | 0..1 | code
coordinateSystem | 1..1 | integer
patient | 0..1 | Reference(Patient)
specimen | 0..1 | Reference(Specimen)
device | 0..1 | Reference(Device)
performer | 0..1 | Reference(Organization)
quantity | 0..1 | Quantity
referenceSeq | 0..1 | BackboneElement
chromosome | 0..1 | CodeableConcept
genomeBuild | 0..1 | string
orientation | 0..1 | code
referenceSeqId | 0..1 | CodeableConcept
referenceSeqPointer | 0..1 | Reference(MolecularSequence)
referenceSeqString | 0..1 | string
strand | 0..1 | code
windowStart | 0..1 | integer
windowEnd | 0..1 | integer
variant | 0..* | BackboneElement
start | 0..1 | integer
end | 0..1 | integer
observedAllele | 0..1 | string
referenceAllele | 0..1 | string
cigar | 0..1 | string
variantPointer | 0..1 | Reference(Observation)
observedSeq | 0..1 | string
quality | 0..* | BackboneElement
type | 1..1 | code
standardSequence | 0..1 | CodeableConcept
start | 0..1 | integer
end | 0..1 | integer
score | 0..1 | Quantity
method | 0..1 | CodeableConcept
truthTP | 0..1 | decimal
queryTP | 0..1 | decimal
truthFN | 0..1 | decimal
queryFP | 0..1 | decimal
gtFP | 0..1 | decimal
precision | 0..1 | decimal
recall | 0..1 | decimal
fScore | 0..1 | decimal
roc | 0..1 | BackboneElement
score | 0..* | integer
numTP | 0..* | integer
numFP | 0..* | integer
numFN | 0..* | integer

precision | 0..* | decimal
sensitivity | 0..* | decimal
fMeasure | 0..* | decimal
readCoverage | 0..1 | integer
repository | 0..* | BackboneElement
type | 1..1 | code
url | 0..1 | uri
name | 0..1 | string
datasetId | 0..1 | string
variantsetId | 0..1 | string
readsetId | 0..1 | string
pointer | 0..* | Reference(MolecularSequence)
structureVariant | 0..* | BackboneElement
variantType | 0..1 | CodeableConcept | Binding Error: Unable to resolve vs 'http://loinc.org/vs/LL379-9' to check dependencies
exact | 0..1 | boolean
length | 0..1 | integer
outer | 0..1 | BackboneElement
start | 0..1 | integer
end | 0..1 | integer
inner | 0..1 | BackboneElement
start | 0..1 | integer
end | 0..1 | integer

10.7 Resource Specimen - Content

URL: https://hl7.org/fhir/R4/specimen.html
Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

A sample to be used for analysis.

10.7.1 Scope and Usage

Any material sample:

• taken from a biological entity, living or dead

• taken from a physical object or the environment

Some specimens are biological and can contain one or more components including but not limited to cellular molecules, cells, tissues, organs, body fluids, embryos, and body excretory products (source: NCI Thesaurus , modified).

The specimen resource covers substances used for diagnostic and environmental testing. The focus of the specimen resource is the process for gathering, maintaining and processing the specimen as well as where the specimen originated. This is distinct from the use of Substance which is only used when these other aspects are not relevant.

10.7.2 Background and Context

The current definition of the specimen resource contains only basic information about specimen containers. It does not address the recursive nature of containers or the tracking of the location of a container within its parent container (for instance: a tube in a tray in a rack in a freezer). The frequency with which these elements are tracked may depend on the context of use; general lab, bio-banking, etc. Comments from reviewers on the appropriate scope for this resource, and the need for tracking related specimen management attributes, are welcomed.

This resource is referenced by DiagnosticReport, ImagingStudy, Media, MolecularSequence, Observation, ServiceRequest and itself

10.7.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Specimen | TU | | DomainResource | Sample for analysis Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | External Identifier
accessionIdentifier | Σ | 0..1 | Identifier | Identifier assigned by the lab
status | ?!Σ | 0..1 | code | available | unavailable | unsatisfactory | entered-in-error SpecimenStatus (Required)
type | Σ | 0..1 | CodeableConcept | Kind of material that forms the specimen v2 Specimen Type (Example)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Substance | Location) | Where the specimen came from. This may be from patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device
receivedTime | Σ | 0..1 | dateTime | The time when specimen was received for processing
parent | | 0..* | Reference(Specimen) | Specimen from which this specimen originated
request | | 0..* | Reference(ServiceRequest) | Why the specimen was collected
collection | | 0..1 | BackboneElement | Collection details
collector | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Who collected the specimen
collected[x] | Σ | 0..1 | | Collection time
collectedDateTime | | | dateTime
collectedPeriod | | | Period
duration | Σ | 0..1 | Duration | How long it took to collect specimen
quantity | | 0..1 | SimpleQuantity | The quantity of specimen collected
method | | 0..1 | CodeableConcept | Technique used to perform collection FHIR Specimen Collection Method (Example)
bodySite | | 0..1 | CodeableConcept | Anatomical collection site SNOMED CT Body Structures (Example)
fastingStatus[x] | Σ | 0..1 | | Whether or how long patient abstained from food and/or drink v2 Relevant Clincial Information (Extensible)
fastingStatusCodeableConcept | | | CodeableConcept
fastingStatusDuration | | | Duration
processing | | 0..* | BackboneElement | Processing and processing step details
description | | 0..1 | string | Textual description of procedure
procedure | | 0..1 | CodeableConcept | Indicates the treatment step applied to the specimen Specimen processing procedure (Example)
additive | | 0..* | Reference(Substance) | Material used in the processing step
time[x] | | 0..1 | | Date and time of specimen processing
timeDateTime | | | dateTime
timePeriod | | | Period
container | | 0..* | BackboneElement | Direct container of specimen (tube/slide, etc.)
identifier | Σ | 0..* | Identifier | Id for the container
description | | 0..1 | string | Textual description of the container
type | | 0..1 | CodeableConcept | Kind of container directly associated with specimen Specimen container (Example)
capacity | | 0..1 | SimpleQuantity | Container volume or size
specimenQuantity | | 0..1 | SimpleQuantity | Quantity of specimen within container
additive[x] | | 0..1 | | Additive associated with container v2 Additive (Example)
additiveCodeableConcept | | | CodeableConcept
additiveReference | | | Reference(Substance)
condition | Σ | 0..* | CodeableConcept | State of the specimen v2 Specimen Condition (Extensible)
note | | 0..* | Annotation | Comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Specimen xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifier --></identifier>
 <accessionIdentifier><!-- 0..1 Identifier Identifier assigned by the lab --></accessionIdentifier>
 <status value="[code]"/><!-- 0..1 available | unavailable | unsatisfactory | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of material that forms the specimen --></type>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Substance|Location) Where the specimen came from. This may be from
    patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device
    --></subject>
 <receivedTime value="[dateTime]"/><!-- 0..1 The time when specimen was received for processing -->
 <parent><!-- 0..* Reference(Specimen) Specimen from which this specimen originated --></parent>
 <request><!-- 0..* Reference(ServiceRequest) Why the specimen was collected --></request>
 <collection>  <!-- 0..1 Collection details -->
  <collector><!-- 0..1 Reference(Practitioner|PractitionerRole) Who collected the specimen --></collector>
  <collected[x]><!-- 0..1 dateTime|Period Collection time --></collected[x]>
  <duration><!-- 0..1 Duration How long it took to collect specimen --></duration>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) The quantity of specimen collected --></quantity>
  <method><!-- 0..1 CodeableConcept Technique used to perform collection --></method>
  <bodySite><!-- 0..1 CodeableConcept Anatomical collection site --></bodySite>
  <fastingStatus[x]><!-- 0..1 CodeableConcept|Duration Whether or how long patient abstained from food and/or drink
    --></fastingStatus[x]>
 </collection>
 <processing>  <!-- 0..* Processing and processing step details -->
  <description value="[string]"/><!-- 0..1 Textual description of procedure -->
  <procedure><!-- 0..1 CodeableConcept Indicates the treatment step  applied to the specimen --></procedure>
  <additive><!-- 0..* Reference(Substance) Material used in the processing step --></additive>
  <time[x]><!-- 0..1 dateTime|Period Date and time of specimen processing --></time[x]>
 </processing>
 <container>  <!-- 0..* Direct container of specimen (tube/slide, etc.) -->
  <identifier><!-- 0..* Identifier Id for the container --></identifier>
  <description value="[string]"/><!-- 0..1 Textual description of the container -->
  <type><!-- 0..1 CodeableConcept Kind of container directly associated with specimen --></type>
  <capacity><!-- 0..1 Quantity(SimpleQuantity) Container volume or size --></capacity>
```

```
  <specimenQuantity><!-- 0..1 Quantity(SimpleQuantity) Quantity of specimen within container --></specimenQuantity>
  <additive[x]><!-- 0..1 CodeableConcept|Reference(Substance) Additive associated with container --></additive[x]>
 </container>
 <condition><!-- 0..* CodeableConcept State of the specimen --></condition>
 <note><!-- 0..* Annotation Comments --></note>
</Specimen>
```

JSON Template

```
{
  "resourceType" : "Specimen",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifier
  "accessionIdentifier" : { Identifier }, // Identifier assigned by the lab
  "status" : "<code>", // available | unavailable | unsatisfactory | entered-in-error
  "type" : { CodeableConcept }, // Kind of material that forms the specimen
  "subject" : { Reference(Patient|Group|Device|Substance|Location) }, // Where the specimen came from. This may be from
    patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device
  "receivedTime" : "<dateTime>", // The time when specimen was received for processing
  "parent" : [{ Reference(Specimen) }], // Specimen from which this specimen originated
  "request" : [{ Reference(ServiceRequest) }], // Why the specimen was collected
  "collection" : { // Collection details
    "collector" : { Reference(Practitioner|PractitionerRole) }, // Who collected the specimen
    // collected[x]: Collection time. One of these 2:
    "collectedDateTime" : "<dateTime>",
    "collectedPeriod" : { Period },
    "duration" : { Duration }, // How long it took to collect specimen
    "quantity" : { Quantity(SimpleQuantity) }, // The quantity of specimen collected
    "method" : { CodeableConcept }, // Technique used to perform collection
    "bodySite" : { CodeableConcept }, // Anatomical collection site
    // fastingStatus[x]: Whether or how long patient abstained from food and/or drink. One of these 2:
    "fastingStatusCodeableConcept" : { CodeableConcept }
    "fastingStatusDuration" : { Duration }
  },
  "processing" : [{ // Processing and processing step details
    "description" : "<string>", // Textual description of procedure
    "procedure" : { CodeableConcept }, // Indicates the treatment step  applied to the specimen
    "additive" : [{ Reference(Substance) }], // Material used in the processing step
    // time[x]: Date and time of specimen processing. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  }],
  "container" : [{ // Direct container of specimen (tube/slide, etc.)
    "identifier" : [{ Identifier }], // Id for the container
    "description" : "<string>", // Textual description of the container
    "type" : { CodeableConcept }, // Kind of container directly associated with specimen
    "capacity" : { Quantity(SimpleQuantity) }, // Container volume or size
    "specimenQuantity" : { Quantity(SimpleQuantity) }, // Quantity of specimen within container
    // additive[x]: Additive associated with container. One of these 2:
    "additiveCodeableConcept" : { CodeableConcept }
    "additiveReference" : { Reference(Substance) }
  }],
  "condition" : [{ CodeableConcept }], // State of the specimen
  "note" : [{ Annotation }] // Comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Specimen;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Specimen.identifier [ Identifier ], ... ; # 0..* External Identifier
  fhir:Specimen.accessionIdentifier [ Identifier ]; # 0..1 Identifier assigned by the lab
  fhir:Specimen.status [ code ]; # 0..1 available | unavailable | unsatisfactory | entered-in-error
  fhir:Specimen.type [ CodeableConcept ]; # 0..1 Kind of material that forms the specimen
  fhir:Specimen.subject [ Reference(Patient|Group|Device|Substance|Location) ]; # 0..1 Where the specimen came from.
    This may be from patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a
    substance or a device
  fhir:Specimen.receivedTime [ dateTime ]; # 0..1 The time when specimen was received for processing
  fhir:Specimen.parent [ Reference(Specimen) ], ... ; # 0..* Specimen from which this specimen originated
  fhir:Specimen.request [ Reference(ServiceRequest) ], ... ; # 0..* Why the specimen was collected
  fhir:Specimen.collection [ # 0..1 Collection details
    fhir:Specimen.collection.collector [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Who collected the specimen
    # Specimen.collection.collected[x] : 0..1 Collection time. One of these 2
      fhir:Specimen.collection.collectedDateTime [ dateTime ]
      fhir:Specimen.collection.collectedPeriod [ Period ]
    fhir:Specimen.collection.duration [ Duration ]; # 0..1 How long it took to collect specimen
    fhir:Specimen.collection.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The quantity of specimen collected
    fhir:Specimen.collection.method [ CodeableConcept ]; # 0..1 Technique used to perform collection
    fhir:Specimen.collection.bodySite [ CodeableConcept ]; # 0..1 Anatomical collection site
    # Specimen.collection.fastingStatus[x] : 0..1 Whether or how long patient abstained from food and/or drink. One of
    these 2
      fhir:Specimen.collection.fastingStatusCodeableConcept [ CodeableConcept ]
      fhir:Specimen.collection.fastingStatusDuration [ Duration ]
  ];
```

```
  fhir:Specimen.processing [ # 0..* Processing and processing step details
    fhir:Specimen.processing.description [ string ]; # 0..1 Textual description of procedure
    fhir:Specimen.processing.procedure [ CodeableConcept ]; # 0..1 Indicates the treatment step  applied to the specimen
    fhir:Specimen.processing.additive [ Reference(Substance) ], ... ; # 0..* Material used in the processing step
    # Specimen.processing.time[x] : 0..1 Date and time of specimen processing. One of these 2
      fhir:Specimen.processing.timeDateTime [ dateTime ]
      fhir:Specimen.processing.timePeriod [ Period ]
  ], ...;
  fhir:Specimen.container [ # 0..* Direct container of specimen (tube/slide, etc.)
    fhir:Specimen.container.identifier [ Identifier ], ... ; # 0..* Id for the container
    fhir:Specimen.container.description [ string ]; # 0..1 Textual description of the container
    fhir:Specimen.container.type [ CodeableConcept ]; # 0..1 Kind of container directly associated with specimen
    fhir:Specimen.container.capacity [ Quantity(SimpleQuantity) ]; # 0..1 Container volume or size
    fhir:Specimen.container.specimenQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of specimen within container
    # Specimen.container.additive[x] : 0..1 Additive associated with container. One of these 2
      fhir:Specimen.container.additiveCodeableConcept [ CodeableConcept ]
      fhir:Specimen.container.additiveReference [ Reference(Substance) ]
  ], ...;
  fhir:Specimen.condition [ CodeableConcept ], ... ; # 0..* State of the specimen
  fhir:Specimen.note [ Annotation ], ... ; # 0..* Comments
]
```

Changes since R3

Specimen
Specimen.status | Change value set from http://hl7.org/fhir/ValueSet/specimen-status to http://hl7.org/fhir/ValueSet/specimen-status|4.0.1
Specimen.subject | Min Cardinality changed from 1 to 0 Type Reference: Added Target Type Location
Specimen.request | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ProcedureRequest
Specimen.collection.collector | Type Reference: Added Target Type PractitionerRole
Specimen.collection.duration | Added Element
Specimen.collection.fastingStatus[x] | Added Element
Specimen.condition | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Specimen | TU | | DomainResource | Sample for analysis Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Identifier
accessionIdentifier | Σ | 0..1 | Identifier | Identifier assigned by the lab
status | ?!Σ | 0..1 | code | available | unavailable | unsatisfactory | entered-in-error SpecimenStatus (Required)
type | Σ | 0..1 | CodeableConcept | Kind of material that forms the specimen v2 Specimen Type (Example)
subject | Σ | 0..1 | Reference(Patient | Group | Device | Substance | Location) | Where the specimen came from. This may be from patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device
receivedTime | Σ | 0..1 | dateTime | The time when specimen was received for processing
parent | | 0..* | Reference(Specimen) | Specimen from which this specimen originated
request | | 0..* | Reference(ServiceRequest) | Why the specimen was collected
collection | | 0..1 | BackboneElement | Collection details
collector | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | Who collected the specimen
collected[x] | Σ | 0..1 | | Collection time
collectedDateTime | | | dateTime
collectedPeriod | | | Period
duration | Σ | 0..1 | Duration | How long it took to collect specimen
quantity | | 0..1 | SimpleQuantity | The quantity of specimen collected
method | | 0..1 | CodeableConcept | Technique used to perform collection FHIR Specimen Collection Method (Example)
bodySite | | 0..1 | CodeableConcept | Anatomical collection site SNOMED CT Body Structures (Example)
fastingStatus[x] | Σ | 0..1 | | Whether or how long patient abstained from food and/or drink v2 Relevant Clincial Information (Extensible)
fastingStatusCodeableConcept | | | CodeableConcept
fastingStatusDuration | | | Duration
processing | | 0..* | BackboneElement | Processing and processing step details
description | | 0..1 | string | Textual description of procedure
procedure | | 0..1 | CodeableConcept | Indicates the treatment step applied to the specimen Specimen processing procedure (Example)
additive | | 0..* | Reference(Substance) | Material used in the processing step
time[x] | | 0..1 | | Date and time of specimen processing
timeDateTime | | | dateTime
timePeriod | | | Period
container | | 0..* | BackboneElement | Direct container of specimen (tube/slide, etc.)
identifier | Σ | 0..* | Identifier | Id for the container
description | | 0..1 | string | Textual description of the container
type | | 0..1 | CodeableConcept | Kind of container directly associated with specimen Specimen container (Example)
capacity | | 0..1 | SimpleQuantity | Container volume or size

specimenQuantity | | 0..1 | SimpleQuantity | Quantity of specimen within container
additive[x] | | 0..1 | | Additive associated with container v2 Additive (Example)
additiveCodeableConcept | | | CodeableConcept
additiveReference | | | Reference(Substance)
condition | Σ | 0..* | CodeableConcept | State of the specimen v2 Specimen Condition (Extensible)
note | | 0..* | Annotation | Comments
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Specimen xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifier --></identifier>
 <accessionIdentifier><!-- 0..1 Identifier Identifier assigned by the lab --></accessionIdentifier>
 <status value="[code]"/><!-- 0..1 available | unavailable | unsatisfactory | entered-in-error -->
 <type><!-- 0..1 CodeableConcept Kind of material that forms the specimen --></type>
 <subject><!-- 0..1 Reference(Patient|Group|Device|Substance|Location) Where the specimen came from. This may be from
    patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device
    --></subject>
 <receivedTime value="[dateTime]"/><!-- 0..1 The time when specimen was received for processing -->
 <parent><!-- 0..* Reference(Specimen) Specimen from which this specimen originated --></parent>
 <request><!-- 0..* Reference(ServiceRequest) Why the specimen was collected --></request>
 <collection>  <!-- 0..1 Collection details -->
  <collector><!-- 0..1 Reference(Practitioner|PractitionerRole) Who collected the specimen --></collector>
  <collected[x]><!-- 0..1 dateTime|Period Collection time --></collected[x]>
  <duration><!-- 0..1 Duration How long it took to collect specimen --></duration>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) The quantity of specimen collected --></quantity>
  <method><!-- 0..1 CodeableConcept Technique used to perform collection --></method>
  <bodySite><!-- 0..1 CodeableConcept Anatomical collection site --></bodySite>
  <fastingStatus[x]><!-- 0..1 CodeableConcept|Duration Whether or how long patient abstained from food and/or drink
    --></fastingStatus[x]>
 </collection>
 <processing>  <!-- 0..* Processing and processing step details -->
  <description value="[string]"/><!-- 0..1 Textual description of procedure -->
  <procedure><!-- 0..1 CodeableConcept Indicates the treatment step  applied to the specimen --></procedure>
  <additive><!-- 0..* Reference(Substance) Material used in the processing step --></additive>
  <time[x]><!-- 0..1 dateTime|Period Date and time of specimen processing --></time[x]>
 </processing>
 <container>  <!-- 0..* Direct container of specimen (tube/slide, etc.) -->
  <identifier><!-- 0..* Identifier Id for the container --></identifier>
  <description value="[string]"/><!-- 0..1 Textual description of the container -->
  <type><!-- 0..1 CodeableConcept Kind of container directly associated with specimen --></type>
  <capacity><!-- 0..1 Quantity(SimpleQuantity) Container volume or size --></capacity>
  <specimenQuantity><!-- 0..1 Quantity(SimpleQuantity) Quantity of specimen within container --></specimenQuantity>
  <additive[x]><!-- 0..1 CodeableConcept|Reference(Substance) Additive associated with container --></additive[x]>
 </container>
 <condition><!-- 0..* CodeableConcept State of the specimen --></condition>
 <note><!-- 0..* Annotation Comments --></note>
</Specimen>
```

JSON Template

```
{
  "resourceType" : "Specimen",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifier
  "accessionIdentifier" : { Identifier }, // Identifier assigned by the lab
  "status" : "<code>", // available | unavailable | unsatisfactory | entered-in-error
  "type" : { CodeableConcept }, // Kind of material that forms the specimen
  "subject" : { Reference(Patient|Group|Device|Substance|Location) }, // Where the specimen came from. This may be from
    patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device
  "receivedTime" : "<dateTime>", // The time when specimen was received for processing
  "parent" : [{ Reference(Specimen) }], // Specimen from which this specimen originated
  "request" : [{ Reference(ServiceRequest) }], // Why the specimen was collected
  "collection" : { // Collection details
    "collector" : { Reference(Practitioner|PractitionerRole) }, // Who collected the specimen
    // collected[x]: Collection time. One of these 2:
    "collectedDateTime" : "<dateTime>",
    "collectedPeriod" : { Period },
    "duration" : { Duration }, // How long it took to collect specimen
    "quantity" : { Quantity(SimpleQuantity) }, // The quantity of specimen collected
    "method" : { CodeableConcept }, // Technique used to perform collection
    "bodySite" : { CodeableConcept }, // Anatomical collection site
    // fastingStatus[x]: Whether or how long patient abstained from food and/or drink. One of these 2:
    "fastingStatusCodeableConcept" : { CodeableConcept }
    "fastingStatusDuration" : { Duration }
  },
  "processing" : [{ // Processing and processing step details
    "description" : "<string>", // Textual description of procedure
    "procedure" : { CodeableConcept }, // Indicates the treatment step  applied to the specimen
    "additive" : [{ Reference(Substance) }], // Material used in the processing step
    // time[x]: Date and time of specimen processing. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  }],
  "container" : [{ // Direct container of specimen (tube/slide, etc.)
```

```
    "identifier" : [{ Identifier }], // Id for the container
    "description" : "<string>", // Textual description of the container
    "type" : { CodeableConcept }, // Kind of container directly associated with specimen
    "capacity" : { Quantity(SimpleQuantity) }, // Container volume or size
    "specimenQuantity" : { Quantity(SimpleQuantity) }, // Quantity of specimen within container
    // additive[x]: Additive associated with container. One of these 2:
    "additiveCodeableConcept" : { CodeableConcept }
    "additiveReference" : { Reference(Substance) }
  }],
  "condition" : [{ CodeableConcept }], // State of the specimen
  "note" : [{ Annotation }] // Comments
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Specimen;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Specimen.identifier [ Identifier ], ... ; # 0..* External Identifier
  fhir:Specimen.accessionIdentifier [ Identifier ]; # 0..1 Identifier assigned by the lab
  fhir:Specimen.status [ code ]; # 0..1 available | unavailable | unsatisfactory | entered-in-error
  fhir:Specimen.type [ CodeableConcept ]; # 0..1 Kind of material that forms the specimen
  fhir:Specimen.subject [ Reference(Patient|Group|Device|Substance|Location) ]; # 0..1 Where the specimen came from.
    This may be from patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a
    substance or a device
  fhir:Specimen.receivedTime [ dateTime ]; # 0..1 The time when specimen was received for processing
  fhir:Specimen.parent [ Reference(Specimen) ], ... ; # 0..* Specimen from which this specimen originated
  fhir:Specimen.request [ Reference(ServiceRequest) ], ... ; # 0..* Why the specimen was collected
  fhir:Specimen.collection [ # 0..1 Collection details
    fhir:Specimen.collection.collector [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Who collected the specimen
    # Specimen.collection.collected[x] : 0..1 Collection time. One of these 2
      fhir:Specimen.collection.collectedDateTime [ dateTime ]
      fhir:Specimen.collection.collectedPeriod [ Period ]
    fhir:Specimen.collection.duration [ Duration ]; # 0..1 How long it took to collect specimen
    fhir:Specimen.collection.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The quantity of specimen collected
    fhir:Specimen.collection.method [ CodeableConcept ]; # 0..1 Technique used to perform collection
    fhir:Specimen.collection.bodySite [ CodeableConcept ]; # 0..1 Anatomical collection site
    # Specimen.collection.fastingStatus[x] : 0..1 Whether or how long patient abstained from food and/or drink. One of
    these 2
      fhir:Specimen.collection.fastingStatusCodeableConcept [ CodeableConcept ]
      fhir:Specimen.collection.fastingStatusDuration [ Duration ]
  ];
  fhir:Specimen.processing [ # 0..* Processing and processing step details
    fhir:Specimen.processing.description [ string ]; # 0..1 Textual description of procedure
    fhir:Specimen.processing.procedure [ CodeableConcept ]; # 0..1 Indicates the treatment step  applied to the specimen
    fhir:Specimen.processing.additive [ Reference(Substance) ], ... ; # 0..* Material used in the processing step
    # Specimen.processing.time[x] : 0..1 Date and time of specimen processing. One of these 2
      fhir:Specimen.processing.timeDateTime [ dateTime ]
      fhir:Specimen.processing.timePeriod [ Period ]
  ], ...;
  fhir:Specimen.container [ # 0..* Direct container of specimen (tube/slide, etc.)
    fhir:Specimen.container.identifier [ Identifier ], ... ; # 0..* Id for the container
    fhir:Specimen.container.description [ string ]; # 0..1 Textual description of the container
    fhir:Specimen.container.type [ CodeableConcept ]; # 0..1 Kind of container directly associated with specimen
    fhir:Specimen.container.capacity [ Quantity(SimpleQuantity) ]; # 0..1 Container volume or size
    fhir:Specimen.container.specimenQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Quantity of specimen within container
    # Specimen.container.additive[x] : 0..1 Additive associated with container. One of these 2
      fhir:Specimen.container.additiveCodeableConcept [ CodeableConcept ]
      fhir:Specimen.container.additiveReference [ Reference(Substance) ]
  ], ...;
  fhir:Specimen.condition [ CodeableConcept ], ... ; # 0..* State of the specimen
  fhir:Specimen.note [ Annotation ], ... ; # 0..* Comments
]
```

Changes since Release 3

Specimen
Specimen.status | Change value set from http://hl7.org/fhir/ValueSet/specimen-status to http://hl7.org/fhir/ValueSet/specimen-status|4.0.1
Specimen.subject | Min Cardinality changed from 1 to 0 Type Reference: Added Target Type Location
Specimen.request | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ProcedureRequest
Specimen.collection.collector | Type Reference: Added Target Type PractitionerRole
Specimen.collection.duration | Added Element
Specimen.collection.fastingStatus[x] | Added Element
Specimen.condition | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.7.3.1 Terminology Bindings

Path | Definition | Type | Reference
Specimen.status | Codes providing the status/availability of a specimen. | Required | SpecimenStatus
Specimen.type | The type of the specimen. | Example | v2.0487
Specimen.collection.method | The technique that is used to perform the procedure. | Example | FHIRSpecimenCollectionMethod
Specimen.collection.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures
Specimen.collection.fastingStatus[x] | Codes describing the fasting status of the patient. | Extensible | v2.0916
Specimen.processing.procedure | Type indicating the technique used to process the specimen. | Example | SpecimenProcessingProcedure
Specimen.container.type | Type of specimen container. | Example | SpecimenContainer
Specimen.container.additive[x] | Substance added to specimen container. | Example | v2.0371
Specimen.condition | Codes describing the state of the specimen. | Extensible | v2.0493

10.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
accession | token | The accession number associated with the specimen | Specimen.accessionIdentifier
bodysite | token | The code for the body site from where the specimen originated | Specimen.collection.bodySite
collected | date | The date the specimen was collected | Specimen.collection.collected
collector | reference | Who collected the specimen | Specimen.collection.collector (Practitioner, PractitionerRole)
container | token | The kind of specimen container | Specimen.container.type
container-id | token | The unique identifier associated with the specimen container | Specimen.container.identifier
identifier | token | The unique identifier associated with the specimen | Specimen.identifier
parent | reference | The parent of the specimen | Specimen.parent (Specimen)
patient | reference | The patient the specimen comes from | Specimen.subject.where(resolve() is Patient) (Patient)
status | token | available | unavailable | unsatisfactory | entered-in-error | Specimen.status
subject | reference | The subject of the specimen | Specimen.subject (Group, Device, Patient, Substance, Location)
type | token | The specimen type | Specimen.type

10.7.5 Resource Specimen - Examples

URL: https://hl7.org/fhir/R4/specimen-examples.html

10.7.5 Resource Specimen - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Example Name | id | Format
General Specimen Example | 101 | XML | JSON | Turtle
Bacterial Isolate from Stool sample | isolate | XML | JSON | Turtle
VMA, Random Urine Sample | vma-urine | XML | JSON | Turtle
Serum Sample | sst | XML | JSON | Turtle
Pooled serum sample | pooled-serum | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.7.6 Resource Specimen - Detailed Descriptions

URL: https://hl7.org/fhir/R4/specimen-definitions.html

10.7.6 Resource Specimen - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Specimen resource.

Specimen
Element Id | Specimen
Definition | A sample to be used for analysis.
Cardinality | 0..*
Type | DomainResource
Specimen.identifier
Element Id | Specimen.identifier
Definition | Id for specimen.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*

Type | Identifier
Summary | true
Specimen.accessionIdentifier
Element Id | Specimen.accessionIdentifier
Definition | The identifier assigned by the lab when accessioning specimen(s). This is not necessarily the same as the specimen identifier, depending on local lab procedures.
Cardinality | 0..1
Type | Identifier
Summary | true
Specimen.status
Element Id | Specimen.status
Definition | The availability of the specimen.
Cardinality | 0..1
Terminology Binding | SpecimenStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Specimen.type
Element Id | Specimen.type
Definition | The kind of material that forms the specimen.
Cardinality | 0..1
Terminology Binding | v2 Specimen Type (Example)
Type | CodeableConcept
Summary | true
Comments | The type can change the way that a specimen is handled and drives what kind of analyses can properly be performed on the specimen. It is frequently used in diagnostic work flow decision making systems.
Specimen.subject
Element Id | Specimen.subject
Definition | Where the specimen came from. This may be from patient(s), from a location (e.g., the source of an environmental sample), or a sampling of a substance or a device.
Cardinality | 0..1
Type | Reference(Patient | Group | Device | Substance | Location)
Requirements | Must know the subject context.
Summary | true
Specimen.receivedTime
Element Id | Specimen.receivedTime
Definition | Time when specimen was received for processing or testing.
Cardinality | 0..1
Type | dateTime
Summary | true
Specimen.parent
Element Id | Specimen.parent
Definition | Reference to the parent (source) specimen which is used when the specimen was either derived from or a component of another specimen.
Cardinality | 0..*
Type | Reference(Specimen)
Comments | The parent specimen could be the source from which the current specimen is derived by some processing step (e.g. an aliquot or isolate or extracted nucleic acids from clinical samples) or one of many specimens that were combined to create a pooled sample.
Specimen.request
Element Id | Specimen.request
Definition | Details concerning a service request that required a specimen to be collected.
Cardinality | 0..*
Type | Reference(ServiceRequest)
Comments | The request may be explicit or implied such with a ServiceRequest that requires a blood draw.
Specimen.collection
Element Id | Specimen.collection
Definition | Details concerning the specimen collection.
Cardinality | 0..1
Specimen.collection.collector
Element Id | Specimen.collection.collector
Definition | Person who collected the specimen.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Summary | true
Specimen.collection.collected[x]
Element Id | Specimen.collection.collected[x]
Definition | Time when specimen was collected from subject - the physiologically relevant time.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]

Summary | true
Specimen.collection.duration
Element Id | Specimen.collection.duration
Definition | The span of time over which the collection of a specimen occurred.
Cardinality | 0..1
Type | Duration
Summary | true
Specimen.collection.quantity
Element Id | Specimen.collection.quantity
Definition | The quantity of specimen collected; for instance the volume of a blood sample, or the physical measurement of an anatomic pathology sample.
Cardinality | 0..1
Type | SimpleQuantity
Specimen.collection.method
Element Id | Specimen.collection.method
Definition | A coded value specifying the technique that is used to perform the procedure.
Cardinality | 0..1
Terminology Binding | FHIR Specimen Collection Method (Example)
Type | CodeableConcept
Specimen.collection.bodySite
Element Id | Specimen.collection.bodySite
Definition | Anatomical location from which the specimen was collected (if subject is a patient). This is the target site. This element is not used for environmental specimens.
Cardinality | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Comments | If the use case requires BodySite to be handled as a separate resource instead of an inline coded element (e.g. to identify and track separately) then use the standard extension bodySite.
Specimen.collection.fastingStatus[x]
Element Id | Specimen.collection.fastingStatus[x]
Definition | Abstinence or reduction from some or all food, drink, or both, for a period of time prior to sample collection.
Cardinality | 0..1
Terminology Binding | v2 Relevant Clincial Information (Extensible)
Type | CodeableConcept|Duration
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Many diagnostic tests require fasting to facilitate accurate interpretation.
Summary | true
Comments | Representing fasting status using this element is preferred to representing it with an observation using a 'pre-coordinated code' such as LOINC 2005-7 (Calcium [Moles/time] in 2 hour Urine --12 hours fasting), or using a component observation such asObservation.component code` = LOINC 49541-6 (Fasting status - Reported).
Specimen.processing
Element Id | Specimen.processing
Definition | Details concerning processing and processing steps for the specimen.
Cardinality | 0..*
Specimen.processing.description
Element Id | Specimen.processing.description
Definition | Textual description of procedure.
Cardinality | 0..1
Type | string
Specimen.processing.procedure
Element Id | Specimen.processing.procedure
Definition | A coded value specifying the procedure used to process the specimen.
Cardinality | 0..1
Terminology Binding | Specimen processing procedure (Example)
Type | CodeableConcept
Specimen.processing.additive
Element Id | Specimen.processing.additive
Definition | Material used in the processing step.
Cardinality | 0..*
Type | Reference(Substance)
Specimen.processing.time[x]
Element Id | Specimen.processing.time[x]
Definition | A record of the time or period when the specimen processing occurred. For example the time of sample fixation or the period of time the sample was in formalin.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Specimen.container
Element Id | Specimen.container
Definition | The container holding the specimen. The recursive nature of containers; i.e. blood in tube in tray in rack is not addressed here.
Cardinality | 0..*
Specimen.container.identifier

Element Id | Specimen.container.identifier
Definition | Id for container. There may be multiple; a manufacturer's bar code, lab assigned identifier, etc. The container ID may differ from the specimen id in some circumstances.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Specimen.container.description
Element Id | Specimen.container.description
Definition | Textual description of the container.
Cardinality | 0..1
Type | string
Specimen.container.type
Element Id | Specimen.container.type
Definition | The type of container associated with the specimen (e.g. slide, aliquot, etc.).
Cardinality | 0..1
Terminology Binding | Specimen container (Example)
Type | CodeableConcept
Specimen.container.capacity
Element Id | Specimen.container.capacity
Definition | The capacity (volume or other measure) the container may contain.
Cardinality | 0..1
Type | SimpleQuantity
Specimen.container.specimenQuantity
Element Id | Specimen.container.specimenQuantity
Definition | The quantity of specimen in the container; may be volume, dimensions, or other appropriate measurements, depending on the specimen type.
Cardinality | 0..1
Type | SimpleQuantity
Specimen.container.additive[x]
Element Id | Specimen.container.additive[x]
Definition | Introduced substance to preserve, maintain or enhance the specimen. Examples: Formalin, Citrate, EDTA.
Cardinality | 0..1
Terminology Binding | v2 Additive (Example)
Type | CodeableConcept|Reference(Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
Specimen.condition
Element Id | Specimen.condition
Definition | A mode or state of being that describes the nature of the specimen.
Cardinality | 0..*
Terminology Binding | v2 Specimen Condition (Extensible)
Type | CodeableConcept
Requirements | The specimen condition can be used to assess its quality or appropriateness for a specific test.
Summary | true
Comments | Specimen condition is an observation made about the specimen. It's a point-in-time assessment. It can be used to assess its quality or appropriateness for a specific test.
Specimen.note
Element Id | Specimen.note
Definition | To communicate any details or issues about the specimen or during the specimen collection. (for example: broken vial, sent with patient, frozen).
Cardinality | 0..*
Type | Annotation

10.7.7 Resource Specimen - Mappings

URL: https://hl7.org/fhir/R4/specimen-mappings.html

10.7.7 Resource Specimen - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the specimen resource (see Mappings to Other Standards for further information & status).

10.7.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Specimen
identifier | FiveWs.identifier
accessionIdentifier | FiveWs.identifier

status | FiveWs.status
type | FiveWs.what[x]
subject | FiveWs.subject[x]
receivedTime | FiveWs.done[x]
request | FiveWs.why[x]
collector | FiveWs.actor
collected[x] | FiveWs.init
duration | FiveWs.init

10.7.7.2 HL7 v2 Mapping (http://hl7.org/v2)

Specimen
identifier | SPM-2
accessionIdentifier | SPM-30 (v2.7+)
status | SPM-20
type | SPM-4 and possibly SPM-5
subject
receivedTime | SPM-18
parent
request | ORC? OBR-2/3?
collection | SPM-14
collector | PRT
collected[x] | SPM-17
duration
quantity | SPM-12
method | SPM-7
bodySite | SPM-8 and SPM-9
fastingStatus[x] | OBR-
processing
description
procedure
additive | SPM-6
time[x]
container
identifier | SAC-3
description
type | SPM-27
capacity | one of: SAC 16, SAC17 and SAC 20, or SAC 21 and SAC 20
specimenQuantity | SAC-23 (value) SAC-24 (units
additive[x] | SAC-27
condition | SPM-24
note | OBX

10.7.7.3 RIM Mapping (http://hl7.org/v3)

Specimen | Role[classCode=SPEC]
identifier | .id
accessionIdentifier | .participation[typeCode=SBJ].act[classCode=ACSN, moodCode=EVN].id
status | status
type | .code
subject | .scoper or .scoper.playsRole
receivedTime | .participation[typeCode=SBJ].act[code=SPCREC, moodCode=EVN].effectiveTime
parent | .scoper (if parent) .player.scopesRole[classCode=SPEC].player (if child)
request | outboundRelationship[typeCode=FLFS].target
collection | .participation[typeCode=SBJ].act[classCode=SPECCOLLECT, moodCode=EVN]
collector | .participation[typeCode=PFM].role
collected[x] | .effectiveTime
quantity | .participation[typeCode=SBJ].role[classCode=SPEC].player.quantity
method | .methodCode
bodySite | .targetSiteCode
processing | .participation[typeCode=SBJ].act[code=SPCTRT, moodCode=EVN]
description | .text
procedure | .code
additive | .participation[typeCode=CSM].role[classCode=ADTV].code
time[x] | .effectiveTime
container | .player.scopingRole[classCode=CONT].scoper
identifier | .id
description | .desc
type | .code
capacity | .quantity
specimenQuantity | .playedRole[classCode=CONT].quantity
additive[x] | .scopesRole[classCode=ADTV].player
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

10.7.8 Resource Specimen - Extensions & Profiles

URL: https://hl7.org/fhir/R4/specimen-profiles.html

10.7.8 Resource Specimen - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the Specimen resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.7.8.1 Profiles

No Profiles defined for this resource

10.7.8.2 Extensions

observation-bodyPosition | Specimen.collection | bodyPosition | for Observation HL7 Extensions
specimen-collectionPriority | Specimen.collection | collectionPriority | for Specimen HL7 Extensions
specimen-isDryWeight | Specimen.collection.quantity | isDryWeight | for Specimen HL7 Extensions
specimen-processingTime | Specimen.processing | processingTime | for Specimen HL7 Extensions
specimen-sequenceNumber | Specimen.container | sequenceNumber | for Specimen HL7 Extensions
specimen-specialHandling | Specimen.collection | specialHandling | for Specimen HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

servicerequest-geneticsItem | Item | for ServiceRequest-Genetics
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.7.8.3 Search Extensions

No Search Extensions defined for this resource

10.7.9 Resource Specimen - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/specimen-version-maps.html

No validation errors - all conversions are clean

10.7.9 Resource Specimen - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for Specimen.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

10.7.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Specimen3to4" = "R3 to R4 Conversions for Specimen"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Specimen" alias SpecimenR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Specimen" alias Specimen as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Specimen(source src : SpecimenR3, target tgt : Specimen) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.accessionIdentifier -> tgt.accessionIdentifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.subject -> tgt.subject;
  src.receivedTime -> tgt.receivedTime;
  src.parent -> tgt.parent;
  src.request -> tgt.request;
  src.collection as vs0 -> tgt.collection as vt0 then collection(vs0, vt0);
  src.processing as vs0 -> tgt.processing as vt0 then processing(vs0, vt0);
  src.container as vs0 -> tgt.container as vt0 then container(vs0, vt0);
  src.note -> tgt.note;
}
```

```
group collection(source src, target tgt) extends BackboneElement {
  src.collector -> tgt.collector;
  src.collected : dateTime as vs -> tgt.collected = create('dateTime') as vt then dateTime(vs, vt);
  src.collected : Period as vs -> tgt.collected = create('Period') as vt then Period(vs, vt);
  src.quantity -> tgt.quantity;
  src.method -> tgt.method;
  src.bodySite -> tgt.bodySite;
}

group processing(source src, target tgt) extends BackboneElement {
  src.description -> tgt.description;
  src.procedure -> tgt.procedure;
  src.additive -> tgt.additive;
  src.time : dateTime as vs -> tgt.time = create('dateTime') as vt then dateTime(vs, vt);
  src.time : Period as vs -> tgt.time = create('Period') as vt then Period(vs, vt);
}

group container(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.description -> tgt.description;
  src.type -> tgt.type;
  src.capacity -> tgt.capacity;
  src.specimenQuantity -> tgt.specimenQuantity;
  src.additive : CodeableConcept as vs -> tgt.additive = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.additive : Reference as vs -> tgt.additive = create('Reference') as vt then Reference(vs, vt);
}
```

10.7.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Specimen4to3" = "R4 to R3 Conversion for Specimen"

uses "http://hl7.org/fhir/StructureDefinition/Specimen" alias Specimen as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Specimen" alias SpecimenR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Specimen(source src : Specimen, target tgt : SpecimenR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.accessionIdentifier -> tgt.accessionIdentifier;
  src.status -> tgt.status;
  src.type -> tgt.type;
  src.subject -> tgt.subject;
  src.receivedTime -> tgt.receivedTime;
  src.parent -> tgt.parent;
  src.request -> tgt.request;
  src.collection as vs0 -> tgt.collection as vt0 then collection(vs0, vt0);
  src.processing as vs0 -> tgt.processing as vt0 then processing(vs0, vt0);
  src.container as vs0 -> tgt.container as vt0 then container(vs0, vt0);
  src.note -> tgt.note;
}

group collection(source src, target tgt) extends BackboneElement {
  src.collector -> tgt.collector;
  src.collected : dateTime as vs -> tgt.collected = create('dateTime') as vt then dateTime(vs, vt);
  src.collected : Period as vs -> tgt.collected = create('Period') as vt then Period(vs, vt);
  src.quantity -> tgt.quantity;
  src.method -> tgt.method;
  src.bodySite -> tgt.bodySite;
}

group processing(source src, target tgt) extends BackboneElement {
  src.description -> tgt.description;
  src.procedure -> tgt.procedure;
  src.additive -> tgt.additive;
  src.time : dateTime as vs -> tgt.time = create('dateTime') as vt then dateTime(vs, vt);
  src.time : Period as vs -> tgt.time = create('Period') as vt then Period(vs, vt);
}

group container(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.description -> tgt.description;
  src.type -> tgt.type;
  src.capacity -> tgt.capacity;
  src.specimenQuantity -> tgt.specimenQuantity;
  src.additive : CodeableConcept as vs -> tgt.additive = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.additive : Reference as vs -> tgt.additive = create('Reference') as vt then Reference(vs, vt);
}
```

10.7.9.3 R4 Validation Errors

10.7.10 Specimen HL7 Extensions

URL: https://hl7.org/fhir/R4/specimen-extensions.html

Defines common extensions used with or related to the Specimen resource

10.7.10 Specimen HL7 Extensions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative

10.7.10.1 Content

Extensions:
specimen-sequenceNumber | sequenceNumber : An assigned number on the specimen denoting the order of collection.
specimen-collectionPriority | collectionPriority : The urgency of sample collection, such as STAT, ASAP, ASAP-ED, ROUTINE, ROUTINE-AM, etc....
specimen-specialHandling | specialHandling : Special handling during the collection, transport, or storage of the specimen.
specimen-isDryWeight | isDryWeight : If the recorded quantity of the specimen is reported as a weight, if it is a dry weight.
specimen-processingTime | processingTime : Period or duration of processing.

10.7.11 JSON Schema for Specimen

URL: https://hl7.org/fhir/R4/specimen.schema.json.html

10.7.11 JSON Schema for Specimen

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for Specimen, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Specimen",
  "$ref": "#/definitions/Specimen",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Specimen": {
      "description": "A sample to be used for analysis.",
      "properties": {
        "resourceType": {
          "description": "This is a Specimen resource",
          "const": "Specimen"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
```

```
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Id for specimen.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "accessionIdentifier": {
      "description": "The identifier assigned by the lab when accessioning specimen(s). This is not necessarily the
same as the specimen identifier, depending on local lab procedures.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "status": {
      "description": "The availability of the specimen.",
      "enum": [
        "available",
        "unavailable",
        "unsatisfactory",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The kind of material that forms the specimen.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "Where the specimen came from. This may be from patient(s), from a location (e.g., the source
of an environmental sample), or a sampling of a substance or a device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "receivedTime": {
      "description": "Time when specimen was received for processing or testing.",
      "$ref": "#/definitions/dateTime"
    },
    "_receivedTime": {
      "description": "Extensions for receivedTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "parent": {
      "description": "Reference to the parent (source) specimen which is used when the specimen was either derived
from or a component of another specimen.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "request": {
      "description": "Details concerning a service request that required a specimen to be collected.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "collection": {
      "description": "Details concerning the specimen collection.",
      "$ref": "#/definitions/Specimen_Collection"
    },
    "processing": {
      "description": "Details concerning processing and processing steps for the specimen.",
      "items": {
        "$ref": "#/definitions/Specimen_Processing"
      },
      "type": "array"
    },
    "container": {
      "description": "The container holding the specimen.  The recursive nature of containers; i.e. blood in tube in
 tray in rack is not addressed here.",
      "items": {
        "$ref": "#/definitions/Specimen_Container"
      },
```

```
      "type": "array"
    },
    "condition": {
      "description": "A mode or state of being that describes the nature of the specimen.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "note": {
      "description": "To communicate any details or issues about the specimen or during the specimen collection.
(for example: broken vial, sent with patient, frozen).",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Specimen_Collection": {
  "description": "A sample to be used for analysis.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "collector": {
      "description": "Person who collected the specimen.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "collectedDateTime": {
      "description": "Time when specimen was collected from subject - the physiologically relevant time.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_collectedDateTime": {
      "description": "Extensions for collectedDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "collectedPeriod": {
      "description": "Time when specimen was collected from subject - the physiologically relevant time.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "duration": {
      "description": "The span of time over which the collection of a specimen occurred.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "quantity": {
      "description": "The quantity of specimen collected; for instance the volume of a blood sample, or the physical
 measurement of an anatomic pathology sample.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "method": {
      "description": "A coded value specifying the technique that is used to perform the procedure.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "bodySite": {
      "description": "Anatomical location from which the specimen was collected (if subject is a patient). This is
the target site.  This element is not used for environmental specimens.",
```

```
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "fastingStatusCodeableConcept": {
      "description": "Abstinence or reduction from some or all food, drink, or both, for a period of time prior to
sample collection.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "fastingStatusDuration": {
      "description": "Abstinence or reduction from some or all food, drink, or both, for a period of time prior to
sample collection.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    }
  },
  "additionalProperties": false
},
"Specimen_Processing": {
  "description": "A sample to be used for analysis.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Textual description of procedure.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "procedure": {
      "description": "A coded value specifying the procedure used to process the specimen.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "additive": {
      "description": "Material used in the processing step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "timeDateTime": {
      "description": "A record of the time or period when the specimen processing occurred.  For example the time of
 sample fixation or the period of time the sample was in formalin.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timeDateTime": {
      "description": "Extensions for timeDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timePeriod": {
      "description": "A record of the time or period when the specimen processing occurred.  For example the time of
 sample fixation or the period of time the sample was in formalin.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
},
"Specimen_Container": {
  "description": "A sample to be used for analysis.",
  "properties": {
    "id": {
```

```
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Id for container. There may be multiple; a manufacturer\u0027s bar code, lab assigned identifier, etc. The container ID may differ from the specimen id in some circumstances.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "description": {
      "description": "Textual description of the container.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of container associated with the specimen (e.g. slide, aliquot, etc.).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "capacity": {
      "description": "The capacity (volume or other measure) the container may contain.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "specimenQuantity": {
      "description": "The quantity of specimen in the container; may be volume, dimensions, or other appropriate measurements, depending on the specimen type.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "additiveCodeableConcept": {
      "description": "Introduced substance to preserve, maintain or enhance the specimen. Examples: Formalin, Citrate, EDTA.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "additiveReference": {
      "description": "Introduced substance to preserve, maintain or enhance the specimen. Examples: Formalin, Citrate, EDTA.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
 }
 }
}
```


10.7.12 Resource Specimen - Maturity Dependency View

URL: https://hl7.org/fhir/R4/specimen-dependencies.html

10.7.12 Resource Specimen - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for Specimen FMM level 2

Name | Card. | Type | Dependency Analysis
Specimen | | DomainResource
identifier | 0..* | Identifier
accessionIdentifier | 0..1 | Identifier

status | 0..1 | code
type | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group | Device | Substance | Location) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
receivedTime | 0..1 | dateTime
parent | 0..* | Reference(Specimen)
request | 0..* | Reference(ServiceRequest)
collection | 0..1 | BackboneElement
collector | 0..1 | Reference(Practitioner | PractitionerRole)
collected[x] | 0..1
collectedDateTime | | dateTime
collectedPeriod | | Period
duration | 0..1 | Duration
quantity | 0..1 | SimpleQuantity
method | 0..1 | CodeableConcept
bodySite | 0..1 | CodeableConcept
fastingStatus[x] | 0..1
fastingStatusCodeableConcept | | CodeableConcept
fastingStatusDuration | | Duration
processing | 0..* | BackboneElement
description | 0..1 | string
procedure | 0..1 | CodeableConcept
additive | 0..* | Reference(Substance)
time[x] | 0..1
timeDateTime | | dateTime
timePeriod | | Period
container | 0..* | BackboneElement
identifier | 0..* | Identifier
description | 0..1 | string
type | 0..1 | CodeableConcept
capacity | 0..1 | SimpleQuantity
specimenQuantity | 0..1 | SimpleQuantity
additive[x] | 0..1
additiveCodeableConcept | | CodeableConcept
additiveReference | | Reference(Substance
condition | 0..* | CodeableConcept
note | 0..* | Annotation

10.7.13 Value Set Sc-valueset-specimen-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-specimen-status.html

10.7.13 Value Set Sc-valueset-specimen-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

10.7.14 SpecimenStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-specimen-status)

Mapping from http://hl7.org/fhir/ValueSet/specimen-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes providing the status/availability of a specimen."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
available | equivalent | active
unsatisfactory | equivalent | failed
unavailable | equivalent | inactive

10.8 Resource SpecimenDefinition - Content

URL: https://hl7.org/fhir/R4/specimendefinition.html
Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A kind of specimen with associated set of requirements.

10.8.1 Scope and Usage

SpecimenDefinition will define a kind of specimen, which can be associated with in vitro diagnostic procedures described in a catalog of orderable services. SpecimenDefinition describes the kind of specimen to be collected from the subject of these procedures as well as the requirements applying to the collection activity, the preparation of the subject for this collection. This resource also describes the associated type(s) of specimen conditioned for testing, which are the output of the specimen collection activity. A specimen conditioned for testing is described with the type(s) of container and possible additive to be used, the minimal and normal volumes of collection, the conditions of storage, transportation and handling for the specimen once collected and conditioned. This resource is a necessary building block of a sharable catalog of orderable in vitro diagnostic services. The subjects of these orderable services may be human patients, non-human living subjects or non-living materials such as water, surfaces, medical devices, etc. All sub-specialties of clinical and anatomic pathology laboratories are concerned, as well as all care services prone to order in vitro diagnostic services to those laboratories. A catalog of orderable services generally belongs to a specific laboratory or facility. Nonetheless, the data items used to build this catalog are of universal meaning and interest.

10.8.2 Boundaries and Relationships

This resource relates to these other resources:

• The SpecimenDefinition resource is referenced by the resource CatalogEntry, when it carries the definition of an in vitro diagnostic testing activity.

• The SpecimenDefinition resource is referenced by the Catalog resource, which represents a whole catalog, to enable the creation and maintenance of kinds of specimens within a catalog of orderable diagnostic services.

• The SpecimenDefinition resource references the Substance resource, in order to describe the type of substance (body fluid, other material) concerned by a kind of specimen. It also references the Substance resource to describe additives to be used in the production of specimens of some kind.

This resource is referenced by ActivityDefinition and CatalogEntry

10.8.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
SpecimenDefinition | TU | | DomainResource | Kind of specimen Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Business identifier of a kind of specimen
typeCollected | Σ | 0..1 | CodeableConcept | Kind of material to collect v2 Specimen Type (Example)
patientPreparation | Σ | 0..* | CodeableConcept | Patient preparation for collection Patient preparation prior specimen collection (Example)
timeAspect | Σ | 0..1 | string | Time aspect for collection
collection | Σ | 0..* | CodeableConcept | Specimen collection procedure Specimen collection methods (Example)
typeTested | | 0..* | BackboneElement | Specimen in container intended for testing by lab
isDerived | | 0..1 | boolean | Primary or secondary specimen
type | | 0..1 | CodeableConcept | Type of intended specimen v2 Specimen Type (Example)
preference | | 1..1 | code | preferred | alternate SpecimenContainedPreference (Required)
container | | 0..1 | BackboneElement | The specimen's container
material | | 0..1 | CodeableConcept | Container material Types of material for specimen containers (Example)
type | | 0..1 | CodeableConcept | Kind of container associated with the kind of specimen Specimen Container Type (Example)
cap | | 0..1 | CodeableConcept | Color of container cap ContainerCap (Example)
description | | 0..1 | string | Container description
capacity | | 0..1 | SimpleQuantity | Container capacity
minimumVolume[x] | | 0..1 | | Minimum volume
minimumVolumeQuantity | | | SimpleQuantity
minimumVolumeString | | | string
additive | | 0..* | BackboneElement | Additive associated with container
additive[x] | | 1..1 | | Additive associated with container v2 Additive (Example)
additiveCodeableConcept | | | CodeableConcept
additiveReference | | | Reference(Substance)
preparation | | 0..1 | string | Specimen container preparation
requirement | | 0..1 | string | Specimen requirements
retentionTime | | 0..1 | Duration | Specimen retention time

rejectionCriterion | | 0..* | CodeableConcept | Rejection criterion RejectionCriterion (Example)
handling | | 0..* | BackboneElement | Specimen handling before testing
temperatureQualifier | | 0..1 | CodeableConcept | Temperature qualifier HandlingConditionSet (Example)
temperatureRange | | 0..1 | Range | Temperature range
maxDuration | | 0..1 | Duration | Maximum preservation time
instruction | | 0..1 | string | Preservation instruction
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SpecimenDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business identifier of a kind of specimen --></identifier>
 <typeCollected><!-- 0..1 CodeableConcept Kind of material to collect --></typeCollected>
 <patientPreparation><!-- 0..* CodeableConcept Patient preparation for collection --></patientPreparation>
 <timeAspect value="[string]"/><!-- 0..1 Time aspect for collection -->
 <collection><!-- 0..* CodeableConcept Specimen collection procedure --></collection>
 <typeTested>  <!-- 0..* Specimen in container intended for testing by lab -->
  <isDerived value="[boolean]"/><!-- 0..1 Primary or secondary specimen -->
  <type><!-- 0..1 CodeableConcept Type of intended specimen --></type>
  <preference value="[code]"/><!-- 1..1 preferred | alternate -->
  <container>  <!-- 0..1 The specimen's container -->
   <material><!-- 0..1 CodeableConcept Container material --></material>
   <type><!-- 0..1 CodeableConcept Kind of container associated with the kind of specimen --></type>
   <cap><!-- 0..1 CodeableConcept Color of container cap --></cap>
   <description value="[string]"/><!-- 0..1 Container description -->
   <capacity><!-- 0..1 Quantity(SimpleQuantity) Container capacity --></capacity>
   <minimumVolume[x]><!-- 0..1 Quantity(SimpleQuantity)|string Minimum volume --></minimumVolume[x]>
   <additive>  <!-- 0..* Additive associated with container -->
    <additive[x]><!-- 1..1 CodeableConcept|Reference(Substance) Additive associated with container --></additive[x]>
   </additive>
   <preparation value="[string]"/><!-- 0..1 Specimen container preparation -->
  </container>
  <requirement value="[string]"/><!-- 0..1 Specimen requirements -->
  <retentionTime><!-- 0..1 Duration Specimen retention time --></retentionTime>
  <rejectionCriterion><!-- 0..* CodeableConcept Rejection criterion --></rejectionCriterion>
  <handling>  <!-- 0..* Specimen handling before testing -->
   <temperatureQualifier><!-- 0..1 CodeableConcept Temperature qualifier --></temperatureQualifier>
   <temperatureRange><!-- 0..1 Range Temperature range --></temperatureRange>
   <maxDuration><!-- 0..1 Duration Maximum preservation time --></maxDuration>
   <instruction value="[string]"/><!-- 0..1 Preservation instruction -->
  </handling>
 </typeTested>
</SpecimenDefinition>
```

JSON Template

```
{
  "resourceType" : "SpecimenDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business identifier of a kind of specimen
  "typeCollected" : { CodeableConcept }, // Kind of material to collect
  "patientPreparation" : [{ CodeableConcept }], // Patient preparation for collection
  "timeAspect" : "<string>", // Time aspect for collection
  "collection" : [{ CodeableConcept }], // Specimen collection procedure
  "typeTested" : [{ // Specimen in container intended for testing by lab
    "isDerived" : <boolean>, // Primary or secondary specimen
    "type" : { CodeableConcept }, // Type of intended specimen
    "preference" : "<code>", // R!  preferred | alternate
    "container" : { // The specimen's container
      "material" : { CodeableConcept }, // Container material
      "type" : { CodeableConcept }, // Kind of container associated with the kind of specimen
      "cap" : { CodeableConcept }, // Color of container cap
      "description" : "<string>", // Container description
      "capacity" : { Quantity(SimpleQuantity) }, // Container capacity
      // minimumVolume[x]: Minimum volume. One of these 2:
      "minimumVolumeQuantity" : { Quantity(SimpleQuantity) },
      "minimumVolumeString" : "<string>",
      "additive" : [{ // Additive associated with container
        // additive[x]: Additive associated with container. One of these 2:
        "additiveCodeableConcept" : { CodeableConcept }
        "additiveReference" : { Reference(Substance) }
      }],
      "preparation" : "<string>" // Specimen container preparation
    },
    "requirement" : "<string>", // Specimen requirements
    "retentionTime" : { Duration }, // Specimen retention time
    "rejectionCriterion" : [{ CodeableConcept }], // Rejection criterion
    "handling" : [{ // Specimen handling before testing
      "temperatureQualifier" : { CodeableConcept }, // Temperature qualifier
      "temperatureRange" : { Range }, // Temperature range
      "maxDuration" : { Duration }, // Maximum preservation time
      "instruction" : "<string>" // Preservation instruction
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SpecimenDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SpecimenDefinition.identifier [ Identifier ]; # 0..1 Business identifier of a kind of specimen
  fhir:SpecimenDefinition.typeCollected [ CodeableConcept ]; # 0..1 Kind of material to collect
  fhir:SpecimenDefinition.patientPreparation [ CodeableConcept ], ... ; # 0..* Patient preparation for collection
  fhir:SpecimenDefinition.timeAspect [ string ]; # 0..1 Time aspect for collection
  fhir:SpecimenDefinition.collection [ CodeableConcept ], ... ; # 0..* Specimen collection procedure
  fhir:SpecimenDefinition.typeTested [ # 0..* Specimen in container intended for testing by lab
    fhir:SpecimenDefinition.typeTested.isDerived [ boolean ]; # 0..1 Primary or secondary specimen
    fhir:SpecimenDefinition.typeTested.type [ CodeableConcept ]; # 0..1 Type of intended specimen
    fhir:SpecimenDefinition.typeTested.preference [ code ]; # 1..1 preferred | alternate
    fhir:SpecimenDefinition.typeTested.container [ # 0..1 The specimen's container
      fhir:SpecimenDefinition.typeTested.container.material [ CodeableConcept ]; # 0..1 Container material
      fhir:SpecimenDefinition.typeTested.container.type [ CodeableConcept ]; # 0..1 Kind of container associated with
    the kind of specimen
      fhir:SpecimenDefinition.typeTested.container.cap [ CodeableConcept ]; # 0..1 Color of container cap
      fhir:SpecimenDefinition.typeTested.container.description [ string ]; # 0..1 Container description
      fhir:SpecimenDefinition.typeTested.container.capacity [ Quantity(SimpleQuantity) ]; # 0..1 Container capacity
      # SpecimenDefinition.typeTested.container.minimumVolume[x] : 0..1 Minimum volume. One of these 2
        fhir:SpecimenDefinition.typeTested.container.minimumVolumeSimpleQuantity [ Quantity(SimpleQuantity) ]
        fhir:SpecimenDefinition.typeTested.container.minimumVolumeString [ string ]
      fhir:SpecimenDefinition.typeTested.container.additive [ # 0..* Additive associated with container
        # SpecimenDefinition.typeTested.container.additive.additive[x] : 1..1 Additive associated with container. One of
     these 2
          fhir:SpecimenDefinition.typeTested.container.additive.additiveCodeableConcept [ CodeableConcept ]
          fhir:SpecimenDefinition.typeTested.container.additive.additiveReference [ Reference(Substance) ]
      ], ...;
      fhir:SpecimenDefinition.typeTested.container.preparation [ string ]; # 0..1 Specimen container preparation
    ];
    fhir:SpecimenDefinition.typeTested.requirement [ string ]; # 0..1 Specimen requirements
    fhir:SpecimenDefinition.typeTested.retentionTime [ Duration ]; # 0..1 Specimen retention time
    fhir:SpecimenDefinition.typeTested.rejectionCriterion [ CodeableConcept ], ... ; # 0..* Rejection criterion
    fhir:SpecimenDefinition.typeTested.handling [ # 0..* Specimen handling before testing
      fhir:SpecimenDefinition.typeTested.handling.temperatureQualifier [ CodeableConcept ]; # 0..1 Temperature qualifier
      fhir:SpecimenDefinition.typeTested.handling.temperatureRange [ Range ]; # 0..1 Temperature range
      fhir:SpecimenDefinition.typeTested.handling.maxDuration [ Duration ]; # 0..1 Maximum preservation time
      fhir:SpecimenDefinition.typeTested.handling.instruction [ string ]; # 0..1 Preservation instruction
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = Not Mapped)

Structure

Name | Flags | Card. | Type | Description & Constraints
SpecimenDefinition | TU | | DomainResource | Kind of specimen Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Business identifier of a kind of specimen
typeCollected | Σ | 0..1 | CodeableConcept | Kind of material to collect v2 Specimen Type (Example)
patientPreparation | Σ | 0..* | CodeableConcept | Patient preparation for collection Patient preparation prior specimen collection (Example)
timeAspect | Σ | 0..1 | string | Time aspect for collection
collection | Σ | 0..* | CodeableConcept | Specimen collection procedure Specimen collection methods (Example)
typeTested | | 0..* | BackboneElement | Specimen in container intended for testing by lab
isDerived | | 0..1 | boolean | Primary or secondary specimen
type | | 0..1 | CodeableConcept | Type of intended specimen v2 Specimen Type (Example)
preference | | 1..1 | code | preferred | alternate SpecimenContainedPreference (Required)
container | | 0..1 | BackboneElement | The specimen's container
material | | 0..1 | CodeableConcept | Container material Types of material for specimen containers (Example)
type | | 0..1 | CodeableConcept | Kind of container associated with the kind of specimen Specimen Container Type (Example)
cap | | 0..1 | CodeableConcept | Color of container cap ContainerCap (Example)
description | | 0..1 | string | Container description
capacity | | 0..1 | SimpleQuantity | Container capacity
minimumVolume[x] | | 0..1 | | Minimum volume
minimumVolumeQuantity | | | SimpleQuantity
minimumVolumeString | | | string
additive | | 0..* | BackboneElement | Additive associated with container
additive[x] | | 1..1 | | Additive associated with container v2 Additive (Example)
additiveCodeableConcept | | | CodeableConcept
additiveReference | | | Reference(Substance)

preparation | | 0..1 | string | Specimen container preparation
requirement | | 0..1 | string | Specimen requirements
retentionTime | | 0..1 | Duration | Specimen retention time
rejectionCriterion | | 0..* | CodeableConcept | Rejection criterion RejectionCriterion (Example)
handling | | 0..* | BackboneElement | Specimen handling before testing
temperatureQualifier | | 0..1 | CodeableConcept | Temperature qualifier HandlingConditionSet (Example)
temperatureRange | | 0..1 | Range | Temperature range
maxDuration | | 0..1 | Duration | Maximum preservation time
instruction | | 0..1 | string | Preservation instruction
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SpecimenDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business identifier of a kind of specimen --></identifier>
 <typeCollected><!-- 0..1 CodeableConcept Kind of material to collect --></typeCollected>
 <patientPreparation><!-- 0..* CodeableConcept Patient preparation for collection --></patientPreparation>
 <timeAspect value="[string]"/><!-- 0..1 Time aspect for collection -->
 <collection><!-- 0..* CodeableConcept Specimen collection procedure --></collection>
 <typeTested>  <!-- 0..* Specimen in container intended for testing by lab -->
  <isDerived value="[boolean]"/><!-- 0..1 Primary or secondary specimen -->
  <type><!-- 0..1 CodeableConcept Type of intended specimen --></type>
  <preference value="[code]"/><!-- 1..1 preferred | alternate -->
  <container>  <!-- 0..1 The specimen's container -->
   <material><!-- 0..1 CodeableConcept Container material --></material>
   <type><!-- 0..1 CodeableConcept Kind of container associated with the kind of specimen --></type>
   <cap><!-- 0..1 CodeableConcept Color of container cap --></cap>
   <description value="[string]"/><!-- 0..1 Container description -->
   <capacity><!-- 0..1 Quantity(SimpleQuantity) Container capacity --></capacity>
   <minimumVolume[x]><!-- 0..1 Quantity(SimpleQuantity)|string Minimum volume --></minimumVolume[x]>
   <additive>  <!-- 0..* Additive associated with container -->
    <additive[x]><!-- 1..1 CodeableConcept|Reference(Substance) Additive associated with container --></additive[x]>
   </additive>
   <preparation value="[string]"/><!-- 0..1 Specimen container preparation -->
  </container>
  <requirement value="[string]"/><!-- 0..1 Specimen requirements -->
  <retentionTime><!-- 0..1 Duration Specimen retention time --></retentionTime>
  <rejectionCriterion><!-- 0..* CodeableConcept Rejection criterion --></rejectionCriterion>
  <handling>  <!-- 0..* Specimen handling before testing -->
   <temperatureQualifier><!-- 0..1 CodeableConcept Temperature qualifier --></temperatureQualifier>
   <temperatureRange><!-- 0..1 Range Temperature range --></temperatureRange>
   <maxDuration><!-- 0..1 Duration Maximum preservation time --></maxDuration>
   <instruction value="[string]"/><!-- 0..1 Preservation instruction -->
  </handling>
 </typeTested>
</SpecimenDefinition>
```

JSON Template

```
{
  "resourceType" : "SpecimenDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business identifier of a kind of specimen
  "typeCollected" : { CodeableConcept }, // Kind of material to collect
  "patientPreparation" : [{ CodeableConcept }], // Patient preparation for collection
  "timeAspect" : "<string>", // Time aspect for collection
  "collection" : [{ CodeableConcept }], // Specimen collection procedure
  "typeTested" : [{ // Specimen in container intended for testing by lab
    "isDerived" : <boolean>, // Primary or secondary specimen
    "type" : { CodeableConcept }, // Type of intended specimen
    "preference" : "<code>", // R!  preferred | alternate
    "container" : { // The specimen's container
      "material" : { CodeableConcept }, // Container material
      "type" : { CodeableConcept }, // Kind of container associated with the kind of specimen
      "cap" : { CodeableConcept }, // Color of container cap
      "description" : "<string>", // Container description
      "capacity" : { Quantity(SimpleQuantity) }, // Container capacity
      // minimumVolume[x]: Minimum volume. One of these 2:
      "minimumVolumeQuantity" : { Quantity(SimpleQuantity) },
      "minimumVolumeString" : "<string>",
      "additive" : [{ // Additive associated with container
        // additive[x]: Additive associated with container. One of these 2:
        "additiveCodeableConcept" : { CodeableConcept }
        "additiveReference" : { Reference(Substance) }
      }],
      "preparation" : "<string>" // Specimen container preparation
    },
    "requirement" : "<string>", // Specimen requirements
    "retentionTime" : { Duration }, // Specimen retention time
    "rejectionCriterion" : [{ CodeableConcept }], // Rejection criterion
    "handling" : [{ // Specimen handling before testing
      "temperatureQualifier" : { CodeableConcept }, // Temperature qualifier
      "temperatureRange" : { Range }, // Temperature range
      "maxDuration" : { Duration }, // Maximum preservation time
```

```
      "instruction" : "<string>" // Preservation instruction
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SpecimenDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SpecimenDefinition.identifier [ Identifier ]; # 0..1 Business identifier of a kind of specimen
  fhir:SpecimenDefinition.typeCollected [ CodeableConcept ]; # 0..1 Kind of material to collect
  fhir:SpecimenDefinition.patientPreparation [ CodeableConcept ], ... ; # 0..* Patient preparation for collection
  fhir:SpecimenDefinition.timeAspect [ string ]; # 0..1 Time aspect for collection
  fhir:SpecimenDefinition.collection [ CodeableConcept ], ... ; # 0..* Specimen collection procedure
  fhir:SpecimenDefinition.typeTested [ # 0..* Specimen in container intended for testing by lab
    fhir:SpecimenDefinition.typeTested.isDerived [ boolean ]; # 0..1 Primary or secondary specimen
    fhir:SpecimenDefinition.typeTested.type [ CodeableConcept ]; # 0..1 Type of intended specimen
    fhir:SpecimenDefinition.typeTested.preference [ code ]; # 1..1 preferred | alternate
    fhir:SpecimenDefinition.typeTested.container [ # 0..1 The specimen's container
      fhir:SpecimenDefinition.typeTested.container.material [ CodeableConcept ]; # 0..1 Container material
      fhir:SpecimenDefinition.typeTested.container.type [ CodeableConcept ]; # 0..1 Kind of container associated with
    the kind of specimen
      fhir:SpecimenDefinition.typeTested.container.cap [ CodeableConcept ]; # 0..1 Color of container cap
      fhir:SpecimenDefinition.typeTested.container.description [ string ]; # 0..1 Container description
      fhir:SpecimenDefinition.typeTested.container.capacity [ Quantity(SimpleQuantity) ]; # 0..1 Container capacity
      # SpecimenDefinition.typeTested.container.minimumVolume[x] : 0..1 Minimum volume. One of these 2
        fhir:SpecimenDefinition.typeTested.container.minimumVolumeSimpleQuantity [ Quantity(SimpleQuantity) ]
        fhir:SpecimenDefinition.typeTested.container.minimumVolumeString [ string ]
      fhir:SpecimenDefinition.typeTested.container.additive [ # 0..* Additive associated with container
        # SpecimenDefinition.typeTested.container.additive.additive[x] : 1..1 Additive associated with container. One of
     these 2
          fhir:SpecimenDefinition.typeTested.container.additive.additiveCodeableConcept [ CodeableConcept ]
          fhir:SpecimenDefinition.typeTested.container.additive.additiveReference [ Reference(Substance) ]
      ], ...;
      fhir:SpecimenDefinition.typeTested.container.preparation [ string ]; # 0..1 Specimen container preparation
    ];
    fhir:SpecimenDefinition.typeTested.requirement [ string ]; # 0..1 Specimen requirements
    fhir:SpecimenDefinition.typeTested.retentionTime [ Duration ]; # 0..1 Specimen retention time
    fhir:SpecimenDefinition.typeTested.rejectionCriterion [ CodeableConcept ], ... ; # 0..* Rejection criterion
    fhir:SpecimenDefinition.typeTested.handling [ # 0..* Specimen handling before testing
      fhir:SpecimenDefinition.typeTested.handling.temperatureQualifier [ CodeableConcept ]; # 0..1 Temperature qualifier
      fhir:SpecimenDefinition.typeTested.handling.temperatureRange [ Range ]; # 0..1 Temperature range
      fhir:SpecimenDefinition.typeTested.handling.maxDuration [ Duration ]; # 0..1 Maximum preservation time
      fhir:SpecimenDefinition.typeTested.handling.instruction [ string ]; # 0..1 Preservation instruction
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = Not Mapped)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.8.3.1 Terminology Bindings

Path	Definition	Type	Reference
SpecimenDefinition.typeCollected	The type of the specimen to be collected.	Example	v2.0487
SpecimenDefinition.patientPreparation	Checks on the patient prior specimen collection.	Example	PreparePatient
SpecimenDefinition.collection	The action to collect a type of specimen.	Example	SpecimenCollection
SpecimenDefinition.typeTested.type	The type of specimen conditioned in a container for lab testing.	Example	v2.0487
SpecimenDefinition.typeTested.preference	Degree of preference of a type of conditioned specimen.	Required	SpecimenContainedPreference
SpecimenDefinition.typeTested.container.material	Types of material for specimen containers.	Example	ContainerMaterials
SpecimenDefinition.typeTested.container.type	Type of specimen container.	Example	SpecimenContainerType
SpecimenDefinition.typeTested.container.cap	Color of the container cap.	Example	ContainerCap
SpecimenDefinition.typeTested.container.additive.additive[x]	Substance added to specimen container.	Example	v2.0371
SpecimenDefinition.typeTested.rejectionCriterion	Criterion for rejection of the specimen by laboratory.	Example	RejectionCriterion
SpecimenDefinition.typeTested.handling.temperatureQualifier	Set of handling instructions prior testing of the specimen.	Example	HandlingConditionSet

10.8.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
container | token | The type of specimen conditioned in container expected by the lab | SpecimenDefinition.typeTested.container.type
identifier | token | The unique identifier associated with the specimen | SpecimenDefinition.identifier
type | token | The type of collected specimen | SpecimenDefinition.typeCollected

10.8.5 Resource SpecimenDefinition - Examples

URL: https://hl7.org/fhir/R4/specimendefinition-examples.html

10.8.5 Resource SpecimenDefinition - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example definition of serum or plasma specimen. | 2364 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.8.6 Resource SpecimenDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/specimendefinition-definitions.html

10.8.6 Resource SpecimenDefinition - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SpecimenDefinition resource.

SpecimenDefinition
Element Id | SpecimenDefinition
Definition | A kind of specimen with associated set of requirements.
Cardinality | 0..*
Type | DomainResource
SpecimenDefinition.identifier
Element Id | SpecimenDefinition.identifier
Definition | A business identifier associated with the kind of specimen.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
SpecimenDefinition.typeCollected
Element Id | SpecimenDefinition.typeCollected
Definition | The kind of material to be collected.
Cardinality | 0..1
Terminology Binding | v2 Specimen Type (Example)
Type | CodeableConcept
Summary | true
SpecimenDefinition.patientPreparation
Element Id | SpecimenDefinition.patientPreparation
Definition | Preparation of the patient for specimen collection.
Cardinality | 0..*
Terminology Binding | Patient preparation prior specimen collection (Example)
Type | CodeableConcept
Summary | true
SpecimenDefinition.timeAspect
Element Id | SpecimenDefinition.timeAspect
Definition | Time aspect of specimen collection (duration or offset).
Cardinality | 0..1
Type | string
Summary | true
SpecimenDefinition.collection
Element Id | SpecimenDefinition.collection
Definition | The action to be performed for collecting the specimen.
Cardinality | 0..*
Terminology Binding | Specimen collection methods (Example)
Type | CodeableConcept
Summary | true
SpecimenDefinition.typeTested
Element Id | SpecimenDefinition.typeTested
Definition | Specimen conditioned in a container as expected by the testing laboratory.
Cardinality | 0..*
SpecimenDefinition.typeTested.isDerived

Element Id | SpecimenDefinition.typeTested.isDerived
Definition | Primary of secondary specimen.
Cardinality | 0..1
Type | boolean
SpecimenDefinition.typeTested.type
Element Id | SpecimenDefinition.typeTested.type
Definition | The kind of specimen conditioned for testing expected by lab.
Cardinality | 0..1
Terminology Binding | v2 Specimen Type (Example)
Type | CodeableConcept
SpecimenDefinition.typeTested.preference
Element Id | SpecimenDefinition.typeTested.preference
Definition | The preference for this type of conditioned specimen.
Cardinality | 1..1
Terminology Binding | SpecimenContainedPreference (Required)
Type | code
SpecimenDefinition.typeTested.container
Element Id | SpecimenDefinition.typeTested.container
Definition | The specimen's container.
Cardinality | 0..1
SpecimenDefinition.typeTested.container.material
Element Id | SpecimenDefinition.typeTested.container.material
Definition | The type of material of the container.
Cardinality | 0..1
Terminology Binding | Types of material for specimen containers (Example)
Type | CodeableConcept
SpecimenDefinition.typeTested.container.type
Element Id | SpecimenDefinition.typeTested.container.type
Definition | The type of container used to contain this kind of specimen.
Cardinality | 0..1
Terminology Binding | Specimen Container Type (Example)
Type | CodeableConcept
SpecimenDefinition.typeTested.container.cap
Element Id | SpecimenDefinition.typeTested.container.cap
Definition | Color of container cap.
Cardinality | 0..1
Terminology Binding | ContainerCap (Example)
Type | CodeableConcept
Requirements | From ISO 6710-2017 Table F.1 Recommended colour codes.
SpecimenDefinition.typeTested.container.description
Element Id | SpecimenDefinition.typeTested.container.description
Definition | The textual description of the kind of container.
Cardinality | 0..1
Type | string
SpecimenDefinition.typeTested.container.capacity
Element Id | SpecimenDefinition.typeTested.container.capacity
Definition | The capacity (volume or other measure) of this kind of container.
Cardinality | 0..1
Type | SimpleQuantity
SpecimenDefinition.typeTested.container.minimumVolume[x]
Element Id | SpecimenDefinition.typeTested.container.minimumVolume[x]
Definition | The minimum volume to be conditioned in the container.
Cardinality | 0..1
Type | SimpleQuantity|string
[x] Note | See Choice of Data Types for further information about how to use [x]
SpecimenDefinition.typeTested.container.additive
Element Id | SpecimenDefinition.typeTested.container.additive
Definition | Substance introduced in the kind of container to preserve, maintain or enhance the specimen. Examples: Formalin, Citrate, EDTA.
Cardinality | 0..*
SpecimenDefinition.typeTested.container.additive.additive[x]
Element Id | SpecimenDefinition.typeTested.container.additive.additive[x]
Definition | Substance introduced in the kind of container to preserve, maintain or enhance the specimen. Examples: Formalin, Citrate, EDTA.
Cardinality | 1..1
Terminology Binding | v2 Additive (Example)
Type | CodeableConcept|Reference(Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
SpecimenDefinition.typeTested.container.preparation
Element Id | SpecimenDefinition.typeTested.container.preparation
Definition | Special processing that should be applied to the container for this kind of specimen.
Cardinality | 0..1

Type | string
SpecimenDefinition.typeTested.requirement
Element Id | SpecimenDefinition.typeTested.requirement
Definition | Requirements for delivery and special handling of this kind of conditioned specimen.
Cardinality | 0..1
Type | string
SpecimenDefinition.typeTested.retentionTime
Element Id | SpecimenDefinition.typeTested.retentionTime
Definition | The usual time that a specimen of this kind is retained after the ordered tests are completed, for the purpose of additional testing.
Cardinality | 0..1
Type | Duration
SpecimenDefinition.typeTested.rejectionCriterion
Element Id | SpecimenDefinition.typeTested.rejectionCriterion
Definition | Criterion for rejection of the specimen in its container by the laboratory.
Cardinality | 0..*
Terminology Binding | RejectionCriterion (Example)
Type | CodeableConcept
SpecimenDefinition.typeTested.handling
Element Id | SpecimenDefinition.typeTested.handling
Definition | Set of instructions for preservation/transport of the specimen at a defined temperature interval, prior the testing process.
Cardinality | 0..*
SpecimenDefinition.typeTested.handling.temperatureQualifier
Element Id | SpecimenDefinition.typeTested.handling.temperatureQualifier
Definition | It qualifies the interval of temperature, which characterizes an occurrence of handling. Conditions that are not related to temperature may be handled in the instruction element.
Cardinality | 0..1
Terminology Binding | HandlingConditionSet (Example)
Type | CodeableConcept
SpecimenDefinition.typeTested.handling.temperatureRange
Element Id | SpecimenDefinition.typeTested.handling.temperatureRange
Definition | The temperature interval for this set of handling instructions.
Cardinality | 0..1
Type | Range
SpecimenDefinition.typeTested.handling.maxDuration
Element Id | SpecimenDefinition.typeTested.handling.maxDuration
Definition | The maximum time interval of preservation of the specimen with these conditions.
Cardinality | 0..1
Type | Duration
SpecimenDefinition.typeTested.handling.instruction
Element Id | SpecimenDefinition.typeTested.handling.instruction
Definition | Additional textual instructions for the preservation or transport of the specimen. For instance, 'Protect from light exposure'.
Cardinality | 0..1
Type | string

10.8.7 Resource SpecimenDefinition - Mappings

URL: https://hl7.org/fhir/R4/specimendefinition-mappings.html

10.8.7 Resource SpecimenDefinition - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

HL7 v2 Mapping

RIM Mapping

Mappings for the specimendefinition resource (see Mappings to Other Standards for further information & status).

10.8.7.1 HL7 v2 Mapping (http://hl7.org/v2)

SpecimenDefinition | OM4
identifier
typeCollected | OM4-6
patientPreparation | OM1-37
timeAspect | OM1-43, OM1-44
collection
typeTested
isDerived | OM4-2
type

preference | OM4-16
container
material
type
cap
description | OM4-3
capacity | OM4-4, OM4-5
minimumVolume[x] | OM4-11
additive | OM4-7
additive[x] | OM4-7
preparation | OM4-8
requirement | OM4-12
retentionTime | OM4-14
rejectionCriterion | OM1-39
handling
temperatureQualifier | OM4-15
temperatureRange
maxDuration
instruction

10.8.7.2 RIM Mapping (http://hl7.org/v3)

SpecimenDefinition | Role[classCode=SPEC]

10.8.8 Resource SpecimenDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/specimendefinition-profiles.html

10.8.8 Resource SpecimenDefinition - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SpecimenDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.8.8.1 Profiles

No Profiles defined for this resource

10.8.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.8.8.3 Search Extensions

No Search Extensions defined for this resource

10.8.9 Resource SpecimenDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/specimendefinition-version-maps.html

n/a

10.8.9 Resource SpecimenDefinition - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for SpecimenDefinition.

Functional status for this map: Not Mapped (see documentation)

10.8.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/SpecimenDefinition3to4" = "R3 to R4 Conversions for SpecimenDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/SpecimenDefinition" alias SpecimenDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/SpecimenDefinition" alias SpecimenDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group SpecimenDefinition(source src : SpecimenDefinitionR3, target tgt : SpecimenDefinition) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.typeCollected -> tgt.typeCollected;
  src.patientPreparation -> tgt.patientPreparation;
  src.timeAspect -> tgt.timeAspect;
  src.collection -> tgt.collection;
  src.specimenToLab as s -> tgt.specimenToLab as t then SpecimenDefinitionSpecimenToLab(s, t);
}

group SpecimenDefinitionSpecimenToLab(source src, target tgt) extends BackboneElement {
  src.isDerived -> tgt.isDerived;
  src.type -> tgt.type;
  src.preference -> tgt.preference;
  src.containerMaterial -> tgt.containerMaterial;
  src.containerType -> tgt.containerType;
  src.containerCap -> tgt.containerCap;
  src.containerDescription -> tgt.containerDescription;
  src.containerCapacity -> tgt.containerCapacity;
  src.containerMinimumVolume -> tgt.containerMinimumVolume;
  src.containerAdditive as s -> tgt.containerAdditive as t then SpecimenDefinitionSpecimenToLabContainerAdditive(s, t);
  src.containerPreparation -> tgt.containerPreparation;
  src.requirement -> tgt.requirement;
  src.retentionTime -> tgt.retentionTime;
  src.rejectionCriterion -> tgt.rejectionCriterion;
  src.handling as s -> tgt.handling as t then SpecimenDefinitionSpecimenToLabHandling(s, t);
}

group SpecimenDefinitionSpecimenToLabContainerAdditive(source src, target tgt) extends BackboneElement {
  src.additive -> tgt.additive;
}

group SpecimenDefinitionSpecimenToLabHandling(source src, target tgt) extends BackboneElement {
  src.conditionSet -> tgt.conditionSet;
  src.tempRange -> tgt.tempRange;
  src.maxDuration -> tgt.maxDuration;
  src.lightExposure -> tgt.lightExposure;
  src.instruction -> tgt.instruction;
}
```

10.8.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/SpecimenDefinition4to3" = "R4 to R3 Conversion for SpecimenDefinition"

uses "http://hl7.org/fhir/StructureDefinition/SpecimenDefinition" alias SpecimenDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/SpecimenDefinition" alias SpecimenDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group SpecimenDefinition(source src : SpecimenDefinitionR3, target tgt : SpecimenDefinition) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.typeCollected -> tgt.typeCollected;
  src.patientPreparation -> tgt.patientPreparation;
  src.timeAspect -> tgt.timeAspect;
  src.collection -> tgt.collection;
  src.specimenToLab as s -> tgt.specimenToLab as t then SpecimenDefinitionSpecimenToLab(s, t);
}

group SpecimenDefinitionSpecimenToLab(source src, target tgt) extends BackboneElement {
  src.isDerived -> tgt.isDerived;
  src.type -> tgt.type;
  src.preference -> tgt.preference;
  src.containerMaterial -> tgt.containerMaterial;
  src.containerType -> tgt.containerType;
  src.containerCap -> tgt.containerCap;
  src.containerDescription -> tgt.containerDescription;
  src.containerCapacity -> tgt.containerCapacity;
  src.containerMinimumVolume -> tgt.containerMinimumVolume;
  src.containerAdditive as s -> tgt.containerAdditive as t then SpecimenDefinitionSpecimenToLabContainerAdditive(s, t);
  src.containerPreparation -> tgt.containerPreparation;
  src.requirement -> tgt.requirement;
  src.retentionTime -> tgt.retentionTime;
  src.rejectionCriterion -> tgt.rejectionCriterion;
  src.handling as s -> tgt.handling as t then SpecimenDefinitionSpecimenToLabHandling(s, t);
}

group SpecimenDefinitionSpecimenToLabContainerAdditive(source src, target tgt) extends BackboneElement {
  src.additive -> tgt.additive;
}

group SpecimenDefinitionSpecimenToLabHandling(source src, target tgt) extends BackboneElement {
  src.conditionSet -> tgt.conditionSet;
```

```
  src.tempRange -> tgt.tempRange;
  src.maxDuration -> tgt.maxDuration;
  src.lightExposure -> tgt.lightExposure;
  src.instruction -> tgt.instruction;
}
```

10.8.9.3 R4 Validation Errors

10.8.10 JSON Schema for SpecimenDefinition

URL: https://hl7.org/fhir/R4/specimendefinition.schema.json.html

10.8.10 JSON Schema for SpecimenDefinition

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SpecimenDefinition, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SpecimenDefinition",
  "$ref": "#/definitions/SpecimenDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SpecimenDefinition": {
      "description": "A kind of specimen with associated set of requirements.",
      "properties": {
        "resourceType": {
          "description": "This is a SpecimenDefinition resource",
          "const": "SpecimenDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A business identifier associated with the kind of specimen.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "typeCollected": {
      "description": "The kind of material to be collected.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "patientPreparation": {
      "description": "Preparation of the patient for specimen collection.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "timeAspect": {
      "description": "Time aspect of specimen collection (duration or offset).",
      "$ref": "#/definitions/string"
    },
    "_timeAspect": {
      "description": "Extensions for timeAspect",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "collection": {
      "description": "The action to be performed for collecting the specimen.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "typeTested": {
      "description": "Specimen conditioned in a container as expected by the testing laboratory.",
      "items": {
        "$ref": "#/definitions/SpecimenDefinition_TypeTested"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SpecimenDefinition_TypeTested": {
  "description": "A kind of specimen with associated set of requirements.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "isDerived": {
      "description": "Primary of secondary specimen.",
```

```
      "$ref": "#/definitions/boolean"
    },
    "_isDerived": {
      "description": "Extensions for isDerived",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The kind of specimen conditioned for testing expected by lab.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "preference": {
      "description": "The preference for this type of conditioned specimen.",
      "enum": [
        "preferred",
        "alternate"
      ]
    },
    "_preference": {
      "description": "Extensions for preference",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "container": {
      "description": "The specimen\u0027s container.",
      "$ref": "#/definitions/SpecimenDefinition_Container"
    },
    "requirement": {
      "description": "Requirements for delivery and special handling of this kind of conditioned specimen.",
      "$ref": "#/definitions/string"
    },
    "_requirement": {
      "description": "Extensions for requirement",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "retentionTime": {
      "description": "The usual time that a specimen of this kind is retained after the ordered tests are completed,
 for the purpose of additional testing.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "rejectionCriterion": {
      "description": "Criterion for rejection of the specimen in its container by the laboratory.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "handling": {
      "description": "Set of instructions for preservation/transport of the specimen at a defined temperature
interval, prior the testing process.",
      "items": {
        "$ref": "#/definitions/SpecimenDefinition_Handling"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SpecimenDefinition_Container": {
  "description": "A kind of specimen with associated set of requirements.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "material": {
```

```
      "description": "The type of material of the container.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "type": {
      "description": "The type of container used to contain this kind of specimen.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "cap": {
      "description": "Color of container cap.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "The textual description of the kind of container.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "capacity": {
      "description": "The capacity (volume or other measure) of this kind of container.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "minimumVolumeQuantity": {
      "description": "The minimum volume to be conditioned in the container.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "minimumVolumeString": {
      "description": "The minimum volume to be conditioned in the container.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_minimumVolumeString": {
      "description": "Extensions for minimumVolumeString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "additive": {
      "description": "Substance introduced in the kind of container to preserve, maintain or enhance the specimen.
Examples: Formalin, Citrate, EDTA.",
      "items": {
        "$ref": "#/definitions/SpecimenDefinition_Additive"
      },
      "type": "array"
    },
    "preparation": {
      "description": "Special processing that should be applied to the container for this kind of specimen.",
      "$ref": "#/definitions/string"
    },
    "_preparation": {
      "description": "Extensions for preparation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"SpecimenDefinition_Additive": {
  "description": "A kind of specimen with associated set of requirements.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "additiveCodeableConcept": {
      "description": "Substance introduced in the kind of container to preserve, maintain or enhance the specimen.
```

```
    Examples: Formalin, Citrate, EDTA.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "additiveReference": {
          "description": "Substance introduced in the kind of container to preserve, maintain or enhance the specimen.
    Examples: Formalin, Citrate, EDTA.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    },
    "SpecimenDefinition_Handling": {
      "description": "A kind of specimen with associated set of requirements.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "temperatureQualifier": {
          "description": "It qualifies the interval of temperature, which characterizes an occurrence of handling.
    Conditions that are not related to temperature may be handled in the instruction element.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "temperatureRange": {
          "description": "The temperature interval for this set of handling instructions.",
          "$ref": "Range.schema.json#/definitions/Range"
        },
        "maxDuration": {
          "description": "The maximum time interval of preservation of the specimen with these conditions.",
          "$ref": "Duration.schema.json#/definitions/Duration"
        },
        "instruction": {
          "description": "Additional textual instructions for the preservation or transport of the specimen. For
    instance, \u0027Protect from light exposure\u0027.",
          "$ref": "#/definitions/string"
        },
        "_instruction": {
          "description": "Extensions for instruction",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


10.8.11 Resource SpecimenDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/specimendefinition-dependencies.html

10.8.11 Resource SpecimenDefinition - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SpecimenDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
SpecimenDefinition | | DomainResource
identifier | 0..1 | Identifier
typeCollected | 0..1 | CodeableConcept

patientPreparation | 0..* | CodeableConcept
timeAspect | 0..1 | string
collection | 0..* | CodeableConcept
typeTested | 0..* | BackboneElement
isDerived | 0..1 | boolean
type | 0..1 | CodeableConcept
preference | 1..1 | code
container | 0..1 | BackboneElement
material | 0..1 | CodeableConcept
type | 0..1 | CodeableConcept
cap | 0..1 | CodeableConcept
description | 0..1 | string
capacity | 0..1 | SimpleQuantity
minimumVolume[x] | 0..1
minimumVolumeQuantity | | SimpleQuantity
minimumVolumeString | | string
additive | 0..* | BackboneElement
additive[x] | 1..1
additiveCodeableConcept | | CodeableConcept
additiveReference | | Reference(Substance
preparation | 0..1 | string
requirement | 0..1 | string
retentionTime | 0..1 | Duration
rejectionCriterion | 0..* | CodeableConcept
handling | 0..* | BackboneElement
temperatureQualifier | 0..1 | CodeableConcept
temperatureRange | 0..1 | Range
maxDuration | 0..1 | Duration
instruction | 0..1 | string

10.9 Resource BodyStructure - Content

URL: https://hl7.org/fhir/R4/bodystructure.html
Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

Record details about an anatomical structure. This resource may be used when a coded concept does not provide the necessary detail needed for the use case.

10.9.1 Scope and Usage

The BodyStructure resource contains details about the anatomical location of a specimen or body part, including patient information, identifiers, as well as text descriptions and images. It provides for the addition of qualifiers such as laterality and directionality to the anatomic location for those use cases where precoordination of codes is not possible. The BodyStructure resource supports recording and tracking of an anatomic location or structure on a patient outside the context of another resource. For example it can be the target of a Procedure resource or Observation resource.

10.9.2 Boundaries and Relationships

The BodyStructure resource is not intended to substitute for precoordination of codes. If precoordination of codes is supported by an implementation the codeableConcept should be used. This resource is not intended for describing the type of anatomical location but rather a specific body site on a specific patient.

10.9.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
BodyStructure | TU | | DomainResource | Specific and identified anatomical structure Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Bodystructure identifier
active | ?!Σ | 0..1 | boolean | Whether this record is in active use
morphology | Σ | 0..1 | CodeableConcept | Kind of Structure SNOMED CT Morphologic Abnormalities (Example)
location | Σ | 0..1 | CodeableConcept | Body site SNOMED CT Body Structures (Example)

locationQualifier | | 0..* | CodeableConcept | Body site modifier Bodystructure Location Qualifier (Example)
description | Σ | 0..1 | string | Text description
image | | 0..* | Attachment | Attached images
patient | Σ | 1..1 | Reference(Patient) | Who this is about
Documentation for this format

UML Diagram (Legend)

XML Template

```
<BodyStructure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Bodystructure identifier --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this record is in active use -->
 <morphology><!-- 0..1 CodeableConcept Kind of Structure --></morphology>
 <location><!-- 0..1 CodeableConcept Body site --></location>
 <locationQualifier><!-- 0..* CodeableConcept Body site modifier --></locationQualifier>
 <description value="[string]"/><!-- 0..1 Text description -->
 <image><!-- 0..* Attachment Attached images --></image>
 <patient><!-- 1..1 Reference(Patient) Who this is about --></patient>
</BodyStructure>
```

JSON Template

```
{
  "resourceType" : "BodyStructure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Bodystructure identifier
  "active" : <boolean>, // Whether this record is in active use
  "morphology" : { CodeableConcept }, // Kind of Structure
  "location" : { CodeableConcept }, // Body site
  "locationQualifier" : [{ CodeableConcept }], // Body site modifier
  "description" : "<string>", // Text description
  "image" : [{ Attachment }], // Attached images
  "patient" : { Reference(Patient) } // R!  Who this is about
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:BodyStructure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:BodyStructure.identifier [ Identifier ], ... ; # 0..* Bodystructure identifier
  fhir:BodyStructure.active [ boolean ]; # 0..1 Whether this record is in active use
  fhir:BodyStructure.morphology [ CodeableConcept ]; # 0..1 Kind of Structure
  fhir:BodyStructure.location [ CodeableConcept ]; # 0..1 Body site
  fhir:BodyStructure.locationQualifier [ CodeableConcept ], ... ; # 0..* Body site modifier
  fhir:BodyStructure.description [ string ]; # 0..1 Text description
  fhir:BodyStructure.image [ Attachment ], ... ; # 0..* Attached images
  fhir:BodyStructure.patient [ Reference(Patient) ]; # 1..1 Who this is about
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
BodyStructure | TU | | DomainResource | Specific and identified anatomical structure Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Bodystructure identifier
active | ?!Σ | 0..1 | boolean | Whether this record is in active use
morphology | Σ | 0..1 | CodeableConcept | Kind of Structure SNOMED CT Morphologic Abnormalities (Example)
location | Σ | 0..1 | CodeableConcept | Body site SNOMED CT Body Structures (Example)
locationQualifier | | 0..* | CodeableConcept | Body site modifier Bodystructure Location Qualifier (Example)
description | Σ | 0..1 | string | Text description
image | | 0..* | Attachment | Attached images
patient | Σ | 1..1 | Reference(Patient) | Who this is about
Documentation for this format

UML Diagram (Legend)

XML Template

```
<BodyStructure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Bodystructure identifier --></identifier>
```

```
 <active value="[boolean]"/><!-- 0..1 Whether this record is in active use -->
 <morphology><!-- 0..1 CodeableConcept Kind of Structure --></morphology>
 <location><!-- 0..1 CodeableConcept Body site --></location>
 <locationQualifier><!-- 0..* CodeableConcept Body site modifier --></locationQualifier>
 <description value="[string]"/><!-- 0..1 Text description -->
 <image><!-- 0..* Attachment Attached images --></image>
 <patient><!-- 1..1 Reference(Patient) Who this is about --></patient>
</BodyStructure>
```

JSON Template

```
{
  "resourceType" : "BodyStructure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Bodystructure identifier
  "active" : <boolean>, // Whether this record is in active use
  "morphology" : { CodeableConcept }, // Kind of Structure
  "location" : { CodeableConcept }, // Body site
  "locationQualifier" : [{ CodeableConcept }], // Body site modifier
  "description" : "<string>", // Text description
  "image" : [{ Attachment }], // Attached images
  "patient" : { Reference(Patient) } // R!  Who this is about
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:BodyStructure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:BodyStructure.identifier [ Identifier ], ... ; # 0..* Bodystructure identifier
  fhir:BodyStructure.active [ boolean ]; # 0..1 Whether this record is in active use
  fhir:BodyStructure.morphology [ CodeableConcept ]; # 0..1 Kind of Structure
  fhir:BodyStructure.location [ CodeableConcept ]; # 0..1 Body site
  fhir:BodyStructure.locationQualifier [ CodeableConcept ], ... ; # 0..* Body site modifier
  fhir:BodyStructure.description [ string ]; # 0..1 Text description
  fhir:BodyStructure.image [ Attachment ], ... ; # 0..* Attached images
  fhir:BodyStructure.patient [ Reference(Patient) ]; # 1..1 Who this is about
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.9.3.1 Terminology Bindings

Path | Definition | Type | Reference
BodyStructure.morphology | Codes describing anatomic morphology. | Example | SNOMEDCTMorphologicAbnormalities
BodyStructure.location | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures
BodyStructure.locationQualifier | Concepts modifying the anatomic location. | Example | BodystructureLocationQualifier

10.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | Bodystructure identifier | BodyStructure.identifier
location | token | Body site | BodyStructure.location
morphology | token | Kind of Structure | BodyStructure.morphology
patient | reference | Who this is about | BodyStructure.patient (Patient)

10.9.5 Resource BodyStructure - Examples

URL: https://hl7.org/fhir/R4/bodystructure-examples.html

10.9.5 Resource BodyStructure - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
This example demonstrates using the bodystructure resource to identify a fetus within systems that choose not to treat a fetus as a Patient | fetus | XML | JSON | Turtle

This example demonstrates using the bodystructure resource to identify a specific tumor or lesion that will have multiple observations and/or procedures performed on it over time | tumor | XML | JSON | Turtle
This example demonstrates using the bodystructure resource to identify skin patches or other portions of a person or animal that are a focus of a clinical trial and that are subject to repeated observations and/or procedures over time | skin-patch | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.9.6 Resource BodyStructure - Detailed Descriptions

URL: https://hl7.org/fhir/R4/bodystructure-definitions.html

10.9.6 Resource BodyStructure - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the BodyStructure resource.

BodyStructure
Element Id | BodyStructure
Definition | Record details about an anatomical structure. This resource may be used when a coded concept does not provide the necessary detail needed for the use case.
Cardinality | 0..*
Type | DomainResource
Alternate Names | anatomical location
BodyStructure.identifier
Element Id | BodyStructure.identifier
Definition | Identifier for this instance of the anatomical structure.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
BodyStructure.active
Element Id | BodyStructure.active
Definition | Whether this body site is in active use.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that can indicate that a record should not be treated as valid)
Meaning if Missing | This resource is generally assumed to be active if no value is provided for the active element
Requirements | Need to be able to mark a body site entry as not to be used because it was created in error.
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
BodyStructure.morphology
Element Id | BodyStructure.morphology
Definition | The kind of structure being represented by the body structure at BodyStructure.location. This can define both normal and abnormal morphologies.
Cardinality | 0..1
Terminology Binding | SNOMED CT Morphologic Abnormalities (Example)
Type | CodeableConcept
Summary | true
Comments | The minimum cardinality of 0 supports the use case of specifying a location without defining a morphology.
BodyStructure.location
Element Id | BodyStructure.location
Definition | The anatomical location or region of the specimen, lesion, or body structure.
Cardinality | 0..1
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
BodyStructure.locationQualifier
Element Id | BodyStructure.locationQualifier
Definition | Qualifier to refine the anatomical location. These include qualifiers for laterality, relative location, directionality, number, and plane.
Cardinality | 0..*
Terminology Binding | Bodystructure Location Qualifier (Example)
Type | CodeableConcept
BodyStructure.description
Element Id | BodyStructure.description
Definition | A summary, characterization or explanation of the body structure.
Cardinality | 0..1
Type | string
Summary | true
Comments | This description could include any visual markings used to orientate the viewer e.g. external reference points, special sutures, ink markings.

BodyStructure.image
Element Id | BodyStructure.image
Definition | Image or images used to identify a location.
Cardinality | 0..*
Type | Attachment
BodyStructure.patient
Element Id | BodyStructure.patient
Definition | The person to which the body site belongs.
Cardinality | 1..1
Type | Reference(Patient)
Summary | true

10.9.7 Resource BodyStructure - Mappings

URL: https://hl7.org/fhir/R4/bodystructure-mappings.html

10.9.7 Resource BodyStructure - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Open EHR Archetype Mapping

Mappings for the bodystructure resource (see Mappings to Other Standards for further information & status).

10.9.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

BodyStructure
identifier | FiveWs.identifier
active | FiveWs.status
morphology | FiveWs.what[x]
location | FiveWs.what[x]
locationQualifier | FiveWs.what[x]
description | FiveWs.what[x]
image | FiveWs.what[x]
patient | FiveWs.subject[x]

10.9.7.2 HL7 v2 Mapping (http://hl7.org/v2)

BodyStructure
identifier
active
morphology | SPM-4/SPM-5
location | OBX-20/SPM-8/SPM-10
locationQualifier | No distinct element defined in V2.x. The IHE Radiology Technical Framework directs the use of OBR-15-specimen source to contain laterality for imaging studies./SPM-9
description | NTE
image | OBX
patient | PID-3

10.9.7.3 RIM Mapping (http://hl7.org/v3)

BodyStructure | ActSIte
identifier | .id
active | statusCode
location | .targetSiteCode
locationQualifier | .targetSiteCode
description | .text
image | outboundRelationship[typeCode=COMP].target
patient | .scoper or .scoper.playsRole

10.9.7.4 Open EHR Archetype Mapping (http://openehr.org)

BodyStructure | Archetype ID openEHR-EHR-CLUSTER.anatomical_location.v1 Concept name Anatomical Location Concept description Record details about an anatomical location. Keywords location, site, anatomical Copyright © openEHR Foundation Purpose To record details about anatomical location. Misuse Not for specifiying unilateral/bilateral occurrence - this is related to an evaluation which perhaps includes multiple locations. Author Author name: Heather Leslie Organization: Ocean Informatics Email: heather.leslie@oceaninformatics.com Date of Origination: 2008-11-10 Translator Norwegian Bokmål: Lars Bitsch-Larsen, Haukeland University Hospital of Bergen, Norway, MD, DEAA, MBA, spec in anesthesia, spec in tropical medicine. Arabic (Syria): Mona Saleh Other contributors Heather Leslie, Ocean Informatics, Australia (Editor) Ian McNicoll, freshEHR Clinical Informatics, United Kingdom (Editor) Dunmail Hodkinson, Black Pear Software Ltd, UK References Use cases: - 5LICS, midclavicular line - L forearm, 2cm anterior to the cubital fossa - Right iliac fossa - 5 cm below L tibial tuberosity - 2 cm medial to R nipple - 4 cm below

R costal margin in midclavicular line - R femur - midline nose - medial aspect of R great toe nail.
identifier | Specific location Cluster Optional (0..1) Cardinality: Mandatory, repeating, unordered (1..*)
active
morphology
location | Name of location
locationQualifier | Combines Side, Numerical identifier, Anatomical plane,and Aspect
description | Description of anatomical location/Visual markings/orientation
image | Image
patient

10.9.8 Resource BodyStructure - Extensions & Profiles

URL: https://hl7.org/fhir/R4/bodystructure-profiles.html

10.9.8 Resource BodyStructure - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the BodyStructure resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.9.8.1 Profiles

No Profiles defined for this resource

10.9.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

bodySite | BodyStructure Reference | for General Extensions for use by FHIR Implementers
procedure-approachBodyStructure | approachBodyStructure | for Procedure HL7 Extensions
procedure-targetBodyStructure | targetBodyStructure | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.9.8.3 Search Extensions

No Search Extensions defined for this resource

10.9.9 Resource BodyStructure - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/bodystructure-version-maps.html

No validation errors - all conversions are clean

10.9.9 Resource BodyStructure - R3/R4 Conversions

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for BodyStructure.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

10.9.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/BodySite3to4" = "R3 to R4 Conversions for BodySite"

uses "http://hl7.org/fhir/3.0/StructureDefinition/BodySite" alias BodySiteR3 as source
uses "http://hl7.org/fhir/StructureDefinition/BodyStructure" alias BodyStructure as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group BodySite(source src : BodySiteR3, target tgt : BodyStructure) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.code -> tgt.location;
  src.qualifier -> tgt.locationQualifier;
```

```
  src.description -> tgt.description;
  src.image -> tgt.image;
  src.patient -> tgt.patient;
}
```

10.9.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/BodyStructure4to3" = "R4 to R3 Conversion for BodyStructure"

uses "http://hl7.org/fhir/StructureDefinition/BodyStructure" alias BodyStructure as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/BodySite" alias BodySiteR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group BodyStructure(source src : BodyStructure, target tgt : BodySiteR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.active -> tgt.active;
  src.location -> tgt.code;
  src.locationQualifier -> tgt.qualifier;
  src.description -> tgt.description;
  src.image -> tgt.image;
  src.patient -> tgt.patient;
}
```

10.9.9.3 R4 Validation Errors

10.9.10 JSON Schema for BodyStructure

URL: https://hl7.org/fhir/R4/bodystructure.schema.json.html

10.9.10 JSON Schema for BodyStructure

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for BodyStructure, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/BodyStructure",
  "$ref": "#/definitions/BodyStructure",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "BodyStructure": {
      "description": "Record details about an anatomical structure.  This resource may be used when a coded concept does
     not provide the necessary detail needed for the use case.",
      "properties": {
        "resourceType": {
          "description": "This is a BodyStructure resource",
          "const": "BodyStructure"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
```

```
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "identifier": {
        "description": "Identifier for this instance of the anatomical structure.",
        "items": {
          "$ref": "Identifier.schema.json#/definitions/Identifier"
        },
        "type": "array"
      },
      "active": {
        "description": "Whether this body site is in active use.",
        "$ref": "#/definitions/boolean"
      },
      "_active": {
        "description": "Extensions for active",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "morphology": {
        "description": "The kind of structure being represented by the body structure at `BodyStructure.location`.
    This can define both normal and abnormal morphologies.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "location": {
        "description": "The anatomical location or region of the specimen, lesion, or body structure.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "locationQualifier": {
        "description": "Qualifier to refine the anatomical location.  These include qualifiers for laterality,
    relative location, directionality, number, and plane.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "description": {
        "description": "A summary, characterization or explanation of the body structure.",
        "$ref": "#/definitions/string"
      },
      "_description": {
        "description": "Extensions for description",
        "$ref": "Element.schema.json#/definitions/Element"

      },
      "image": {
        "description": "Image or images used to identify a location.",
        "items": {
          "$ref": "Attachment.schema.json#/definitions/Attachment"
        },
        "type": "array"
      },
      "patient": {
        "description": "The person to which the body site belongs.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      }
    },
    "additionalProperties": false,
    "required": [
      "patient",
      "resourceType"
    ]
  }
 }
}
```

10.9.11 Resource BodyStructure - Maturity Dependency View

URL: https://hl7.org/fhir/R4/bodystructure-dependencies.html

10.9.11 Resource BodyStructure - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Dependency Graph for BodyStructure FMM level 1

Name | Card. | Type | Dependency Analysis
BodyStructure | | DomainResource
identifier | 0..* | Identifier
active | 0..1 | boolean
morphology | 0..1 | CodeableConcept
location | 0..1 | CodeableConcept
locationQualifier | 0..* | CodeableConcept
description | 0..1 | string
image | 0..* | Attachment
patient | 1..1 | Reference(Patient)

10.10 Genomics Implementation Guidance

URL: https://hl7.org/fhir/R4/genomics.html

(see the publication above for details about each)

10.10 Genomics Implementation Guidance

Clinical Genomics Work Group | Maturity Level: 1 | Standards Status: Trial Use

Table of Contents

• Background
• Overview
• MolecularSequence Resource
• Observation-genetics Profile
• DiagnosticReport-genetics Profile
• ServiceRequest-genetics Profile
• HLA genotyping results Profile
• Relationship among MolecularSequence resource and genetics profiles
• HL7 Domain Analysis Model Use Cases
• Other Genomics Use Cases
• Clinical Genomics Apps
• Appendix 1: From DSTU2 Genomics to STU3

Trial-Use Note:

The material on this page is currently undergoing work to be refactored in a future release as further analysis is done, for example as part of the Genomics Reporting Implementation Guide .

10.10 Background

Rapid advances in sequencing technologies, clinical genetics tests for whole genome and exome sequencing are allowing sophisticated genetics testing to be used by providers and patients in clinical decisions. Results from such tests are used to identify distinct genetic variants that may contribute to syndromes, conditions and/or predictive responses to treatments. The implementation of precision medicine will depend upon having such data to diagnose patients, choose medications, and predict the course of disease and care, but will require standards and effective user interfaces.

A current technical challenge exists in interoperability, the ability to access and share clinical and genetics data. The challenges of interoperability includes collection, coding, and retrieval to scale. An individual's genetic data set is large, complex and requires curation. Unfortunately, incompatible systems and nomenclatures are already in use. A standards-based ontology that could be adopted to integrate both genetic data and clinical information systems will be crucial to accelerating the integration of precision medicine and to make sense of genetic testing results in a complete clinical context.

One approach for collecting, coding, and retrieving genetics data comes from the Global Alliance for Genomics and Health (GA4GH). The GA4GH organization has built and is refining an API and data model for the exchange of full sequence genomic information across multiple research organizations and platforms. The GA4GH focuses on the needs of researchers.

A second approach is evolving from HL7 through FHIR. FHIR is attractive because it is relatively easy to implement because it is comprised of a set of modular components called resources, which can be easily and

incrementally assembled into working systems. The clinical requirements for genetics data is, relative to genomics research needs, utilitarian and reductive because it is about distilling and extracting particular genetics data produced by ever more sophisticated testing for use at the point-of-care. This has made FHIR a very functional framework to initiate an interoperable clinical genetics data standardization to which multiple stakeholders have contributed to this guide.

FHIR DSTU2 introduced a standard genetics profile that applies to the FHIR Observation resource. Using this profile, Observation payloads can return genetic testing results in a standardized manner.

Extending Observations rather than creating a new, dedicated FHIR genetics data resource is consistent with the FHIR community mandate because it only adds new resources to an existing resource library when there is a compelling case to do so. We provide the case to add a MolecularSequence resource in this specification.

10.10 Overview

STU3 moves beyond FHIR DSTU2 Standard Genetics profile on Observation allowing increased granularity and less ambiguity by creating a new resource to be called MolecularSequence. This resource will be used to hold clinically relevant sequence data in a manner that is both efficient and versatile integrating new and as yet undefined types of genomic and other -omics data that will soon be commonly entered into health records for clinical use. MolecularSequence will be leveraged by other FHIR resources, including Observation. This is consistent with how all FHIR resources are designed and used.

The September 2014 Informative Ballot (“HL7 Clinical Genomics, Domain Analysis Model: Clinical Sequencing Release 1”) provided guiding use cases, which initially informed development of the initial Standard Genetics profile that is found in FHIR DSTU2. The same use cases also led to a second Project to develop a MolecularSequence resource (“Develop FHIR sequence resource for Clinical Genomics”). A preliminary effort to address these issues has been explored and published in context of the Substitutable Medical Applications and Reusable Technologies (SMART) Platforms Project and described in an article (“SMART on FHIR Genomics: Facilitating standardized clinico-genomic apps ”).

MolecularSequence is designed to hold genetic sequences in blocks relevant to actionable clinical decision-making. Extensions to MolecularSequence address complex cases and can associate it with repositories for retrieving a patient’s full sequence data, such as those defined by GA4GH. Other changes include a suite of genetics profiles for other FHIR resources. In addition, the Observation-genetics profile adds new references so that an Observation can report genetics test results to be integrated into the EHR. There are also new genetics-extension profiles for DiagnosticReport, ServiceRequest and FamilyMemberHistory, respectively, to extend them to report genetics results. We have given all of these FHIR genetics profiles the suffix “-genetics” (e.g. “DiagnosticReport-genetics profile”). New profiles on top of DiagnosticReport have been created for reporting HLA genotyping results.

On the following pages, we elaborate upon the rationale for the proposed design, introducing in some detail the following resource and profiles:

• A MolecularSequence resource

• An Observation-genetics profile on Observation

• A DiagnosticReport-genetics profile on DiagnosticReport

• A ServiceRequest-genetics profile on ServiceRequest

• An HLA-genotyping-results profile on DiagnosticReport

With these resource and profiles, FHIR can support a large set of clinical use cases (see Section 9, 10 , and 11) and is thus positioned to address all emergent -omics use cases, including Next-Generation Sequencing (NGS). These tools are simple to implement, will optimize payload sizes, and help developers avoid redundant retrieval of data. Appendix 1 of this document shows how DSTU 2.0 can be mapped to the new additions to the resource.

10.10 MolecularSequence Resource

10.10.1 Structure Diagram

Name | Flags | Card. | Type | Description & Constraints
MolecularSequence | ΣITU | | DomainResource | Information about a biological sequence + Rule: Only 0 and 1 are valid for coordinateSystem Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique ID for this particular sequence. This is a FHIR-defined id
type | Σ | 0..1 | code | aa | dna | rna sequenceType (Required)
coordinateSystem | Σ | 1..1 | integer | Base number of coordinate system (0 for 0-based numbering or coordinates, inclusive start, exclusive end, 1 for 1-based numbering, inclusive start, inclusive end)
patient | Σ | 0..1 | Reference(Patient) | Who and/or what this is about
specimen | Σ | 0..1 | Reference(Specimen) | Specimen used for sequencing
device | Σ | 0..1 | Reference(Device) | The method for sequencing
performer | Σ | 0..1 | Reference(Organization) | Who should be responsible for test result
quantity | Σ | 0..1 | Quantity | The number of copies of the sequence of interest. (RNASeq)
referenceSeq | ΣI | 0..1 | BackboneElement | A sequence used as reference + Rule: GenomeBuild and chromosome must be both contained if either one of them is contained + Rule: Have and only have one of the following elements in referenceSeq : 1. genomeBuild ; 2 referenceSeqId; 3. referenceSeqPointer; 4. referenceSeqString;

chromosome | Σ | 0..1 | CodeableConcept | Chromosome containing genetic finding chromosome-human (Example)
genomeBuild | Σ | 0..1 | string | The Genome Build used for reference, following GRCh build versions e.g. 'GRCh 37'
orientation | Σ | 0..1 | code | sense | antisense orientationType (Required)
referenceSeqId | Σ | 0..1 | CodeableConcept | Reference identifier E n s e m b l (Example)
referenceSeqPointer | Σ | 0..1 | Reference(MolecularSequence) | A pointer to another MolecularSequence entity as reference sequence
referenceSeqString | Σ | 0..1 | string | A string to represent reference sequence
strand | Σ | 0..1 | code | watson | crick strandType (Required)
windowStart | Σ | 0..1 | integer | Start position of the window on the reference sequence
windowEnd | Σ | 0..1 | integer | End position of the window on the reference sequence
variant | Σ | 0..* | BackboneElement | Variant in sequence
start | Σ | 0..1 | integer | Start position of the variant on the reference sequence
end | Σ | 0..1 | integer | End position of the variant on the reference sequence
observedAllele | Σ | 0..1 | string | Allele that was observed
referenceAllele | Σ | 0..1 | string | Allele in the reference sequence
cigar | Σ | 0..1 | string | Extended CIGAR string for aligning the sequence with reference bases
variantPointer | Σ | 0..1 | Reference(Observation) | Pointer to observed variant information
observedSeq | Σ | 0..1 | string | Sequence that was observed
quality | Σ | 0..* | BackboneElement | An set of value as quality of sequence
type | Σ | 1..1 | code | indel | snp | unknown qualityType (Required)
standardSequence | Σ | 0..1 | CodeableConcept | Standard sequence for comparison F d a- standard sequence (Example)
start | Σ | 0..1 | integer | Start position of the sequence
end | Σ | 0..1 | integer | End position of the sequence
score | Σ | 0..1 | Quantity | Quality score for the comparison
method | Σ | 0..1 | CodeableConcept | Method to get quality F d a- method (Example)
truthTP | Σ | 0..1 | decimal | True positives from the perspective of the truth data
queryTP | Σ | 0..1 | decimal | True positives from the perspective of the query data
truthFN | Σ | 0..1 | decimal | False negatives
queryFP | Σ | 0..1 | decimal | False positives
gtFP | Σ | 0..1 | decimal | False positives where the non-REF alleles in the Truth and Query Call Sets match
precision | Σ | 0..1 | decimal | Precision of comparison
recall | Σ | 0..1 | decimal | Recall of comparison
fScore | Σ | 0..1 | decimal | F-score
roc | Σ | 0..1 | BackboneElement | Receiver Operator Characteristic (ROC) Curve
score | Σ | 0..* | integer | Genotype quality score
numTP | Σ | 0..* | integer | Roc score true positive numbers
numFP | Σ | 0..* | integer | Roc score false positive numbers
numFN | Σ | 0..* | integer | Roc score false negative numbers
precision | Σ | 0..* | decimal | Precision of the GQ score
sensitivity | Σ | 0..* | decimal | Sensitivity of the GQ score
fMeasure | Σ | 0..* | decimal | FScore of the GQ score
readCoverage | Σ | 0..1 | integer | Average number of reads representing a given nucleotide in the reconstructed sequence
repository | Σ | 0..* | BackboneElement | External repository which contains detailed report related with observedSeq in this resource
type | Σ | 1..1 | code | directlink | openapi | login | oauth | other repositoryType (Required)
url | Σ | 0..1 | uri | URI of the repository
name | Σ | 0..1 | string | Repository's name
datasetId | Σ | 0..1 | string | Id of the dataset that used to call for dataset in repository
variantsetId | Σ | 0..1 | string | Id of the variantset that used to call for variantset in repository
readsetId | Σ | 0..1 | string | Id of the read
pointer | Σ | 0..* | Reference(MolecularSequence) | Pointer to next atomic sequence
structureVariant | Σ | 0..* | BackboneElement | Structural variant
variantType | Σ | 0..1 | CodeableConcept | Structural variant change type LOINC LL379-9 answerlist (Required)
exact | Σ | 0..1 | boolean | Does the structural variant have base pair resolution breakpoints?
length | Σ | 0..1 | integer | Structural variant length
outer | Σ | 0..1 | BackboneElement | Structural variant outer
start | Σ | 0..1 | integer | Structural variant outer start
end | Σ | 0..1 | integer | Structural variant outer end
inner | Σ | 0..1 | BackboneElement | Structural variant inner
start | Σ | 0..1 | integer | Structural variant inner start
end | Σ | 0..1 | integer | Structural variant inner end
Documentation for this format

10.10.2 Description

The MolecularSequence resource is designed for next-generation sequencing data. Patients' observed sequences should be represented by recording reference sequence id/string and detected variants. To specify how it proceed, here is a picture below:

MolecularSequence.coordinateSystem: This element shall be constrained into only two possible values: 0 for 0-based system and 1 for 1-based system. Below is the picture that could explain what's the difference between these two systems:

Here are two examples that clarify the usage in both cases (they represent same segment part):

• 0-based example: here Source:XML,JSON

• 1-based example: here Source:XML,JSON

MolecularSequence.referenceSeq: Four optional ways are provided to represent reference sequence in MolecularSequence resource:

• MolecularSequence.referenceSeq.referenceSeqId: Reference sequence id in public database;

• MolecularSequence.referenceSeq.referenceSeqString: Reference sequence string;

• MolecularSequence.referenceSeq.referenceSeqPointer: Reference to observed sequence stored in another sequence entity;

• MolecularSequence.referenceSeq.genomeBuild, MolecularSequence.reference.chromosome: The combination of genome build and chromosome.

The window selects a range from the reference sequence (or genome) that is used to define building block of a current sequence (e.g. MolecularSequence resource instance 1).

MolecularSequence.referenceSeq.strand: Only two possible values can be made by strand, +1 for plus strand while -1 for minus strand. Since the directionality of the sequence string might be represented in different word in different omics scenario, below are simple example of how to map other expressions into its correlated value:

Map to +1	Map to -1
5′-to-3′ direction	3′-to-5′ direction
Watson	Crick
Sense	Antisense
Positive	Negative

MolecularSequence.quality: Quality scores for bases in the sequence. It is intended to be compliant with emerging regulatory needs needs (eg: those found at PrecisionFDA).

MolecularSequence.variant: This complex element is used for encoding sequence. When the information of reference sequence and variants are provided, the observed sequence will be derived.

10.10.2.1 Internal Pointers

MolecularSequence.patient: This element points to a Patient identifier to show that this sequence is related to the same patient.

MolecularSequence.specimen: A pointer to specimen identifier, if needed.

MolecularSequence.device: A pointer to Device identifier which is used for describing sequencing method (such as chip id, chip manufacturer etc.)

MolecularSequence.pointer: A pointer to a MolecularSequence instance for the next sequence block to build a sequence graph.

10.10.2.2 External Pointers

MolecularSequence.repository: This complex element is used to provide a clarifying structure, a base URL, and/or relevant IDs when referring to an external repository.

GA4GH Repository Example. If the MolecularSequence resource refers to a GA4GH repository for read info, references to a GA4GH full sequence dataset should conform to GA4GH data models and accessed via the GA4GH API. The URL of a GA4GH repository, ids of a GA4GH variant and read group are contained in the MolecularSequence resource. The URL of a GA4GH repository is an api_base of a GA4GH server that could be called for sequence data. The GA4GH variant set is a collection of call sets and the GA4GH call set is a collection of variant calls, typically for one sample. A variant call represents a determination of genotype with respect to that variant.

VariantSet definition: A VariantSet is a collection of variants and variant calls intended to be analyzed together.

CallSet definition: A CallSet is a collection of calls that were generated by the same analysis of the same sample.

A read group is a collection of reads produced by a sequencer. A read group set typically models reads corresponding to one sample, sequenced one way, and aligned one way. The API reference of Google Genomics is a GA4GH repository built by Google and provides details of the data models, such as the resource representations.

10.10.3 Usage Examples

10.10.3.1 Different way to represent sequence record with variations

We provide a detailed example to show how sequence resource can be used to represent record of observed sequence by different method.

The diagram above describes 4 optional ways provided in the MolecularSequence resource to encode sequencing data. Here are the corresponding examples

• By referenceSeqId Source:XML,JSON,TURTLE

• By referenceSeqString Source:XML,JSON,TURTLE

• By referenceSeqPointer Source:XML,JSON,TURTLE

• By Genomebuild+chromosome Source:XML,JSON,TURTLE

10.10.3.2 Example from precision FDA

We provide one example to show how precision FDA vcf data can be upload and comform FHIR specification, and how quality in sequence resource is represented by comparison between a reference sequence and the observed sequence.

Source: XML, JSON, Turtle.

10.10.3.3 Complex Variation example

We provide one example to show how a complex variant can be represented with the help of cigar. The deletion, insertion and mutation is represented in characters along with the number of repetition.

Source: XML, JSON, Turtle.

10.10 Observation-genetics Profile

10.10.1 Structure Diagram

Name | Flags | Card. | Type | Description & Constraints
Observation | | 0..*
observation-geneticsGene | | 0..1 | CodeableConcept | HGNC gene symbol URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGene Binding: H g n c (preferred)
observation-geneticsDNARegionName | | 0..1 | string | DNA region name URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsDNARegionName
observation-geneticsCopyNumberEvent | | 0..1 | CodeableConcept | Copy number variation URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsCopyNumberEvent
observation-geneticsGenomicSourceClass | | 0..1 | CodeableConcept | Genomic source class URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsGenomicSourceClass
observation-geneticsInterpretation | | 0..1 | Reference(Observation) | Clinical interpretations for variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsInterpretation
observation-geneticsVariant | | 0..1 | (Complex) | Variant URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant
observation-geneticsAminoAcidChange | | 0..1 | (Complex) | AminoAcidChange URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAminoAcidChange
observation-geneticsAllele | | 0..1 | (Complex) | Allele URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAllele
observation-geneticsAncestry | | 0..1 | (Complex) | Ancestry URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsAncestry
observation-geneticsPhaseSet | | 0..* | (Complex) | Phase set URL: http://hl7.org/fhir/StructureDefinition/observation-geneticsPhaseSet
Documentation for this format

10.10.2 Description

The Observation-genetics profile Observation-genetics profile is used to interpret variants from sequence resource. Clinical usage may need more specific representation of variant at locus or structural variant in whole genome.

Some of the attributes of the profile follow:

• The observation-geneticsSequence extension will refer to the MolecularSequence resource for sequence information related to this variant.

• The observation-geneticsInterpretation extension will refer to an Observation instance which contains clinical interpretations for the variant described.

• The code, effective[x], issued, performer, method, specimen elements can be used to describe how the genetic observation (variant and sequence data) is obtained.

• Other extensions are used to describe attributes of this variant such as Genomics Source Class, Amino Acid Change Type, etc. These are mappings from v2 and lonic code reference with details can be found in this list .

Additional Observations instance will be created for variant's further analysis. For example, Observation.component element will be used for knowledge-based interpretations of the sequence variant. Here are some examples for the component.code.

LOINC Code | LOINC Element Name | Comments
51963-7 | Medication Assessed | A coded medication accessed in a pharmacogenetic test (recommend RxNorm).
51967-8 | Genetic disease assessed | A coded disease that is associated with the region of DNA covered by the genetic test (recommend SNOMED).
53037-8 | Genetic Disease Sequence Variant Interpretation | Interpretation of the pathogenicity of the DNA Sequence Variant in the context of the assessed genetic disease.
53040-2 | Drug Metabolism Sequence Variant Interpretation | Predicted phenotype for drug efficacy. A sequence variant interpretation value known to allow (responsive) or prevent (resistant) the drug to perform.
51961-1 | Drug Efficacy Sequence Variant Interpretation | Predicted phenotype for ability of drug to bind to intended site in order to deliver intended effect. A Sequence Variant interpretation value known to allow (responsive) or prevent (resistant) the drug to perform.

In the meantime, the related element in this Observation instance will point to Observaiton-genetics profile to show these clinical interpretations are further analysis for the variant. (For example, sequence variant has its pointer back to the observation, see PGx example)

10.10.3 Usage Examples

10.10.3.1 An Observation-genetics instance and its two related Observation instances for interpretations

We provide an example of an Observation-genetics instance which records a variant detected in the patient. (we call this example A)

Source: XML, JSON, Turtle.

An example of an Observation instance which records knowledge-based clinical interpretations for the variant represented in A.

Source: XML, JSON, Turtle.

An example of an Observation instance which records PCR validation test for the variant in A.

Source: XML, JSON, Turtle.

10.10.3.2 Phenotype, Diplotype and Haplotye

We provide the following examples to reveal how PGx data can be harmonized within the FHIR specification. Detailed discussion will be put here.

• Phenotype Source:XML,JSON,TURTLE

• Diplotype Source:XML,JSON,TURTLE

• Haplotype Source:XML,JSON,TURTLE

10.10 DiagnosticReport-genetics Profile

10.10.1 Structure Diagram

Name | Flags | Card. | Type | Description & Constraints
DiagnosticReport | | 0..*
DiagnosticReport-geneticsAssessedCondition | | 0..* | Reference(Condition) | AssessedCondition URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAssessedCondition
DiagnosticReport-geneticsFamilyMemberHistory | | 0..* | Reference(FamilyMemberHistory) | FamilyHistory URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsFamilyMemberHistory
DiagnosticReport-geneticsAnalysis | | 0..* | (Complex) | Analysis URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsAnalysis
DiagnosticReport-geneticsReferences | | 0..* | (Complex) | References URL: http://hl7.org/fhir/StructureDefinition/DiagnosticReport-geneticsReferences
conclusionCode | | 0..0
Documentation for this format

This DiagnosticReport-genetics is built on top of DiagnosticReport. The new profile is used to describe a genetics test report. The result element in DiagnosticReport will refer to the Observation resource that can lead to a bundle of genetic observations. The element of code, effective[x], issued, performer, request, specimen are be used to describe the details of the genetic test. Extensions about AssessedCondition and FamilyMemberHistory are added. Overall, this profile extends the DiagnosticReport resource to enable reporting of structured genetic test results. In addition, it denotes condition context for genetic testing, which may influence reported variants and interpretations for large genomic testing panels.

• The DiagnosticReport-genetics profile contains two extensions referring to other resources/profiles and one complex extension.

• AssessedCondition is used to denote condition context for genetic testing, which may influence reported variants and interpretation for large genomic testing panels. It refers to the Condition resource.

• FamilyMemberHistory are significant health events and conditions for one or more persons related to the patient. It refers to the FamilyMemberHistory-Genetic profile.

• Analysis provides knowledge-based interpretation of overall results of tests performed on patients.

Examples for analysis.code:

LOINC Code | LOINC Element Name | Comments
51968-6 | Genetic Disease Analysis Overall Interpretation | Interpretation of all identified DNA Sequence variants along with any known clinical information for the benefit of aiding clinicians in understanding the results overall in either the context of diagnosis or increased risk of disease.
51964-5 | Drug Efficacy Analysis Overall Interpretation | Overall predicted phenotype for drug efficacy for all DNA Sequence Variants identified in a single case.
51971-0 | Drug metabolism analysis overall interpretation | Overall predicted phenotype for drug metabolism for all DNA Sequence Variants identified in a single case.

10.10.2 Usage Examples

An example of a genetic test report for a patient with FamilyHistory.

Source: XML, JSON, Turtle.

An example of a comprehensive bone marrow report.

Source: XML, JSON, Turtle.

10.10 ServiceRequest-genetics profile

10.10.1 Structure Diagram

A complex extension is added on top of the ServiceRequest resource. Here is the structure of the extension:

Name | Flags | Card. | Type | Description & Constraints
ServiceRequest | | 0..*
servicerequest-geneticsItem | | 0..* | (Complex) | The items the orderer requested URL: http://hl7.org/fhir/StructureDefinition/servicerequest-geneticsItem
Documentation for this format

10.10.2 Description

To describe an order requested sequence variants detection. User must set up the code for the request and they can also refer to the corresponding sequence instance for that variant.

10.10.3 Usage Example

10.10.3.1 An order for genetics test

Here is a diagnostic request for testing 185delAG variant. The mother of the patient received results from a mutation panel (eg. MyRisk from Myriad) and she has a BRCA1 185delAG mutation. The clinician the would like to request to test the patient only for an 185delAG mutation. In this case, the diagnostic request for the patient will specify the sequence variant - 185delAG.

10.10 HLA genotyping results Profile

Human leukocyte antigen (HLA) genotyping is fundamental for research and clinical practice in immunogenetics and histocompatibility. Pointers to external locations refer to registered methods, raw NGS reads, and reference standards can be conveyed in this profile. Information about allele assignment including ambiguous results and the allele database used for assignments is stored in extensions.

The structure of the HLA typing report in this profile attempts to follow the principles outlined in the Minimum Information for ReportIng Next-generation sequence Genotyping (MIRING). These principles were identified through a series of meetings with international group of stakeholders in the application of Next Generation Sequencing (NGS) technology for genotyping the HLA and KIR loci as well as other immune-related loci (http://igdawg.org/ngs.html). MIRING describes eight principles, described in detail in Human Immunology. 2015 Dec; 76(12):954-962 . These include detailed metadata about:

• MIRING Annotation
• Reference Context
• Full Genotype
• Consensus Sequence
• Novel Polymorphisms
• Platform Documentation
• Read Processing Documentation
• Primary Data

These principles were implemented in a technical specification by extending an existing XML based format for exchanging histocompatibility and immunogenetic genotyping data called Histoimmunogenetics Markup Language (HML) to include results from NGS methodologies (https://bioinformatics.bethematchclinical.org/hla-resources/hml/). The resulting schema may be found in https://schemas.nmdp.org/. The National Marrow Donor Program (NMDP)/Be The Match uses this format for reporting HLA genotyping from potential donors and for patients needing stem cell transplants.

Both MIRING and HML were used to inform mapping data elements to FHIR resources such as Patient, Specimen, MolecularSequence, Observation-Genetic Profile, and DiagnosticReport where possible. Several additional data elements were needed specifically for this use case, resulting in the development of a specific profile for reporting HLA genotyping results (Diagnostic Report Profile for HLA Genotyping Results).

• Minimum information for reporting next generation sequence genotyping (MIRING): Guidelines for reporting HLA and KIR genotyping via next generation sequencing (see here)

• Histoimmunogenetics Markup Language 1.0: Reporting Next Generation Sequencing-based HLA and KIR Genotyping (see here)

10.10.1 Structure Diagram of HLA genotyping results Profile

Name | Flags | Card. | Type | Description & Constraints

haploid | | 0..* | Extension | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid haploid: haploid. Use on Element ID DiagnosticReport
locus | | 0..1 | CodeableConcept | gene region.
type | | 0..1 | CodeableConcept | haploid type.
method | | 0..1 | CodeableConcept | haploid method.
Documentation for this format

Both hla-genotyping-results-glstring and hla-genotyping-results-haploid are complex extensions.

The development of these extensions were informed from the allele-assignment structure found in HML. While allele assignment for individual loci can be reported in an Observation, here they are used to summarize the assignments at a report level. Two methods may be used for reporting HLA allele-assignments: Haploid and GL String.

Structure of hla-genotyping-results-haploid:

Name | Flags | Card. | Type | Description & Constraints
haploid | | 0..* | Extension | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid haploid: haploid. Use on Element ID DiagnosticReport
locus | | 0..1 | CodeableConcept | gene region.
type | | 0..1 | CodeableConcept | haploid type.
method | | 0..1 | CodeableConcept | haploid method.
Documentation for this format

Haploid reporting is a method that is used to report an individual allele, or a list of possible alleles if the results are ambiguous, using NMDP Multiple Allele Codes (MAC) which is a shorthand for describing allele ambiguity (https://bioinformatics.bethematchclinical.org/hla-resources/allele-codes/). MACs are widely used for reporting HLA typing results because they are able to report long allele lists in a small amount of space, but the system has a number of shortcomings (Tissue Antigens. 2013 Aug;82(2):106-12). When reporting data using haploid, typical use is one or two haploid elements for a particular locus, but possibly more if multiple loci are covered (ex: two HLA-DRB1 haploids + one HLA-DRB3 haploid).

Within Haploid, Method indicates whether the general methodology is DNA based typing (e.g., Sequence Specific Primers (SSP), Sequence Specific Oligonucleotide Probes (SSOP), or Sequence Based Typing (SBT)), or SER which indicates serology based methods.

Example of using two hla-genotyping-results-haploid extensions to report a HLA-A genotype of HLA-A*01:AB and HLA-A*02:MN which expands to HLA-A*01:01/HLA-A*01:02 and HLA-A:02:01/HLA-A:02:02/HLA-A:02:03

```
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid" >
  <extension url=”locus”>
      <valueCodeableConcept>
          <coding>
              <system value="http://www.genenames.org"/>
              <code value="4931"/>
              <display value="HLA-A"/>
          </coding>
          <text value="HLA-A"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”type”>
      <valueCodeableConcept>
           <coding>
              <system value="https://bioinformatics.bethematchclinical.org/hla-resources/allele-codes/"/>
              <code value="AB"/>
              <display value="01/02"/>
          </coding>
          <text value="HLA-A*01:AB"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”method”>
      <valueCodeableConcept>
          <text value=”DNA”/>
      </valueCodeableConcept>
  </extension>
</extension>
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-haploid" >
  <extension url=”locus”>
      <valueCodeableConcept>
          <coding>
              <system value="http://www.genenames.org"/>
              <code value="4931"/>
              <display value="HLA-A"/>
          </coding>
          <text value="HLA-A"/>
      </valueCodeableConcept>
  </extension>
  <extension url=”type”>
      <valueCodeableConcept>
           <coding>
              <system value="https://bioinformatics.bethematchclinical.org/hla-resources/allele-codes/"/>
              <code value="AB"/>
              <display value="01/02/03"/>
          </coding>
          <text value="HLA-A*02:MN"/>
```

```
        </valueCodeableConcept>
    </extension>
    <extension url=”method”>
        <valueCodeableConcept>
            <text value=”DNA”/>
        </valueCodeableConcept>
    </extension>
</extension>
```

Because of limitations with MAC, another method called GL Strings was developed that encodes the results in a text string with hierarchical set of operators to describe the relationships between alleles, lists of possible alleles, phased alleles, genotypes, lists of possible genotypes, and multilocus unphased genotypes, without losing typing information or increasing typing ambiguity. (Tissue Antigens. 2013 Aug;82(2):106-12).

The structure of HLA-genotyping-results-glstring:

Name | Flags | Card. | Type | Description & Constraints
glstring | | 0..1 | Extension | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring glstring: glstring. Use on Element ID DiagnosticReport
url | | 0..1 | uri | glstring using a URI reference.
text | | 0..1 | string | glstring using inline data.
Documentation for this format

While the string format is easily parsed into separate components to be rendered for user viewing, GL Strings by themselves are potentially quite long and difficult to read. It often advantageous to point to a URI which may return the GL String on demand to avoid manual data entry. This is available through the URI element in this extension.

Example of using the hla-genotyping-results-glstring to report a GL String in both text and URI formats

```
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-glstring">
  <extension url=”text”>
      <valueString value="HLA-A*01:01:01:01/HLA-A*01:02+HLA-A*02:01:02/HLA-A*02:02:02/HLA-A*02:03:01"/>
  </extension>
  <extension url=”uri”>
      <valueUri value="https://gl.nmdp.org/imgt-hla/3.23.0/genotype/1h"/>
  </extension>
</extension>
```

Whether reporting in Haploid or GL String formats, it is important to identify the version of the IMGT/HLA allele database, as new HLA alleles are constantly being discovered and allele assignment is based on the known alleles at the time, and so results may need to be reinterpreted later. This is done through the Allele Database element.

Structure of hla-genotyping-results-allele-database

Name | Flags | Card. | Type | Description & Constraints
allele-database | | 0..1 | CodeableConcept | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database allele-database: Allele Database. Use on Element ID DiagnosticReport
Documentation for this format

Example of using hla-genotyping-results-allele-database to report using the IMGT/HLA database, version 3.23

```
<extension url="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-allele-database">
  <valueCodeableConcept>
      <coding>
          <system value="http://www.ebi.ac.uk/ipd/imgt/hla"/>
          <version value="3.23"/>
      </coding>
      <text value="IMGT/HLA 3.23"/>
  </valueCodeableConcept>
</extension>
```

Overall methodology may be reported using hla-genotyping-results-method. Here a codeable concept may be used to refer to method entered into a public registry, such as the NCBI Genetic Test Registry, or a local private registry.

Structure of hla-genotyping-results-method

Name | Flags | Card. | Type | Description & Constraints
method | | 0..1 | CodeableConcept | URL = http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method method: The platform, methodology and software applied at the time of the genotyping. Use on Element ID DiagnosticReport
Documentation for this format

Example of hla-genotyping-results-method to report a lab test registered in the NCBI Genetic Test Registry

```
<extension  uri="http://hl7.org/fhir/StructureDefinition/hla-genotyping-results-method">
  <valueCodeableConcept>
        <coding>
            <system value="http://www.ncbi.nlm.nih.gov/gtr/"/>
            <code value="GTR000000000.0"/>
        </coding>
        <text value=”NGS based HLA-A genotyping”
  </valueCodeableConcept>
</extension>
```

10.10.2 Usage Example

Example of a HLA genotyping report

• that was requested in a Diagnostic Request,
• with results for HLA-A, HLA-B, HLA-C loci,
• by Sequence Based Typing (SBT) of exons 2 and 3,
• using Next Generation Sequencing technologies,
• allele assignment based on the IMGT/HLA database version 3.23,
• reported as a multilocus unphased genotype using GL Strings,
• and with genotyping results for individual loci available in separate Observations

Source: XML, JSON, Turtle.

In the above example, three Observations are referenced, each containing the details surrounding the genotyping of each gene/locus. Those observations would further refer to other observations (using Observation.related) that point to phased exon sequences which are the basis for individual allele assignment. This strategy is illustrated in the following figure for one gene, and sequencing two exons for each allele. Note that references from DiagnosticReport and Observation back to Patient/Donor and Specimen are not shown for clarity, although we would include them in the resource instances.

10.10 Relationship among MolecularSequence resource and genetics profiles

10.10.1 Pointers to MolecularSequence resource

MolecularSequence resource is used to transmit and represent sequencing data. In FHIR Genomics, there are several genetics profiles containing internal pointers to MolecularSequence resource for genetic data reporting. Here is the summary:

MolecularSequence.pointer: A bundle of MolecularSequence instances can be connected by MolecularSequence.pointer to build sequence graph described in Section 3. MolecularSequence.pointer will point to the next sequence block.

MolecularSequence extension in Observation-genetics profile: Observation-genetics profile is used to report a genetic variant found in patients. MolecularSequence extension contains a pointer to MolecularSequence identifier which has related sequencing read info. It will provide reference/observed allele information and quality scores for each base/sequence block.

10.10.2 Pointer to Observation resource (based on genetics profile)

Observation-genetics based Observation resource is used for interpretative genetic data. MolecularSequence resource and genetics profiles will use internal pointers to Observation-genetics-profile based Observation instance for variant report.

MolecularSequence.observation: A pointer to genetics Observation instance which is used for interpretations of this sequence block. Interpretations are generally about genetic variant found in this sequence block.

DiagnosticReport.result: A pointer to Observation instance. This element is used for reporting genetic result.

Item extension in ServiceRequest-genetics profile: It is used to describe the genetic test order.

FamilyMemberHistory: A pointer to genetics Observation instance to report genetic test results of family member

10.10.3 Observation vs. MolecularSequence

MolecularSequence is used for raw sequencing data while genetic-profile-based Observation resource is used for reporting interpretative genetic information (eg. DNA/amino acid variant detected in the patient). Although both MolecularSequence and Observation-genetics profiles contain variant info, they are used differently: MolecularSequence.variant is used for encoding the sequence block; the Observation-genetics profile is mainly used for a variants test result or interpretations of raw data in MolecularSequence resource.

10.10.4 Usage example

This picture depicts the logical relationship among these resources. The definition of pointers are described above:

Here are the details and examples mentioned above.

Resource	Details of data structures and pointers	Links of resources example
MolecularSequence	MolecularSequence Data Structure	Example Here
Observation	Observation Data Structure	Example Here
ServiceRequest	ServiceRequest Data Structure	Example Here
DiagnosticReport	DiagnosticReport Data Structure	Example Here

Here is a specific use case to depict the whole story of FHIR in the clinical genomics setting: a ServiceRequest is requested, then a Diagnosticreport are reported. Two MolecularSequence instance, two Observation-haplotype instance, a Observation-diplotype and a Observation-phenotype for further interpretation are also created. (You can see the xml code and json code if you click the link.)

• Haplotype: A haplotype is a set of DNA variations, or polymorphisms, that tend to be inherited together. A haplotype can refer to a combination of alleles or to a set of single nucleotide polymorphisms (SNPs) found on the same chromosome.

• Diplotype: A diplotype is defined as a specific combination of two haplotypes.

• Genotype: A genotype is an individual's collection of genes.

• Phenotype: A phenotype is the composite of an organism's observable characteristics or traits, such as its morphology, development, biochemical or physiological properties, behavior and products of behavior.

You can see the relationship among these resources in following picture:

10.10 HL7 Domain Analysis Use Cases

Simulated use of the proposed MolecularSequence resource and profiles on other FHIR resources are shown for many of the examples in Section 5 – Use Case Scenarios in the HL7 Domain Analysis Model (DAM): Clinical Genomics, Release 1, September 2014 Informative Ballot .

We show use of the FHIR design we are proposing for these cases:

• Specimen Identification Get references to all variants obtained from germline analysis. Explore antibacterial drug resistance over time (non-patient DNA)

• Clinical Sequencing – Germline Testing

• Cancer Profiling

• Decision Making Tools – Family History and Drug Dosage Calculators Today clinicians translate (i.e. manually re-enter) genetic data into tools for decision making. This includes family history tools and drug dosage calculators. In the future, this data will automatically be incorporated into clinical decision making tools. Get pharmacogenomic reporting and disease risk or diagnosis associated with variant

• Public Health

• Clinical and Research Data Warehouses

10.10.1 Specimen Identification

• Germline testing for biomarkers/mutations (usually inherited)

• Tumor testing for biomarkers/mutations (somatic / tumor specific)

• Pediatric Testing

• Infectious Disease Testing

10.10.1.1 Get references to all variants obtained from germline analysis

```
GET   /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    GenomicSourceClass=http://loinc.org|LA6683-2
```

*Return a bundle of genetics-profile-based Observation instances

10.10.2 Clinical Sequencing – Germline Testing

For a chart review, get references to all DNA sequences related to mutations with an interpretation "Unknown Significance"

```
GET /Observation?
    subject=123&
    component-code-value-[x]= http://loinc.org|53037-8$LA6682-4
```

53037-8: LOINC code for "Genetic disease sequence variant interpretation"

LA6682-4: LOINC answer code for "Unknown Significance".

10.10.3 Cancer Profiling

The goal of this profile methodology is to get references from all variants obtained from somatic analysis. Changes in the population of cells with particular mutations will change overtime as well as in conjunction with events such as therapy. For instance, targeted chemotherapy may kill a specific population of cancer cells with specific mutations and other cancer cell populations may survive and continue to divide. Therefore, clearly annotating these specimens as somatic variants and capturing annotations related to a time relevant to a treatment timeline may be critical for analysis.

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    GenomicsSourceClass=http://loinc.org|LA6684-0&
    date=2015-07-04&
    subject=123
```

10.10.4 Decision Making Tools – Family History and Drug Dosage Calculators

Today clinicians translate (i.e. manually re-enter) genetic data into tools for decision making. This includes family history tools and drug dosage calculators. In the future, this data will automatically be incorporated into clinical decision making tools.

Get family history related to one observation:

```
GET /DiagnosticReport?
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
    subject=123
```

10.10.4.1 Get pharmacogenomic reporting and disease risk or diagnosis associated with variant

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    subject=123&
    Interpretation.component-code= http://loinc.org|51963-7
```

51963-7 : LOINC code for “Medication Assessed ”

10.10.5 Public Health

Today Registrars manually translate clinical data into public health reporting systems. This data is used to monitor and improve public health (e.g. surveillance and clinical research). In the future, this data will be extracted from the EHR in an automated (or semi-automated) fashion.

For a breast cancer clinical genomic study, get all genetic-profile-based observations of patients with breast cancer:

```
GET /DiagnosticReport?
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
    AssessedCondition.code=http://snomed.info/sct|254837009
```

10.10.6 Clinical and Research Data Warehouses

Health data warehousing should persist data in its standardized formats, while allowing users to export subsets of the data in the warehouse into multiple ‘data marts’, optimized for specific use cases, analysis type or reporting needs.

Get all genetic-profile-based observations of patients with the variant c.181T>G

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observation-genetics&
    DNAVariantID=http://www.ncbi.nlm.nih.gov/projects/SNP|rs58238560
```

10.10 Other Genomics Use Cases

During the development of the FHIR Genomics design, CGWG participants have commented on the earlier use cases and/or proposed new use cases. In this section, we list describe of these new cases and demonstrate how the proposed design will address them.

10.10.1 Get a patient’s family members genetics report

This example is proposed by Kevin Hughes. Family history is useful for clinicians to know more about the condition of the patient.

```
Get /FamilyMemberHistory?
    _profile=http://hl7.org/fhir/StructureDefinition/familymemberhistory-genetic&
    patient=123
```

10.10.2 Panel

Search for results from nephrotic syndrome panel

```
Get /DiagnosticReport&
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
    code=N0336&
    patient=123
```

10.10.3 Basic Queries

Find patients by condition and affected status

```
GET /Condition?
    component-code-value-[x]=http://snomed.info/sct|439401001
```

Find patients by phenotypical attribute

```
Get /Observation?
    =http://hl7.org/fhir/StructureDefinition/observation-geneticsGene&
    CodeableConcept=http://loinc.org|79716-7
```

79716-7: LOINC code for “CYP2C9 gene product metabolic activity interpretation”

Find patients by molecular data type available

```
Get /Observation?
    =http://hl7.org/fhir/StructureDefinition/observation-genetics&
    Interpretation.component-code=http://loinc.org|34193-3
```

34193-3: LOINC code for “SMPD1 gene mutations tested for in Blood or Tissue by Molecular genetics method Nominal"

Find patients by specific molecular marker (genomic region, position, gene, SNP)

```
Get /Observation?
    =http://hl7.org/fhir/StructureDefinition/observation-genetics&
    DNARegionName=NC_000017.10:g.497469T>A
```

53035-2: LOINC code for “DNA marker assessed"

Find mutation by mutation type and position

```
Get /Observation?
    =http://hl7.org/fhir/StructureDefinition/observation-geneticsVariant&
    Name=NM_000059.3:c.706C;A&
  Type=http://loinc.org|81289-1
```

Find patients based on mode of inheritance (genomic region, position, gene, SNP)

```
Get /FamilyMemberHistory?
    _profile=http://hl7.org/fhir/StructureDefinition/familymemberhistory-genetic&
    reasonCode=http://snomed.info/sct|272398008
```

7272398008: SNOMED code for “ Inheritance types "

Search for patient by ID

```
Get /Patient?
    subject=123
```

Search for diagnostic reports with a given variant

```
GET /DiagnosticReport?
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetics&
    geneticsAnalysis=http://www.ncbi.nlm.nih.gov/projects/SNP|rs58238560
```

Given diagnostic report, return associated family history information

```
GET /DiagnosticReport?
    _profile=http://hl7.org/fhir/StructureDefinition/diagnosticreport-geneticsFamilyMemberHistory&
    id=123
```

Search for variants given gene and variant classification

```
GET /Observation?
    =http://hl7.org/fhir/StructureDefinition/observation-genetics&
    Gene=http://www.genenames.org|3236&
  GenomicSourceClass=http://loinc.org|LA6684-0
```

Search for patients based on genetic ancestry

```
GET /Observation?
    =http://hl7.org/fhir/StructureDefinition/observation-genetics&
    Ancestry=http://www.genenames.org|186044009
  GenomicSourceClass=http://snomed.info/sct|LA6684-0
```

10.10.4 Boolean Queries

Find all patients with a set of variants

```
GET /Observation?
    _profile=http://hl7.org/fhir/StructureDefinition/observaiton-genetics&
    DNAVariantID=http://www.ncbi.nlm.nih.gov/projects/SNP|rs58238560$rs58238559$rs58238565
```

rs58238560: variant: NC_000017.10:g.497469T>A

rs58238559: variant: NC_000007.13:g.87082273T>C

rs58238565: variant: NC_000005.10:g.131929452C>A

10.10.5 Specimen-based Queries

Find specimens collected from specific specimen body sites

```
GET /Specimen?
   bodySite=http://snomed.info/sct|85151006
```

Find specimens from this patient's pedigree

```
Get /FamilyMemberHistory?
    =http://hl7.org/fhir/StructureDefinition/familymemberhistory-genetic&
    patient=123
```

Find specimens based on patient disease status

```
Get /DiagnosticReport?
    =http://hl7.org/fhir/StructureDefinition/diagnosticreport-genetic&
    AssessedCondition.code=http://snomed.info/sct|254837009&
    patient=123
```

Find specimens based on specimen disease status

```
Get /Specimen?
    status=available&
```

Find specimens based on technology platform

```
Get /Specimen?
    =http://hl7.org/fhir/StructureDefinition/shareablecodesystem&
    code=Information Technology
```

Find specimens based on instrument identifier

```
Get /Specimen?
    container.identifier=48736-15394-75465
```

10.10.6 Time-based Queries

Find specimens collected within an absolute date range

```
Get /Specimen?
    receivedTime=2011-03-04T07:03:00Z
```

Find specimens collected within a relative date range

```
Get /Specimen?
    =http://hl7.org/fhir/StructureDefinition/specimen-treatmentTime&
    duration=quot
```

10.10.7 MolecularSequence Test Metrics

10.10 Clinical Genomic Apps

The idea for a MolecularSequence resource grew out, in part, the SMART Platforms Project, which explored creating clinical genomic apps to integrated traditional EMR clinical data and genomic data to show data visualization and analysis, including CDS that depended upon both types of data. Below are a couple of examples. Several apps have already been designed including Genomics Advisor, SMART Precision Cancer Medicine, and Diabetes Bear EMR. Below, one of these apps will be described. To include other apps in this section, please feel free to add a note on it and how it uses FHIR/Genomics calls.

10.10.1 Genomics Advisor

The SMART on FHIR Genomics Advisor was an app incorporating genomics data to show risk of disease, drug susceptibility, and related conditions based upon genotype. Technically, this app was architected (see below) by combining data from independent data services, a SMART on FHIR clinical server for clinical information and one for a SMART on FHIR Genomics data server for genomic data. The set of FHIR API calls that are necessary to support this app are shown below:

The presentation of the app looks like this:

10.10 Appendix 1: From DSTU2 Genomics to STU3

The table shows how the data elements in the DSTU2 Observation resource as extended by the Standard Genetics Profile would be mapped to the new MolecularSequence resource.

DSTU2 Standard Genetics profile on Observation | Observation-genetics profile | Maps to MolecularSequence (Proposed)
geneticsGenomeBuild | MolecularSequence extension | referenceSeq.genomicsBuild
geneticsChromosome | MolecularSequence extension | referenceSeq.chromosome
geneticsGenomicsStart | MolecularSequence extension | variant.start
geneticsGenomicsStop | MolecularSequence extension | variant.end
geneticsReferenceAllele | MolecularSequence extension | variant.referenceAllele
geneticsObservedAllele | MolecularSequence extension | variant.observedAllele
geneticsSpecies | MolecularSequence extension | specimen
geneticsAllelicState | Allelic State extension
geneticsAllelicFrequency | Allelic Frequency extension
geneticsReadCoverage | MolecularSequence extension | readCoverage
geneticsCopyNumberEvent | Copy number event extension
geneticsTranscriptReferenceSequenceId | Transcript reference sequence identifier extension
geneticsProteinReferenceSequenceId | Protein reference sequence identifier
geneticsCIGAR | MolecularSequence extension | variant.cigar
geneticsVariationId | DNA variation id extension
geneticsVariationType | DNA sequence variation type extension
geneticsAminoAcidChange | Amino acid change extension
geneticsAminoAcidChangeType | Amino acid change type extension
geneticsGene | Gene extension
geneticsDNARegionName | DNA region name extension
geneticsGenomicSourceClass | Genomic source class extension

10.11 Resource BiologicallyDerivedProduct - Content

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct.html
Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Not linked to any defined compartments

A material substance originating from a biological entity intended to be transplanted or infused into another (possibly the same) biological entity.

10.11.1 Scope and Usage

Trial-Use Note: Note that this content is preliminary has not undergone proper review by the appropriate Workgroups.

A material substance originating from a biological entity intended to be transplanted or infused into another (possibly the same) biological entity.

Examples include:

• hematopoietic stem cells (bone marrow, peripheral blood, or cord blood extraction)

• blood (whole, extracted cells, plasma, etc.)
• organs
• tissues (porcine valves, skin, bovine cardiac tissue, etc.)
• manipulated cells (e.g. CAR T-cells)

The workflow using this resource (e.g., request, administration) should be discussed and implemented in a consistent way as other similar resources are handled (e.g., device, medication)

10.11.2 Boundaries and Relationships

This resource relates to these other resource:

• ProcedureRequest (for collection)
• Patient ("receiver" and "source")
• Practitioner (who collected product)
• Substance (product processing)
• DiagnosticReport (containing HLA-typing)
• BiologicallyDerivedProduct ("parent" product for multi-day collections)
• Procedure (one for collection and one for transplantation, will need to add BiologicallyDerivedProduct to the "usedReference")

This resource is referenced by itself

10.11.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
BiologicallyDerivedProduct | TU | | DomainResource | A material substance originating from a biological entity Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External ids for this item
productCategory | | 0..1 | code | organ | tissue | fluid | cells | biologicalAgent BiologicallyDerivedProductCategory (Required)
productCode | | 0..1 | CodeableConcept | What this biologically derived product is
status | | 0..1 | code | available | unavailable BiologicallyDerivedProductStatus (Required)
request | | 0..* | Reference(ServiceRequest) | Procedure request
quantity | | 0..1 | integer | The amount of this biologically derived product
parent | | 0..* | Reference(BiologicallyDerivedProduct) | BiologicallyDerivedProduct parent
collection | | 0..1 | BackboneElement | How this product was collected
collector | | 0..1 | Reference(Practitioner | PractitionerRole) | Individual performing collection
source | | 0..1 | Reference(Patient | Organization) | Who is product from
collected[x] | | 0..1 | | Time of product collection
collectedDateTime | | | dateTime
collectedPeriod | | | Period
processing | | 0..* | BackboneElement | Any processing of the product during collection
description | | 0..1 | string | Description of of processing
procedure | | 0..1 | CodeableConcept | Procesing code Procedure Codes (SNOMED CT) (Example)
additive | | 0..1 | Reference(Substance) | Substance added during processing
time[x] | | 0..1 | | Time of processing
timeDateTime | | | dateTime
timePeriod | | | Period
manipulation | | 0..1 | BackboneElement | Any manipulation of product post-collection
description | | 0..1 | string | Description of manipulation
time[x] | | 0..1 | | Time of manipulation
timeDateTime | | | dateTime
timePeriod | | | Period
storage | | 0..* | BackboneElement | Product storage
description | | 0..1 | string | Description of storage
temperature | | 0..1 | decimal | Storage temperature
scale | | 0..1 | code | farenheit | celsius | kelvin BiologicallyDerivedProductStorageScale (Required)
duration | | 0..1 | Period | Storage timeperiod
Documentation for this format

UML Diagram (Legend)

XML Template

```
<BiologicallyDerivedProduct xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this item --></identifier>
 <productCategory value="[code]"/><!-- 0..1 organ | tissue | fluid | cells | biologicalAgent -->
 <productCode><!-- 0..1 CodeableConcept What this biologically derived product is --></productCode>
 <status value="[code]"/><!-- 0..1 available | unavailable -->
 <request><!-- 0..* Reference(ServiceRequest) Procedure request --></request>
 <quantity value="[integer]"/><!-- 0..1 The amount of this biologically derived product -->
 <parent><!-- 0..* Reference(BiologicallyDerivedProduct) BiologicallyDerivedProduct parent --></parent>
 <collection>  <!-- 0..1 How this product was collected -->
  <collector><!-- 0..1 Reference(Practitioner|PractitionerRole) Individual performing collection --></collector>
  <source><!-- 0..1 Reference(Patient|Organization) Who is product from --></source>
  <collected[x]><!-- 0..1 dateTime|Period Time of product collection --></collected[x]>
 </collection>
 <processing>  <!-- 0..* Any processing of the product during collection -->
  <description value="[string]"/><!-- 0..1 Description of of processing -->
  <procedure><!-- 0..1 CodeableConcept Procesing code --></procedure>
  <additive><!-- 0..1 Reference(Substance) Substance added during processing --></additive>
  <time[x]><!-- 0..1 dateTime|Period Time of processing --></time[x]>
 </processing>
 <manipulation>  <!-- 0..1 Any manipulation of product post-collection -->
  <description value="[string]"/><!-- 0..1 Description of manipulation -->
  <time[x]><!-- 0..1 dateTime|Period Time of manipulation --></time[x]>
 </manipulation>
 <storage>  <!-- 0..* Product storage -->
  <description value="[string]"/><!-- 0..1 Description of storage -->
  <temperature value="[decimal]"/><!-- 0..1 Storage temperature -->
  <scale value="[code]"/><!-- 0..1 farenheit | celsius | kelvin -->
  <duration><!-- 0..1 Period Storage timeperiod --></duration>
 </storage>
</BiologicallyDerivedProduct>
```

JSON Template

```
{
  "resourceType" : "BiologicallyDerivedProduct",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this item
  "productCategory" : "<code>", // organ | tissue | fluid | cells | biologicalAgent
  "productCode" : { CodeableConcept }, // What this biologically derived product is
  "status" : "<code>", // available | unavailable
  "request" : [{ Reference(ServiceRequest) }], // Procedure request
  "quantity" : <integer>, // The amount of this biologically derived product
  "parent" : [{ Reference(BiologicallyDerivedProduct) }], // BiologicallyDerivedProduct parent
  "collection" : { // How this product was collected
    "collector" : { Reference(Practitioner|PractitionerRole) }, // Individual performing collection
    "source" : { Reference(Patient|Organization) }, // Who is product from
    // collected[x]: Time of product collection. One of these 2:
    "collectedDateTime" : "<dateTime>"
    "collectedPeriod" : { Period }
  },
  "processing" : [{ // Any processing of the product during collection
    "description" : "<string>", // Description of of processing
    "procedure" : { CodeableConcept }, // Procesing code
    "additive" : { Reference(Substance) }, // Substance added during processing
    // time[x]: Time of processing. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  }],
  "manipulation" : { // Any manipulation of product post-collection
    "description" : "<string>", // Description of manipulation
    // time[x]: Time of manipulation. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  },
  "storage" : [{ // Product storage
    "description" : "<string>", // Description of storage
    "temperature" : <decimal>, // Storage temperature
    "scale" : "<code>", // farenheit | celsius | kelvin
    "duration" : { Period } // Storage timeperiod
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:BiologicallyDerivedProduct;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:BiologicallyDerivedProduct.identifier [ Identifier ], ... ; # 0..* External ids for this item
  fhir:BiologicallyDerivedProduct.productCategory [ code ]; # 0..1 organ | tissue | fluid | cells | biologicalAgent
  fhir:BiologicallyDerivedProduct.productCode [ CodeableConcept ]; # 0..1 What this biologically derived product is
  fhir:BiologicallyDerivedProduct.status [ code ]; # 0..1 available | unavailable
```

```
  fhir:BiologicallyDerivedProduct.request [ Reference(ServiceRequest) ], ... ; # 0..* Procedure request
  fhir:BiologicallyDerivedProduct.quantity [ integer ]; # 0..1 The amount of this biologically derived product
  fhir:BiologicallyDerivedProduct.parent [ Reference(BiologicallyDerivedProduct) ], ... ; # 0..*
    BiologicallyDerivedProduct parent
  fhir:BiologicallyDerivedProduct.collection [ # 0..1 How this product was collected
    fhir:BiologicallyDerivedProduct.collection.collector [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Individual
     performing collection
    fhir:BiologicallyDerivedProduct.collection.source [ Reference(Patient|Organization) ]; # 0..1 Who is product from
    # BiologicallyDerivedProduct.collection.collected[x] : 0..1 Time of product collection. One of these 2
      fhir:BiologicallyDerivedProduct.collection.collectedDateTime [ dateTime ]
      fhir:BiologicallyDerivedProduct.collection.collectedPeriod [ Period ]
  ];
  fhir:BiologicallyDerivedProduct.processing [ # 0..* Any processing of the product during collection
    fhir:BiologicallyDerivedProduct.processing.description [ string ]; # 0..1 Description of of processing
    fhir:BiologicallyDerivedProduct.processing.procedure [ CodeableConcept ]; # 0..1 Procesing code
    fhir:BiologicallyDerivedProduct.processing.additive [ Reference(Substance) ]; # 0..1 Substance added during
    processing
    # BiologicallyDerivedProduct.processing.time[x] : 0..1 Time of processing. One of these 2
      fhir:BiologicallyDerivedProduct.processing.timeDateTime [ dateTime ]
      fhir:BiologicallyDerivedProduct.processing.timePeriod [ Period ]
  ], ...;
  fhir:BiologicallyDerivedProduct.manipulation [ # 0..1 Any manipulation of product post-collection
    fhir:BiologicallyDerivedProduct.manipulation.description [ string ]; # 0..1 Description of manipulation
    # BiologicallyDerivedProduct.manipulation.time[x] : 0..1 Time of manipulation. One of these 2
      fhir:BiologicallyDerivedProduct.manipulation.timeDateTime [ dateTime ]
      fhir:BiologicallyDerivedProduct.manipulation.timePeriod [ Period ]
  ];
  fhir:BiologicallyDerivedProduct.storage [ # 0..* Product storage
    fhir:BiologicallyDerivedProduct.storage.description [ string ]; # 0..1 Description of storage
    fhir:BiologicallyDerivedProduct.storage.temperature [ decimal ]; # 0..1 Storage temperature
    fhir:BiologicallyDerivedProduct.storage.scale [ code ]; # 0..1 farenheit | celsius | kelvin
    fhir:BiologicallyDerivedProduct.storage.duration [ Period ]; # 0..1 Storage timeperiod
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
BiologicallyDerivedProduct | TU | | DomainResource | A material substance originating from a biological entity Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External ids for this item
productCategory | | 0..1 | code | organ | tissue | fluid | cells | biologicalAgent BiologicallyDerivedProductCategory (Required)
productCode | | 0..1 | CodeableConcept | What this biologically derived product is
status | | 0..1 | code | available | unavailable BiologicallyDerivedProductStatus (Required)
request | | 0..* | Reference(ServiceRequest) | Procedure request
quantity | | 0..1 | integer | The amount of this biologically derived product
parent | | 0..* | Reference(BiologicallyDerivedProduct) | BiologicallyDerivedProduct parent
collection | | 0..1 | BackboneElement | How this product was collected
collector | | 0..1 | Reference(Practitioner | PractitionerRole) | Individual performing collection
source | | 0..1 | Reference(Patient | Organization) | Who is product from
collected[x] | | 0..1 | | Time of product collection
collectedDateTime | | | dateTime
collectedPeriod | | | Period
processing | | 0..* | BackboneElement | Any processing of the product during collection
description | | 0..1 | string | Description of of processing
procedure | | 0..1 | CodeableConcept | Procesing code Procedure Codes (SNOMED CT) (Example)
additive | | 0..1 | Reference(Substance) | Substance added during processing
time[x] | | 0..1 | | Time of processing
timeDateTime | | | dateTime
timePeriod | | | Period
manipulation | | 0..1 | BackboneElement | Any manipulation of product post-collection
description | | 0..1 | string | Description of manipulation
time[x] | | 0..1 | | Time of manipulation
timeDateTime | | | dateTime
timePeriod | | | Period
storage | | 0..* | BackboneElement | Product storage
description | | 0..1 | string | Description of storage
temperature | | 0..1 | decimal | Storage temperature
scale | | 0..1 | code | farenheit | celsius | kelvin BiologicallyDerivedProductStorageScale (Required)
duration | | 0..1 | Period | Storage timeperiod
Documentation for this format

UML Diagram (Legend)

XML Template

```
<BiologicallyDerivedProduct xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
```

```
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this item --></identifier>
 <productCategory value="[code]"/><!-- 0..1 organ | tissue | fluid | cells | biologicalAgent -->
 <productCode><!-- 0..1 CodeableConcept What this biologically derived product is --></productCode>
 <status value="[code]"/><!-- 0..1 available | unavailable -->
 <request><!-- 0..* Reference(ServiceRequest) Procedure request --></request>
 <quantity value="[integer]"/><!-- 0..1 The amount of this biologically derived product -->
 <parent><!-- 0..* Reference(BiologicallyDerivedProduct) BiologicallyDerivedProduct parent --></parent>
 <collection>  <!-- 0..1 How this product was collected -->
  <collector><!-- 0..1 Reference(Practitioner|PractitionerRole) Individual performing collection --></collector>
  <source><!-- 0..1 Reference(Patient|Organization) Who is product from --></source>
  <collected[x]><!-- 0..1 dateTime|Period Time of product collection --></collected[x]>
 </collection>
 <processing>  <!-- 0..* Any processing of the product during collection -->
  <description value="[string]"/><!-- 0..1 Description of of processing -->
  <procedure><!-- 0..1 CodeableConcept Procesing code --></procedure>
  <additive><!-- 0..1 Reference(Substance) Substance added during processing --></additive>
  <time[x]><!-- 0..1 dateTime|Period Time of processing --></time[x]>
 </processing>
 <manipulation>  <!-- 0..1 Any manipulation of product post-collection -->
  <description value="[string]"/><!-- 0..1 Description of manipulation -->
  <time[x]><!-- 0..1 dateTime|Period Time of manipulation --></time[x]>
 </manipulation>
 <storage>  <!-- 0..* Product storage -->
  <description value="[string]"/><!-- 0..1 Description of storage -->
  <temperature value="[decimal]"/><!-- 0..1 Storage temperature -->
  <scale value="[code]"/><!-- 0..1 farenheit | celsius | kelvin -->
  <duration><!-- 0..1 Period Storage timeperiod --></duration>
 </storage>
</BiologicallyDerivedProduct>
```

JSON Template

```
{
  "resourceType" : "BiologicallyDerivedProduct",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this item
  "productCategory" : "<code>", // organ | tissue | fluid | cells | biologicalAgent
  "productCode" : { CodeableConcept }, // What this biologically derived product is
  "status" : "<code>", // available | unavailable
  "request" : [{ Reference(ServiceRequest) }], // Procedure request
  "quantity" : <integer>, // The amount of this biologically derived product
  "parent" : [{ Reference(BiologicallyDerivedProduct) }], // BiologicallyDerivedProduct parent
  "collection" : { // How this product was collected
    "collector" : { Reference(Practitioner|PractitionerRole) }, // Individual performing collection
    "source" : { Reference(Patient|Organization) }, // Who is product from
    // collected[x]: Time of product collection. One of these 2:
    "collectedDateTime" : "<dateTime>"
    "collectedPeriod" : { Period }
  },
  "processing" : [{ // Any processing of the product during collection
    "description" : "<string>", // Description of of processing
    "procedure" : { CodeableConcept }, // Procesing code
    "additive" : { Reference(Substance) }, // Substance added during processing
    // time[x]: Time of processing. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  }],
  "manipulation" : { // Any manipulation of product post-collection
    "description" : "<string>", // Description of manipulation
    // time[x]: Time of manipulation. One of these 2:
    "timeDateTime" : "<dateTime>"
    "timePeriod" : { Period }
  },
  "storage" : [{ // Product storage
    "description" : "<string>", // Description of storage
    "temperature" : <decimal>, // Storage temperature
    "scale" : "<code>", // farenheit | celsius | kelvin
    "duration" : { Period } // Storage timeperiod
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:BiologicallyDerivedProduct;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:BiologicallyDerivedProduct.identifier [ Identifier ], ... ; # 0..* External ids for this item
  fhir:BiologicallyDerivedProduct.productCategory [ code ]; # 0..1 organ | tissue | fluid | cells | biologicalAgent
  fhir:BiologicallyDerivedProduct.productCode [ CodeableConcept ]; # 0..1 What this biologically derived product is
  fhir:BiologicallyDerivedProduct.status [ code ]; # 0..1 available | unavailable
  fhir:BiologicallyDerivedProduct.request [ Reference(ServiceRequest) ], ... ; # 0..* Procedure request
  fhir:BiologicallyDerivedProduct.quantity [ integer ]; # 0..1 The amount of this biologically derived product
  fhir:BiologicallyDerivedProduct.parent [ Reference(BiologicallyDerivedProduct) ], ... ; # 0..*
    BiologicallyDerivedProduct parent
```

```
  fhir:BiologicallyDerivedProduct.collection [ # 0..1 How this product was collected
    fhir:BiologicallyDerivedProduct.collection.collector [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Individual
     performing collection
    fhir:BiologicallyDerivedProduct.collection.source [ Reference(Patient|Organization) ]; # 0..1 Who is product from
    # BiologicallyDerivedProduct.collection.collected[x] : 0..1 Time of product collection. One of these 2
      fhir:BiologicallyDerivedProduct.collection.collectedDateTime [ dateTime ]
      fhir:BiologicallyDerivedProduct.collection.collectedPeriod [ Period ]
  ];
  fhir:BiologicallyDerivedProduct.processing [ # 0..* Any processing of the product during collection
    fhir:BiologicallyDerivedProduct.processing.description [ string ]; # 0..1 Description of of processing
    fhir:BiologicallyDerivedProduct.processing.procedure [ CodeableConcept ]; # 0..1 Procesing code
    fhir:BiologicallyDerivedProduct.processing.additive [ Reference(Substance) ]; # 0..1 Substance added during
    processing
    # BiologicallyDerivedProduct.processing.time[x] : 0..1 Time of processing. One of these 2
      fhir:BiologicallyDerivedProduct.processing.timeDateTime [ dateTime ]
      fhir:BiologicallyDerivedProduct.processing.timePeriod [ Period ]
  ], ...;
  fhir:BiologicallyDerivedProduct.manipulation [ # 0..1 Any manipulation of product post-collection
    fhir:BiologicallyDerivedProduct.manipulation.description [ string ]; # 0..1 Description of manipulation
    # BiologicallyDerivedProduct.manipulation.time[x] : 0..1 Time of manipulation. One of these 2
      fhir:BiologicallyDerivedProduct.manipulation.timeDateTime [ dateTime ]
      fhir:BiologicallyDerivedProduct.manipulation.timePeriod [ Period ]
  ];
  fhir:BiologicallyDerivedProduct.storage [ # 0..* Product storage
    fhir:BiologicallyDerivedProduct.storage.description [ string ]; # 0..1 Description of storage
    fhir:BiologicallyDerivedProduct.storage.temperature [ decimal ]; # 0..1 Storage temperature
    fhir:BiologicallyDerivedProduct.storage.scale [ code ]; # 0..1 farenheit | celsius | kelvin
    fhir:BiologicallyDerivedProduct.storage.duration [ Period ]; # 0..1 Storage timeperiod
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

10.11.3.1 Terminology Bindings

Path | Definition | Type | Reference

BiologicallyDerivedProduct.productCategory | Biologically Derived Product Category. | Required | BiologicallyDerivedProductCategory

BiologicallyDerivedProduct.productCode | Biologically Derived Product Code. | Unknown | No details provided yet

BiologicallyDerivedProduct.status | Biologically Derived Product Status. | Required | BiologicallyDerivedProductStatus

BiologicallyDerivedProduct.processing.procedure | Biologically Derived Product Procedure. | Example | ProcedureCodes(SNOMEDCT)

BiologicallyDerivedProduct.storage.scale | BiologicallyDerived Product Storage Scale. | Required | BiologicallyDerivedProductStorageScale

10.11.4 Resource BiologicallyDerivedProduct - Examples

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct-examples.html

10.11.4 Resource BiologicallyDerivedProduct - Examples

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example of biologicallyderivedproduct | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

10.11.5 Resource BiologicallyDerivedProduct - Detailed Descriptions

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct-definitions.html

10.11.5 Resource BiologicallyDerivedProduct - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the BiologicallyDerivedProduct resource.

BiologicallyDerivedProduct

Element Id | BiologicallyDerivedProduct

Definition | A material substance originating from a biological entity intended to be transplanted or infused into another (possibly the same) biological entity.

Cardinality | 0..*

Type | DomainResource
Comments | Substances include, but are not limited to: whole blood, bone marrow, organs, and manipulated blood cells.
BiologicallyDerivedProduct.identifier
Element Id | BiologicallyDerivedProduct.identifier
Definition | This records identifiers associated with this biologically derived product instance that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
BiologicallyDerivedProduct.productCategory
Element Id | BiologicallyDerivedProduct.productCategory
Definition | Broad category of this product.
Cardinality | 0..1
Terminology Binding | BiologicallyDerivedProductCategory (Required)
Type | code
BiologicallyDerivedProduct.productCode
Element Id | BiologicallyDerivedProduct.productCode
Definition | A code that identifies the kind of this biologically derived product (SNOMED Ctcode).
Cardinality | 0..1
Terminology Binding | BiologicallyDerivedProductCode:
Type | CodeableConcept
BiologicallyDerivedProduct.status
Element Id | BiologicallyDerivedProduct.status
Definition | Whether the product is currently available.
Cardinality | 0..1
Terminology Binding | BiologicallyDerivedProductStatus (Required)
Type | code
BiologicallyDerivedProduct.request
Element Id | BiologicallyDerivedProduct.request
Definition | Procedure request to obtain this biologically derived product.
Cardinality | 0..*
Type | Reference(ServiceRequest)
BiologicallyDerivedProduct.quantity
Element Id | BiologicallyDerivedProduct.quantity
Definition | Number of discrete units within this product.
Cardinality | 0..1
Type | integer
BiologicallyDerivedProduct.parent
Element Id | BiologicallyDerivedProduct.parent
Definition | Parent product (if any).
Cardinality | 0..*
Type | Reference(BiologicallyDerivedProduct)
Comments | For products that have multiple collections. For example Peripheral Blood Stem Cells may be collected over several days from a single donor and the donation split into in multiple containers which must be linked to the parent donation.
BiologicallyDerivedProduct.collection
Element Id | BiologicallyDerivedProduct.collection
Definition | How this product was collected.
Cardinality | 0..1
BiologicallyDerivedProduct.collection.collector
Element Id | BiologicallyDerivedProduct.collection.collector
Definition | Healthcare professional who is performing the collection.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
BiologicallyDerivedProduct.collection.source
Element Id | BiologicallyDerivedProduct.collection.source
Definition | The patient or entity, such as a hospital or vendor in the case of a processed/manipulated/manufactured product, providing the product.
Cardinality | 0..1
Type | Reference(Patient | Organization)
BiologicallyDerivedProduct.collection.collected[x]
Element Id | BiologicallyDerivedProduct.collection.collected[x]
Definition | Time of product collection.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
BiologicallyDerivedProduct.processing
Element Id | BiologicallyDerivedProduct.processing
Definition | Any processing of the product during collection that does not change the fundamental nature of the product. For example adding anti-coagulants during the collection of Peripheral Blood Stem Cells.
Cardinality | 0..*
BiologicallyDerivedProduct.processing.description

Element Id | BiologicallyDerivedProduct.processing.description
Definition | Description of of processing.
Cardinality | 0..1
Type | string
BiologicallyDerivedProduct.processing.procedure
Element Id | BiologicallyDerivedProduct.processing.procedure
Definition | Procesing code.
Cardinality | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
BiologicallyDerivedProduct.processing.additive
Element Id | BiologicallyDerivedProduct.processing.additive
Definition | Substance added during processing.
Cardinality | 0..1
Type | Reference(Substance)
BiologicallyDerivedProduct.processing.time[x]
Element Id | BiologicallyDerivedProduct.processing.time[x]
Definition | Time of processing.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
BiologicallyDerivedProduct.manipulation
Element Id | BiologicallyDerivedProduct.manipulation
Definition | Any manipulation of product post-collection that is intended to alter the product. For example a buffy-coat enrichment or CD8 reduction of Peripheral Blood Stem Cells to make it more suitable for infusion.
Cardinality | 0..1
BiologicallyDerivedProduct.manipulation.description
Element Id | BiologicallyDerivedProduct.manipulation.description
Definition | Description of manipulation.
Cardinality | 0..1
Type | string
BiologicallyDerivedProduct.manipulation.time[x]
Element Id | BiologicallyDerivedProduct.manipulation.time[x]
Definition | Time of manipulation.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
BiologicallyDerivedProduct.storage
Element Id | BiologicallyDerivedProduct.storage
Definition | Product storage.
Cardinality | 0..*
BiologicallyDerivedProduct.storage.description
Element Id | BiologicallyDerivedProduct.storage.description
Definition | Description of storage.
Cardinality | 0..1
Type | string
BiologicallyDerivedProduct.storage.temperature
Element Id | BiologicallyDerivedProduct.storage.temperature
Definition | Storage temperature.
Cardinality | 0..1
Type | decimal
BiologicallyDerivedProduct.storage.scale
Element Id | BiologicallyDerivedProduct.storage.scale
Definition | Temperature scale used.
Cardinality | 0..1
Terminology Binding | BiologicallyDerivedProductStorageScale (Required)
Type | code
BiologicallyDerivedProduct.storage.duration
Element Id | BiologicallyDerivedProduct.storage.duration
Definition | Storage timeperiod.
Cardinality | 0..1
Type | Period

10.11.6 Resource BiologicallyDerivedProduct - Mappings

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct-mappings.html

10.11.6 Resource BiologicallyDerivedProduct - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the biologicallyderivedproduct resource (see Mappings to Other Standards for further information & status).

10.11.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

BiologicallyDerivedProduct
identifier | FiveWs.identifier

10.11.6.2 HL7 v2 Mapping (http://hl7.org/v2)

BiologicallyDerivedProduct
identifier | IAM-7
productCategory
productCode
status
request
quantity
parent
collection
collector
source
collected[x]
processing
description
procedure
additive
time[x]
manipulation
description
time[x]
storage
description
temperature
scale
duration

10.11.6.3 RIM Mapping (http://hl7.org/v3)

BiologicallyDerivedProduct | Material[classCode= Observation[classCode=OBS, moodCode=EVN]
identifier | id

10.11.7 Resource BiologicallyDerivedProduct - Extensions & Profiles

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct-profiles.html

10.11.7 Resource BiologicallyDerivedProduct - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the BiologicallyDerivedProduct resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

10.11.7.1 Profiles

No Profiles defined for this resource

10.11.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions

relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

10.11.7.3 Search Extensions

No Search Extensions defined for this resource

10.11.8 JSON Schema for BiologicallyDerivedProduct

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct.schema.json.html

10.11.8 JSON Schema for BiologicallyDerivedProduct

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for BiologicallyDerivedProduct, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/BiologicallyDerivedProduct",
  "$ref": "#/definitions/BiologicallyDerivedProduct",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "BiologicallyDerivedProduct": {
      "description": "A material substance originating from a biological entity intended to be transplanted or
    infused\ninto another (possibly the same) biological entity.",
      "properties": {
        "resourceType": {
          "description": "This is a BiologicallyDerivedProduct resource",
          "const": "BiologicallyDerivedProduct"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
```

```
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "This records identifiers associated with this biologically derived product instance that are
defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not
appropriate (e.g. in CDA documents, or in written / printed documentation).",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "productCategory": {
      "description": "Broad category of this product.",
      "enum": [
        "organ",
        "tissue",
        "fluid",
        "cells",
        "biologicalAgent"
      ]
    },
    "_productCategory": {
      "description": "Extensions for productCategory",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "productCode": {
      "description": "A code that identifies the kind of this biologically derived product (SNOMED Ctcode).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "Whether the product is currently available.",
      "enum": [
        "available",
        "unavailable"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "request": {
      "description": "Procedure request to obtain this biologically derived product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "quantity": {
      "description": "Number of discrete units within this product.",
      "$ref": "#/definitions/integer"
    },
    "_quantity": {
      "description": "Extensions for quantity",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "parent": {
      "description": "Parent product (if any).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "collection": {
      "description": "How this product was collected.",
      "$ref": "#/definitions/BiologicallyDerivedProduct_Collection"
    },
    "processing": {
      "description": "Any processing of the product during collection that does not change the fundamental nature of
 the product. For example adding anti-coagulants during the collection of Peripheral Blood Stem Cells.",
      "items": {
        "$ref": "#/definitions/BiologicallyDerivedProduct_Processing"
      },
      "type": "array"
    },
    "manipulation": {
      "description": "Any manipulation of product post-collection that is intended to alter the product.  For
```

```
example a buffy-coat enrichment or CD8 reduction of Peripheral Blood Stem Cells to make it more suitable for
infusion.",
      "$ref": "#/definitions/BiologicallyDerivedProduct_Manipulation"
    },
    "storage": {
      "description": "Product storage.",
      "items": {
        "$ref": "#/definitions/BiologicallyDerivedProduct_Storage"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"BiologicallyDerivedProduct_Collection": {
  "description": "A material substance originating from a biological entity intended to be transplanted or
infused\ninto another (possibly the same) biological entity.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "collector": {
      "description": "Healthcare professional who is performing the collection.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "source": {
      "description": "The patient or entity, such as a hospital or vendor in the case of a
processed/manipulated/manufactured product, providing the product.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "collectedDateTime": {
      "description": "Time of product collection.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_collectedDateTime": {
      "description": "Extensions for collectedDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "collectedPeriod": {
      "description": "Time of product collection.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
},
"BiologicallyDerivedProduct_Processing": {
  "description": "A material substance originating from a biological entity intended to be transplanted or
infused\ninto another (possibly the same) biological entity.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Description of of processing.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "procedure": {
      "description": "Procesing code.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "additive": {
      "description": "Substance added during processing.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "timeDateTime": {
      "description": "Time of processing.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timeDateTime": {
      "description": "Extensions for timeDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timePeriod": {
      "description": "Time of processing.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
},
"BiologicallyDerivedProduct_Manipulation": {
  "description": "A material substance originating from a biological entity intended to be transplanted or
infused\ninto another (possibly the same) biological entity.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Description of manipulation.",
      "$ref": "#/definitions/string"
```

```
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timeDateTime": {
      "description": "Time of manipulation.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timeDateTime": {
      "description": "Extensions for timeDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timePeriod": {
      "description": "Time of manipulation.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
},
"BiologicallyDerivedProduct_Storage": {
  "description": "A material substance originating from a biological entity intended to be transplanted or
infused\ninto another (possibly the same) biological entity.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Description of storage.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "temperature": {
      "description": "Storage temperature.",
      "$ref": "#/definitions/decimal"
    },
    "_temperature": {
      "description": "Extensions for temperature",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "scale": {
      "description": "Temperature scale used.",
      "enum": [
        "farenheit",
        "celsius",
        "kelvin"
      ]
    },
    "_scale": {
      "description": "Extensions for scale",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "duration": {
      "description": "Storage timeperiod.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
```

```
    }
  }
}
```

10.11.9 Resource BiologicallyDerivedProduct - Maturity Dependency View

URL: https://hl7.org/fhir/R4/biologicallyderivedproduct-dependencies.html

10.11.9 Resource BiologicallyDerivedProduct - Maturity Dependency View

Orders and Observations Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for BiologicallyDerivedProduct FMM level 0

Name | Card. | Type | Dependency Analysis
BiologicallyDerivedProduct | | DomainResource
identifier | 0..* | Identifier
productCategory | 0..1 | code
productCode | 0..1 | CodeableConcept
status | 0..1 | code
request | 0..* | Reference(ServiceRequest)
quantity | 0..1 | integer
parent | 0..* | Reference(BiologicallyDerivedProduct)
collection | 0..1 | BackboneElement
collector | 0..1 | Reference(Practitioner | PractitionerRole)
source | 0..1 | Reference(Patient | Organization)
collected[x] | 0..1
collectedDateTime | | dateTime
collectedPeriod | | Period
processing | 0..* | BackboneElement
description | 0..1 | string
procedure | 0..1 | CodeableConcept
additive | 0..1 | Reference(Substance)
time[x] | 0..1
timeDateTime | | dateTime
timePeriod | | Period
manipulation | 0..1 | BackboneElement
description | 0..1 | string
time[x] | 0..1
timeDateTime | | dateTime
timePeriod | | Period
storage | 0..* | BackboneElement
description | 0..1 | string
temperature | 0..1 | decimal
scale | 0..1 | code
duration | 0..1 | Period

10.11.10 Value Set Sc-valueset-product-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-product-status.html

10.11.10 Value Set Sc-valueset-product-status - Status Code Mapping

Orders and Observations Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

10.11.11 BiologicallyDerivedProductStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-product-status)

Mapping from http://hl7.org/fhir/ValueSet/product-status to http://hl7.org/fhir/ValueSet/resource-status

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Canonical Mapping for "Biologically Derived Product Status."

Source Code | Equivalence | Destination Code
available | equivalent | active
unavailable | equivalent | busy-unavailable